

Accounting Simulation and Capital Structure

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Part I

Literature Reviews

1 Introduction

This literature review is structured around three core areas: accounting simulation, corporate finance theory, and their integration. The first part of the review focuses on Accounting Simulation. This field provides the essential tools for creating dynamic and behaviorally-grounded forecasts of a firm’s financial statements, moving beyond simple statistical methods. We will survey various simulation techniques, with a particular focus on System Dynamics (SD), whose “stock-and-flow” structure naturally maps to accounting statements. However, the existing literature faces significant challenges, including the over-simplification of accounting rules, the difficulty of implementing accrual-basis accounting accurately, and a fundamental paradigm conflict with modern, principle-based standards like IFRS. This section will explore these methodological gaps and survey the leading-edge efforts to bridge them. The goal is to establish a clear picture of the current state and future direction of simulating a firm’s complex reporting system.

While accounting simulation provides the framework for modeling a firm’s reporting process, corporate finance theory offers the economic rationale that drives its strategic decisions. To create a realistic simulation, we must therefore model the firm’s behavior. A logical starting point is to formulate the firm’s central optimization problem.

A firm’s target can be either maximizing shareholder’s value or stakeholder’s value. Suppose a firm’s target at time t is to maximize its enterprises value, so we can write the problem as

$$V(x_t) = \max_{\{a_{t+i}\}} V_t^L$$

s.t.

$$V_t^L = S_t + D_t$$

where $V(x)$ is the firm manager’s value function subject to a vector of states x_t . V_t^L is the levered enterprises value, S_t is the market value of shareholder’s value and D_t is the market value of debt. $\{a_{t+i}\}$ is a series of future decisions a firm will make. Aside from maximizing enterprises value V_t^L , the manager can also choose maximizing shareholders value such that

$$V(x_t) = \max_{\{a_{t+i}\}} V_t^L - D_t$$

Different choices of optimal target does not change the core of the problem. We need to find a precise form of V_t^L , define elements in the decisions $\{a_{t+i}\}$ that a firm makes, and connect V_t^L and $\{a_{t+i}\}$ to accounting identities. However, conflicts exist when we involve the growth opportunity as a part of V_t^L as in Myers (1977). When managers are maximizing shareholder’s value, they will give up some investment opportunities while these opportunities are remained if they maximize enterprises value, and details of which will be discussed in later sections.

The ultimate goal of this literature review is to explore how these two fields—accounting simulation and corporate finance—can be effectively integrated. The third section, therefore, focuses on this intersection. A central theme is how corporate finance theories, such as the pecking order theory, trade-off theory, and the agency cost theory of free cash flow, can be operationalized into concrete decision rules within a simulation model. This allows a firm’s capital structure and payout policy to become endogenous outcomes of its cumulative investment and financing history, rather than exogenous assumptions. Furthermore, we will explore the reverse linkage: how accounting information itself can inform financial models. As demonstrated in works like Penman’s ‘Accounting for Risk,’ financial statement data are not merely records of past activity but also contain vital information about firm risk, which can be used to more

accurately estimate the risk premia and discount rates required by corporate finance models. This dual integration is crucial for creating a comprehensive simulation framework that is both compliant with accounting principles and behaviorally realistic in its financial strategy. In summary, this review will examine the interconnected branches of accounting simulation, dynamic corporate finance, and their synthesis, aiming to clarify the latest advancements and outline a path toward building more holistic and realistic corporate models.

2 Accounting Simulation

2.1 Simulation Methods in Accounting

Hertz (1964) has been credited with introducing simulation methods to finance. Since then the method has been applied broadly, notably in asset pricing (Boyle, 1977). At first, simulation was applied in corporate finance before being introduced to capital markets where it was broadly adopted to simulate stochastic assets (Boyle et al., 1997). Similarly, the merits of simulation for accounting were recognized early, e.g. to simulate budgeting spreadsheets. However, neither management (Labro, 2015; Grisar and Meyer, 2016) nor financial accounting appear to have embraced simulation methods despite some use cases (Balakrishnan and Penno, 2014). Barriers to usage include the lack of familiarity of many researchers and their readers with simulation methods (Labro, 2015) and the absence of universally agreed methodological standards (Lorscheid et al., 2012). Thus, there appears to be a discrepancy between simulation's adoption in accounting and finance research with a clear application in finance though not in accounting.

However, there are still some methods being used for accounting simulation worth discussion. An examination of the literature reveals several distinct simulation approaches that have been applied to various accounting domains, highlighting both established practices and areas for future development.

Foundational quantitative methods, such as Stochastic Simulation, have a long history in the discipline. Early applications demonstrated their utility for modeling uncertainty in tools like budgeting spreadsheets (Mattessich, 1961), and their potential continues to be recognized in areas like costing and financial analysis. Similarly, Discrete Event Simulation has proven effective for evaluating complex operational processes, such as the application of activity-based costing in manufacturing environments (Spedding and Sun, 1999). These methods are particularly adept at capturing and quantifying the effects of uncertainty on financial outcomes.

More recent and behavior-focused simulation approaches, particularly Agent-Based Models (ABM), are identified as having substantial, though largely unrealized, potential. ABM is well-suited for modeling the "complex interactions" and human behaviors inherent in organizational settings, making it a promising tool for research in management accounting and corporate governance. For instance, Ouksel et al. (1997) utilized an agent-based simulation to investigate how accounting information systems influence organizational learning. Despite this potential, the diffusion of ABM in accounting research has been limited, and it has yet to cross the "chasm" into mainstream use (Polhill et al., 2019).

Despite the limited overall adoption, a review of the literature reveals several pioneering studies where simulation was applied to specific, complex accounting problems. In the domain of costing, for example, research has been conducted to simulate entire costing systems to analyze the interaction of errors (Labro and Vanhoucke, 2007) and to model target costing processes (Kee and Matherly, 2013) of the American market.

In the field of auditing, simulation has been employed to assess the effectiveness of these procedures in detecting errors (Chen and Leitch, 1998). These examples demonstrate that while simulation is not a dominant methodology, it has served as a critical tool for providing unique insights into specific and challenging research questions across various sub-fields of accounting.

2.2 System Dynamics

A formal introduction of system dynamics can be referred to [System Dynamics Society](#). The field was invented in the late 1950s by Jay W. Forrester, a pioneer in engineering and computer design. His seminal book *Industrial Dynamics* in 1961 ([Forrester, 1997](#)) is still a significant statement of philosophy and methodology in the field. Within ten years of its publication, the span of applications grew from corporate and industrial problems to include the management of research and development, urban stagnation and decay, commodity cycles, and the dynamics of growth in a finite world. It is now applied in economics, public policy, environmental studies, defense, theory-building in social science, and other areas, as well as its home field, management. The name industrial dynamics no longer does justice to the breadth of the field, so it has become generalized to System Dynamics.

The core strength of System Dynamics lies in its ability to model the feedback loops that are central to value creation but are difficult to represent in traditional spreadsheets or stochastic simulations. For example, a capital investment decision (a cash outflow) increases fixed assets (a stock), which in turn enhances production capacity to generate more revenue (a flow), and the resulting profit replenishes cash (a stock) to fund future investments. SD models explicitly map these reinforcing and balancing loops, enabling a deeper understanding of the long-term, non-linear consequences of different policies.

Literatures in System Dynamics usually employs systems of nonlinear ordinary differential equations, solved numerically for simulation. [Sterman \(2000\)](#) defines the stock as an integral equation:

$$Stock(t) = Stock(t_0) + \int_{t_0}^t Inflow(s) - Outflow(s)ds$$

with differential form

$$\frac{dS}{dt} = Inflow(t) - Outflow(t)$$

Although it appears continuous, system dynamics models can be implemented using various simulation architectures. These architectures differ in their representation of time (continuous or discrete), state variables (continuous or discrete), and uncertainty (stochastic or deterministic). Ordinary differential equations, stochastic differential equations, discrete event simulation, agent-based models, and dynamic network models are common computational architectures, offering different choices across these dimensions. ([Sterman, 2018](#))

2.2.1 System Dynamics for Accounting and Management

Of the various simulation methods applied to accounting, System Dynamics (SD) is particularly well-suited to capture the complex, dynamic, and interconnected causal feedback relationships between a firm's operational activities and its financial performance. Traditional accounting reports, such as the balance sheet and income statement, inherently provide static snapshots of financial stocks at a point in time or summaries of flows over a period. This format makes it difficult to fully represent the continuous operational drivers and decision-making feedback that produce these outcomes.

To address this limitation, research has focused on operationalizing the accounting reporting system within SD models ([Kunc et al., 2023](#)). A key contribution of this work is demonstrating how standard accounting items can be mapped to the foundational structures of SD: balance sheet accounts like cash, inventory, and debt are effectively modeled as "stocks," as they represent accumulated value. Correspondingly, income statement and cash flow items such as revenue, costs, and payments are modeled as "flows," as they describe the activities that cause the stocks to change over time. This approach allows for the construction of fully integrated and dynamic financial statements where every change is driven by explicit operational decisions within the model.

Models of businesses and business processes are some of the earliest applications of system dynamics (Forrester, 1997, 1968) and continue to be very influential today (Lyneis, 1999; Sterman, 2000, 2014; Warren, 2015). While these models often include structures that are intended to represent financial accounting reports, the way that the accounting reporting process is represented is usually highly simplistic. In some cases, a simplistic treatment of accounting is all that is needed for a model’s purpose, but it is also likely that the lack of a standard structure (Warren, 2014) for operationalizing accounting inside of system dynamics models leads some modelers to over-simplify what can be an opaque and detail-oriented area of management.

Aside from the over-simplicity, another issue with current system dynamics models in accounting is the mis-identification of flow variables. Most system dynamics models in management accurately represent basic accounting definitions such as net income being the difference between revenue and expenses, but they almost always model the instantaneous value of these flows rather than considering the way that the process of generating accounting reports aggregates the flows over time and reports them during a single time step. The instantaneous value of the flow of revenue at the end of the year is not the same as the total revenue earned over the year, yet because it is so easy to create a flow of revenue, most system dynamics models rely on simple flows. It is rare to see modelers use the correct accounting reporting structure (Warren, 2005; Strohhecker and Größler, 2012; Cosenz, 2017), even when their express purpose is to represent accounting reports (Bianchi, 2002; Yamaguchi, 2003). Put in other words, they failed to consider the accrual basis of accounting. In almost all cases where authors create their own accounting structures, they treat the instantaneous values of revenue, expenses, and cash flow as if they were identical to the reported values of those concepts (Yamaguchi, 2013).

To take accrual basis into consideration, we need to replicate the income statement, balance sheet, statement of cash flows, and owners’ equity statement, which now lacks a standard structure. According to Pierson (2020), the most highly operational representations of accounting reporting inside of system dynamics models are the work of John Sterman and his co-authors. Correct accounting reporting structures are a common thread in the work of Sterman’s coauthors and students (Oliva et al., 2003; Pierson and Sterman, 2013). While Repenning and Sterman (1994) provide an excellent foundation for such a model, other authors often approach accounting by re-inventing the wheel rather than incorporating previously published models such as Repenning and Sterman (1994), even when accounting is an important focus of a system dynamics modeling project (Bianchi, 2002; Nguyen et al., 2017; Sana et al., 2018; Mohammadifardi et al., 2019).

The lack of standard is partially because of there are some incremental improvements that can be made in current studies. For example, Repenning and Sterman do not extend the logic of the reporting process to the cash flow statement, and so their model’s reporting of those results relies on instantaneous values of flows. They also fail to represent straight line depreciation, opting instead to model depreciation as a first order delay. These decisions are understandable given their purpose, but these omissions will stand out as incorrect to people familiar with accounting conventions.

To correct this issue, Pierson (2020) presents a comprehensive and standardized system dynamics model designed to correct a fundamental flaw in many business simulations: the misrepresentation of the accounting reporting process. The core innovation of the model lies in its detailed operationalization of accrual accounting principles. It explicitly models key accrual accounts—such as Accounts Receivable, Accounts Payable, and Accrued Wages—as stocks that integrate the difference between flows of value and flows of cash. For instance, Accounts Receivable accumulates the difference between the value of credit sales and the subsequent cash collections from customers, correctly separating the timing of revenue recognition from cash receipts. Similarly, the model implements a sophisticated structure for straight-line depreciation by separating the historical cost of an asset from its accumulated depreciation, ensuring that value is lost at a constant rate over its useful life, which is a more accurate representation than

the exponential delays often used in system dynamics.

Crucially, the model is the first of its kind in the literature to generate an indirect Statement of Cash Flows. This is a definitive feature of accrual-based reporting, as it reconciles the accrual-based Net Income figure with the actual Net Cash Flow from Operations. It achieves this by systematically adjusting for non-cash expenses, like depreciation, and accounting for the changes in all the accrual balance sheet accounts. By building these mechanisms, the model moves beyond simple accounting identities to replicate the actual reporting process.

2.2.2 System Dynamics in Other Fields

System Dynamics is not specifically designed for accounting but can be applied to a wide range of fields, including business, environmental, health, and public policy domains.

Business Domain System Dynamics is widely used in the business domain, covering multiple aspects from corporate performance to supply chain management. In business strategy, [Cosenz and Noto \(2018\)](#) utilized SD to experiment with new business venture strategies. Their simulation model could predict the cost of customer acquisition and customer lifetime value, providing quantitative support for startup decision-making. In project management, [Abbaspour and Dabirian \(2020\)](#) used SD models to assess different labor hiring policies for construction projects. The model revealed how policies based on time versus labor composition impacted project costs and progress, offering decision support for effective human resource allocation. In supply chain analysis, [Armendáriz et al. \(2016\)](#) applied SD to analyze food supply chains with the goal of achieving a sustainable agricultural system. The study identified a set of critical factors that have a significant impact on the sustainability and resilience of agricultural production and distribution systems.

Environmental Domain Environmental science is another core area for SD applications, particularly in issues like pollution, resource management, and climate change. To address carbon emissions, [Liu and Xiao \(2018\)](#) constructed an SD model to conduct a scenario analysis of China's carbon emission peak by 2025. Their simulation results forecasted the consumption of coal, oil, and natural gas, as well as the corresponding carbon emissions under various development paths. For the sustainable development of water resources, a study by [Jin et al. \(2019\)](#) focused on the synergistic relationship between the water, energy, and food systems in China. By creating a stock-and-flow model that included these three subsystems, they predicted the consumption of water, energy, and food through 2026 under several different scenarios. In agricultural ecosystems, [Dizyee et al. \(2019\)](#) studied the impact of policy interventions on the smallholder dairy value chain in Tanzania. The model simulations indicated that due to the higher cost of artificial insemination for local-breed cows, both short-term (1 year) and medium-term (5 year) policy interventions had a negative impact on farmer income. The model projected milk production, profit, and consumption rates up to the year 2034.

Health Care Domain In the health care sector, System Dynamics is used to understand disease transmission, the operation of health care systems, and the effects of public health policies. In disease prevention and control, [Jalali et al. \(2019\)](#) used SD to study the dynamics of obesity prevention intervention programs. Their research focused on analyzing the dynamic mechanisms of program adoption, implementation, and maintenance, combining strategy and organizational behavior theories to evaluate the effectiveness of these interventions. Regarding health care service quality, [Farid et al. \(2020\)](#) built a model to show the effect of nurse workload on their health and the quality of care. The simulation results concluded that long nurse shifts and workweeks doubled fatigue levels, increased burnout and absenteeism by several times, and led to a significant increase in medical errors.

Social and Public Policy The application of System Dynamics also extends to many social and urban problems, informing the creation of public policy. In the area of social justice, [Ford and White III \(2020\)](#) applied SD to prison rehabilitation programs. Their model was used for the prediction of the prison population and the number of prisoners returning to prison over 105 months, which helps in assessing the risks of programs like Social Impact Bonds. In urban planning and policy, [Bao and He \(2019\)](#) used CLDs (Causal Loop Diagrams) and SFDs (Stock and Flow Diagrams) to analyze urbanization development and water scarcity in a case study of the Beijing-Tianjin-Hebei region. Through scenario modeling, the study predicted the area's urbanization development and water supply status under different development patterns, providing insights for regional policy-making.

In summary, as a powerful modeling tool, System Dynamics has proven its cross-disciplinary potential in fields such as business, the environment, health care, and public policy. By revealing the internal structures and dynamic behaviors of complex systems, it provides crucial scientific support for developing more effective and forward-looking policies and strategies.

2.2.3 Challenges in System Dynamics Methodology

In this section, I mainly summarize the challenges of SD presented in [Sterman \(2018\)](#)'s article.

Over-reliance on Traditional Compartmental Models

- **The Root of the Problem:** In the early days of system dynamics, computing power was extremely limited. To simulate and analyze complex problems, researchers had to adopt high-level aggregation, grouping system elements into large "compartments" or "stocks." This was a necessary compromise given the technological constraints of the time.
- **The Current Fallacy:** Although today's computing power allows us to build and run highly disaggregated and more granular models, many practitioners still teach and use these highly aggregated compartmental models as the sole method.

Focus on Continuous-Time Models

- **The False Equation:** A very common and harmful misconception is that many people incorrectly equate "system dynamics" with "deterministic, continuous-time, compartmental models based on differential equations."
- **The Ignored Dimensions:** In reality, as a philosophy and methodology, the core of system dynamics lies in understanding how feedback, accumulation, and time delays drive system behavior. It can and should incorporate **Discrete Time/Events**, because many decisions and actions occur discretely, not in a smooth, continuous flow.

Neglect of Stochasticity

- **A Historical Irony:** The article points out that the myth that system dynamics models must be deterministic is particularly perplexing. The field's founder, Jay Forrester, created a stochastic model as his very first one and dedicated significant effort to studying the proper representation of "noise" and randomness in models.
- **The Problem in Practice:** Despite this history, stochasticity has been marginalized in subsequent practice and teaching. Most models are simplified into deterministic forms, making them unable to assess a system's robustness and resilience in the face of external shocks and internal fluctuations.

Misreading the Relationship Between "Continuity" and "Discreteness"

- **A Suggestion Mistaken for a Golden Rule:** Forrester did suggest that starting with a continuous flow model is a good way to learn and understand system dynamics, as its behavior is relatively smooth and easier to grasp. However, this was meant only as a "starting point," an exploratory step before introducing more complex elements like discreteness and noise.
- **Forrester's Warning:** He explicitly warned that one must not ignore the "micro-level discrete events" that constitute the continuous flow. These discrete events (like a single purchase, a single machine failure, a single policy enactment) are the richest sources of information for understanding and building macro-level flows. He never implied that discreteness was difficult to represent or should be excluded.

Lack of Model Testing and Validation

- **Prioritizing "Building" Over "Validating":** In many educational and practical projects, the vast majority of effort and focus is placed on the conceptualization and construction of the model. The crucial stage of rigorous testing and validation is severely neglected or even skipped entirely.
- **A Harmful Notion:** The article criticizes the harmful argument that rigorous scientific validation methods (such as sensitivity analysis, extreme conditions testing, and statistical validation) are only for academic work aimed at "getting published," and that "doers" solving practical problems do not need them.

Rigid Understanding of "Complete System Endogeneity"

- **The Core of the Endogeneity Principle:** System dynamics emphasizes an "endogenous" perspective—that a system's behavior is generated by its internal feedback structure, not as a simple result of external shocks. This is a profound and important insight.
- **The Danger of Dogmatism:** However, making this principle absolute and dogmatic is problematic. It leads some practitioners to incorrectly believe that no dynamic influences from outside the model boundary should be included.
- **The Complexity of the Real World:** The article brilliantly illustrates this issue with the example of global warming. Changes in atmospheric carbon concentration are largely endogenous to the feedback system of human activity (e.g., fossil fuel use). However, changes in solar cycles act as an external driving force (exogenous forcing) that also has a non-negligible impact on global temperature. A good model should be able to clearly define its boundary while acknowledging and properly incorporating important exogenous variables, rather than ignoring key external realities in pursuit of so-called "pure endogeneity."

2.3 Simulation in Business Applications

While academic literature has extensively explored the use of simulation in accounting, the application of these techniques in a commercial context is a less formally documented area, primarily detailed in practitioner-focused guides and software-centric literature. Two seminal works that bridge this gap are John Charnes's "Financial Modeling with Crystal Ball and Excel" ([Charnes, 2012](#)) and Michael Rees's "Business Risk and Simulation Modelling in Practice: Using Excel, VBA and @Risk" ([Rees, 2015](#)).

Charnes (2012)’s work provides a conceptual framework for transforming static spreadsheet models into dynamic risk models using Monte Carlo simulation within Oracle’s Crystal Ball software. Notably, Charnes demonstrates the use of simulation for complex tasks such as hedging with derivatives to comply with specific accounting regulations like Financial Accounting Standard (FAS), providing a direct link between advanced modeling and mandatory commercial reporting requirements.

Rees (2015)’s book serves as a comprehensive guide for professionals on the practical application of quantitative risk analysis in financial modeling. The book’s primary focus is not on simulating the accounting process itself, but rather on enhancing traditional financial models—such as those for project valuation, cash flow forecasting, and budgeting—with a robust framework for uncertainty. Rees demonstrates how to move beyond deterministic inputs by implementing Monte Carlo simulation techniques using common commercial tools like @RISK, Excel, and VBA.

2.4 Simulation under IFRS

2.4.1 Introduction to IFRS

International Financial Reporting Standards (IFRS) represent a globally recognized set of accounting principles developed and maintained by the International Accounting Standards Board (IASB). These standards aim to enhance transparency, comparability, and consistency in financial reporting across international borders, enabling investors, regulators, and other stakeholders to better understand and compare the financial performance of companies worldwide. IFRS covers a wide range of topics, including revenue recognition, asset valuation, and financial statement presentation, and has been designed to adapt to the complexities of modern global business environments.

IFRS has achieved widespread adoption globally, with approximately 167 jurisdictions committing to its use and 147 requiring it for all or most domestic publicly listed companies. This includes major economies such as the European Union member states, Canada, Australia, Brazil, South Africa, Korea, India (through its IFRS-converged Ind AS), and Hong Kong. In terms of the proportion based on the number of companies, the global total of listed companies stands at around 53,795. Of these, an estimated 70-80% (or about 37,000 to 43,000 companies) use IFRS, primarily in those 147 jurisdictions. In contrast, US GAAP, which is predominant in the United States, accounts for about 8-9% (roughly 4,300 to 4,600 companies), while other local standards make up the remaining 12-20% (around 6,000 to 10,000 companies), such as those in China and Japan. This distribution highlights IFRS as the dominant framework for the majority of listed companies by count, though US GAAP holds significant influence due to the size of the American market.

Therefore, combining accounting simulation with IFRS is particularly important.

2.4.2 Literatures on Accounting Simulation and IFRS

Despite the clear benefits of combining IFRS with accounting simulation, a review of existing academic literature reveals a notable research gap. There is a limited body of scholarly work that directly employs accounting simulation as a core methodology to investigate the application of specific IFRS standards. However, foundational work exists in the realm of the preceding International Accounting Standards (IAS).

An example is Amen (2007)’s study, which utilized Monte-Carlo simulation to compare the long-term accounting effects of different treatments for actuarial gains and losses under IAS 19, Employee Benefits. The research demonstrated that due to complexity and probabilistic elements, traditional analytical methods often fall short, whereas simulation can systematically reveal the long-term financial statement impacts of different accounting choices. This study

provides a vital precedent and methodological framework for applying simulation to analyze complex IFRS standards.

In fact, academia has begun to explore the integration of IFRS with accounting simulations, yielding some valuable research findings. For instance, Müller (2020) investigates the impact of transitioning from IAS 39 to IFRS 9 on hedge accounting. Her simulation-based study quantitatively illustrates how this shift affects portfolio earnings, demonstrating that earnings are impacted differently under the two standards, particularly in response to deviations between foreign and domestic interest rates and changes in market volatility. A notable finding from her simulation is the counter-intuitive conclusion that a partially ineffective hedging relationship does not necessarily lead to lower earnings compared to a fully effective one.

In a similar vein, Seitz et al. (2018) utilizes a simulation-based approach in "Understanding Loan Loss Reserves under IFRS 9" to examine the forward-looking nature of the new Expected Credit Loss (ECL) model compared to the incurred loss model of IAS 39. Their simulation, calibrated with historical European banking data, reveals that while IFRS 9's loan loss reserves tend to exceed those under IAS 39 during crises, they are not consistently higher in normal economic conditions and exhibit significant volatility. The study also highlights the high sensitivity of the ECL calculation to the specific model chosen for estimating the probability of default, ultimately suggesting that the new standard may not fully achieve the desired goal of increasing countercyclical loan loss provisioning.

However, these two studies collectively reveal a characteristic of the current literature: they are not designed to simulate a complete balance sheet or income statement. Instead, they use simulation as a highly focused tool to address complex issues arising from specific IFRS standards.

2.4.3 Existing Issue with Simulation under IFRS

With the evolution of global accounting standards, particularly the widespread adoption of the principles-based International Financial Reporting Standards (IFRS), the application of traditional System Dynamics in the field of accounting simulation has encountered a fundamental conceptual and methodological challenge. Unlike past rule-based standards that focused on historical cost, the modern IFRS framework—encompassing standards like **IFRS 9** "Financial Instruments", **IFRS 15** "Revenue from Contracts with Customers", **IFRS 16** "Leases", and **IFRS 17** "Insurance Contracts"—introduces a significant number of principles based on management judgment, forward-looking information, and complex estimations. These principles require companies to reflect economic substance and future expectations in their financial reports, rather than merely recording transactions that have already occurred. Previous System Dynamics accounting models, while capable of handling fundamental accounting identities, were built in a different era and employ simplifications that make it difficult for them to fully capture the depth and complexity inherent in modern IFRS.

There is a profound structural mismatch between traditional System Dynamics and modern IFRS. This mismatch is not merely a technical implementation issue but a fundamental paradigm conflict. System Dynamics is a top-down systems that aims to understand the overall dynamic behavior of a system by identifying key feedback structures and cumulative effects. It tends to abstract away from details to capture the macro-structures that govern the system's behavior. In contrast, the foundation of modern IFRS is bottom-up and entity-specific (Weaver and Woods, 2015). It emphasizes the application of professional judgment in specific contexts and requires reasonable expectations about future events. This philosophy demands that models delve into the micro-level, focusing on heterogeneity, specific contract terms, and management's subjective assessments (Perera and Chand, 2015).

The Gap Between Qualitative Concepts and Quantitative Flows IFRS standards are replete with qualitative, principle-based concepts that are central to accounting treatment

but are difficult to directly quantify into the flow equations of an SD model. For example:

- **Materiality:** This is a cornerstone of all financial reporting. Determining whether information is material depends on whether it could influence the decisions of financial statement users. This is a relative and context-dependent judgment that cannot be expressed by a fixed mathematical formula. (Collings, 2013)
- **Significant Judgment:** In standards such as **IFRS 15** and **IFRS 16**, entities are required to disclose the significant judgments made in applying their accounting policies. These judgments, such as identifying performance obligations or determining the lease term, are based on qualitative logic rather than continuous change. (Deloitte, 2025; HouseofControl, 2025)
- **Reasonable Certainty:** When assessing whether to exercise an extension or termination option in a lease contract, **IFRS 16** requires the lessee to assess whether it is "reasonably certain" to exercise that option. This involves forecasting future economic incentives and corporate strategy, representing a subjective probability assessment rather than a smooth function. (HouseofControl, 2025)
- **Control:** The core principle for revenue recognition in **IFRS 15** is the transfer of "control." Determining whether control has been transferred requires evaluating a series of indicators, but ultimately relies on a comprehensive judgment of the contract's substance, not merely physical delivery. (Deloitte, 2025)

Fitting these qualitative concepts—which are highly dependent on interpretation, experience, and context—into SD flow equations driven by deterministic or simple stochastic processes is difficult.

The Conflict Between Forward-Looking Information and Endogenous Feedback A core idea of system dynamics is that a system's behavior is endogenously determined by its internal feedback structure (Scholl, 2001). The model generates future dynamics by simulating the causal relationships and feedback loops among its internal variables. However, modern IFRS, particularly **IFRS 9** and **IFRS 17**, mandates that companies incorporate exogenous, forward-looking information into their accounting models. For instance, the Expected Credit Loss (ECL) model in **IFRS 9** must reflect reasonable and supportable forecasts of future macroeconomic conditions, such as GDP growth rates and unemployment rates (KPMG, 2025). Similarly, when measuring insurance liabilities under **IFRS 17**, it is necessary to use discount rates and risk scenarios that reflect current market conditions (Moody's, 2020).

This forward-looking information does not originate from the feedback loops within the company's internal operating system but comes from external economic forecasts. It is very difficult for a pure SD model to endogenously generate multiple, reasonable, and consistent macroeconomic scenarios and seamlessly link them to the company's internal operational dynamics. The model must either rely on external inputs, thereby undermining its core advantage of endogeneity, or attempt to endogenously model the entire macro-economy, which vastly increases the model's complexity and uncertainty. Therefore, a new modeling approach is urgently needed. For example, we can combine corporate finance and system dynamics (Khan et al., 2021).

The Conflict Between Macro-Aggregation and Micro-Level Detail The strength of a system dynamics model lies in its ability to capture the overall behavior of a system at a macro and strategic level. It simplifies complexity through aggregation to reveal key feedback structures (Hettinger et al., 2015). However, many IFRS requirements demand accounting treatment at a very fine level of granularity. For example, **IFRS 17** requires insurance companies to aggregate and measure insurance contracts in "cohorts," with each cohort requiring a

separate calculation of its Contractual Service Margin (CSM) (Yousuf et al., 2021). Similarly, the Expected Credit Loss model under **IFRS 9** requires assessments to be made for individual financial instruments or for portfolios of assets that share similar credit risk characteristics.

2.4.4 Detailed Discussion on IFRS 15, 16 and 17

IFRS 15 Revenue from Contracts with Customers IFRS 15 introduced a unified five-step model intended to more accurately reflect the pattern of an entity's transfer of goods or services to customers. However, this seemingly structured model is filled with "soft" variables at every step, each requiring qualitative judgment and posing significant challenges for quantitative modeling. (Deloitte, 2025)

- **Challenge 1: Identifying Performance Obligations.** This is the first and most challenging step of the model. An entity needs to determine whether the promised goods or services in a contract are "distinct." To be distinct, a promise must meet two criteria: can the customer benefit from the good or service on its own? Is the promise separately identifiable in the context of the contract? The latter is particularly difficult, as it requires assessing whether the nature of the entity's promise is to deliver individual items or a combined output. For example, in the software industry, a single contract might include a software license, installation services, and subsequent technical support. Whether these services constitute separate performance obligations depends on whether they are highly interrelated or interdependent. This judgment is based on an interpretation of the contract's substance, not on simple mathematical calculations, and SD models lack the intrinsic mechanisms to handle such semantic and commercial logic.
- **Challenge 2: Allocating the Transaction Price.** If a contract contains multiple performance obligations, the entity must allocate the total transaction price to each obligation based on its relative standalone selling price. However, standalone selling prices are often not readily available, especially for customized services, requiring the entity to use estimates and judgment to determine them. This estimation process itself introduces uncertainty and subjectivity.
- **Challenge 3: Timing of Recognition.** An entity must determine whether revenue should be recognized "over time" or "at a point in time." This depends on whether the process of transferring control meets specific criteria, such as whether the customer simultaneously receives and consumes the economic benefits as the entity performs. This determination also requires a deep analysis of the nature of the performance activities, rather than being a simple linear process.

IFRS 16 Leases: Reflecting Strategic Intent IFRS 16 fundamentally changed lease accounting by requiring lessees to recognize most leases on the balance sheet, thereby recognizing a right-of-use asset and a lease liability. At the core of this change is a reinterpretation of lease contracts, which includes judgments about a company's future strategic intentions. (House of Control, 2025)

- **Challenge 1: Determining the Lease Term.** The basis for calculating the lease liability is the lease term. According to the standard, the lease term includes not only the non-cancellable period but also periods covered by an option to extend the lease if the lessee is "reasonably certain" to exercise that option, or an option to terminate the lease if the lessee is "reasonably certain" not to exercise it. Assessing "reasonable certainty" is a highly subjective process that requires considering the economic incentives associated with exercising the option, such as significant leasehold improvements, the strategic importance of the asset, and relocation costs. This essentially requires the model to be able to capture

management's forecasts of future operational strategy, a task for which traditional SD models are ill-suited.

- **Challenge 2: Determining the Discount Rate.** A discount rate is needed to calculate the present value of the lease liability. If the interest rate implicit in the lease cannot be readily determined, the lessee should use its "incremental borrowing rate." This is the rate of interest that a lessee would have to pay to borrow over a similar term, and with a similar security, the funds necessary to obtain an asset of a similar value to the right-of-use asset in a similar economic environment. This rate is specific to the lessee, the leased asset, and the lease term, and is therefore highly heterogeneous. In an SD model that includes multiple companies, dynamically determining and updating an appropriate discount rate for each lease is an extremely complex task.
- **Challenge 3: Separating Components.** Many contracts (such as service contracts) may contain "embedded leases," or a single lease contract may include both lease components (e.g., the right to use equipment) and non-lease components (e.g., maintenance services). IFRS 16 requires entities to identify and separate these components and account for them differently. This process also requires judgment and estimation.

IFRS 17 Insurance Contracts: Modeling the Core of the Business Model IFRS 17 is widely recognized as one of the most complex accounting standards ever issued, completely reshaping the measurement and presentation of insurance contracts. Its core concept—the Contractual Service Margin (CSM)—imposes unprecedented demands on modeling ([Moody's, 2020](#)).

- **The Complexity of the Contractual Service Margin (CSM):** The CSM represents the unearned expected profit in an insurance contract. Its initial recognition and subsequent amortization is a complex process that involves: grouping contracts by year of issue and profitability (i.e., "cohorts"); estimating future contract-related cash flows (premiums, claims, expenses, etc.); discounting these cash flows using rates that reflect current market interest rates; and making a risk adjustment for non-financial risks (such as mortality and catastrophe risk).
- **The Need for Stochastic Simulation:** Because many insurance products (especially life insurance and annuities) contain complex payout options and guarantees (such as minimum guaranteed interest rates), their valuation cannot be accomplished through simple deterministic cash flow discounting. The principle-based nature of the standard effectively requires insurance companies to adopt market-consistent valuation methods.

Looking across IFRS 15, 16, and 17, a unifying theme emerges: modern IFRS increasingly demands that accounting systems do more than just record historical transactions; they must model and reflect the company's own business model and strategic intent. IFRS 15 requires a deconstruction of the company's value proposition, IFRS 16 requires a reflection of its asset utilization strategy, and IFRS 17 requires a full life-cycle simulation of the core profitability model of the insurance business.

2.5 Further Challenges and Solutions

In above sections, I discussed the over-simplicity, failure to consider accrual basis, lack of heterogeneity, and falling behind current popular accounting standards as main issues of existing accounting simulation literatures. I also discussed the challenges in combining IFRS and system dynamics in accounting. In this section, I will first point out further potential challenges in system dynamics in accounting simulation, and then I will give solutions to these challenges.

2.5.1 Challenges from Uncertainty

A traditional system dynamics model is essentially a system of coupled, non-linear, first-order ordinary differential equations (or difference equations). This mathematical structure is deterministic: given a set of initial conditions and parameters, the model will generate a single, unique output trajectory. This creates a fundamental conflict with the inherent stochastic nature of financial markets and economic variables. As a result, the model provides a "point estimate" forecast, which can only tell us the most likely outcome but fails to provide information about the confidence interval, variance, or probability distribution of potential outcomes. For risk management or accounting applications that require the evaluation of extreme events (tail risks), such as stress testing, this single-path prediction is far from sufficient.

Uncertainty within a model can be divided into two categories. The first is parameter uncertainty, which arises from the difficulty of accurately estimating model parameters from noisy historical data or expert judgment. For example, in an accounting model, parameters such as the elasticity of sales to marketing expenditure or the average accounts receivable collection period are subject to estimation errors. The second category is structural uncertainty (or process noise), which refers to the inherent randomness in the system's processes themselves, such as random fluctuations in sales volume, unexpected changes in raw material costs, or macroeconomic shocks. Traditional system dynamics models have difficulty rigorously representing either of these two types of uncertainty.

2.5.2 Luca's Critique (Structural Change)

The critique proposed by economist [Lucas Jr \(1976\)](#) (the Lucas Critique) has had a profound impact on macroeconomic modeling. Its core argument is that it is naive to predict the effects of economic policy changes based solely on relationships observed in historical data, because rational agents will adjust their behavior in response to the new policy, thereby changing the very "structural" relationships upon which the model depends. For example, historical data may show a negative correlation between inflation and unemployment (the Phillips Curve). However, if the government attempts to permanently lower unemployment by increasing inflation, firms and workers will adjust their inflation expectations. This, in turn, will alter their hiring and wage-setting decisions, ultimately causing the original negative correlation to break down.

The Lucas Critique was aimed at the large-scale macroeconometric models popular at the time, and standard system dynamics business models are structurally very similar to them. In a system dynamics model, the equations that define managerial decisions (such as production decisions, pricing strategies, and inventory management) are essentially "decision rules." These rules are typically set as fixed functions of other variables within the model. The Lucas Critique points out that these decision rules are not the "deep parameters" of an economic system—such as preferences, technology, and resource constraints—but are rather the optimal responses of economic agents within a specific policy environment. When the policy environment changes (for instance, due to new tax policies, a shift in the competitive landscape, or new accounting standards), these decision rules will themselves change, rendering the model's fixed structure invalid.

This is critically important in the context of accounting simulation. For example, a model might assume a fixed inventory management strategy. If a new accounting standard changes the way inventory is valued and expensed (e.g., a shift from the Last-In, First-Out method to the First-In, First-Out method), a rational manager would adjust their inventory strategy to optimize reported profits and cash flow. A system dynamics model with a fixed equation for inventory management would fail to capture this adaptive behavior and, as a result, would produce misleading predictions.

2.5.3 Bootstrapping

To address the problem of parameter uncertainty, two powerful statistical techniques can be employed. The first is bootstrapping. Bootstrapping is a non-parametric resampling method. It works by repeatedly sampling with replacement from the original sample to generate a large number of new datasets. By estimating the model parameters for each new dataset, it constructs an empirical distribution of the parameters. The greatest advantage of this method is that it does not make strong assumptions about the data's distribution (such as normality), making it very suitable for system dynamics models, where the output data often exhibits autocorrelation.

2.5.4 Bayesian Estimation

The second method is Bayesian inference. Unlike the traditional frequentist approach (which produces point estimates), the Bayesian paradigm treats the model parameters themselves as random variables. The modeler begins by setting a "prior distribution" for the parameters, which is then updated using observed data via Bayes' theorem to arrive at a "posterior distribution." This posterior distribution provides a complete description of the probability of all possible values the parameters could take, given the data. By propagating this parameter distribution through the model, one can ultimately obtain a full predictive distribution for the model's output, rather than just a single point estimate.

In fact, Bayesian methods have already been applied in both accounting and system dynamics, demonstrating a significant step forward in handling model uncertainty. Instead of yielding a single point estimate for discretionary accruals, [Breuer and Schütt \(2023\)](#) uses a Bayesian framework to generate a full posterior distribution, thereby allowing auditors and researchers to assess the statistical probability of earnings management rather than relying on a simple, and potentially misleading, numerical value.

Within the field of system dynamics, [Andrade and Duggan \(2021\)](#) address the critical challenge of model calibration. Traditional calibration relies on optimization algorithms to find a single set of "best-fit" parameters, a process that risks getting trapped in local optima and fails to quantify parameter uncertainty. This paper replaces that method with a full Bayesian framework that employs a specific, highly efficient algorithm: Hamiltonian Monte Carlo (HMC). HMC uses principles from physics to navigate the model's complex, high-dimensional parameter space, ensuring a more thorough exploration than older sampling methods. The output is not a single parameter set, but the full joint posterior distribution of all parameters.

2.5.5 State Space Model

As previously mentioned, Accurately simulating a firm's financial statements presents a series of challenges. The simulation must consider:

- Accounting for the influence of external macroeconomic factors while simultaneously capturing the unique characteristics and heterogeneity of the individual firm.
- Large number of variables across the three financial statements, which creates a high-dimensional system where accounts are intricately linked through accounting principles like the clean surplus identity.
- Distinguishing between the rigid, deterministic rules of the IFRS framework and the discretionary, often stochastic, decision-making of corporate management.
- Moving the model beyond deterministic forecasts and formally capture the inherent uncertainty in business operations and financial reporting, which is best achieved through a Bayesian estimation.

The State-Space Model (SSM) framework offers a unified and elegant solution to all these challenges. Its flexible state vector is designed to incorporate external variables to model macroeconomic influences, as well as latent states to represent firm-specific heterogeneity. The SSM's structure is fundamentally built for high-dimensional time-series analysis, allowing it to coherently manage the complex interdependencies between all financial accounts through its state transition function. This function can use deterministic equations to enforce clean surplus accounting identities while employing stochastic equations to model managerial behaviors like investment and dividend policies. By integrating a Bayesian approach for estimation, the SSM directly confronts the issue of uncertainty. Instead of producing a single point forecast, it yields full posterior probability distributions for all parameters and future states, thus providing a comprehensive view of potential outcomes.

2.5.6 Simple Example of SSM

Consider a simple state space model below, let $x_t = \hat{f}(x_t, x_{t-1}, u_t | \Theta) + w_t$ which implicitly equals $x_t = f(x_{t-1}, u_t | \Theta) + w_t$:

$$\begin{aligned} x_t &= f(x_{t-1}, u_t | \Theta) + w_t \\ y_t &= g(x_t, u_t | \Theta) + v_t \end{aligned}$$

with initial condition

$$x_1 \sim p(x_1 | \Theta)$$

where

- x_t is unobserved latent states
- y_t is observed data that is subject to errors
- u_t is the control variables
- w_t is the process uncertainty with respect to latent states
- v_t is the measurement error and Θ is model parameters that is subject to uncertainty.

Latent states x_t can be further decomposed into three components:

$$x_t = \begin{bmatrix} x_t^{acct} \\ x_t^{macro} \\ x_t^{firm} \end{bmatrix}$$

where x_t^{acct} is the firm's real information on financial statements that are subject to no measurement error, x_t^{macro} reflects macro conditions and x_t^{firm} reflects a firm's heterogeneity, including the firm's profitability, productivity, etc. y_t can be the observed macro economy conditions such as GDP and inflation, as well as reported numbers on the firm's seasonally financial statements. x_t^{acct} can be further decomposed into

$$x_t^{acct} = \begin{bmatrix} x_t^{flow} \\ x_t^{stock} \end{bmatrix}$$

and

$$\begin{aligned} x_{t+1}^{stock} &= x_t^{stock} + x_{t+1}^{flow} \\ x_{t+1}^{flow} &= h(x_{t+1}^{stock}, x_t^{stock}, x_{t+1}^{flow}, \epsilon_{t+1} | \Theta) \end{aligned}$$

Various types of errors in accounting simulation are discussed in [Leitner and Wall \(2015\)](#), which can be mapped to the components of a state space model.

A Measurement Error, which the paper describes as inaccuracies in quantifying costs or drivers, corresponds directly to the observation error term v_t in an SSM. When this inaccuracy is not random, such as in the case of **Biased Raw Data**, it is modeled as a biased observation error where the expected value of the noise is non-zero $\mathbb{E}(v_t) \neq 0$.

The general concept of **Noise** or random errors canceling each other out can be attributed to the two primary stochastic components of an SSM: the process noise w_t , which affects the system’s evolution, and the observation error v_t .

Aggregation Error, or information loss from summarizing data, is not treated as a random noise but rather as a structural of $g(\cdot|\Theta)$, which dictates how underlying states x_t are combined into observable y_t . It leads to information lost by design but is not a noise or a wrong in specifying DGP. **A specification error**, on the other hand, means that the model’s assumed relationship between the states and observations $g(\cdot|\Theta)$ is fundamentally wrong.

Furthermore, the State-Space Model provides a robust framework for addressing the Lucas Critique. By incorporating regime-switching mechanisms, as detailed in works like [Kim and Nelson \(2017\)](#), the model can explicitly account for structural breaks and changes in policy environments. Instead of relying on fixed decision rules, a regime-switching SSM allows the parameters of the system—and thus the agents’ behaviors—to vary across different states or regimes.

How to apply Bayesian method to estimate parameters in SSM is listed in the appendix. Details on how SSM for the accounting simulation purpose is built, solved when circularity is considered, and how parameters are estimated will be presented in the research proposal.

3 Literatures with Complete Accounting Simulation Process

This section shifts from a general survey of simulation literature to a focused analysis of specific academic papers and textbooks that present a comprehensive process for accounting simulation. Unlike the preceding review, the emphasis in this section is on the mechanics of constructing a dynamic accounting simulator. This includes the individual dynamics of each account, the intricate accounting linkages between them, the identification of accounts that require significant managerial judgment and estimation, and the fundamental constraints that govern the model’s integrity. These works provide both a theoretical foundation and a practical guide for building a dynamic simulation model that is consistent with accounting principles and reflective of corporate operational realities.

3.1 Articles

3.1.1 Shahnazarian (2004): A baseline simulator

A foundational study in this area is presented by [Shahnazarian \(2004\)](#). This work is notable for developing a complete stock-flow consistent (SFC) model that dynamically links a company’s financial statements. A key contribution of their approach is the meticulous specification of the constraints for each variable, ensuring that all accounting identities have economic meaning throughout the simulation. The model provides a clear framework to distinguish between the accounting identities and the assumed and estimated variables. A more detailed examination of this model’s structure and its implications for dynamic financial analysis is provided in [Section 7](#).

3.1.2 Vélez-Pareja (2010, 2011): No-plugs, no-circularity dynamic forecasting

Vélez-Pareja (2010); Vélez-Pareja (2011) propose a rigorous forecasting engine that eliminates balance-sheet plugs and circular references by: (i) deriving every account from explicit operating assumptions and managerial policies; (ii) computing financing needs through a cash budget that targets a minimum cash balance; and (iii) using an end-of-period convention so that interest in period t depends on balances at the end of $t - 1$. The approach improves transparency, error-detection, and economic interpretability.

Dynamics (drivers and account roll-forwards). Sales are decomposed into price and volume, with nominal rates obtained via a Fisher relation; operating working capital scales with activity via policy ratios; fixed assets evolve through the standard accounting identity with a capex policy that separates maintenance and growth. Illustrative equations include:

$$\begin{aligned}
\text{Nominal Rate}_t &= (1 + \text{Real Rate}_t)(1 + \text{Inflation}_t) - 1, \\
\text{Sales}_t &= P_t Q_t, \\
\text{Ending Inv.}_t(\text{units}) &= \text{Sales}_{t+1}(\text{units}) \times \text{InvPolicy}\%, \\
\text{Purchases}_t(\text{units}) &= \text{Sales}_t(\text{units}) + \text{Ending Inv.}_t(\text{units}) - \text{Beginning Inv.}_t(\text{units}), \\
\text{COGS}_t(\$) &= \text{Beginning Inv.}_t(\$) + \text{Purchases}_t(\$) - \text{Ending Inv.}_t(\$), \\
AR_t &= \text{Sales}_t \times \text{AR Policy}\%, \\
AP_t &= \text{Purchases}_t(\$) \times \text{AP Policy}\%, \\
NFA_t &= NFA_{t-1} + \text{Capex}_t - \text{Depreciation}_t, \\
\text{Capex}_t &= \text{Depreciation}_t + (NFA_{t-1} \times \text{Real Growth}_{t+1}), \\
\text{Interest Expense}_t &= \text{Debt Balance}_{t-1} \times \text{Cost of Debt}_t, \\
\text{Interest Income}_t &= \text{ST Investment Balance}_{t-1} \times \text{Return on ST Inv.}_t, \\
\text{Dividends Paid}_t &= \text{Net Income}_{t-1} \times \text{Payout Ratio}\%, \\
CRE_t &= CRE_{t-1} + \text{Net Income}_t - \text{Dividends}_t.
\end{aligned}$$

Financing logic via the cash budget (no plugs). Cash is not a balancing item. Instead, a minimum cash target is enforced and any shortfall or surplus triggers financing or investment actions consistent with matching maturities and a financing-first rule:

$$\begin{aligned}
\text{ST Loan}_t &= \max\left(0, \text{MinCashTarget}_t - (\text{Cash}_{t-1} + \text{Operating NCB}_t)\right), \\
\text{LT Deficit}_t &= \max\left(0, \text{Capex}_t - \text{Cash after Op.\&ST Fin.}_t\right), \\
\text{LT Loan}_t &= \text{LT Deficit}_t \times \text{Debt Financing}\%, \\
\text{New Equity}_t &= \text{LT Deficit}_t \times (1 - \text{Debt Financing}\%), \\
\text{ST Investment}_t &= \begin{cases} 0, & \text{if New Debt or Equity}_t > 0, \\ \max(0, \text{Final Cash Available}_t - \text{MinCashTarget}_t), & \text{otherwise.} \end{cases}
\end{aligned}$$

This structure ensures that companies do not simultaneously borrow and invest excess cash, and that short-term needs are covered with short-term debt while long-term investment gaps are covered with long-term sources and equity.

3.1.3 Pierson (2020): Operationalizing accrual accounting in SD

Pierson (2020) develops a standardized module that embeds accrual accounting faithfully inside a system dynamics architecture. The model separates value flows from cash flows by representing accrual accounts as stocks, implements straight-line depreciation by splitting historical

cost and accumulated depreciation, and generates an indirect Statement of Cash Flows by reconciling reported net income with changes in accrual stocks. Financing decisions (debt, equity, dividends) are treated as exogenous inputs, making the accounting engine portable across business contexts.

Dynamics (accrual stocks, reporting, and roll-forwards). Accrual stocks integrate the differences between value creation/consumption and their associated cash settlements, while reporting accumulators translate instantaneous flows into period totals:

$$\begin{aligned}
\frac{d AR_t}{dt} &= \underbrace{\text{Credit Sales}_t}_{\text{value}} - \underbrace{\text{Cash Collections}_t}_{\text{cash}}, \\
\text{Cash Collections}_t &= \frac{AR_t}{\text{Collection Period}}, \\
\frac{d AP_t}{dt} &= \underbrace{\text{Credit Purchases}_t}_{\text{value}} - \underbrace{\text{Supplier Payments}_t}_{\text{cash}}, \\
\text{Supplier Payments}_t &= \frac{AP_t}{\text{Payables Days}}, \\
\frac{d AW_t}{dt} &= \underbrace{\text{Wages Earned}_t}_{\text{value}} - \underbrace{\text{Wages Paid}_t}_{\text{cash}}, \\
\text{Wages Paid}_t &= \frac{AW_t}{\text{Payroll Lag}}.
\end{aligned}$$

where AR = Accounts Receivable; AP = Accounts Payable; AW = Accrued Wages; “Collection Period” = average collection period (days); “Payables Days” = average payables period (days); “Payroll Lag” = payroll payment lag. Inventory is tracked in both units and value using average cost:

$$\begin{aligned}
\frac{d \text{InvUnits}_t}{dt} &= \text{Production}_t - \text{Shipments}_t, \\
\frac{d \text{InvVal}_t}{dt} &= \underbrace{\text{Material Cash Out}_t + \text{Direct Labor Cash Out}_t}_{\text{value added}} - \underbrace{\text{COGS}_t}_{\text{value removed}}, \\
\text{COGS}_t &= \text{Units Shipped}_t \times \bar{C}_t, \\
\bar{C}_t &= \frac{\text{InvVal}_t}{\max(\text{InvUnits}_t, \varepsilon)} \quad (\varepsilon > 0).
\end{aligned}$$

where InvUnits = inventory units; InvVal = inventory value; COGS = cost of goods sold; $\bar{C}$ = average unit cost; ε = small positive constant; Production = units produced; Shipments = units shipped; Units Shipped = same as shipments in units; Material Cash Out = cash spent on materials; $\text{Direct Labor Cash Out}$ = cash spent on direct labor. Fixed assets implement straight-line depreciation by splitting historical cost and accumulated depreciation:

$$\begin{aligned}
\frac{d HC_t}{dt} &= \text{Capex}_t - \text{Retirements}_t, \\
\frac{d AD_t}{dt} &= \text{Depreciation}_t - \text{Retired AD}_t, \\
\text{Depreciation}_t &= \frac{HC_t}{\text{Useful Life}}, \\
NFA_t &= HC_t - AD_t.
\end{aligned}$$

where HC = historical cost; AD = accumulated depreciation; NFA = net fixed assets; Capex = capital expenditures; Retirements = asset historical cost retired; Retired AD = accumulated depreciation retired; “Useful Life” = average asset life.

Asset retirement and cohort aging (fixed delay). To strictly match straight-line depreciation with end-of-life removal, the historical cost can be implemented as a fixed-delay (aging-chain) conveyor so that retirements equal past capex delayed by the useful life, and the corresponding accumulated depreciation of that cohort is removed simultaneously:

$$\begin{aligned}\text{Retirements}_t &= \text{Capex}_{t-\text{Useful Life}}, \\ \text{Retired AD}_t &= \text{HC}_{t-\text{Useful Life}}.\end{aligned}$$

This cohort-based formulation preserves the accounting identity that, at retirement, accumulated depreciation equals the cohort's historical cost, and avoids exponential tails that arise from first-order delays. Debt is managed through a long-term/short-term chain consistent with current vs. non-current classification:

$$\begin{aligned}\frac{dLT_t}{dt} &= \text{New Debt}_t - \text{Reclass}_t, \\ \text{Reclass}_t &= \frac{LT_t}{\text{Avg Maturity} - \text{Current Horizon}}, \\ \frac{dST_t}{dt} &= \text{Reclass}_t - \text{Principal Repay}_t, \\ \text{Principal Repay}_t &= \frac{ST_t}{\text{Current Horizon}}.\end{aligned}$$

where LT = long-term debt; ST = short-term debt; Reclass = reclassification from long-term to current; Principal Repay = principal repayment; “Avg Maturity” = average contractual maturity; “Current Horizon” = current classification horizon (e.g., one year). Equity follows clean surplus at instantaneous frequency and separates owner transactions:

$$\begin{aligned}\frac{dAPIC_t}{dt} &= \text{Shares Issued}_t \times \text{Issue Price}_t, \\ \frac{dRE_t}{dt} &= \underbrace{\text{Instantaneous Revenue}_t - \text{Instantaneous Expense}_t - \text{Taxes}_t}_{\text{instantaneous net income}} - \text{Dividends Paid}_t.\end{aligned}$$

where $APIC$ = additional paid-in capital; RE = retained earnings; Shares Issued = number of shares issued; Issue Price = issuance price per share; $\text{Instantaneous Revenue}$ = value inflow rate; $\text{Instantaneous Expense}$ = value outflow rate; Taxes = tax expense; Dividends Paid = cash dividends. This construction enforces clean surplus at the reporting frequency by pairing instantaneous dynamics with period aggregation (see below): $CRE_t = CRE_{t-1} + \text{Net Income}_t - \text{Dividends}_t$. Reporting accumulators convert instantaneous flows into period totals and then reset:

$$\begin{aligned}\text{Reported Revenue}_t &= \int_{t-1}^t \text{Instantaneous Revenue}(s) ds, \\ \text{Reported Expense}_t &= \int_{t-1}^t \text{Instantaneous Expense}(s) ds, \\ \text{Net Income}_t &= \text{Reported Revenue}_t - \text{Reported Expense}_t - \text{Taxes}_t.\end{aligned}$$

where Reported Revenue and Reported Expense = period totals from accumulators; Net Income = reported net income for the period.

Taxes and payment timing. In the base specification, taxes are a constant share of pretax income and are assumed to be paid immediately (no tax payable stock):

$$\begin{aligned}\text{Taxes}_t &= \tau \times \text{Pretax Income}_t, \\ \text{Tax Paid}_t &= \text{Taxes}_t.\end{aligned}$$

If payment lags are material, an optional tax payable stock can be introduced to separate expense recognition from cash settlement:

$$\begin{aligned}\frac{dTP_t}{dt} &= \text{Taxes}_t - \text{Tax Paid}_t, \\ \text{Tax Paid}_t &= \frac{TP_t}{\text{Tax Payment Lag}}.\end{aligned}$$

The immediate-payment assumption simplifies the baseline model and is common in SD implementations; the payable stock restores realism when needed. The indirect cash flow from operations reconciles net income with accrual changes:

$$\begin{aligned}\text{CFO}_t &= \text{Net Income}_t + \text{Depreciation}_t - \Delta AR_t - \Delta \text{InvVal}_t + \Delta AP_t + \Delta AW_t, \\ \text{Cash}_t &= \text{Cash}_{t-1} + \text{CFO}_t + \text{CFI}_t + \text{CFF}_t.\end{aligned}$$

where CFO = cash flow from operations; ΔX_t = change in X over the reporting period; CFI = cash flow from investing; CFF = cash flow from financing; Cash = cash balance. For completeness, investing and financing sections follow standard definitions:

$$\begin{aligned}\text{CFI}_t &= -\text{Capex}_t + \text{Proceeds from Asset Sales}_t, \\ \text{CFF}_t &= \Delta ST_t + \Delta LT_t + (\text{Shares Issued}_t \times \text{Issue Price}_t) - \text{Dividends Paid}_t.\end{aligned}$$

Under the exogenous-financing boundary, debt and equity issuance policies are specified outside the accounting engine; the engine reconciles them mechanically in the statements.

Explicit constraints (accounting integrity).

- **Double-entry and identities:** $\text{Assets}_t = \text{Liabilities}_t + \text{Equity}_t$; Cash reconciliation holds each period: $\text{Cash}_t = \text{Cash}_{t-1} + \text{CFO}_t + \text{CFI}_t + \text{CFF}_t$; Clean surplus at reporting frequency: $\text{CRE}_t = \text{CRE}_{t-1} + \text{Net Income}_t - \text{Dividends}_t$.
- **Accrual vs. cash separation:** AR, AP, and Accrued Wages integrate value–cash timing differences; dividends and equity issuance affect equity only through RE and $APIC$ respectively.
- **Straight-line depreciation discipline:** Depreciation removes value at a constant rate from HC via AD , avoiding exponential delays.
- **Reporting periodization:** Period totals are produced by integrating instantaneous flows over the reporting window and then resetting the accumulators; a constant “held” value can be maintained for managerial decision inputs until the next report.
- **Inventory costing convention:** Average cost is enforced for tractability in SD; FIFO/LIFO are not attempted.
- **Financing boundary:** Debt/equity issuance and payout policy are exogenous inputs; the base accounting engine may allow negative cash, with financing modules supplying remedies in full applications.

Formulas grouped by identities vs. policy-based dynamics.

- **Accounting identities (must always hold):**

$$\begin{aligned}\text{Balance Sheet: } \text{Assets}_t &= \text{Liabilities}_t + \text{Equity}_t, \\ \text{Cash Reconciliation: } \text{Cash}_t &= \text{Cash}_{t-1} + \text{CFO}_t + \text{CFI}_t + \text{CFF}_t, \\ \text{Fixed Assets Roll-forward: } \text{NFA}_t &= \text{NFA}_{t-1} + \text{Capex}_t - \text{Depreciation}_t, \\ \text{Inventory–COGS (value): } \text{COGS}_t &= \text{BegInv}_t + \text{Purchases}_t - \text{EndInv}_t, \\ \text{Clean Surplus (Retained Earnings): } \text{CRE}_t &= \text{CRE}_{t-1} + \text{Net Income}_t - \text{Dividends}_t.\end{aligned}$$

- **Policy-based forecasting (model assumptions):**

Nominal Rate _t	$= (1 + \text{Real Rate}_t)(1 + \text{Inflation}_t) - 1,$
P_t	$= P_{t-1}(1 + \text{Nominal Price Increase}_t),$
Q_t	$= Q_{t-1}(1 + \text{Volume Growth}_t),$
Sales _t	$= P_t Q_t,$
EndInv _t (units)	$= \text{Sales}_{t+1}(\text{units}) \times \text{InvPolicy}\%,$
Purchases _t (units)	$= \text{Sales}_t(\text{units}) + \text{EndInv}_t(\text{units}) - \text{BegInv}_t(\text{units}),$
AP_t	$= \text{Purchases}_t(\$) \times \text{AP Policy}\%,$
AR_t	$= \text{Sales}_t \times \text{AR Policy}\%,$
Capex _t	$= \text{Depreciation}_t + (NFA_{t-1} \times \text{Real Growth}_{t+1}),$
Dividends Paid _t	$= \text{Net Income}_{t-1} \times \text{Payout Ratio}\%.$

Explicit constraints (governance). Beyond the no-plugs, no-circularity, and double-entry principles, the framework enforces: (i) **Matching maturities** (operating shortfalls with short-term debt; investment gaps with long-term debt/equity); (ii) **Financing-first** (no new short-term investments in periods with fresh borrowing/equity); (iii) **Minimum cash target** (cash is a policy constraint, not a residual); (iv) **Dividend cap** (dividends do not exceed prior-year net income under the stated policy); (v) **Buyback discipline** (illustratively, buybacks constrained not to exceed current-year depreciation in some implementations), and (vi) **Stable working-capital ratios** to activity. These constraints collectively yield a transparent, auditable forecasting engine suitable for valuation and risk analysis.

3.1.4 Montesi and Papiro (2018): Stochastic top-down simulation for banks

Unlike the corporate-forecasting literature summarized above, [Montesi and Papiro \(2018\)](#) develop a top-down stochastic simulation framework tailored to banks. The accounting engine projects full financial statements across thousands of Monte Carlo scenarios by treating the principal drivers—asset and liability rates, credit losses, deposit and loan growth, market/trading returns, and risk weights—as bounded random variables calibrated to historical statistics and peer benchmarks. Two pillars structure the model: the statement **dynamics** and the **constraints** that ensure coherence of accounting identities and prudential rules.

Dynamics (P&L structure, credit-risk flows, balance-sheet roll-forwards). The profit-and-loss is organized as a ladder from net interest income to net income:

Net Interest Income _t	$= \text{Interest Income}_t - \text{Interest Expense}_t,$
Net Revenues _t	$= \text{Net Interest Income}_t + \text{Net Commission Income}_t$ $+ \text{Net Financial \& Trading Income}_t + \text{Other Operating Income (Expense)}_t,$
Operating Income _t	$= \text{Net Revenues}_t$ $+ \text{Net Adjustments for Impairment on Loans}_t - \text{Non-Interest Expense}_t,$
Pretax Income _t	$= \text{Operating Income}_t$ $+ \text{Other Non Operating Income (Losses)}_t,$
Net Income _t	$= \text{Pretax Income}_t - \text{Taxes}_t.$

Net interest margins arise from stochastic rates applied to earning assets and interest-bearing liabilities; trading and other operating results are modeled as stochastic returns on the relevant stocks.

Credit-risk dynamics drive impairments and loan stocks via an expected-loss logic:

$$\begin{aligned}
\text{Defaulted Credit Flow}_t &= PD_t \times \text{Gross Performing Loans}_{t-1}, \\
NPL_t &= NPL_{t-1} + \text{Defaulted Credit Flow}_t - NPL_{t-1}(w_t + r_t + b_t), \\
\text{Net Adjustments}_t &= \text{Defaulted Credit Flow}_t LGD_t \\
&\quad + NPL_{t-1}(1 - w_t - r_t - b_t)(LGD_t - LGD_{t-1}) \\
&\quad + NPL_{t-1} b_t LGD_{t-1}, \\
\text{Reserve for Loan Losses}_t &= \text{Reserve}_{t-1} + \text{Net Adjustments}_t - NPL_{t-1} w_t, \\
\text{Credit Growth}_t &= \text{Gross Performing Loans}_{t-1} g_t, \\
\text{Gross Performing Loans}_t &= \text{Gross Performing Loans}_{t-1} \\
&\quad + \text{Credit Growth}_t - \text{Defaulted Credit Flow}_t + NPL_{t-1} b_t, \\
\text{Net Total Loans}_t &= \text{Gross Performing Loans}_t + NPL_t - \text{Reserve for Loan Losses}_t.
\end{aligned}$$

where PD = default rate; LGD = loss given default; w, r, b = write-off, recovery, and cure rates; g = loan growth rate. Customers' deposits and other core items follow stochastic growth rules, while market/trading income is modeled as random returns on financial assets. Regulatory capital is built from equity via the standard transformations:

$$\begin{aligned}
\text{Tangible Common Equity}_t &= \text{Equity Book Value}_t - \text{Intangible Assets}_t, \\
CET1_t &= \text{Tangible Common Equity}_t - \text{CET1 Adjustments}_t.
\end{aligned}$$

Stochastic drivers and calibration. Key P&L and balance-sheet drivers are modeled as bounded random variables calibrated to the bank's own history and peer distributions. In particular: earning-asset and interest-bearing liability rates are drawn from Beta distributions whose means and bounds reflect 5-year historical averages and variability; customers' deposits and gross performing loans grow with stochastic rates whose ranges are anchored to consensus GDP growth; net financial and trading income is modeled as stochastic returns on financial assets with a Logistic distribution, with the left tail (e.g., 1st percentile around -4.26%) used to cap severe losses; other non-operating income/losses proxy operational risk via a Beta-distributed loss rate bounded by the bank's reported operational risk capital; taxes are a deterministic effective rate.

Credit-risk measurement proxies (estimating PD). Where direct historical PD series are unavailable, default flows are proxied via NPL stocks and provisioning. The NPL generation time is first computed as a ratio of the reserve stock to a 5-year average of loan impairment charges, and then mapped into a PD estimate relative to performing loans:

$$\begin{aligned}
\text{NPL Generation Time}_t &= \frac{\text{Reserve for Loan Losses}_t}{\frac{1}{5} \sum_{i=t-4}^t \text{Net Adjustments}_i}, \\
PD_t &= \frac{\left(\frac{NPL_t}{\text{NPL Generation Time}_t} \right)}{\text{Gross Performing Loans}_{t-1}}.
\end{aligned}$$

In the simulation, PD follows a right-skewed Weibull distribution truncated to scenario-appropriate bounds (from benign to extreme), while LGD follows a Beta distribution with lower/upper bounds aligned to stress references. Write-off (w), recovery (r), and cure (b) rates are random within min/max ranges derived from G-SIB peer histories. NPLs are assumed not to accrue interest.

Risk-weighted assets (RWA) dynamics. Risk density $RW = \frac{RWA}{\text{Net Risk Assets}}$ is treated as a random variable (e.g., Beta) whose admissible range widens over the horizon to reflect

prudential tightening under stress. A simple rule-of-thumb governs the annual increment to the range based on the bank's initial risk density:

$$\Delta RW = \begin{cases} 5\% & \text{if } RW_{\text{base}} < 20\% \\ 4\% & \text{if } 20\% \leq RW_{\text{base}} < 30\% \\ 3\% & \text{if } 30\% \leq RW_{\text{base}} < 40\% \\ 2\% & \text{if } 40\% \leq RW_{\text{base}} < 50\% \\ 1\% & \text{if } RW_{\text{base}} \geq 50\% \end{cases}$$

so that both the mean and support of RW_t can drift upward across scenarios.

Constraints (capital policy and financial balance). Payouts are endogenized by a capital-adequacy rule that links dividends to the attainment of a target CET1 ratio on risk-weighted assets:

$$\begin{aligned} \text{Target CET1 Capital}_t &= \text{Net Risk Assets}_t \times RW_t \times CET1_{\text{target}}, \\ \text{Target Equity Book Value}_t &= \text{Target CET1 Capital}_t + \text{CET1 Adjustments}_t, \\ \text{Dividend}_t &= \max \left(\text{Equity Book Value}_{t-1} + \text{Net Income}_t \right. \\ &\quad \left. - \text{Target Equity Book Value}_t, 0 \right), \\ \text{Equity Book Value}_t &= \text{Equity Book Value}_{t-1} + \text{Net Income}_t - \text{Dividend}_t. \end{aligned}$$

Financial balance is enforced each period by assigning the funding gap (AFN) to financial liabilities, thereby maintaining the accounting identity:

$$\begin{aligned} AFN_t &= \Delta \text{Net Risk Assets}_t + \Delta \text{Net No Risk Assets}_t \\ &\quad - \Delta \text{Deposits}_t - \text{Net Income}_t - \Delta \text{Other Liabilities}_t + \text{Dividend}_t, \\ AFN_t &= \Delta \text{Financial Liabilities}_t, \\ \text{Financial Liabilities}_t &= \text{Net Risk Assets}_t + \text{Net No Risk Assets}_t \\ &\quad - \text{Deposits}_t - \text{Other Liabilities}_t - \text{Equity Book Value}_t. \end{aligned}$$

However, this is an example of using plugs to enforce financial balance, which is not a good practice. In regulatory stress testing, treating financial liabilities as the balancing item is an explicit design choice to guarantee stock-flow consistency under capital constraints; nonetheless, analysts should examine the implied funding path and liquidity feasibility.

3.2 Textbooks

Academic literature rarely provides fully specified, account-by-account models that operationalize the complete set of financial statements in a stock-flow consistent manner. Most published studies concentrate on one or two building blocks (for example, inventory and COGS mechanics, working-capital policies, or depreciation) or employ simplified, pedagogical structures to demonstrate a particular modeling technique (for example, no plugs, no circularity). When a comprehensive and transparent treatment is required—covering roll-forwards, recognition and measurement conventions, linkages across statements, and auditable identities for each major account—the most dependable references are modern textbooks in financial modeling, corporate finance, and valuation. The following titles are representative starting points rather than an exhaustive list. [Pignataro \(2022\)](#) [Benninga \(2014\)](#) [Koller et al. \(2010\)](#) [Sawyer and Dalglish \(2015\)](#) [Rees \(2011\)](#)

3.2.1 Benninga (2014): Sales-driven financial modeling framework

Benninga (2014) presents a comprehensive approach to pro forma financial statement modeling that exemplifies the systematic construction of dynamic accounting simulations. The textbook's methodology is built on the fundamental principle of **sales-driven modeling**, where sales revenue serves as the primary driver for most other financial variables across the income statement and balance sheet. This approach assumes that the level of sales is the key determinant of a company's operational scale and financing requirements, making it a natural starting point for integrated financial forecasting.

Income statement dynamics and forecasting rules. The model constructs the income statement through a cascading series of relationships anchored by sales growth assumptions. Sales revenue is projected using a simple growth formula, while most expense categories are modeled as fixed percentages of sales, reflecting the assumption of stable operating margins. The treatment of depreciation follows accounting conventions by applying a depreciation rate to the average fixed asset cost during the year, acknowledging that capital investments occur throughout the period. Interest calculations employ a dynamic approach: debt interest expense is computed as the debt rate multiplied by the average debt balance, while cash interest income uses the cash rate applied to average cash and marketable securities balances. This creates circular references in the model since interest income affects net income, which in turn impacts retained earnings and ultimately the cash balance. In addition, operating expenses such as SG&A can be parameterized as fixed or variable percentages of sales, consistent with policy assumptions.

The framework incorporates sophisticated tax modeling that extends beyond simple tax rate applications. Advanced versions include tax loss carryforward mechanisms, where prior losses can offset future taxable income, potentially reducing the effective tax rate to zero until all carryforward benefits are exhausted. Dividend policy is typically modeled as a fixed payout ratio applied to net income, although alternative approaches such as targeted dividend per share growth are also discussed.

Balance sheet architecture and the "plug" mechanism. The balance sheet forecasting employs a systematic approach where most accounts are projected independently, with a designated "plug" item ensuring the fundamental accounting identity $Assets = Liabilities + Equity$ holds in each period. This plug mechanism represents a critical financial policy choice and typically takes one of three forms: (1) **Cash as plug**: excess cash is retained while funding gaps are covered by drawing down cash reserves; (2) **Debt as plug**: cash is maintained at minimum levels with financing needs met through borrowing and surpluses used for debt repayment; (3) **Equity as plug**: debt levels are determined by target capital structure ratios, with stock issuance or repurchases maintaining the desired debt-to-equity balance. In the baseline specification, both debt and stock are assumed to remain constant (no new issuances, repayments beyond scheduled, or buybacks), with alternative formulations introducing plug behavior or target capital-structure rules.

Current assets and current liabilities are modeled as fixed percentages of sales, reflecting the proportional relationship between business activity and working capital requirements. Net fixed assets can be projected using several approaches: as a fixed percentage of sales (implying constant asset turnover), with gross fixed assets scaling with sales (treating depreciation as a timing rather than economic phenomenon), or holding net fixed assets constant (implying capital expenditures exactly offset depreciation). The model maintains dynamic consistency through accounting identities: fixed assets at cost equal net fixed assets plus accumulated depreciation, while accumulated depreciation represents the cumulative sum of annual depreciation expenses.

Cash flow statement integration and model validation. Rather than being independently forecasted, the cash flow statement is derived from the projected income statement and balance sheet changes, serving as a crucial validation mechanism for model integrity. Operating cash flow begins with net income, adds back non-cash expenses like depreciation, and adjusts for working capital changes (year-over-year changes in current assets and current liabilities). Investing cash flow primarily reflects capital expenditures represented by increases in fixed assets at cost, while financing cash flow captures changes in debt and equity balances along with dividend payments. The net change in cash from these three activities must reconcile exactly with the change in the cash and marketable securities account on the balance sheet, providing a fundamental consistency check.

Key formulations and mathematical structure. The core forecasting relationships can be expressed as follows:

$$\begin{aligned}
\text{Sales}_t &= \text{Sales}_{t-1} \times (1 + g_{\text{sales}}), \\
\text{COGS}_t &= \text{Sales}_t \times r_{\text{COGS}}, \\
\text{Depreciation}_t &= r_{\text{dep}} \times \frac{(\text{FA}_{\text{cost},t-1} + \text{FA}_{\text{cost},t})}{2}, \\
\text{Interest Expense}_t &= r_{\text{debt}} \times \frac{(\text{Debt}_{t-1} + \text{Debt}_t)}{2}, \\
\text{Interest Income}_t &= r_{\text{cash}} \times \frac{(\text{Cash}_{t-1} + \text{Cash}_t)}{2}, \\
\text{Taxes}_t &= \max(\text{Taxable Profit}_t \times r_{\text{tax}}, 0), \\
\text{Dividends}_t &= \text{Net Income}_t \times p_{\text{div}}, \\
\text{Retained Earnings}_t &= \text{Net Income}_t - \text{Dividends}_t.
\end{aligned}$$

Balance sheet dynamics follow proportional and identity-based rules:

$$\begin{aligned}
\text{Current Assets}_t &= \text{Sales}_t \times r_{\text{CA}}, \\
\text{Current Liabilities}_t &= \text{Sales}_t \times r_{\text{CL}}, \\
\text{Net Fixed Assets}_t &= \text{Sales}_t \times r_{\text{NFA}}, \\
\text{Accumulated Depreciation}_t &= \text{Accumulated Depreciation}_{t-1} + \text{Depreciation}_t, \\
\text{Fixed Assets at Cost}_t &= \text{Net Fixed Assets}_t + \text{Accumulated Depreciation}_t, \\
\text{Accumulated Retained Earnings}_t &= \text{Accumulated Retained Earnings}_{t-1} + \text{Retained Earnings}_t.
\end{aligned}$$

Explicit constraints and governance mechanisms. The framework incorporates several critical constraints that ensure model realism and financial feasibility. The **balance sheet identity** represents the fundamental constraint requiring total assets to equal total liabilities plus equity in each period, enforced through the plug mechanism. **Circular reference resolution** addresses the technical challenge where variables are mutually dependent (e.g., interest income depends on cash balance, which depends on net income, which includes interest income), typically resolved through iterative calculation procedures. In practice, spreadsheet implementations handle these mutual dependencies by enabling iterative calculation.

The model may incorporate a **non-negative cash constraint** to prevent economically meaningless negative cash positions, automatically triggering debt issuance when financing needs arise. In specialized applications such as project finance, additional **covenant constraints** may apply, including dividend restrictions until debt retirement, prohibitions on new equity issuance, and adherence to fixed debt amortization schedules. For valuation applications, the framework recognizes that **terminal value calculations** using the Gordon growth model

are only valid when the long-term growth rate is less than the weighted average cost of capital, ensuring finite and economically meaningful valuations.

The plug mechanism formulations vary by financial policy choice:

$$\begin{aligned}
\text{Cash as Plug: } \text{Cash}_t &= (\text{CL}_t + \text{Debt}_t + \text{Stock}_t + \text{ARE}_t) \\
&\quad - \text{CA}_t - \text{NFA}_t, \\
\text{Debt as Plug: } \text{Debt}_t &= \max((\text{CA}_t + \text{NFA}_t) - (\text{CL}_t + \text{Stock}_t + \text{ARE}_t), \\
&\quad \text{Debt}_{t-1}), \\
\text{Equity as Plug: } \text{Debt}_t &= r_{D/E} \times (\text{Stock}_t + \text{ARE}_t), \\
\text{Stock}_t &= \text{Total Assets}_t - \text{CL}_t - \text{Debt}_t - \text{ARE}_t.
\end{aligned}$$

The built of dynamic system of [Benninga \(2014\)](#) can serve as a reference though the use of "plug" should be avoided.

3.2.2 McKinsey (2010): Business Valuation

Though the book does not provide a detailed accounting simulation framework, it provides a valuable perspective on building blocks for accounting simulations.

Revenue Forecasting with Growth Decay: Evidence and Modeling Implications A realistic revenue forecast is foundational for any accounting simulation and valuation exercise because most downstream items depend directly or indirectly on sales. Empirical evidence summarized in [Koller et al. \(2010\)](#) shows that high growth is typically short-lived and decays rapidly toward a sustainable, economy-consistent level. Incorporating this decay into the forecast avoids overly optimistic long-run projections and yields financially coherent statements.

Empirical regularities (stylized facts). The historical evidence indicates a rapid decay of high growth: firms starting with very high real growth typically converge within a few years to single-digit rates and, over roughly a decade, to low single digits ([Koller et al., 2010](#)). Over time, cross-sectional dispersion narrows markedly—the gap between top and bottom growth cohorts shrinks over 5–10 years. Scale also acts as a headwind, making it increasingly difficult for larger firms to sustain elevated growth. In sum, transitions from high to low growth are common, and multi-year persistence of very high real growth is rare.

Forecasting approaches (drivers). Two complementary approaches embed these regularities in simulations. In a top-down framing, the analyst projects total addressable market, market share, and price; in mature markets, firm growth is assumed to converge to macro trends (e.g., real GDP plus inflation), whereas in emerging categories a diffusion/penetration logic (S-curves) implies early acceleration followed by decay as saturation approaches. In a bottom-up framing, forecasts aggregate customer-cohort and unit-economics dynamics—new-customer acquisition, churn/retention, and expansion—recognizing that acquisition generally becomes harder as markets saturate and that spend-per-customer growth also decays, thereby reducing the contribution to aggregate growth.

Implementation with an explicit decay path. Use a short explicit forecast horizon (e.g., 5–7 years) with potentially elevated near-term growth, then transition to a sustainable long-run rate tied to macro fundamentals. A simple continuous decay law that is easy to calibrate is:

$$\begin{aligned}
\text{Sales}_t &= \text{Sales}_{t-1} (1 + g_t), \\
g_t &= g_\infty + (g_0 - g_\infty) \exp(-\kappa(t-1)),
\end{aligned}$$

where g_0 is the initial growth rate, g_∞ is the sustainable long-run rate (e.g., inflation plus real GDP growth), and $\kappa > 0$ governs the speed of convergence so that the model can match the stylized facts (e.g., high growth decays to single digits within 5 years and to low single digits within about 10 years). The decay law can be used under either top-down (market/price/share) or bottom-up (cohorts/ARPU/churn) drivers by mapping driver-level assumptions into the implied g_t path.

Ratio-driven income statement forecasting (COGS as an example). After revenue is forecast, operating items in the income statement can be projected with a ratio-driven method that ties each line to its primary economic driver—typically revenue. The workflow is straightforward: identify the economic driver; estimate a forecast ratio using historical data (for example, the historical cost-of-goods-sold-to-revenue ratio); and compute the forecast by multiplying the ratio by the projected driver. For COGS, a baseline specification is:

$$\text{COGS}_t = \text{Sales}_t \times r^{\text{COGS}}$$

where $r^{\text{COGS}} = \frac{\text{Historical COGS}}{\text{Historical Sales}}$ is calibrated from recent history and can later be adjusted to reflect expected efficiency gains or cost pressures. This ratio formulation is preferable to naively growing last year's COGS by the sales growth rate because it preserves a clear linkage to the driver and enables sensitivity tests on operating efficiency independent of top-line assumptions.

Interpretation, calibration, and scenario analysis. The ratio method confers three practical benefits. First, it automatically keeps operating items synchronized with any change in the revenue path, maintaining coherence across statements. Second, it supports scenario analysis targeted to efficiency: analysts can vary the COGS-to-sales ratio to explore supply-chain improvements or inflationary stress without rebuilding the model. Third, the structure generalizes to other operating expenses such as SG&A by replacing r^{COGS} with the corresponding historical ratio and applying the same linkage to Sales_t . In all cases, the implied margins and working-capital needs should be checked for plausibility and consistency with prior sections.

Working capital forecasting: ratio and days methods. Balance-sheet operating items are projected by linking them to revenue or cost drivers through stable historical relationships. A direct ratio method specifies, for example, accounts receivable as a percent of sales or inventory as a percent of COGS. To improve operational realism, a days-based formulation can be used. With Days Sales Outstanding (DSO), Inventory Days, and Days Payable Outstanding (DPO), the canonical specifications are:

$$\begin{aligned} \text{AR}_t &= \text{Sales}_t \times \frac{\text{DSO}}{365}, \\ \text{Inventory}_t &= \text{COGS}_t \times \frac{\text{Inventory Days}}{365}, \\ \text{AP}_t &= \text{COGS}_t \times \frac{\text{DPO}}{365}. \end{aligned}$$

These formulations preserve proportionality to the relevant drivers while embedding timing discipline consistent with operating cycles.

PP&E and CAPEX via roll-forward identities. Instead of directly forecasting capital expenditures, it is often more consistent to target capital intensity through a Net PP&E-to-sales ratio, derive the Net PP&E path from sales, and then back out CAPEX using the accounting roll-forward. The core identity is:

$$\begin{aligned} \text{Net PP\&E}_t &= \text{Net PP\&E}_{t-1} + \text{CAPEX}_t - \text{Depreciation}_t, \\ \text{CAPEX}_t &= (\text{Net PP\&E}_t - \text{Net PP\&E}_{t-1}) + \text{Depreciation}_t. \end{aligned}$$

This approach aligns investment with the revenue-driven scale of operations and keeps depreciation and capital intensity internally consistent.

3.2.3 Sawyer (2015): From operational blocks to an integrated Company Business Model

Unlike abbreviated or template-driven approaches, Part II of [Sawyer and Dalglish \(2015\)](#) is organized as a deliberately sequenced “financial modeling practice.” Each chapter builds on the previous one by first modeling the firm’s operational components and then integrating them into the primary financial statements to form a comprehensive Company Business Model (CBM). The treatment of accounting mechanics is explicitly operational—especially for cost of goods sold (COGS) and inventory—rather than a simple application of fixed percentages of revenue.

Chapter 4-9 of this book are the core accounting simulations, with Chapter 11 integrating all the previous chapters into the income statement and cash flow statement.

The workflow begins with Chapter 4, a staffing model. It is foundational because personnel costs are typically the largest operating outlay and they drive secondary resource needs such as office space, telecommunications, and equipment. The organizational structure established here provides the scaffolding for subsequent modules.

Next is Chapter 5, the sales and revenue model, which serves as the primary driver for the CBM. The sales forecast determines the company’s revenue trajectory, the associated cost-of-goods-sold requirements, and the overall scale of sales and marketing activities.

Chapter 6 is built directly on the sales plan. The COGS and inventory module computes the variable costs tied to each sale and projects inventory needs with discipline consistent with the sales model.

Department-level operating costs are then layered in Chapter 7 and 8. The sales and marketing cost model quantifies the resources required to achieve the sales targets specified in Chapter 5, using staffing inputs from Chapter 4. The product development cost model forecasts expenses tied to the product lifecycle, again anchored by the staffing plan.

Chapter 9 is an operating-expenditure (OPEX) model captures general operating costs not covered by the departmental modules. A capital-expenditure (CAPEX) model consolidates planned capital purchases and computes straight-line depreciation, a key non-cash expense that feeds into the financial statements.

Chapter 11 is the integration of all prior modules to construct the Profit and Loss (P&L) statement and the Statement of Cash Flows. These statements summarize operating performance and actual cash dynamics, providing an auditable, end-to-end simulation of the firm’s financial position. Finally, the outputs from all prior modules are integrated to construct the Profit and Loss (P&L) statement and the Statement of Cash Flows. These statements summarize operating performance and actual cash dynamics, providing an auditable, end-to-end simulation of the firm’s financial position.

Summing up, the workflow begins with a Staffing Model to plan human resources and forecast core personnel costs; proceeds to a Sales and Revenue Model to project the top line; then estimates sale-linked variable costs through a Cost of Goods Sold Model. It next quantifies departmental operating spend via a Cost of Sales and Marketing Model and a Cost of Product Development Model, and captures company-level Operating and Capital Expenditures Models (with straight-line depreciation feeding the statements). Finally, by integrating the income and expense outputs from all of these operational modules, the model produces the Profit and Loss Statement and the Statement of Cash Flows, thereby fully simulating the company’s financial performance.

Below is a detailed explanation of each chapter, how they are connected, and how they are used to construct a complete financial statements.

Chapter 4: Staffing Model (inputs, formulas, outputs)

- **Inputs:** FTE by role/department; target salary or hourly rate; salary adjustment factor; payroll tax rate; benefits rates; flags for phones/computers/work spaces.
- **Core formulas:**
 - Adjusted salary per role: $\text{AdjComp}_t = \text{FTE}_t \times \text{TargetSalary}_t \times (1 + \text{AdjFactor}_t)$.
 - Payroll taxes and benefits: $\text{Tax}_t = \text{AdjComp}_t \times r_{\text{tax}}$, $\text{Benefits}_t = \text{AdjComp}_t \times r_{\text{ben}}$.
 - Equipment/space counts: $\text{Phones}_t = \sum_r 1\{r \text{ needs phone}\} \times \text{FTE}_{r,t}$ (similarly for computers, work spaces).
- **Outputs:** Total payroll expense to P&L; departmental headcount/costs to Chapters 7&8; asset-count drivers (phones, computers, work spaces) to Chapter 9.

Chapter 5: Sales and Revenue Model

- **Inputs:** Annual sales targets; seasonal spread; pricing/discounts; recurring-revenue parameters; credit terms (DSO); opening AR.
- **Core formulas:**
 - Units/month: $\text{Units}_m = \text{AnnualUnits} \times \text{SeasonalShare}_m$; Revenue: $\text{Sales}_m = \text{Units}_m \times \text{Price}_m \times (1 - \text{Discount}_m)$.
 - Recurring revenue cohort build as specified (maintenance/subscription carryover).
 - AR roll-forward: $\text{AR}_t = \text{AR}_{t-1} + \text{CreditSales}_t - \text{CashCollections}_t$; collections based on DSO or an aging schedule.
- **Outputs:** Total revenue to P&L; unit forecast to Chapter 6; ΔAR to cash flow (indirect) in Chapter 11.

Chapter 6: COGS and Inventory

- **Inputs:** Unit forecast from Chapter 5; unit COGS by component (e.g., hardware, software); inventory policy (Inventory Days); supplier terms (DPO); opening inventory and AP.
- **Core formulas:**
 - COGS: $\text{COGS}_t = \sum_k \text{Units}_{k,t} \times \text{UnitCOGS}_{k,t}$.
 - Inventory target: $\text{Inv}_t = \text{COGS}_t \times \frac{\text{Inventory Days}}{365}$.
 - Purchases: $\text{Purch}_t = \text{COGS}_t + (\text{Inv}_t - \text{Inv}_{t-1})$.
 - AP roll-forward: $\text{AP}_t = \text{AP}_{t-1} + \text{Purch}_t - \text{Payments}_t$; payments scheduled per DPO.
- **Outputs:** Total COGS to P&L; pricing/unit-cost back to Chapter 5 where referenced; $\Delta\text{Inventory}$ and ΔAP to cash flow in Chapter 11.

Chapters 7–8: Departmental Operating Costs

- **Inputs:** Headcount and compensation from Chapter 4; sales targets from Chapter 5; policy rates (commissions/bonuses); vendor rates (software, hosting, marketing).
- **Core formulas:**
 - Commissions/bonuses: $\text{Commission}_t = r_{\text{comm}} \times \text{Sales}_t$ (allocated by headcount/policy as needed).

- Travel/marketing/tech spend: quantity $\times$ unit cost or ratio-of-sales where specified.
- Departmental OPEX: sum across line items; optional CAPEX flags accumulate asset purchases.
- **Outputs:** Departmental OPEX to P&L; flagged CAPEX needs to Chapter 9.

Chapter 9: OPEX & CAPEX

- **Inputs:** Company-level OPEX assumptions; CAPEX requests from Chapters 4, 7, 8; useful lives; opening fixed assets (cost) and accumulated depreciation.
- **Core formulas:**
 - OPEX master schedule: direct amounts or drivers; totals feed P&L.
 - Straight-line depreciation: for asset class i , $\text{Dep}_{i,t} = \frac{\text{Cost}_i}{\text{Life}_i}$ (adjusted for in-period additions per policy).
 - Fixed assets roll-forward: $\text{FA}_{\text{cost},t} = \text{FA}_{\text{cost},t-1} + \text{CAPEX}_t - \text{Disposals}_t$; $\text{AD}_t = \text{AD}_{t-1} + \text{Dep}_t$; $\text{NetPPE}_t = \text{FA}_{\text{cost},t} - \text{AD}_t$.
- **Outputs:** OPEX and Depreciation to P&L; CAPEX to investing cash flow; Net PP&E to balance sheet.

Chapter 11: P&L and Cash Flow (indirect)

- **Inputs:** Revenue from Chapter 5; COGS from Chapter 6; OPEX from Chapters 7–9; Depreciation from Chapter 9; working-capital changes (AR, Inventory, AP) from Chapters 5–6; CAPEX from Chapter 9; financing flows per policy.
- **Core formulas:**
 - P&L: $\text{NI}_t = \text{Sales}_t - \text{COGS}_t - \text{OPEX}_t - \text{Dep}_t - \text{Interest}_t - \text{Taxes}_t$.
 - Cash flow (indirect): $\text{CFO}_t = \text{NI}_t + \text{Dep}_t - \Delta \text{AR}_t - \Delta \text{Inv}_t + \Delta \text{AP}_t$; $\text{CFI}_t = -\text{CAPEX}_t + \text{Proceeds}_t$; $\text{CFF}_t = \Delta \text{Debt}_t + \Delta \text{Equity}_t - \text{Dividends}_t$.
 - Cash reconciliation: $\Delta \text{Cash}_t = \text{CFO}_t + \text{CFI}_t + \text{CFF}_t$; $\text{Cash}_t = \text{Cash}_{t-1} + \Delta \text{Cash}_t$.
- **Outputs:** Auditable P&L and Statement of Cash Flows consistent with balance sheet changes and prior modules.

Constraints

- Forecast logic constraint: the sales forecast follows a top-down approach—from total market size to target segment penetration to planned market share—to determine unit targets.
- Seasonal spread constraint: annual sales must be allocated to months using a predefined seasonal spread.
- Product mix and pricing constraint: revenue is constrained by a detailed product mix with component-specific pricing rules, including list price and discount percentage.
- Cash collection cycle constraint: the Accounts Receivable model is governed by an explicit cash collection cycle (e.g., receipts 30 days after sale).
- Component-based COGS constraint: COGS is the sum of variable costs by product component (e.g., hardware procurement, assembly/testing, packaging/shipping).

- Inventory level rule constraint: inventory must be sized to cover a forward-looking period (e.g., the next 90 days of the sales forecast), which determines purchases.
- Inventory payment terms constraint: Accounts Payable timing follows supplier terms (e.g., payment 60 days after purchase).
- Inventory valuation method constraint: the model uses Last-In, First-Out (LIFO) to compute the cost of goods sold for inventory issued.
- Linkage to staffing constraint: both cost models must use headcount and compensation data from the Staffing Model (Chapter 4).
- Bonus and commission logic constraint: bonuses and commissions are computed as a percentage of company revenue (Plan Basis) and allocated across eligible employees via the Bonus Pool.
- Predefined trip-cost constraint: travel expenses follow a predefined Trip Cost model by trip type (e.g., Direct Sales Trip).
- Capital plan consolidation constraint: the Capital Plan is a consolidated schedule that links inputs from STAFF, COSM, and DEV modules rather than direct entry.
- Depreciation method constraint: depreciation must use the straight-line method based on user-defined useful life.

3.2.4 Pignataro (2022): A complete framework under GAAP

Pignataro (Pignataro, 2022) presents a GAAP-consistent, end-to-end framework for the three statements. Chapters 1–4 walk through each major account on the balance sheet, income statement, and cash flow statement; Chapter 4 also compares depreciation methods (straight-line, accelerated, units-of-production) and formalizes the PP&E roll-forward. From Chapter 5, the book operationalizes working capital with explicit AR, inventory, AP, and accrual/deferral policies. Chapter 6 emphasizes that the balance sheet is closed by cash-flow drivers (operations, investing, financing) rather than forecasted independently. Chapter 7 details treasury mechanics for cash and debt (minimum cash, revolver/term-debt draws and repayments, interest on average balances) and highlights circularities requiring iterative resolution. For simulation purposes, Chapters 5–7 are the most relevant, and the following synthesis concentrates on these modules.

Chapter 5 details: Working capital via days-based formulation and CFO link. Operating working capital is specified using days metrics that connect balance-sheet stocks to income-statement flows, using a 360-day convention and average balances ($\text{Avg}X = (X_{t-1} + X_t)/2$).

The canonical specifications are:

$$\begin{aligned}
\text{Inventory Days} &= \frac{\text{Avg Inventory}}{\text{COGS}} \times 360, \\
\text{Inventory}_t &= \frac{\text{Inventory Days}}{360} \times \text{COGS}_t, \\
\text{AR Days} &= \frac{\text{Avg AR}}{\text{Net Sales}} \times 360, \\
\text{AR}_t &= \frac{\text{AR Days}}{360} \times \text{Net Sales}_t, \\
\text{AP Days} &= \frac{\text{Avg AP}}{\text{COGS}} \times 360, \\
\text{AP}_t &= \frac{\text{AP Days}}{360} \times \text{COGS}_t, \\
\text{Accrued Exp Days} &= \frac{\text{Avg Accrued Exp}}{\text{Total Opex}} \times 360, \\
\text{Accrued Exp}_t &= \frac{\text{Accrued Exp Days}}{360} \times \text{Total Opex}_t, \\
\text{Unearned Rev Days} &= \frac{\text{Avg Unearned Rev}}{\text{Net Sales}} \times 360, \\
\text{Unearned Rev}_t &= \frac{\text{Unearned Rev Days}}{360} \times \text{Net Sales}_t.
\end{aligned}$$

Allowances (for example, doubtful accounts) and policy choices (for example, inventory valuation) are layered on top as needed. In the indirect cash flow, year-over-year increases in current asset balances (for example, AR, Inventory) reduce cash from operations, while increases in current liabilities (for example, AP, Accrued Expenses, Unearned Revenue) increase cash from operations. These deltas reconcile net income to operating cash flow and close the cash-flow-driven balance sheet.

Core principle: let the cash flow statement drive the balance sheet. This chapter's central modeling principle is that balance sheet projections should be driven by the statement of cash flows rather than backing into cash from two forecast balance sheets. Forecasting two balance sheets and inferring cash is ambiguous and error-prone (for example, an increase in Net PP&E could arise from many CAPEX and depreciation combinations). Treat the balance sheet as the accumulation of operating, investing, and financing cash movements; period-end balances are the result of beginning balances plus the period's relevant cash-flow changes.

Generic dynamic formula for balance-sheet items. Across accounts, the stock-flow logic can be summarized as:

$$\text{Ending Balance}_t = \text{Beginning Balance}_{t-1} \pm \text{Relevant Cash-Flow Change}_t.$$

The signs depend on the account class:

$$\begin{aligned}
\text{Cash}_t &= \text{Cash}_{t-1} + \Delta\text{Cash}_t, \\
\text{Other Assets}_t &= \text{Other Assets}_{t-1} - \Delta\text{Asset}_t^{\text{CFS}}, \\
\text{Liabilities/Equity}_t &= \text{Liabilities/Equity}_{t-1} + \Delta\text{Liab/Equity}_t^{\text{CFS}}.
\end{aligned}$$

Account-by-account simulation mechanics. Applying the structure to specific accounts yields:

$$\begin{aligned}
\text{Inventory}_t &= \text{Inventory}_{t-1} - \Delta \text{Inventory}_t^{\text{CFS}}, \\
\text{AR}_t &= \text{AR}_{t-1} - \Delta \text{AR}_t^{\text{CFS}}, \\
\text{PP\&E}_t &= \text{PP\&E}_{t-1} - \text{CAPEX}_t - \text{Depreciation/Amortization}_t - \text{Asset Disposition Proceeds}_t, \\
\text{AP}_t &= \text{AP}_{t-1} + \Delta \text{AP}_t^{\text{CFS}}, \\
\text{Accrued Expenses}_t &= \text{Accrued Expenses}_{t-1} + \Delta \text{Accrued Expenses}_t^{\text{CFS}}, \\
\text{Long-term Debt}_t &= \text{Long-term Debt}_{t-1} + \Delta \text{Long-term Borrowings}_t^{\text{CFS}}, \\
\text{Retained Earnings}_t &= \text{Retained Earnings}_{t-1} + \text{Net Income}_t.
\end{aligned}$$

NYSF method: a disciplined balance-sheet reconciliation. When the balance sheet does not balance, a practical diagnostic is the “NYSF” method:

- Build a difference column: compute the change in every balance-sheet line between the first unbalanced period and the last balanced period.
- One-to-one matching: map each difference to the corresponding unique line in the cash flow statement for that period.
- Cross-off control: cross off each cash-flow line once it is matched to ensure complete and non-duplicative coverage.

This process typically surfaces four error classes:

- Unlinked cash-flow item: a cash-flow line is not linked anywhere on the balance sheet.
- Double-use: a cash-flow line is linked to the balance sheet more than once.
- Wrong sign or period: a link is reversed in sign or pointed to the wrong year.
- Summation error: an incorrect subtotal or total on either statement.

Chapter 7: Debt schedule, structure and purpose. The debt schedule tracks all major debt instruments, their interest, and repayment plans with three goals: (1) precisely roll forward each instrument from beginning balance through issuances/(retirements) to ending balance; (2) compute interest expense and interest income based on average balances and instrument rates, linking results to the income statement; and (3) govern cash dynamics by determining whether excess cash should be used for discretionary paydowns or whether shortfalls should be covered by discretionary borrowings.

Cash management within the debt schedule. Cash is managed via a structured set of bullet points that directly mirror the debt schedule’s logic:

- **Cash at beginning of year:** link directly from the prior period’s ending cash balance.
- **Cash flow before debt paydown:** a dedicated metric excluding all debt-related financing items; purpose is to measure cash generated by operating and investing activities before any discretionary debt actions.
 - Formula: $\text{CFO}_t + \text{CFI}_t + \text{FX Impact on Cash}_t$.
- **Minimum cash cushion:** a management assumption (and often a covenant requirement) representing the minimum cash level that must be retained; deduct this “locked” cash when computing availability.

- **Total Cash Available to Pay Down Debt:** the core decision variable for discretionary paydowns/draws.
 - Formula: $\text{CashAvail}_t = \text{Cash}_{t-1} + \text{CF}_t^{\text{pre-debt}} - \text{MinCash}_t$.
 - Interpretation: if positive, there is a cash surplus for non-mandatory repayments; if negative, there is a cash deficit requiring non-mandatory borrowings.
- **Cash at the end of the year:** summarizes all in-period cash movements including debt activity aggregated within the schedule.
 - Formula: $\text{Cash}_t = \text{Cash}_{t-1} + \text{CF}_t^{\text{pre-debt}} + \text{Total Issuances}/(\text{Retirements})_t$.

Debt mechanics: mandatory vs non-mandatory flows.

- **Mandatory issuances/(retirements):** changes that must occur under contractual terms (for example, scheduled principal amortization); typically hardcoded from the debt agreement.
- **Non-mandatory issuances/(retirements):** discretionary changes driven by cash availability; this component acts as the plug linked to CashAvail_t . Positive availability implies additional paydowns; negative availability implies draws (for example, revolver utilization).
- **Interest calculation:** use average balances to reflect intra-period changes.

$$- \text{Interest Expense}_t = r_{\text{debt}} \times \frac{\text{Debt}_{t-1} + \text{Debt}_t}{2}, \quad \text{Interest Income}_t = r_{\text{cash}} \times \frac{\text{Cash}_{t-1} + \text{Cash}_t}{2}.$$

Circular references: the feedback core.

- Paying down debt → lower average debt balance.
- Lower average debt balance → lower interest expense.
- Lower interest expense → higher pre-tax income and net income (income statement).
- Higher net income → higher cash from operations (cash flow statement).
- Higher CFO → higher cash available to pay down debt (debt schedule top section).
- Higher availability → more discretionary debt repayment, reinforcing the loop.

Closing the model: final links.

- Link total interest expense and interest income from the debt schedule back to the income statement.
- Link each instrument's issuances/(retirements) to the financing section of the cash flow statement.
- Reconcile ending cash computed in the debt schedule to the cash balance on the balance sheet.

Modeling constraints: methodology and structure. These constraints govern build order and structure to preserve logic, auditability, and balance:

- **Cash-flow-driven balance sheet:** forecast each balance-sheet line via its specific link to the statement of cash flows; avoid backing into cash from two forecast balance sheets.
- **Debt schedule comes last:** construct the debt schedule only after the balance sheet already balances, because it introduces circular references that complicate debugging.
- **CFO via the indirect method:** start from net income and adjust for non-cash items and working-capital changes to obtain cash from operations; maintain internal consistency across statements.

Modeling constraints: data and assumptions. These constraints reflect real-world data limitations and the need for disciplined assumptions:

- **Limits of depreciation forecasting:** precise forecasting would require asset-level details (cost, in-service date, life) that are rarely available; instead, calibrate a rough average life against historical PP&E and depreciation trends.
- **Ambiguity of “Other” lines:** “Other” in the income statement, cash-flow statement, and balance sheet aggregates heterogeneous sub-items; forecasting typically uses simplified approaches (for example, conservative method, prior-year carry, or cycle repeats) rather than granular drivers.
- **Hard-to-predict cash-flow items:** investing items such as acquisitions or marketable-securities activity depend on available cash and are difficult to predict; set to zero in the base case and revisit via scenarios.
- **Minimum cash balance assumption:** a key input in the debt schedule representing management risk tolerance and covenant discipline; it directly influences discretionary debt draws/repayments and interest.

3.2.5 Summary of current flaws in accounting simulation

A persistent issue is the overuse of fixed ratios. A common example is projecting cost of goods sold as a constant historical COGS-to-revenue ratio.

Another weakness is the reliance on a *plug* to close the balance sheet in influential references such as [Koller et al. \(2010\)](#) and [Benninga \(2014\)](#). Using cash, debt, or equity as an all-purpose balancing residual hides structural gaps, weakens causal interpretability, and complicates validation of cash-flow consistency across statements.

Relatedly, some workflows derive income-statement flows by first prescribing balance-sheet levels. For instance, [Koller et al. \(2010\)](#) often targets a PP&E path and then backs out CAPEX from the roll-forward. Although expedient, this reverses causality.

Under the indirect method for cash from operations, one must first determine changes in working capital such as accounts receivable and accounts payable. [Sawyer and Dalgleish \(2015\)](#) proceeds by computing next period’s inventory and then backing out the implied change in accounts payable. By contrast, [Koller et al. \(2010\)](#) obtains both AP and inventory via target ratios and then uses cash or debt as a plug. Despite starting from balance-sheet levels, Sawyer’s approach is more robust.

Finally, [Pignataro \(2022\)](#) advances two important disciplines. First, he rejects the use of plugs: cash is computed fully, after which short-term debt is repaid when there is a surplus and drawn when there is a shortfall, making financing a policy response rather than a balancing

artifact. Second, although future AP and inventory levels are forecasted rather than constructed dynamically, this structure yields the implied purchases via the identity

$$\text{Purchases}_t = (\text{Inventory}_t - \text{Inventory}_{t-1}) + \text{COGS}_t,$$

and then cash paid to suppliers is computed as

$$\text{Cash Paid}_t = \text{Purchases}_t - (\text{AP}_t - \text{AP}_{t-1}).$$

4 IFRS Dynamics

This section details the IFRS that forms the backbone of our financial simulation. We present a comprehensive set of dynamic equations and stock-flow constraints that govern the relationships between financial statement items. These formulations are grounded in International Financial Reporting Standards (IFRS) and are systematically organized by accounting area—from working capital to equity, to ensure a structurally sound model.

General Assumptions This document outlines the accounting dynamics based on a set of simplifying assumptions that apply globally unless otherwise specified in a subsection.

I. Basis of Accounting

- **Going Concern:** The model assumes the entity is a going concern, with financial statements prepared on that basis (IAS 1.25).
- **Measurement Basis:** All non-financial assets are measured at the cost model (e.g., IAS 16.30), avoiding the revaluation model. Financial instruments are measured at amortised cost, avoiding complex fair value hierarchy judgements (IFRS 13).
- **Reporting:** The statement of cash flows is prepared using the indirect method (IAS 7.18(b)). The profit and loss statement is prepared by classifying expenses by function (IAS 1.99).

II. Excluded Scenarios & Transactions

- **Macro Environment:** The model excludes the financial impact of foreign currency (FX) fluctuations (IAS 21), hyperinflationary environments (IAS 29), and climate-related risks.
- **Corporate Structure:** The model assumes no business combinations (IFRS 3), discontinued operations (IFRS 5), or loss of control events.
- **Complex Items:** The model excludes government grants (IAS 20), uncertain tax positions (IFRIC 23), most provisions (IAS 37), share-based payments (IFRS 2), and the capitalisation of borrowing costs (IAS 23.8).
- **Employee Benefits:** All post-employment benefit plans are treated as simple defined contribution plans (IAS 19.50-54); the complexities of defined benefit obligations are excluded.

4.1 Working-capital

WC 1 *Trade receivables*

Assumptions: Trade receivables are measured at amortised cost (IFRS 9.4.1.2). The model applies the IFRS 9 simplified approach for expected credit losses (ECL), where the loss allowance

is always measured at lifetime ECL (IFRS 9.5.5.15). It is assumed there is no significant financing component, as permitted for short-term receivables (IFRS 15.63), and that complexities from variable consideration like returns or rebates are excluded for simplicity (IFRS 15.50). Derecognition from transfers is limited to scenarios where substantially all risks and rewards are transferred (IFRS 9.3.2.6), excluding complex continuing involvement.

The roll-forward equations are as follows:

$$AR_t^{\text{gross}} = AR_{t-1}^{\text{gross}} + \text{CreditSales}_t - \text{CashCollected}_t - \text{WriteOff}_t - \text{TransDerecog}_t \quad (1)$$

$$\text{LossAllow}_t = \text{LossAllow}_{t-1} + \text{ImpairmentExp}_t - \text{WriteOff}_t - \text{Recover}_t \quad (2)$$

$$AR_t^{\text{net}} = AR_t^{\text{gross}} - \text{LossAllow}_t \quad (3)$$

where

- **WriteOff_t**: derecognise the *gross* receivable when there is no reasonable expectation of recovery. (IFRS 9.5.4.4)
- **Recover_t**: cash recovered on amounts previously written off; in this simplified model, recoveries are recorded as a movement in the loss allowance (no separate P&L credit), which is a permitted presentation (IFRS 7.35F(e)).
- **TransDerecog_t**: receivables sold/assigned that are derecognised because substantially all risks and rewards were transferred (e.g., true non-recourse). If there is continuing involvement (e.g., with recourse), do *not* derecognise—present a linked liability and give transfer disclosures. (IFRS 9.3.2.6; IFRS 7.42D)
- **LossAllow_t**: loss allowance under the ECL model; for amortised cost assets the allowance is deducted from the *gross* carrying amount, so $AR^{\text{net}} = AR^{\text{gross}} - \text{LossAllow}$. (IFRS 7.35H; plus IFRS 7 guidance that FVOCI debt is *not* reduced by the allowance)
- **CreditSales_t**: Revenue from contracts with customers recognised on credit terms during the period.
- **CashCollected_t**: Cash receipts from customers against outstanding trade receivables.
- **ImpairmentExp_t**: The charge to profit or loss for the period reflecting the change in the ECL allowance.

Note. Policy simplification: recoveries are captured in Recover_t within the allowance roll-forward and do not appear as a separate P&L credit; apply consistently (IFRS 7.35F(e)).

WC 2 Inventories and COGS

Assumptions: Inventories are measured at cost, which simplifies the lower of cost or net realisable value (NRV) principle by assuming cost is always lower than NRV (IAS 2.9); this excludes initial write-downs or subsequent reversals (IAS 2.33). The model assumes all inventories are finished goods, excluding raw materials and work-in-progress (IAS 2.6). The cost formula is a policy choice between FIFO or weighted-average (IAS 2.25). Complexities related to sales with right of return are also excluded (IFRS 15).

The key inventory and cost of sales dynamics are defined as follows:

$$\text{Inv}_t = \text{Inv}_{t-1} + \text{Purchases}_t - \text{COGS}_t \quad (4)$$

$$\text{COGS}_t = \text{Inv}_{t-1} + \text{Purchases}_t - \text{Inv}_t \quad (5)$$

where

- **Purchases_t**: Represents the cost of inventory acquired or produced...
- **COGS_t**: Represents the cost of inventories recognised as an expense...

Notes. Inventories are presented as a current asset on the statement of financial position (IAS 1.54(g)).

WC 3 Trade Payables & Prepayments

Assumptions: Trade payables are financial liabilities measured at amortised cost (IFRS 9.4.2.1). The model assumes no significant financing component given their short-term nature. Prepayments for goods are recognised as other current assets until the goods are received (IAS 1.66).

The dynamics of payables, prepayments, and total purchases are defined as follows:

$$AP_t = AP_{t-1} + \text{CreditPurchases}_t - \text{CashSettlements}_t \quad (6)$$

$$\text{Prepay}_t = \text{Prepay}_{t-1} + \text{Advances}_t - \text{PrepayApplied}_t \quad (7)$$

$$\text{Purchases}_t = \text{CreditPurchases}_t + \text{CODPurchases}_t + \text{PrepayApplied}_t \quad (8)$$

where

- **CreditPurchases_t**: Goods received on supplier credit, which increases trade payables.
- **CashSettlements_t**: Cash paid to suppliers to settle outstanding trade payables.
- **Advances_t**: Cash paid to suppliers *before* goods are received, which increases prepayments.
- **PrepayApplied_t**: The value of goods received that were settled by using a prepayment balance.
- **CODPurchases_t**: Cash-on-delivery purchases, where goods are paid for upon receipt.

Notes. Trade payables are presented under "Trade and other payables" (IAS 1.54(k)). Prepayments are presented under other current assets.

WC 4 Prepaid & Accrued Opex

Assumptions: This section models operating expenses (e.g., salaries, services) that are not inventoriable. Accrued expenses are recognised as liabilities based on the entity's obligation for services received but not yet paid. Prepaid expenses are recognised as assets for services paid for in advance. The model excludes complex liabilities like provisions (IAS 37).

The relationship between opex cash payments and the recognised accounting expense is as follows:

$$\text{OpexExp}_t = \text{AccruedOpexIncurred}_t + \text{PrepayOpexApplied}_t + \text{OpexCOD}_t \quad (9)$$

$$\text{AccruedOpex}_t = \text{AccruedOpex}_{t-1} + \text{AccruedOpexIncurred}_t - \text{CashSettledOnAccruals}_t \quad (10)$$

$$\text{PrepaidOpex}_t = \text{PrepaidOpex}_{t-1} + \text{OpexAdvances}_t - \text{PrepayOpexApplied}_t \quad (11)$$

where

- **OpexExp_t**: The total operating expense recognised in profit or loss for the period.

- **AccruedOpexIncurred_t**: Services received on credit, increasing accrued expenses.
- **PrepayOpexApplied_t**: Expenses recognised by consuming a prepayment balance.
- **OpexCOD_t**: Services paid for in cash in the same period they are received.
- **CashSettledOnAccruals_t**: Cash paid to settle expenses that were previously accrued.
- **OpexAdvances_t**: Cash paid for services before they are received, increasing prepaid expenses.

Notes. Accrued opex is presented within "Trade and other payables" (IAS 1.54(k)). Prepaid opex is presented within other current assets.

4.2 Long-lived Assets & Leases

LL 1 *Property, Plant and Equipment (PPE)*

Assumptions: The model uses the cost model for PPE (IAS 16.30), excluding the revaluation model. Depreciation is calculated on a straight-line basis. Additions are assumed to be available for use, and thus begin depreciating, in the subsequent period ($t + 1$), a policy simplification on IAS 16.55. The model initially excludes impairment (IAS 36) and capitalised borrowing costs (IAS 23). Right-of-use assets under IFRS 16 are accounted for separately.

The roll-forward dynamics for PPE are as follows:

$$\text{PPE}_t^{\text{gross}} = \text{PPE}_{t-1}^{\text{gross}} + \text{Capex}_t - \text{GrossDisposed}_t \quad (12)$$

$$\text{AccDep}_t = \text{AccDep}_{t-1} + \text{DepExp}_t - \text{AccDepDisposed}_t \quad (13)$$

$$\text{PPE}_t^{\text{net}} = \text{PPE}_t^{\text{gross}} - \text{AccDep}_t \quad (14)$$

$$\text{DepExp}_t = \frac{\text{PPE}_{t-1}^{\text{gross}} - \text{ResidualValue}_{t-1}}{\text{UsefulLife}} \quad (15)$$

$$\text{GainLossDisp}_t = \text{Proceeds}_t - (\text{GrossDisposed}_t - \text{AccDepDisposed}_t) \quad (16)$$

where

- **Capex_t**: Cost of assets acquired in the period that meet recognition criteria (IAS 16.7).
- **GrossDisposed_t**: The gross cost of assets derecognised upon disposal (IAS 16.67).
- **AccDepDisposed_t**: The accumulated depreciation of assets derecognised upon disposal (IAS 16.67).
- **Proceeds_t**: Cash received from the disposal of assets.
- **UsefulLife, ResidualValue**: Key estimates for calculating depreciation.

Note. The depreciation method, useful life, and residual value are subject to annual review, with any changes treated prospectively (IAS 16.51, 16.61). The gain or loss on disposal is recognised in profit or loss and is not classified as revenue (IAS 16.68).

LL 2 *Intangible Assets*

Assumptions: The model uses the cost model for intangible assets (IAS 38.74), excluding the revaluation model. The scope is limited to finite-lived intangibles amortised on a straight-line basis; indefinite-lived intangibles are excluded (IAS 38.107). Internally generated brands, customer lists, and similar items are not recognised as assets (IAS 38.63). The model initially

excludes impairment testing (IAS 36).

The dynamics for intangible assets, including the treatment of development costs, are as follows:

$$IA_t^{\text{gross}} = IA_{t-1}^{\text{gross}} + \text{AcqExternal}_t + \text{CapDev}_t - \text{GrossDisposed}_t \quad (17)$$

$$\text{AccAmort}_t = \text{AccAmort}_{t-1} + \text{AmortExp}_t - \text{AccAmortDisposed}_t \quad (18)$$

$$IA_t^{\text{net}} = IA_t^{\text{gross}} - \text{AccAmort}_t \quad (19)$$

$$\text{AmortExp}_t = \frac{IA_{t-1}^{\text{gross}} - \text{ResidualValue}_{t-1}}{\text{UsefulLife}} \quad (20)$$

$$\text{GainLossDisp}_t = \text{Proceeds}_t - (\text{GrossDisposed}_t - \text{AccAmortDisposed}_t) \quad (21)$$

$$\text{DevSpend}_t = \text{CapDev}_t + \text{DevExp}_t \quad (22)$$

where

- **DevSpend_t**: Total expenditure on development activities in the period.
- **CapDev_t**: The portion of development spend that is capitalised as an intangible asset.
- **DevExp_t**: The portion of development spend that is expensed to profit or loss.
- **AcqExternal_t**: Cost of separately acquired intangible assets in the period (IAS 38.24).
- **AmortExp_t**: Amortisation expense for the period.
- **GrossDisposed_t**: The gross cost of intangible assets derecognised upon disposal (IAS 38.113).
- **AccAmortDisposed_t**: The accumulated amortisation of assets derecognised (IAS 38.113).
- **Proceeds_t**: Cash received from the disposal of intangible assets.

Notes. Research expenditure is always recognised as an expense when incurred (IAS 38.54). Development expenditure is only capitalised if all six criteria in IAS 38.57 are met. The residual value of an intangible asset is assumed to be zero unless specific criteria are met (IAS 38.100).

LL 3 Leases (Lessee Accounting)

Assumptions: The model considers only lessee accounting under IFRS 16. It excludes complexities such as lease modifications, most variable lease payments, and residual value guarantees. The recognition exemptions for short-term leases (IFRS 16.5(a)) and leases of low-value assets (IFRS 16.5(b)) are assumed to be applied. The interest rate implicit in the lease is assumed not to be readily determinable; therefore, the lessee's incremental borrowing rate is used (IFRS 16.26).

The dynamics of the lease liability and right-of-use (ROU) asset are as follows:

$$\text{NewLeaseLiab}_t = \text{PV}(\text{Lease Payments}) \quad (23)$$

$$\text{NewROU}_t = \text{NewLeaseLiab}_t \quad (24)$$

$$\text{ROU}_t = \text{ROU}_{t-1} + \text{NewROU}_t - \text{ROUDep}_t \quad (25)$$

$$\text{LeaseLiab}_t = \text{LeaseLiab}_{t-1} + \text{NewLeaseLiab}_t + \text{LeaseInterestExp}_t \quad (26)$$

$$- \text{LeasePrincipalPaid}_t \quad (27)$$

$$\text{LeaseInterestExp}_t = r_t \cdot \text{LeaseLiab}_{t-1} \quad (28)$$

$$\text{ROUDep}_t = \frac{\text{ROU}_{t-1}}{\text{LeaseTermRemaining}} \quad (29)$$

where

- **NewLeaseLiab_t**: The present value of lease payments for a new lease commencing in the period.
- **NewROU_t**: The initial value of the right-of-use asset, measured at an amount equal to the initial lease liability (a simplification of IFRS 16.24).
- **LeasePrincipalPaid_t**: The portion of the lease payment that reduces the lease liability.
- **LeaseInterestExp_t**: The interest expense on the lease liability for the period.
- **ROUDep_t**: Depreciation expense on the ROU asset.
- **r_t**: The discount rate for the lease, which is the lessee's incremental borrowing rate.

Notes. The principal portion of lease payments is classified as a financing cash flow. The interest portion follows the entity's policy for interest paid (IFRS 16.50). ROU assets may be presented within PPE or as a separate line item (IFRS 16.47).

4.3 Financing & Financial Instruments

FF 1 *Financial Assets*

Assumptions: To generate finance income, the model includes a simple class of financial assets measured at amortised cost, which dictates the use of the effective interest method for income recognition. The model excludes impairment on these assets for simplicity.

The dynamics are as follows:

$$\begin{aligned} \text{FinAsset}_t &= \text{FinAsset}_{t-1} + \text{NewInvestments}_t - \text{Maturities}_t \\ &\quad + \text{FinanceIncome}_t - \text{InterestReceived}_t \end{aligned} \quad (30)$$

$$\text{FinanceIncome}_t = \text{EIR}_{\text{asset}} \cdot \text{FinAsset}_{t-1} \quad (31)$$

where

- **FinAsset_t**: The carrying amount of interest-bearing financial assets.
- **NewInvestments_t**: Cash used to acquire new financial assets.
- **Maturities_t**: Proceeds from financial assets that have matured or been sold.
- **InterestReceived_t**: Cash interest received from financial assets during the period.
- **EIR_{asset}**: The effective interest rate on the portfolio of financial assets.

FF 2 *Borrowings*

Assumptions: Borrowings are financial liabilities measured at amortised cost using the effective interest method (IFRS 9.4.2.1). The model excludes complex scenarios such as debt modifications or extinguishments (IFRS 9.3.3). Transaction costs are included in the initial carrying amount and amortised via the effective interest rate (EIR). Lease liabilities are excluded as they are treated separately under IFRS 16.

The dynamics of the carrying amount of debt are as follows:

$$\text{NewBorrowCA}_t = \text{Proceeds}_t - \text{DebtIssueCosts}_t \quad (32)$$

$$\begin{aligned} \text{DebtCA}_t &= \text{DebtCA}_{t-1} + \text{NewBorrowCA}_t + \text{InterestExp}_t \\ &\quad - \text{InterestPaid}_t - \text{PrincipalRepaid}_t \end{aligned} \quad (33)$$

$$\text{InterestExp}_t = \text{EIR} \cdot \text{DebtCA}_{t-1} \quad (34)$$

where

- **Proceeds_t**: Cash received from issuing new debt instruments.
- **DebtIssueCosts_t**: Transaction costs directly attributable to the new debt.
- **InterestPaid_t**: Cash interest paid during the period.
- **PrincipalRepaid_t**: Contractual repayments of principal.
- **EIR**: The effective interest rate; the rate that exactly discounts estimated future cash payments or receipts through the expected life of the financial instrument.

Notes. For presentation on the statement of financial position, the carrying amount (DebtCA_t) must be split between current and non-current portions (IAS 1.69). The model currently excludes the detailed mechanics of covenant breaches and their classification effects.

4.4 Taxes

T 1 *Income Taxes*

Assumptions: The model assumes all income taxes are recognised in profit or loss (IAS 12.58). Deferred taxes are excluded. Complexities from uncertain tax positions are also excluded.

The dynamics for current taxes are as follows:

$$\text{TaxExp}_t = \text{CurrTaxExp}_t \quad (35)$$

$$\text{CurrTaxPay}_t = \text{CurrTaxPay}_{t-1} + \text{CurrTaxExp}_t - \text{IncomeTaxesPaid}_t \quad (36)$$

where

- **TaxExp_t**: The total income tax expense for the period (equals current tax only).
- **CurrTaxExp_t** = $\tau \cdot \text{PBT}_t$: Current tax expense at statutory rate τ .
- **IncomeTaxesPaid_t**: Cash paid for income taxes during the period.
- **CurrTaxPay_t**: Current tax payable at period end.

Notes. Deferred tax assets and liabilities, and timing differences, are not modelled in this simplified version.

4.5 Equity & OCI

E 1 *Equity*

Assumptions: The model assumes no changes in ownership that result in a loss of control. All transactions are with owners in their capacity as owners. The model excludes complexities from cash-settled share-based payments (IFRS 2) and certain treasury share transactions.

The roll-forward dynamics for the primary components of equity are as follows:

$$\text{RE}_t = \text{RE}_{t-1} + \text{Profit}_t - \text{DividendsToOwners}_t \pm \text{TransfersToFromRE}_t \quad (37)$$

$$\text{ContribEq}_t = \text{ContribEq}_{t-1} + \text{ShareIssues}_t - \text{Buybacks}_t \quad (38)$$

$$\text{Res}_t^{(k)} = \text{Res}_{t-1}^{(k)} + \text{OCIIn}_t^{(k)} - \text{OCIReclassToPL}_t^{(k)} \quad (39)$$

$$\text{NCI}_t = \text{NCI}_{t-1} + \text{NCIShareOfProfit}_t + \text{NCIShareOfOCI}_t - \text{DividendsToNCI}_t \quad (40)$$

where

- **RE_t**: Retained Earnings at period end.
- **ContribEq_t**: Contributed Equity (e.g., share capital, share premium).
- **Res_t^(k)**: The closing balance of a specific Other Comprehensive Income (OCI) reserve.
- **NCI_t**: The closing balance of Non-Controlling Interests.
- **Profit_t**: Profit or loss for the period attributable to the owners of the parent.
- **ShareIssues_t**: Proceeds from the issuance of new shares.
- **Buybacks_t**: Cost of shares repurchased from owners.
- **OCIIn_t^(k)**: OCI generated during the period for reserve k .
- **OCIREclassToPL_t^(k)**: OCI from reserve k that is reclassified ('recycled') to profit or loss.

Notes. A full Statement of Changes in Equity would be prepared based on these roll-forwards (IAS 1.106). Total equity is the sum of its components: ContribEq + RE + $\sum_k \text{Res}^{(k)}$ + NCI.

4.6 Statement of Profit or Loss

Assumptions: The statement is prepared using the 'function of expense' method (IAS 1.99). The model excludes discontinued operations (IFRS 5).

For clarity, certain P&L lines are aggregated using the following building blocks:

$$\text{DA}_t = \text{DepExp}_t + \text{ROUDep}_t + \text{AmortExp}_t \quad (41)$$

$$\text{FinanceCosts}_t = \text{InterestExp}_t + \text{LeaseInterestExp}_t \quad (42)$$

The calculation of profit for the period is then as follows:

$$\begin{aligned} \text{PBT}_t &= \text{Revenue}_t - \text{COGS}_t - \text{OpexExp}_t - \text{DA}_t - \text{ImpairmentECL}_t \\ &\quad + \text{OtherIncome}_t + \text{FinanceIncome}_t - \text{FinanceCosts}_t \end{aligned} \quad (43)$$

$$\text{Profit}_t = \text{PBT}_t - \text{TaxExp}_t \quad (44)$$

where

- **Revenue_t**: Revenue from contracts with customers, per IFRS 15.
- **COGS_t**: Cost of Goods Sold, derived from the inventory dynamics.
- **OpexExp_t**: Operating expenses from services and other period costs.
- **DA_t**: Total depreciation and amortisation, aggregating the expense from PPE, ROU assets, and Intangibles.
- **ImpairmentExp_t**: Total impairment losses. For the current model, this only comprises the Expected Credit Loss expense on trade receivables (ImpairmentECL_t).
- **FinanceCosts_t**: Includes interest on borrowings and leases.
- **TaxExp_t**: Total tax expense for the period, from the tax calculations.
- **PBT_t**: Profit Before Tax.
- **Profit_t**: Profit for the period, the bottom line of the statement.

Notes. An entity may present additional line items or use different descriptions if relevant to an understanding of its financial performance (IAS 1.85). Presenting expenses by function requires allocating costs like depreciation to functional lines (e.g., COGS), though they are shown separately in the equation above for clarity.

4.7 Statement of Cash Flows

Assumptions: The statement is prepared using the indirect method for operating activities (IAS 7.18). The entity classifies interest paid and received as operating cash flows, and dividends paid as financing cash flows (IAS 7.31).

Cash Flow Building Blocks To ensure clarity, certain cash flow items are aggregations of items from other sections:

$$\text{GainLossDisp}_t = \text{GainLossDisp}_t^{\text{PPE}} + \text{GainLossDisp}_t^{\text{IA}} \quad (45)$$

$$\text{Proceeds}_t^{\text{Disposal}} = \text{Proceeds}_t^{\text{PPE}} + \text{Proceeds}_t^{\text{IA}} \quad (46)$$

$$\text{Prepayments}_t = \text{Prepay}_t + \text{PrepaidOpex}_t \quad (47)$$

Cash Flow from Operations (CFO)

$$\begin{aligned} \text{CFO}_t = & \text{Profit}_t + \text{TaxExp}_t - \text{IncomeTaxesPaid}_t \\ & + \text{DA}_t + \text{ImpairmentExp}_t - \text{GainLossDisp}_t \\ & - \Delta \text{AR}_t^{\text{gross}} - \text{WriteOff}_t + \text{Recover}_t \\ & - \Delta \text{Inv}_t - \Delta \text{Prepayments}_t + \Delta \text{AP}_t + \Delta \text{AccruedOpex}_t \\ & + \text{InterestExp}_t + \text{LeaseInterestExp}_t - \text{FinanceIncome}_t \\ & - \text{InterestPaid}_t - \text{LeaseInterestPaid}_t + \text{InterestReceived}_t \end{aligned} \quad (48)$$

Cash Flow from Investing (CFI)

$$\begin{aligned} \text{CFI}_t = & - \text{Capex}_t - \text{AcqExternal}_t - \text{CapDev}_t + \text{Proceeds}_t^{\text{Disposal}} \\ & - \text{NewInvestments}_t + \text{Maturities}_t \end{aligned} \quad (49)$$

Cash Flow from Financing (CFF)

$$\begin{aligned} \text{CFF}_t = & \text{Proceeds}_t^{\text{Debt}} - \text{PrincipalRepaid}_t - \text{DebtIssueCosts}_t \\ & - \text{LeasePrincipalPaid}_t \\ & + \text{ShareIssues}_t - \text{Buybacks}_t - \text{DividendsToOwners}_t - \text{DividendsToNCI}_t \end{aligned} \quad (50)$$

Net Change in Cash

$$\Delta \text{Cash}_t = \text{CFO}_t + \text{CFI}_t + \text{CFF}_t \quad (51)$$

Notes. All variables in these equations are now either primary inputs or are explicitly calculated in a preceding section. This ensures the Statement of Cash Flows fully reconciles the Income Statement with the Statement of Financial Position.

4.8 Statement of Financial Position

The following table presents the structure of a complete Statement of Financial Position based on IFRS presentation standards. It is populated with the variables from our simulation model where applicable. Items that are not yet included in our model are left blank, providing a clear view of the model's current scope and areas for future development. Detailed in Figure 4.8.

4.9 Constraints

To ensure accounting realism, the following inequalities should hold in each period t :

- $\text{AR}_t^{\text{net}} \geq 0$ (net trade receivables cannot be negative)
- $\text{Inv}_t \geq 0$ (inventories are non-negative)

Entity	Note / Variable
Assets	
Non-current assets	
Property, plant and equipment	$PPE_t^{\text{net}} + ROU_t$
Intangible assets and goodwill	IA_t^{net}
Investment property	
Equity-accounted investees	
Other investments, including derivatives	$FinAsset_t$
Biological assets	
Employee benefits	
Deferred tax assets	
Total non-current assets	
Current assets	
Inventories	Inv_t
Trade and other receivables	AR_t^{net}
Contract assets	
Prepayments	$Prepay_t + PrepaidOpex_t$
Other investments, including derivatives	
Biological assets	
Current tax assets	
Assets held for sale	
Cash and cash equivalents	$Cash_t$
Total current assets	
Total assets	
Equity and liabilities	
Equity	
Share capital and premium	$ContribEq_t$
Reserves	$\sum_k Res_t^{(k)}$
Retained earnings	RE_t
Equity attributable to owners	
Non-controlling interests	NCI_t
Total equity	
Non-current liabilities	
Loans and borrowings	Non-current part of $DebtCA_t$
Lease liabilities	Non-current part of $LeaseLiab_t$
Provisions	
Deferred income	
Employee benefits	
Trade and other payables	
Deferred tax liabilities	
Total non-current liabilities	
Current liabilities	
Loans and borrowings	Current part of $DebtCA_t$
Lease liabilities	Current part of $LeaseLiab_t$
Bank overdraft	
Trade and other payables	$AP_t + AccruedOpex_t$
Contract liabilities	
Provisions	
Deferred income	
Employee benefits	
Current tax payable	$CurrTaxPay_t$
Liabilities assoc. with assets held for sale	
Total current liabilities	
Total liabilities	
Total equity and liabilities	

Table 1: Statement of Financial Position

- $PPE_t^{\text{net}} \geq 0$, $IA_t^{\text{net}} \geq 0$, $ROU_t \geq 0$ (long-lived assets are non-negative)
- $AP_t \geq 0$, $\text{AccruedOpex}_t \geq 0$, $\text{CurrTaxPay}_t \geq 0$ (liabilities cannot be negative)
- $\text{Prepay}_t \geq 0$, $\text{PrepaidOpex}_t \geq 0$ (prepayments are non-negative assets)
- $\text{DividendsToOwners}_t \leq \text{RE}_{t-1} + \text{Profit}_t$ (dividends cannot exceed distributable profits)
- $\text{DividendsToOwners}_t \leq \text{Cash}_{t-1} + \text{CFO}_t + \text{CFI}_t + \text{CFF}_t$ (dividends cannot exceed available cash)
- $\text{DepExp}_t \leq PPE_{t-1}^{\text{gross}}$ (depreciation cannot exceed gross PPE)
- $\text{AmortExp}_t \leq IA_{t-1}^{\text{gross}}$ (amortisation cannot exceed gross intangibles)
- $\text{ROUDep}_t \leq \text{ROU}_{t-1}$ (ROU depreciation bounded by asset)
- $\text{LeaseInterestExp}_t \leq r_t \cdot \text{LeaseLiab}_{t-1}$ (interest consistent with liability)
- $\text{Purchases}_t \geq 0$, $\text{OpexExp}_t \geq 0$, $\text{COGS}_t \geq 0$ (expenses non-negative)
- $\text{Equity}_t \geq 0$ (no negative equity in this simplified model)

Notes. These constraints are not strict IFRS requirements but practical checks to ensure the simulation generates plausible and internally consistent outcomes.

5 Corporate Finance

5.1 Overview: The Evolution of Corporate Valuation Theory — From Static to Dynamic Stochastic Frameworks

The valuation of leveraged firms is a central issue in corporate finance, with its complexity in accurately quantifying the impact of capital structure (i.e., the mix of debt and equity) on firm value. Since the inception of modern corporate finance theory, this issue has been a focal point for scholars, driving the continuous evolution of theoretical paradigms from idealized static models to more realistic dynamic stochastic models.

In this literature review I begin with the seminal Modigliani-Miller (MM) theorem, analyzing its core arguments in perfect and tax-inclusive markets. I also explore [Myers \(1974\)](#) Adjusted Present Value (APV) method as a bridge between theory and practice, comparing it with the classical Weighted Average Cost of Capital (WACC) approach. Then, I explore literatures that take firms' investment opportunity as real options and connect the value of such option and firms' financing decision. Next, I delve into dynamic corporate finance models, which incorporate adjustment costs and endogenous decision-making to explain firm behavior in reality. Finally, I will introduce the core of modern asset pricing—the Stochastic Discount Factor (SDF) framework—demonstrating how it provides a unified, rigorous theoretical foundation to integrate previous valuation methods and offers a broad perspective for future research.

5.2 Modigliani and Miller's theorem, and its extensions

[Modigliani and Miller \(1958\)](#) published their seminal paper, *The Cost of Capital, Corporation Finance, and the Theory of Investment*, widely regarded as the foundation of modern corporate finance. The core idea of their theory is that, under certain simplifying assumptions, a firm's capital structure does not affect its value. In [Modigliani and Miller \(1963\)](#), they refined their theory to account for the impact of corporate income taxes, laying a solid foundation for modern capital structure theory.

In the 1958 paper, M&M developed their arguments under the assumption of a “perfect capital market.” These assumptions include: no corporate or personal income taxes, no transaction costs, no bankruptcy costs, equal borrowing rates for individuals and firms, perfect information symmetry, and no agency costs. In this idealized, frictionless world, M&M proposed two core propositions.

First, M&M Proposition I states that a firm’s market value is independent of its capital structure. The total value of a firm is determined solely by the expected returns and risks of its underlying assets, not by how it is financed through debt or equity. This argument rests on the “arbitrage process”: if a leveraged firm’s total value exceeds that of an unleveraged firm with identical risk and operating income, rational investors can sell the leveraged firm’s stock and use personal borrowing (i.e., “personal leverage”) to buy the unleveraged firm’s stock, achieving higher returns without increased risk. This arbitrage continues until the two firms’ values equalize. Conversely, if the unleveraged firm is valued higher, investors can profit by “deleveraging” investments in the unleveraged firm. Thus, under M&M’s framework, altering capital structure cannot increase firm value.

Second, M&M Proposition II states that, in a no-tax world, the cost of equity rises linearly with leverage. As a firm increases debt, shareholders face greater financial risk, demanding a higher return on equity to compensate. This increase in equity costs exactly offsets the benefits of using relatively cheaper debt, keeping the firm’s weighted average cost of capital (WACC) constant and independent of capital structure.

M&M later introduced the significant market friction of corporate income taxes. Since interest on debt is typically tax-deductible, it creates a “tax shield.” The value of a leveraged firm equals the value of an unleveraged firm plus the present value of the tax shield. This is commonly expressed as:

$$V_t^L = V_t^U + \frac{r\tau D_t}{r} = V_t^U + \tau D_t$$

where τ is the tax rate.

This conclusion overturns the “capital structure irrelevance” principle, suggesting firms should maximize debt to maximize value. While the tax-adjusted M&M theory successfully explains the value-creating effect of leverage, its implication of “100% debt as optimal” is clearly at odds with reality, as firms’ leverage ratios fall far short of this extreme which is stated in [Modigliani and Miller \(1963\)](#). This disconnect between theory and practice prompted researchers to critically examine M&M’s assumptions, leading to subsequent theories.

5.3 Adjusted Present Value

5.3.1 Introduction of APV

The Adjusted Present Value (APV) method, proposed by [Myers \(1974\)](#), is a significant valuation approach that is highly consistent with the M&M tax-adjusted framework and provides a more practical tool for application.

The core idea of the APV method is “valuation-by-components,” which breaks down a firm’s total value into two main parts: first, estimating the firm’s operating value without leverage; second, explicitly adding or subtracting the “financing side effects” brought by leverage. The basic formula for the APV method is:

$$V_t^L = V_t^U + PV(TaxShield) - PV(DebtCost)$$

In words, APV = Unlevered Firm Value + Present Value of Debt Tax Shield – Present Value of Financial Distress Costs from issuing more debts. The valuation process of the APV method typically involves the following three steps:

1. Estimating the Unlevered Firm Value: This step involves discounting the firm's expected Free Cash Flow to the Firm (FCFF) using the "unlevered cost of equity" (r_t^u). The unlevered cost of equity represents the business risk of a firm financed entirely by equity and can be calculated using the unlevered beta (β^u), which is derived from the firm's current equity beta, tax rate, and debt-to-equity (D/E) ratio.

2. Estimating the Present Value of the Debt Tax Shield: This step calculates the cash flow benefits from the tax deductibility of interest payments. The interest tax shield is computed as: Interest Expense $\times$ Tax Rate.

3. Estimating the Present Value of Financial Distress Costs: Theoretically, this step involves assessing the increased probability of bankruptcy due to high leverage and the present value of its expected costs. Bankruptcy costs include direct costs (e.g., legal fees) and indirect costs (e.g., loss of customers, deterioration of supplier relationships).

5.3.2 Valuation of taxshield

However, similar to the case of MM, one core issue of the APV method—the appropriate discount rate for the tax shield—remains an unresolved "open question." Traditionally, the tax shield is assumed to carry the same risk as debt and is thus discounted at the pre-tax cost of debt (r_t^d). However, this assumption is not without controversy, as the realization of the tax shield depends on the firm's profitability.

[Fernández \(2004\)](#) argues that value of tax shield is NOT the present value of tax shield, because it is difficult to evaluate which . Instead, it is the difference between two values:

$$VTS = G_u - G_L$$

where G_u is the tax paid by an unlevered firm and G_L is the tax paid by a levered firm. G_u and G_L has different risk so they are discounted at different costs. Therefore, we cannot simply assume

$$VTS = PV(r\tau D)$$

and the methods of [Modigliani and Miller \(1963\)](#), [Myers \(1974\)](#), and [Miles and Ezzell \(1980\)](#) are incorrect.

[Cooper and Nyborg \(2006\)](#) on the other hand argues the value of tax shields IS equal to the present value of tax shields by proving that Fernandez's derivation is incorrect, for Fernandez assumes constant debt for some equations while assume constant debt to value ratio for others. If the mistake is modified, even we still have

$$VTS = G_u - G_L$$

where G_u and G_L are discount at different rate, it can be derived that VTS is a present value of future stream of tax shield $\tau r D$, at a given discount rate.

Moreover, if we take uncertainty into consideration, the value of the tax shield is dependent on the character of the stochastic process of the debt increase and cost of debt ([Valaskova and Bakes, 2017](#)).

Also, [Qi and Han \(2010\)](#) point out, if a firm lacks sufficient taxable income, the tax shield cannot be fully realized. This theoretical uncertainty highlights the stark contrast between the APV method's strength as a theoretical tool and its limitations in precise practical application.

To address these issues, I will introduce stochastic discount factor (SDF) in later sections, before which I will compare WACC and APV, and discuss literatures that further develop the corporate financing models into stochastic dynamic frameworks.

5.4 Comparison between WACC and APV

5.4.1 Weighted Average Cost of Capital (WACC)

Before comparing WACC and APV, I will briefly introduce WACC and its weakness, and why APV is a better method.

To value a firm, we can discount the free cash flow to firms through a combined, single discount rate, the weighted average cost of capital. Specifically,

$$V_t^L = \sum_{i=1} \left(\frac{FCFF_{t+i}}{\prod_{j=1}^i (1 + WACC_{t+j})} \right)$$

where

$$FCFF_t = EBIT \times (1 - \tau) + D\&A_t - CapEx_t - \Delta WC_t$$
$$WACC_{t+1} = \frac{S_t}{V_t^L} \times r_{t+1}^e + \frac{D_t}{V_t^L} \times r_{t+1}^d \times (1 - \tau)$$

The WACC method, one of the most commonly used tools in firm valuation, is based on the core idea of calculating the weighted average cost of all capital sources (including debt and equity) and using it as a single discount rate to estimate the overall value of the firm. The WACC formula is derived by taking a weighted average of the cost of debt (adjusted for taxes) and the cost of equity r_t^e , typically estimated using the Capital Asset Pricing Model, CAPM), based on their respective market value weights in the capital structure. This method assumes that the firm's value equals the present value of its future Free Cash Flows (FCF) discounted at the WACC. The distinctive feature of WACC is that it implicitly incorporates the value of the tax shield from debt financing into the discount rate by reducing the cost of debt to reflect this financial benefit.

The simplicity of the WACC method makes it widely popular in practice, but this simplicity comes with a prerequisite: it relies on the core assumption that the firm maintains a constant target capital structure (debt-to-equity ratio). This assumption implies that the firm continuously adjusts its debt and equity proportions based on their market values to maintain this target ratio. Under this assumption, WACC can be treated as a constant discount rate, simplifying valuation calculations.

However, this core assumption is one of the WACC method's limitations. In reality, especially in certain valuation scenarios, a firm's capital structure is not static.

Of course, modern theory allows for the possibility that the firm does not maintain a fixed debt-to-equity ratio in the WACC framework but can adjust it over time. However, this introduces a circularity problem. In the WACC formula, the weights S_t/V_t^L and D_t/V_t^L require the market values of equity and debt to be determined. Yet, the firm's value $V_t^L = S_t + D_t$ is precisely what we aim to estimate by discounting future cash flows with WACC. This creates a classic "chicken-and-egg" paradox: to calculate WACC, we need the firm's value; but to calculate the firm's value, we need WACC.

To address the circularity problem in WACC valuation, both academia and practice have proposed various solutions. The most mainstream and practical approach is the iterative method, which approximates a balanced solution through repeated calculations. This solution has been widely discussed and formalized in modern financial literature. [Velez-Pareja and Benavides-Franco \(2006\)](#) explored the circularity issue in WACC in depth, proposing both analytical and iterative solutions. They explicitly noted that, while rigorous analytical solutions exist, the simplest practical approach is to leverage the iterative calculation capabilities of modern spreadsheet software (e.g., Excel). By enabling this feature, the software can iteratively calculate WACC and firm value within a preset number of iterations until the difference between successive results falls below a certain threshold, yielding a convergent solution.

5.4.2 Compare WACC and APV

Miles and Ezzell (1980) proved the equivalence between WACC and APV under assumptions of perfect capital market, constant discount rate and constant leverage ratio.

If any of the assumptions above is deviated, WACC and APV would no longer be equal, and many scholars hold an opposite view toward WACC. For example Luehrman (1997) discussed that WACC is obsolete, although it is still taught in standard textbooks, while APV can be applied to value operations and each asset in place, because WACC only considered the effect from tax shield but neglect other side effects. Further, APV does not involve a presumption of debt structure, or there is no need to solve for circularity.

In Hillier et al. (2010), the flexibility of APV is further exhibited with an example of the application of APV in leveraged buyouts. In LBOs, the acquirer typically uses substantial borrowing to complete the transaction, then repays the debt with the company's future cash flows, leading to a rapid decline in debt levels over time. In such cases, the WACC method becomes unsuitable due to its assumption of a constant capital structure, unless addressed through complex annual recalculations of WACC.

In contrast, the APV method is well-suited for such situations. It accurately handles changing debt levels by separately calculating and discounting tax shields for each year. For example, in the RJR Nabisco LBO case, the APV method was used to analyze KKR's acquisition strategy.

5.4.3 Discount rate for tax shield in WACC and APV

In the WACC formula, the cost of debt R_t^d is multiplied by $(1 - \tau)$ to obtain the after-tax cost of debt. This mathematical treatment is not arbitrary; it implicitly incorporates the present value of the tax shield into the discount rate, assuming that the risk of the tax shield is the same as that of the debt. Therefore, the implied discount rate for the tax shield in the WACC method is the pre-tax cost of debt R_t^d , as in Vélez-Pareja and Tham (2009).

This implicit treatment is a key drawback of the WACC method, as it lacks flexibility. It simplifies a complex and contentious theoretical issue—the true risk of the tax shield—through a fixed formulaic assumption. Analysts cannot adjust the assumption about the tax shield's risk based on a firm's specific circumstances, such as when operating risk is high or taxable income is uncertain. The WACC method rigidly uses the cost of debt as its implied discount rate, ignoring the possibility that the tax shield's cash flows may vary with the firm's operating risk.

What is the appropriate discount rate for the interest tax shield in the APV method? The literature notes that this remains “largely an unresolved issue.” Academic debate persists without consensus: Myers (1974) and others argue that the tax shield's risk matches that of the firm's debt, so the cost of debt R_t^d , should be used as the discount rate. In contrast, Harris and Pringle (1985) and others contend that the tax shield's risk aligns with the firm's operating risk, suggesting the unlevered cost of capital R_t^u as the discount rate. Cooper and Nyborg (2006) attempt to reconcile these views, proposing that the tax shield's discount rate depends on the firm's capital structure policy—whether it maintains a fixed debt amount or a fixed leverage ratio.

This flexibility is precisely the strength of the APV model. It allows analysts to model more precisely based on their unique judgment of the tax shield's risk. For example, for mature companies with stable profits and cash flows, the tax shield is considered relatively certain, and a lower discount rate close to the cost of debt can be used. However, for high-growth firms with significant operational uncertainty and volatile profits, the realization of the tax shield carries greater risk, and analysts can choose a higher discount rate that aligns with the operational risk.

5.5 Debt cost, investment opportunities and call options

A third method, closely related to APV, is to add a firm's investment opportunity as part of firm's value and price it as an option.

Modigliani and Miller (1963) noted in their paper that, despite the tax shield advantages of debt, firms still maintain a "reserve borrowing capacity." This can be explained by the cost of debt in the APV framework, with one of the most significant costs being the loss of investment opportunity value, which resembles a real option. Merton (1974) applied option pricing theory to corporate finance, assuming the levered firm value is known, treating equity as a call option, and pricing risky bonds. Myers (1977), on the other hand, directly applied option pricing theory to corporate assets, particularly valuing growth opportunities, still under the assumption of complete and perfect market. Specifically, he assumes that value of a firm V_t^L equals value of asset-in-place V_t^A and present value of future growth opportunity V_t^G such that

$$V_t^L = V_t^A + V_t^G$$

where V_t^G can be deemed as real option. The manager can choose to invest in it if the expected return will exceed the investment cost, otherwise he will simply give up this option. Therefore, part of the value of a firm is accounted for by the present value of options to make further investments on possibly favorable terms.

The firm with risky debt outstanding, and which acts in its stockholders' interest, will follow a different decision rule than one which issues no debt at all. The firm financed with risky debt will give up valuable investment opportunities which could make a positive net contribution to the market value of the firm.

Issuing risky debt reduces the present market value of the firm by inducing a future strategy that is suboptimal in the sense just described.

A conclusion from this paper is that asset-in-place V_t^A should be more financed by debt, so that capital-intensive firms are with high operating leverage, while V_t^G is inversely related to the optimal debt level.

Philippon and Sannikov (2007) extend and inherit from Myers (1977)'s work. In their continuous-time dynamic model, growth opportunities are also regarded as options (e.g., the option to upgrade capital), but moral hazard is incorporated: the manager (agent) may shirk or divert funds, leading to delayed investment (the waiting option). Debt (or agency conflicts) amplifies this issue, similar to Myers' option distortion. Both papers predict that firms with a high proportion of growth opportunities (e.g., young high-tech firms) borrow less, as debt reduces the value of these options.

5.6 Dynamic Framework

5.6.1 Literatures of Dynamic Framework

While the Modigliani-Miller (MM) theorem and subsequent Adjusted Present Value (APV) models provide valuable insights into capital structure, their static frameworks fail to capture the dynamic complexities of real-world financial decisions, necessitating the adoption of dynamic models.

Mauer and Triantis (1994) employ a stochastic dynamic framework to analyze corporate investment and financing decisions, achieving several notable advancements. First, their model endogenizes both investment and financing decisions, allowing these choices to interact dynamically within the model. Second, it incorporates operational uncertainty, with future sales prices modeled as a stochastic process, and firms forming expectations about this uncertainty, which in turn influences their financing behavior. Additionally, the study preserves the option-like nature of corporate investment, as posited by Myers (1977), highlighting that unlevered firms, despite lacking tax shield benefits, can gain from delaying investment. Rather than pursuing

a closed-form solution, Mauer and Triantis adopt a numerical approach to solve the model, enabling greater flexibility in addressing complex dynamics. A notable limitation, however, is the subjective nature of adjustment costs, which may undermine the robustness of the results.

Leland (1994) also utilizes a stochastic dynamic model to examine optimal capital structure and debt valuation under the possibility of bankruptcy. The model assumes that asset values follow a diffusion process and reformulates the debt problem into a time-independent framework, enabling the derivation of closed-form solutions for optimal capital structure and debt value. However, this approach assumes that a firm's operating activities are independent of its financial structure and that the capital structure remains constant.

Building on Leland (1994), Leland and Toft (1996) extends the model to include finite-maturity debt and allows firms to continuously issue new debt to maintain a stationary debt structure. This study endogenizes the bankruptcy trigger value V^B , determined optimally by equity holders along a stochastic path using the smooth-pasting condition, in contrast to static models (e.g., MM or APV) that assume exogenous bankruptcy triggers (such as zero net worth or a fixed threshold).

Specifically, Leland and Toft (1996) demonstrates that the bankruptcy trigger value V^B is contingent on debt maturity T : under short-term debt, V^B may exceed the principal, potentially triggering bankruptcy even with positive net worth, as expected equity appreciation may be insufficient to cover debt servicing costs; under long-term debt, V^B may fall below the principle of debt, permitting operations with negative net worth. This dynamic framework captures realistic scenarios where cash flow shortages do not necessarily lead to immediate bankruptcy, as equity holders may inject funds to sustain operations. Through this approach, Leland and Toft (1996) provides a more nuanced theoretical framework for understanding dynamic capital structures and bankruptcy mechanisms, though they remain less comprehensive than the model developed by Mauer and Triantis (1994).

5.6.2 Tutorials

What we want is a model that can value V_t^L that is under a stochastic dynamic environment, holds the spirit of the additivity in APV, incorporate the value of real option of a firm, and is a result of optimal control problem. To fully get familiar with these fields, two tutorial books serve as reference. One is Dynamic Corporate Finance theory by Flor (2012). The second is Dynamic Models and Structural Estimation in Corporate Finance by Strebulaev et al. (2012). The paper by Strebulaev et al. (2012) provides a comprehensive review of the dynamic corporate finance literature over the past two decades, focusing on capital structure and investment-financing decisions. The paper covers several key areas. It begins with **Dynamic Contingent Claims Models**, which use option-pricing methods to value firms by treating their cash flows as derivatives. This section explores the value of **real options** in investment decisions, explaining why firms' investment thresholds often exceed NPV-based rules. It then applies this framework to **optimal capital structure**, analyzing decisions under both endogenous and exogenous default scenarios, and extends this to **dynamic settings** where firms can adjust leverage over time. Dynamic models, by incorporating refinancing costs, explain the "inaction regions" where firms do not adjust leverage, leading to more realistic predictions of leverage ratios compared to static models. The review also discusses **Discrete-Time Investment Models**, starting with a basic formulation and progressively adding real-world features like financing costs and risky debt. A significant portion of the paper is dedicated to **Structural Estimation**, detailing methods like GMM, SMM, and SML for estimating these dynamic models, and providing practical guidance on their implementation. It further explores the **Cross-Sectional Implications of Dynamic Capital Structure**, arguing that empirical studies must account for the "real dynamics" of infrequent adjustment, which can explain phenomena like the negative leverage-profitability correlation. Finally, the paper briefly touches on the integration of **Macroeconomic Fundamentals** into corporate finance models. A central theme is that dynamic models offer a superior

ability to capture the complexities of real-world corporate decisions and provide quantitative, rather than purely qualitative, predictions.

5.7 Stochastic Discount Factor

The proper discount rate for tax shield is under debate, as previously mentioned. In fact, a correct discount rate for equity and debt are also under debate, because we do not know the exact risk premium. In traditional textbooks, such as [Cochrane \(2009\)](#)'s Asset Pricing, the stochastic discount factor (SDF) is presented as a tool suitable for discounting any cash flow, precisely because it accommodates the reality that discount rates are not constant but vary over time, due to economic fluctuations and market dynamics ([Kozak and Santosh, 2020](#)).

Future changes in discount rates cannot be perfectly predicted, yet investors still form expectations about this uncertainty itself, incorporating it into their valuations. Consequently, the discount rate includes a risk premium to compensate for this exposure, which can be formally expressed through the covariance between the asset's required return and the SDF, as in [Hansen and Renault \(2009\)](#). Building on this foundation, the book Accounting for Risk by [Penman et al. \(2021\)](#) introduces an accounting-based consumption asset pricing model (ACCAPM), in which he decomposes the value of firms into three components: the book value of equity discounted at the risk-free rate, future earnings discounted at the risk-free rate, and a covariance term between future earnings and the stochastic discount factor, effectively integrating accounting data with SDF principles to capture risk more accurately.

As many models in corporate finance are built upon continuous time model, I will use [Brunnermeier and Sannikov \(2016\)](#) and [Huang \(2022\)](#)'s work to briefly illustrate SDF in continuous time.

5.7.1 Setup

The household maximize its utility and can invest in risky assets A_t and risk-free assets with instantaneous return r_t^f . The target of the household is:

$$V_t^{Household} = \max \mathbb{E} \int_t^\infty e^{-\rho s} u(c_s) ds$$

and the law of motion with risky assets

$$\frac{dA_t}{A_t} = r_t^A dt + \sigma_t^A dW_t$$

Firms are owned by households and their cash flow D_t are discounted by household's stochastic discount factor $m_t = e^{-\rho t} u'(c_t)$. Value of a firm is given by

$$m_t V_t = \mathbb{E}_t \left[\int_t^\infty m_s D_s ds \right]$$

subject to

$$\begin{aligned} \frac{dV_t}{V_t} &= \mu_t^V dt + \sigma_t^V dW_t \\ \frac{dm_t}{m_t} &= -r_t^m dt - \sigma_t^m dW_t \end{aligned}$$

5.7.2 Results

With the setup above, we can derive the following:

$$\begin{aligned}
r_t^m &= r_t^f \\
r_t^A &= r_t^f + \sigma_t^A \sigma_t^m \\
\mu_t^V &= -\frac{D_t}{V_t} + \frac{1}{m_t} [r_t^f + \sigma_t^V \sigma_t^m]
\end{aligned}$$

5.7.3 Optimal Control with SDF

When we consider firm's optimizing problem, we can use the HJB equation. The optimal value function is

$$J(t, x) = \max_u E_t \left[\int_t^\infty f(s, x, u) ds \right]$$

where x_t is the state variable with law of motion,

$$dx_t = g(t, x, u)dt + \sigma(t, x, u)dW_t$$

with the discounted cash flow as our target so that

$$f(s, x, u) = m(x_s) \times D(x_s, u)$$

The HJB equation is

$$0 = \max_u \left(f(t, x, u) + J_t + J_x g + \frac{1}{2} J_{xx} \sigma^2 \right)$$

5.7.4 Estimation of SDF

Despite mature theories in SDF, in real application, we still have to empirically estimate it. The SDF is not directly observable in practice, necessitating empirical estimation. This has led to a rich literature focused on identifying and estimating SDFs using various econometric techniques, macroeconomic proxies, or asset returns data alone. Early work emphasized consumption-based models, while recent advances incorporate high-dimensional data and machine learning to address challenges like model misspecification and factor proliferation.

[Hansen and Singleton \(1982\)](#) introduced the generalized method of moments (GMM) to estimate consumption-based asset pricing models, linking the SDF to marginal utility of consumption. This approach tests over-identifying restrictions via the J-statistic but often struggles with the equity premium puzzle, where smooth consumption data implies insufficient SDF volatility to explain high risk premia.

[Campbell \(2000\)](#)'s survey "Asset Pricing at the Millennium" highlights how the SDF generalizes mean-variance efficiency: if the SDF is linear in factors, asset returns follow a linear factor model with risk premia tied to factor betas. The review also discusses other challenges in estimating SDF: low-frequency macroeconomic data lacks heteroskedasticity for time-varying premiums, high-dimensional factor "zoos" risk overfitting, and heterogeneity in investor consumption complicates aggregation.

Estimation techniques have evolved from parametric to more flexible approaches. [Smith and Wickens \(2002\)](#) advocate observable macroeconomic factors (e.g., consumption growth, inflation) over latent ones for interpretability and hedging, proposing multivariate GARCH-in-mean (MGIM) models to capture time-varying covariances essential for risk premia. They also critique vector autoregressions (VARs) for ignoring conditional heteroskedasticity. Empirical applications to equity, bonds, and forex show that factors like dividends and bond yields are significant, with GMM and Efficient Method of Moments (EMM) used for latent variables via simulation-based estimation.

Modern literature addresses high-dimensionality. For example, [Chen et al. \(2024\)](#) use maximum-Sharpe ratio estimation with sparse regression (MAXSER) on 194 portfolios, selecting 20 factors (momentum, earnings-related) for an SDF with Sharpe ratio 0.42, outperforming Fama-French models in explaining anomalies. It shows SDF sparsity in characteristics but not principal components, emphasizing factor interactions.

5.8 Empirical Results

An assumption of empirical reviews is that the observed debt ratios are relatively close to the firm's actual targets.

The study by [Frank and Goyal \(2024\)](#) on empirical corporate capital structure reveals several key observations. Aggregate leverage remains stationary, while public firms increasingly rely on debt relative to their assets over time. Financing needs, after accounting for internal funds, closely align with net debt issues, with net equity issues nearing zero. Interestingly, taxes show no significant impact on capital structure. Factors influencing a firm's capital structure include a positive effect from median industry level, tangibility, and firm size, while profitability and market-to-book ratio have a negative effect.

In their review titled "Empirical Capital Structure: A Review," [Parsons et al. \(2009\)](#) explore factors related to capital structure and deviations from it. Key factors positively affecting leverage include size, asset tangibility, and tax exposure, whereas market-to-book ratio, profitability, product uniqueness, and cash flow volatility exert a negative influence. Factors leading to deviation from a target capital structure include issuing more equity when stock is overvalued, which lowers the debt-to-value ratio, and a high stock return or entrenched manager, both reducing the debt-to-value ratio. Conversely, a unionization threat increases the debt-to-value ratio, while reliance on a specific supplier decreases it.

6 Connecting Corporate Finance to Accounting

In this section, I will focus on the combination of corporate finance and accounting.

6.1 Accounting Risk and Corporate Finance

The first issue is how to relate a firm's risk, which is widely discussed in the field of corporate finance, to its financial statements. The book 'Accounting for Risk' by [Penman et al. \(2021\)](#) well established this relationship. In previous reviews, I have considered intertemporal and endogenous decisions made by firms. I also discuss the selection of required rate of return and estimations of stochastic discount factors. These attributes can all be reflected in financial statements, as exhibited by Penman.

First, he explains how a firm's endogenous strategic decisions are reflected in its financial statements through specific, risk-based accounting principles. When a firm makes a particularly uncertain investment, such as in R&D or brand building, conservative accounting requires that the expenditure be expensed immediately against earnings. Instead, less risky investments are capitalized, like plant and equipment, whose future benefits are more probable.

Second, he details how accounting principles capture the intertemporal trade-offs inherent in corporate investment decisions. Conservative accounting, by expensing risky investments, depresses current earnings and Return on Equity (ROE). However, it creates the potential for higher future ROE, because if the investments succeed, the resulting revenues will not be burdened by subsequent amortization or depreciation charges.

Third, he critiques most methods for estimating the required rate of return and proposes an alternative method based on accounting analysis. For example, he blames implied cost of capital (ICC) for its poor predictive power and failure to properly incorporate the risk information conveyed by the accounting numbers. Instead of seeking a single indicator, he argues

that financial statements provide indicators of risk, which in turn imply the required return. For example, a low current ROE, when driven by conservative investment expensing, signals higher risk and thus justifies a higher required return. In application, he designed a indicator, ER score, which is a composite of accounting-based measures.

Finally, he provides a theoretical and empirical framework for estimating the unobservable Stochastic Discount Factor (SDF) using accounting data. It introduces an Accounting-Based Consumption Asset Pricing Model (ACCAPM), which links the risk to the covariance between earnings and SDF $Cov_t(Earnings_t, m_t)$. When the earning is realized, the covariance becomes close to zero as the risk is released, while unrealized earnings has a higher covariance with SDF. This provides a theoretical basis for the ER score to be a proxy for the SDF.

6.2 Modeling Capital Structure Decisions

One of the central challenges in dynamic financial simulation is translating corporate finance theories into programmable decision rules. The pecking order theory, as proposed by Myers and Majluf (1984), offers a clear, hierarchical logic that is particularly well-suited for algorithmic implementation. The theory posits that due to information asymmetry between managers and external investors, firms follow a distinct pecking order when seeking funds.

- **Financing Hierarchy:** The primary preference is for internal funds (retained earnings), which avoids the scrutiny and potential undervaluation associated with external financing. If internal funds are insufficient, firms turn to debt. Equity is considered the last resort, as issuing new stock is often perceived by the market as a negative signal that the firm's shares are overvalued.
- **Algorithmic Implementation:** This hierarchy can be directly translated into a decision-making algorithm within a simulation model:
 1. At the start of each period, the model calculates the firm's financing gap by subtracting available internal funds from its planned investments.
 2. If the gap is zero or negative, all investments are funded internally.
 3. If the gap is positive, the model checks the firm's debt capacity, which can be defined as a function of its assets, profitability, and current leverage.
 4. If the financing gap is within the debt capacity, the firm issues new debt to cover the shortfall.
 5. Only if the financing gap exceeds the firm's debt capacity will the model resort to issuing new equity.
- **Endogenous Leverage:** By implementing this algorithm, a firm's leverage ratio ceases to be an exogenous target and instead becomes an endogenous outcome of its accumulated investment and financing decisions over time. This allows researchers to dynamically observe how leverage evolves in response to changes in profitability and investment opportunities, providing a powerful method to test the explanatory power of the pecking order theory.

6.3 Modeling Investment and Financing Decisions

While the pecking order theory provides a hierarchical logic for financing choices, recent research suggests that financing decisions are more directly driven by immediate cash needs rather than long-term capital structure targets. Study by Huang and Ritter (2021) posits that corporate financing decisions are primarily a response to cash shortfalls. Their central argument is that most firms seek external financing—whether debt or equity—not to achieve an optimal capital structure or to hoard cash, but to cover current or impending cash deficits.

The nature of the cash shortfall dictates the choice of financing instrument. Firms facing short-term cash needs, for instance due to tangible asset investments or temporary losses, tend to issue debt. In contrast, firms with long-term, persistent cash requirements, such as those driven by R&D expenditures or ongoing operational losses, are more likely to issue equity. This is supported by their finding that 75.0% of net debt issuers and 53.9% of net equity issuers would have run out of cash by year-end without external financing. Among these firms, profitable ones are more inclined to issue debt, while unprofitable firms, which typically face more persistent cash needs, favor equity.

The study also examines how proceeds from financing are retained. On average, debt issuers spend most of the raised funds immediately, retaining only about 18.9 cents per dollar as cash. Equity issuers, however, retain a much larger portion (about 62.9 cents per dollar). This is attributed to higher anticipated future cash needs and the significant fixed costs associated with public equity offerings, which incentivize firms to raise more capital at once. In contrast, the lower fixed costs of debt financing, such as bank loans, allow firms to borrow on an as-needed basis, reducing the need to hold large cash balances.

This perspective supports a “financing horizon theory,” which proposes that financing decisions are primarily driven by the nature and timing of cash needs. Capital structure, in this view, is a path-dependent outcome of sequential financing choices rather than a primary optimization goal. This theory challenges traditional frameworks like the static trade-off theory and the pecking order theory, which, according to the authors, struggle to explain the slow speed of capital structure adjustments and the frequency of financing activities observed in practice. The financing horizon theory aligns with the dynamic models, such as that of [DeAngelo et al. \(2011\)](#), and offers a better explanation for why firms often deviate from their “target” leverage ratios and why some profitable firms do not issue debt even when they have no immediate cash needs.

Further exploring the interplay between investing and financing, research shows that accounting discretion itself is a key determinant. [Batta and Ruchti \(2024\)](#) find that firms over-invest for earnings management purposes and strategically hold excess cash instead of leveraging debt to maintain flexibility for future profitable opportunities. Their structural model demonstrates that incorporating accounting discretion significantly improves the explanation of simultaneous peaks in investment and debt issuance. Through counterfactual simulations, the study reveals that a model accounting for earnings management motives better explains the investment-financing relationship. It shows that firms tend to underutilize debt during normal periods to preserve financial flexibility, which in turn allows them to substantially increase investment when high-productivity opportunities arise. Overall, their work underscores the importance of considering accounting discretion when modeling corporate financing and investment decisions.

6.4 Modeling Payout Policy

Similarly, corporate payout policy can be modeled by operationalizing the agency cost theory of free cash flow, famously articulated by [Jensen \(1986\)](#). This theory provides a behavioral foundation for why and when firms decide to return cash to shareholders.

- **Theoretical Foundation:** The theory addresses the agency conflict that arises when firms generate substantial free cash flow (cash remaining after funding all positive net present value projects) but have limited profitable investment opportunities. In such scenarios, managers may be tempted to use this excess cash for empire-building or perquisite consumption, which does not maximize shareholder value.
- **Payout as a Governance Mechanism:** Dividend payments and share repurchases serve as a crucial governance tool. By committing to return cash to shareholders, the firm

reduces the amount of discretionary funds available to managers, thereby mitigating the risk of overinvestment and value-destroying decisions.

- **Simulation Logic:** This concept can be embedded into a simulation model as a set of decision rules:
 1. The model continuously assesses the level of free cash flow being generated.
 2. It simultaneously evaluates the firm's set of investment opportunities, which can be proxied by metrics such as Tobin's Q.
 3. A payout decision is triggered when the simulation detects a combination of high free cash flow and a low Tobin's Q, characteristic of a mature "cash cow" firm.
 4. The model can further be refined to distinguish between dividends and repurchases. For instance, an increase in dividends might be linked to a perceived permanent increase in cash flow, reflecting their "sticky" nature. In contrast, a share repurchase could be triggered by a temporary cash surplus, reflecting its greater flexibility.

7 PoC: Simulation Design-A Replication of Shahnazarian (2004)

As part of the literature review, we present details of our current simulation design, which replicates the result of [Shahnazarian \(2004\)](#). This part also serves as a proof of concept for our proposed framework.

7.1 Variable Dynamics

In the design of [Shahnazarian \(2004\)](#), variables are of two types, the stock variables and the flow variables, same as the field of system dynamics. Specifically, variables in the balance sheet of financial statements are the stock variables, while investment, dividends, etc. are the flow variables. Their relationship is captured by the following equation:

$$x_{t+1}^{stock} = x_t^{stock} + x_{t+1}^{flow}$$

where x_t^{stock} is the stock variable at the last period, and x_{t+1}^{flow} is the flow variable at the current period.

The other classification of variables described in [Shahnazarian \(2004\)](#) is more important: variables on financial statements are grouped into two categories:

- **Following the accounting identity** These include all variables on balance sheet, and some variables on income statement. For example, Machinery at current time equals to machinery at the last period plus the investment and minus the depreciation expense. While Earning before tax equals to earnings minus depreciations, plus financial income, and minus net increase in reserves. The relationships are captured by the following equation:

$$x_{t+1}^{stock} = x_t^{stock} + x_{t+1}^{flow}$$

$$x_{t+1}^{flow} = h\left(x_{t+1}^{stock}, x_{t+1}^{flow}, x_t^{stock}, x_t^{flow} | \Theta\right)$$

where x_t^{stock} is the stock variable at the last period, and x_{t+1}^{flow} is the flow variable at the current period.

- **Assumed and estimated** These include flow variables that are assumed to be a decision made by the firm, or that are subject to uncertainty of the market environment. For example, investment is assumed to be a decision made by the firm, on which we can have an estimation. While though revenue is not a decision made by the firm, it is assumed

to be subject to other variables, as well as uncertainty of the market environment. The relationships are captured by

$$x_{t+1}^{flow} = g\left(x_{t+1}^{flow}, x_t^{stock}, x_t^{flow}, \epsilon_{t+1} | \Theta\right)$$

which is stochastic with the noise given by ϵ_{t+1} .

Exact variables and dynamics on the assumed and estimated variables are given in Appendix B. In the table below, we summarize the variables and the categories they belong to:

	Follow Accounting Identities	Assumed and Estimated
Balance sheet	Balance sheet items: $MA_t, BU_t, OFA_t, CA_t, SC_t, RR_t,$ $OUR_t, CMA_t, ASD_t,$ $PF_t^{t-5}, PF_t^{t-4}, PF_t^{t-3}, PF_t^{t-2},$ $PF_t^{t-1},$ PF_t, LL_t, CL_t, URE_t	
Cash Flow	$cashfl_t, mcash_t$	
Income statement	Some Income statement items: $EBT_t, TA_t, TAX_t, FTAX_t, NBI_t,$ OL_t	$I_t^{MA}, I_t^{BU},$ $dofa_t, dca_t, dsc_t, drr_t, dour_t, p_t^{allo},$ $TDEP_t^{MA}, zpf_t, dll_t, dcl_t$ $EDEP_t^{MA}, S_t^{MA}, EDEP_t^{BU},$ $OIBD_t, FI_t, FE_t, GC_t, OA_t, TL_t,$ $OTA_t, TDEP_t^{BU}, ROT_t$

To replicate the result of [Shahnazarian \(2004\)](#), we have to first generate synthetic data of all variables, including stock and flow variables, using our simulator in Appendix B.1 with given parameters. and B.3 as well as our construction of noises in B.4. Then, we apply the estimation methods in Appendix B.2 to estimate the parameters of the behaviors in Appendix B.3 and to see how close are the generated parameters to the true parameters.

The machinery and equipment MA_t is used as an illustrative example in this section. The law of motion of MA_t is given by:

$$MA_t = MA_{t-1} + I_t^{MA} - S_t^{MA} - EDEP_t^{MA}$$

where I_t^{MA} is the investment in machinery and equipment, S_t^{MA} is the sale of machinery and equipment, and $EDEP_t^{MA}$ is the depreciation expense of machinery and equipment, from which we can see it is strictly following the accounting identity. However, I_t^{MA} , S_t^{MA} and $EDEP_t^{MA}$ are assumed to be decisions made by the firm, on which we can have an pre-defined equations. $EDEP_t^{MA}$ is the first decision to be considered. The steps of obtaining $EDEP_t^{MA}$ are given by:

Step 1: For the first stage, we use the following equation to estimate the probability that a firm will make a positive depreciation:

$$\begin{aligned} \eta_{EDEPMA} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 TDEP_{t-1}^{MA} + \beta_4 MA_{t-1} + \beta_5 I_{t-1}^{MA} + \beta_6 (I_{t-1}^{MA})^2 \\ & + \beta_7 EDEP_{t-1}^{BU} + \beta_8 (EDEP_{t-1}^{BU})^2 + \beta_9 ddmtdm_{t-1} + \beta_{10} (ddmtdm_{t-1})^2 + \beta_{11} dca_{t-1} \\ & + \beta_{12} ddmpa_{t-1} + \beta_{13} (ddmpa_{t-1})^2 + \beta_{14} dcl_{t-1} + \beta_{15} dgnp_t + \beta_{16} FAAB_t \\ & + \beta_{17} Public_t + \beta_{18} ruralare_t + \beta_{19} largcity_t + \beta_{20} market_t + \beta_{21} marketw_t \end{aligned}$$

Step 2: Define a threshold on whether a firm will make a positive depreciation or not. The threshold is given by:

$$P(EDEPMA > 0 | \eta_{EDEPMA}) = \frac{1}{1 + e^{-\eta_{EDEPMA}}}$$

where $P(EDEPMA > 0|\eta_{EDEPMA})$ is the probability that a firm will make a positive depreciation. Then, we generate a random number U from a uniform distribution $U(0,1)$. If $P(EDEPMA > 0|\eta_{EDEPMA}) > U$, then $EDEPMA = 1$, otherwise $EDEPMA = 0$.

Step 3: If $EDEPMA = 1$, then $EDEP_t^{MA}$ is given by:

$$\begin{aligned} EDEP_t^{MA} = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 TDEP_{t-1}^{MA} + \gamma_4 MA_{t-1} + \gamma_5 I_{t-1}^{MA} + \gamma_6 (I_{t-1}^{MA})^2 \\ & + \gamma_7 EDEP_{t-1}^{BU} + \gamma_8 (EDEP_{t-1}^{BU})^2 + \gamma_9 ddmtdm_{t-1} + \gamma_{10} (ddmtdm_{t-1})^2 + \gamma_{11} dca_{t-1} \\ & + \gamma_{12} ddmpa_{t-1} + \gamma_{13} (ddmpa_{t-1})^2 + \gamma_{14} dcl_{t-1} + \gamma_{15} dgnp_t + \gamma_{16} FAAB_t \\ & + \gamma_{17} Public_t + \gamma_{18} ruralare_t + \gamma_{19} largcity_t + \gamma_{20} market_t + \gamma_{21} marketw_t \\ & + \epsilon_t^{MA} \end{aligned}$$

where ϵ_t^{MA} is the noise in the depreciation expense, details of which are given in Appendix B.4.

Similar steps are taken for I_t^{MA} and S_t^{MA} . After obtaining $EDEP_t^{MA}$, I_t^{MA} and S_t^{MA} , we can obtain MA_t by:

$$MA_t = MA_{t-1} + I_t^{MA} - S_t^{MA} - EDEP_t^{MA}$$

7.2 Constraints on Simulated Variables

To ensure that the variables in the simulated financial statements are economically meaningful, a number of constraints are imposed during the simulation. These constraints are derived from accounting principles, financial logic, and Swedish tax regulations as outlined by [Shahnazarian \(2004\)](#).

7.2.1 Financial and Accounting Constraints

Dividend Payment Constraints. Firms are not permitted to pay negative dividends, which would be equivalent to receiving a subsidy from the government. Additionally, dividend payments cannot exceed the sum of the previous period's unrestricted equity (URE_{t-1}) and the current period's net operating income (after allocations to restricted reserves). This rule is in place to prevent the erosion of the firm's equity base. The maximum amount available for dividend payments in a period is referred to as the cash flow ($cashfl_t$).

External Funding Constraint. Any portion of investment in assets not covered by depreciation, asset sales, reserves, and unrestricted equity must be financed through the issuance of new long-term liabilities, current liabilities, or changes in share capital.

Non-negativity of Balance Sheet Items. All balance sheet items, including machinery and equipment, buildings, current assets, various reserves, and liabilities, must be greater than or equal to zero.

Minimum Share Capital. The firm's share capital must be maintained at or above 10,000 SEK.

7.2.2 Taxation Constraints

Tax Depreciation of Machinery and Equipment. Swedish tax law specifies a maximum amount of tax depreciation that a company can deduct from its taxable income. Firms have the option to choose from three depreciation methods: the declining-balance method (30%), a supplementary straight-line method (20%), and the remaining value method (25%). The company will select the method that provides the highest tax depreciation. The tax depreciation for machinery and equipment is constrained by a variable denoted as $MTDM_t$ which is an aggregation variable of depreciation methods.

Allocations to Periodical Reserves. Tax laws limit the maximum amount a company can allocate to periodical reserves each year. Companies are allowed to make a deduction by allocating up to 25% of their taxable income (20% for the year 1998) to a special reserve account.

This allocation is recorded as an untaxed reserve on the balance sheet and must be reversed back into taxable income no later than the sixth fiscal year following the allocation.

7.2.3 Mathematical Formulation of Constraints

This section provides the mathematical representation of the constraints described above.

Dividend Payment Constraints. Dividends cannot be negative:

$$cashfl_t \geq 0$$

Dividends are capped by the firm's distributable profit:

$$cashfl_t \leq mcash_t$$

where the maximum distributable amount, $mcash_t$, is defined as:

$$mcash_t = URE_{t-1} + NBI_t - drr_t$$

The final dividend payment, DIV_t , is determined by:

$$DIV_t = \max[0, \min(cashfl_t, mcash_t)]$$

Non-negativity Constraints. All balance sheet items must be non-negative:

$$\begin{aligned} MA \geq 0, \quad BU \geq 0, \quad OFA \geq 0, \quad CA \geq 0 \\ RR \geq 0, \quad ASD \geq 0, \quad PF \geq 0, \quad OUR \geq 0 \\ LL \geq 0, \quad CL \geq 0 \end{aligned}$$

The taxable residual value of machinery and equipment must also be non-negative:

$$CMA \geq 0$$

Minimum Share Capital. The firm's share capital, SC , must be at or above 10,000 SEK:

$$SC \geq 10000$$

Tax Depreciation of Machinery and Equipment. The actual tax depreciation, $TDEP_t^{MA}$, cannot exceed the maximum amount allowed by tax law, $MTDM_t$:

$$TDEP_t^{MA} \leq MTDM_t$$

The maximum allowable depreciation, $MTDM_t$, is determined by the following three methods:

Declining-balance method ($TDDb$):

$$TDDb_t^{MA} = \left(\frac{M}{12}\right) \delta^{db} [CMA_{t-1} + I_t^{MA} - S_t^{MA} - IG_t], \quad \text{where } \delta^{db} = 0.3$$

Straight-line method ($TDSL$):

$$TDSL_t^{MA} = CMA_{t-1} + I_t^{MA} - S_t^{MA} - IG_t - [\delta_t^s (I_t^{MA} - IG_t) + \dots]$$

Remaining value method ($TDRV$):

$$TDRV_t^{MA} = \left(\frac{M}{12}\right) \delta^{rv} [CMA_{t-1} + I_t^{MA} + S_t^{MA} - IG_t], \quad \text{where } \delta^{rv} = 0.25$$

Allocations to Periodical Reserves. The annual allocation to periodical reserves, p_t^{allo} , cannot exceed the maximum permissible amount, MPA_t :

$$p_t^{allo} \leq MPA_t$$

The maximum permissible amount, MPA_t , is calculated as follows, with an allocation rate η of 25%:

$$MPA_t = \max[0, (\eta \times pbase_t)]$$

The base for the calculation, $pbase_t$, is:

$$pbase_t = OIBD_t - EDEP_t^{BU} + FI_t - FE_t - TDEP_t^{MA} + zpf_t + OA_t - TL_t + TA_t$$

Part II

Research Proposal

8 IFRS and IAS Standards

To construct a detailed accounting simulator based on International Financial Reporting Standards (IFRS), it is essential first to delineate the specifics of both IFRS and the International Accounting Standards (IAS). In addition to the previously mentioned IFRS, IAS represents the older set of standards issued by the International Accounting Standards Committee (IASC) between 1973 and 2000. These standards formed the initial foundation for a globally harmonized approach to financial reporting before the International Accounting Standards Board (IASB) took over in 2001 and began issuing the new series of standards known as IFRS.

While some of the original IAS standards have now been withdrawn or superseded by newer IFRS pronouncements, a significant number of them remain active and are still mandated. These active IAS standards, in conjunction with the full suite of IFRS, collectively form the complete, authoritative framework for contemporary international financial reporting. Therefore, any comprehensive accounting simulation must integrate the principles and requirements from both sets of standards to accurately reflect the complex and interconnected regulatory landscape that governs how most companies prepare their financial statements today.

8.1 International Accounting Standards

In this section, I will exhibit the main theme of each section in IAS, as well as whether each of them are withdrawn, superseded, or still under application.

- **IAS 1: Presentation of Financial Statements**

This standard sets the overall requirements for the presentation of general purpose financial statements to ensure comparability both with an entity's own financial statements of previous periods and with the financial statements of other entities.

- **IAS 2: Inventories**

This standard provides guidance on the accounting treatment for inventories, prescribing how cost is determined and subsequently recognised as an expense, including any write-down to net realisable value.

- **IAS 3: Consolidated Financial Statements**

Superseded by IAS 27 and IFRS 10. This standard prescribed the requirements for preparing and presenting consolidated financial statements for a group of entities under the control of a parent.

- **IAS 4: Depreciation Accounting**
Superseded by IAS 16 and IAS 38. This standard outlined the accounting treatment for depreciation, which is the systematic allocation of the depreciable amount of an asset over its useful life.
- **IAS 5: Information to be Disclosed in Financial Statements**
Superseded by IAS 1. This standard specified particular items of information that had to be disclosed in financial statements.
- **IAS 6: Accounting Responses to Changing Prices**
Withdrawn. This standard dealt with the accounting treatment and disclosure for entities reporting in economies experiencing significant changes in price levels (inflation).
- **IAS 7: Statement of Cash Flows**
This standard requires an entity to present a statement of cash flows which classifies cash flows during the period from operating, investing, and financing activities.
- **IAS 8: Accounting Policies, Changes in Accounting Estimates and Errors**
This standard prescribes the criteria for selecting and changing accounting policies, together with the accounting treatment and disclosure of changes in accounting estimates and corrections of prior period errors.
- **IAS 9: Accounting for Research and Development Activities**
Superseded by IAS 38. This standard specified the accounting treatment for the costs of research and development activities.
- **IAS 10: Events After the Reporting Period**
This standard prescribes when an entity should adjust its financial statements for events that occur after the reporting period and the disclosures that it should give about such events.
- **IAS 11: Construction Contracts**
Superseded by IFRS 15. This standard provided guidance on the accounting for revenue and costs associated with construction contracts over the life of the contract.
- **IAS 12: Income Taxes**
This standard prescribes the accounting treatment for income taxes, including how to account for the current and future tax consequences of transactions and other events.
- **IAS 13: Presentation of Current Assets and Current Liabilities**
Superseded by IAS 1. This standard governed how assets and liabilities should be classified as either current or non-current in the statement of financial position.
- **IAS 14: Segment Reporting**
Superseded by IFRS 8. This standard required entities to disclose information about their operating segments, products and services, geographical areas, and major customers.
- **IAS 15: Information Reflecting the Effects of Changing Prices**
Withdrawn. This standard provided guidance for entities wishing to present supplementary information reflecting the effects of changing prices during inflationary periods.
- **IAS 16: Property, Plant and Equipment**
This standard prescribes the accounting treatment for tangible fixed assets, including their recognition, initial and subsequent measurement, depreciation, and derecognition.

- **IAS 17: Leases**
Superseded by IFRS 16. This standard prescribed the appropriate accounting policies and disclosures for both lessees and lessors in relation to finance and operating leases.
- **IAS 18: Revenue**
Superseded by IFRS 15. This standard provided accounting guidance for the revenue arising from certain types of transactions and events, such as the sale of goods, rendering of services, and the use by others of entity assets yielding interest, royalties and dividends.
- **IAS 19: Employee Benefits**
 This standard prescribes the accounting and disclosure for all forms of employee benefits, including short-term benefits, post-employment benefits, other long-term benefits, and termination benefits.
- **IAS 20: Accounting for Government Grants and Disclosure of Government Assistance**
 This standard deals with the accounting for government grants and the disclosure requirements for other forms of government assistance.
- **IAS 21: The Effects of Changes in Foreign Exchange Rates**
 This standard prescribes how to include foreign currency transactions and foreign operations in the financial statements of an entity and how to translate financial statements into a presentation currency.
- **IAS 22: Business Combinations**
Superseded by IFRS 3. This standard set out the accounting treatment for business combinations, including the determination and amortisation of goodwill.
- **IAS 23: Borrowing Costs**
 This standard requires the capitalisation of borrowing costs that are directly attributable to the acquisition, construction, or production of a qualifying asset as part of the cost of that asset.
- **IAS 24: Related Party Disclosures**
 This standard requires the disclosure of the existence of related party relationships, transactions, and outstanding balances to ensure that a company's financial statements draw attention to the potential effect of such relationships.
- **IAS 25: Accounting for Investments**
Superseded by IAS 39 and IAS 40. This standard dealt with the accounting treatment for various types of investments.
- **IAS 26: Accounting and Reporting by Retirement Benefit Plans**
 This standard prescribes the measurement and disclosure principles for the financial statements of retirement benefit plans as reporting entities.
- **IAS 27: Separate Financial Statements**
 This standard prescribes the accounting and disclosure requirements for investments in subsidiaries, joint ventures, and associates when an entity prepares separate (non-consolidated) financial statements.
- **IAS 28: Investments in Associates and Joint Ventures**
 This standard prescribes the accounting for investments in associates (where an investor has significant influence) and joint ventures, requiring the use of the equity method.

- **IAS 29: Financial Reporting in Hyperinflationary Economies**
This standard specifies the principles for restating the financial statements of any entity whose functional currency is the currency of a hyperinflationary economy.
- **IAS 30: Disclosures in the Financial Statements of Banks and Similar Financial Institutions**
Superseded by IFRS 7. This standard prescribed specific disclosure requirements for banks and similar financial institutions.
- **IAS 31: Interests in Joint Ventures**
Superseded by IFRS 11. This standard prescribed the accounting treatment and disclosure for a venturer's interests in joint ventures.
- **IAS 32: Financial Instruments: Presentation**
This standard establishes principles for presenting financial instruments as liabilities or equity from the issuer's perspective and for offsetting financial assets and financial liabilities.
- **IAS 33: Earnings per Share**
This standard prescribes principles for the determination and presentation of earnings per share, so as to improve performance comparisons between different entities in the same reporting period and between different reporting periods for the same entity.
- **IAS 34: Interim Financial Reporting**
This standard prescribes the minimum content of an interim financial report and the principles for recognition and measurement in such reports.
- **IAS 35: Discontinuing Operations**
Superseded by IFRS 5. This standard established principles for reporting information about discontinuing operations to better enable users to make projections.
- **IAS 36: Impairment of Assets**
This standard ensures that an entity's assets are not carried at more than their recoverable amount and prescribes how to recognise an impairment loss if an asset's carrying amount exceeds this value.
- **IAS 37: Provisions, Contingent Liabilities and Contingent Assets**
This standard ensures that appropriate recognition criteria and measurement bases are applied to provisions, contingent liabilities and contingent assets and that sufficient information is disclosed to enable users to understand their nature, timing and amount.
- **IAS 38: Intangible Assets**
This standard prescribes the recognition, measurement, and disclosure requirements for all intangible assets that are not specifically dealt with in other standards.
- **IAS 39: Financial Instruments: Recognition and Measurement**
Largely superseded by IFRS 9. This standard established principles for recognising, derecognising, and measuring financial assets and financial liabilities, and for hedge accounting.
- **IAS 40: Investment Property**
This standard prescribes the accounting treatment for property held to earn rentals or for capital appreciation (or both) and its related disclosure requirements.
- **IAS 41: Agriculture**
This standard prescribes the accounting treatment and disclosures related to agricultural activity, covering biological assets and agricultural produce at the point of harvest.

8.2 International Financial Reporting Standards

Similarly, in this section I will give a brief summary of the main theme of IFRS 1-17, as well as if they are currently performed.

- **IFRS 1: First-time Adoption of International Financial Reporting Standards**
This standard aims to ensure that an entity's first IFRS financial statements, and its interim financial reports for part of that period, contain high-quality, transparent information that is comparable over all periods presented and provides a suitable starting point.
- **IFRS 2: Share-based Payment**
This standard requires an entity to reflect in its financial statements the effects of share-based payment transactions, including transactions with employees or other parties.
- **IFRS 3: Business Combinations**
This standard sets out the financial reporting requirements for an acquirer in a business combination, focusing on the application of the acquisition method, which requires the recognition and measurement of identifiable assets acquired, liabilities assumed, any non-controlling interest, and goodwill.
- **IFRS 4: Insurance Contracts**
Superseded by IFRS 17, effective from 1 January 2023. This standard provided temporary guidance for the accounting of insurance contracts before the implementation of IFRS 17, allowing entities to largely continue their existing accounting policies.
- **IFRS 5: Non-current Assets Held for Sale and Discontinued Operations**
This standard specifies the accounting for non-current assets held for sale, and the presentation and disclosure of discontinued operations.
- **IFRS 6: Exploration for and Evaluation of Mineral Resources**
This standard provides specific guidance on financial reporting for entities engaged in the exploration for and evaluation of mineral resources like oil, gas, and similar non-regenerative resources.
- **IFRS 7: Financial Instruments: Disclosures**
This standard requires entities to provide disclosures in their financial statements that enable users to evaluate the significance of financial instruments for the entity's financial position and performance, and the nature and extent of risks arising from those financial instruments.
- **IFRS 8: Operating Segments**
This standard requires an entity to disclose information to enable users of its financial statements to evaluate the nature and financial effects of the business activities in which it engages and the economic environments in which it operates.
- **IFRS 9: Financial Instruments**
This standard comprehensively covers the principles for the classification, measurement, and impairment (using an 'expected credit loss' model) of financial assets and financial liabilities, as well as hedge accounting, largely replacing IAS 39.
- **IFRS 10: Consolidated Financial Statements**
This standard establishes a single control model that applies to all entities, including structured entities, and requires a parent that controls one or more other entities to present consolidated financial statements.

- **IFRS 11: Joint Arrangements**

This standard provides for the financial reporting by parties that have joint control of an arrangement, classifying joint arrangements as either joint operations or joint ventures.

- **IFRS 12: Disclosure of Interests in Other Entities**

This standard requires an entity to disclose information that enables users of its financial statements to evaluate the nature of, and risks associated with, its interests in subsidiaries, joint arrangements, associates, and unconsolidated structured entities.

- **IFRS 13: Fair Value Measurement**

This standard defines fair value, sets out in a single framework the guidelines for measuring fair value, and requires disclosures about fair value measurements.

- **IFRS 14: Regulatory Deferral Accounts**

An interim standard. This standard permits a first-time adopter of IFRS engaged in rate-regulated activities to continue to account for regulatory deferral account balances in accordance with its previous GAAP, pending a more comprehensive review of the topic by the IASB.

- **IFRS 15: Revenue from Contracts with Customers**

This standard establishes a comprehensive framework (the five-step model) for determining when to recognise revenue and how much revenue to recognise, superseding previous revenue guidance in IAS 18 and IAS 11.

- **IFRS 16: Leases**

This standard sets out the principles for the recognition, measurement, presentation and disclosure of leases, requiring lessees to account for most leases on-balance sheet by recognising a right-of-use asset and a lease liability.

- **IFRS 17: Insurance Contracts**

This standard establishes a comprehensive and consistent accounting model for insurance contracts, replacing the interim guidance in IFRS 4 to improve the transparency and comparability of accounting for insurance contracts.

Therefore, the set of currently active and relevant accounting standards from IAS and IFRS that we must consider for the simulator includes IAS 1, 2, 7, 8, 10, 12, 16, 19-21, 23, 24, 26-29, 32-34, 36-38, 40, and 41, alongside the complete series of IFRS 1-3 and 5-17. In the subsequent work, I will discuss the inherent complexities and challenges of implementing each of these standards within the simulator, as well as explore potential solutions and simplifications that can be achieved by applying certain reasoned assumptions.

8.3 Foundational Structure: IAS 1 and the Complete Set of Financial Statements

The simulator's primary output will be a complete set of financial statements as defined by IAS 1. The model will be designed to generate not just the three commonly cited statements, but all components required for a fair presentation. The architectural blueprint for the simulator's reporting module will therefore include:

1. A Statement of Financial Position (as a representation of the system's *stocks* at the end of a period).
2. A Statement of Profit or Loss and Other Comprehensive Income.
3. A Statement of Changes in Equity.

4. A Statement of Cash Flows (to be modelled per IAS 7).
5. Accompanying Notes (initially focusing on accounting policy disclosures).

A fundamental parameter of the simulation model will be the **going concern** assumption. As stipulated by IAS 1, the simulator will operate on the basis that all entities within the model are expected to continue in operation for the foreseeable future. The accounting logic for asset valuation (e.g., historical cost less depreciation) and liability classification (current vs. non-current) will be built upon this premise.

8.3.1 Modelling the Statement of Changes in Equity (SOCIE)

The simulator is a stock-flow system, and the Statement of Changes in Equity (SOCIE) is the perfect embodiment of this concept and will be a critical component of the model.

The simulator will explicitly model the SOCIE to reconcile the opening and closing equity balances. The opening equity balances represent the initial *stock variables*. The changes during the period are the *flow variables*, which alter the stocks to arrive at the closing balances. The columns within the simulator's SOCIE module will, at a minimum, disaggregate equity into the following components:

- Share Capital
- Share Premium
- Retained Earnings
- Treasury Shares (as a deduction from equity)
- Other Reserves, representing the accumulated balance of **Other Comprehensive Income (AOCI)**. This component will be further disaggregated based on the nature of the OCI flows, as detailed in the following section.

8.3.2 Modelling the Statement of Comprehensive Income (P/L and OCI)

The simulator will generate a detailed Statement of Comprehensive Income, strictly adhering to the structure mandated by IAS 1. This includes a clear distinction between the Profit or Loss (P/L) section and the Other Comprehensive Income (OCI) section.

Profit or Loss (P/L) Component The P/L section will present, at a minimum, the line items required by IAS 1, paragraph 82. The simulator will be designed to calculate and present items such as:

- Revenue
- Cost of Sales (COGS)
- Gross Profit
- Other income
- Distribution costs
- Administrative expenses
- Finance costs
- Share of the profit or loss of associates and joint ventures accounted for using the equity method

- Tax expense
- **Profit or loss for the period**

The calculation logic for each of these flows (e.g., COGS derived from IAS 2, Revenue from IFRS 15) will be governed by the specific topical standards, with IAS 1 governing their final presentation.

Other Comprehensive Income (OCI) Component A key feature of the simulator's compliance with IAS 1 will be the correct classification and presentation of OCI items. The model will segregate OCI flows into two distinct categories based on their potential for future reclassification ("recycling") to P/L.

The simulator will include the capacity to model the following OCI items:

(a) OCI items that will *not* be reclassified subsequently to profit or loss:

- Changes in revaluation surplus for property, plant and equipment (governed by IAS 16).
- Remeasurements of a defined benefit plan (actuarial gains and losses) (governed by IAS 19).
- Gains and losses on investments in equity instruments designated at Fair Value Through Other Comprehensive Income (FVOCI) (governed by IFRS 9).

(b) OCI items that *may* be reclassified subsequently to profit or loss:

- Gains and losses on investments in debt instruments measured at FVOCI (governed by IFRS 9).
- The effective portion of gains and losses on hedging instruments in a cash flow hedge (governed by IFRS 9).
- Exchange differences on translating the financial statements of a foreign operation (governed by IAS 21).

By modelling this distinction, the simulator will correctly populate the separate reserves within Accumulated OCI in the Statement of Changes in Equity and ensure that the accounting logic for subsequent events (e.g., the disposal of a foreign operation triggering reclassification) is accurately represented.

8.4 Modelling IAS 2: *Inventories*

The simulator will incorporate a detailed inventory module designed to manage the stock and flow of inventory items in strict accordance with IAS 2 *Inventories*. This module will be fundamental for modelling the core operations of manufacturing and retail entities within the simulation.

8.4.1 Measurement Principle: Lower of Cost and Net Realisable Value

The valuation of inventory stocks at each reporting date will adhere to the core principle of IAS 2: measurement at the **lower of cost and Net Realisable Value (NRV)**.

A key feature of the valuation module will be the estimation of NRV. While the formula is straightforward, the simulator will treat its components as variables that require estimation, reflecting real-world uncertainty. The model will calculate NRV as:

$$\text{NRV} = \text{Estimated Selling Price} - \text{Estimated Costs of Completion} - \text{Estimated Costs to Sell}$$

The simulator’s architecture will allow these estimated variables to be defined and adjusted, enabling the analysis of how changes in market conditions and cost structures impact inventory valuation and profitability.

8.4.2 Cost Formula: First-In, First-Out (FIFO)

For assigning costs to inventory items, the simulator will implement the **First-In, First-Out (FIFO)** method. This approach is one of the benchmark treatments permitted by IAS 2 and is widely used in practice. The model’s logic will assume that the first units purchased are the first ones sold, ensuring that the closing inventory stock is valued at the most recent purchase costs. This design choice explicitly excludes the Last-In, First-Out (LIFO) method, which is prohibited under IFRS.

8.4.3 Impairment and Reversal Logic

The simulator will automatically trigger an inventory write-down (impairment loss) when the calculated cost of an inventory item exceeds its estimated NRV. The amount of the write-down will be recognised as an expense in the period the impairment occurs.

Furthermore, the model will handle the reversal of such write-downs. A subsequent reversal will be modelled if evidence suggests that the NRV has recovered. However, the reversal is capped, ensuring that the new carrying amount of the inventory is the lower of:

- The (new) Net Realisable Value, and
- The inventory’s *original cost*.

This prevents the inventory from being carried at an amount greater than its initial historical cost.

8.4.4 Recognition as an Expense: Cost of Goods Sold (COGS)

The simulator will model the recognition of the Cost of Goods Sold (COGS) based on the principle that when inventories are sold, their *carrying amount* shall be recognised as an expense in the period in which the related revenue is recognised.

The model will clarify the relationship between impairment and the final COGS figure reported in the Statement of Profit or Loss. The logic will be implemented as follows:

- The expense recognised for a unit of inventory sold is always its **carrying amount at the point of sale**.
- If no impairment has occurred, the carrying amount is its original cost (determined via FIFO).
- If the inventory was previously written down, its carrying amount is the lower, impaired value (its NRV at the last reporting date). Selling this item means a lower cost is transferred to COGS for that specific unit.
- However, IAS 2 requires that the amount of any inventory write-down is recognised as an expense in the period the write-down occurs. In practice, this write-down expense is typically presented within the COGS line item.

8.5 Modelling Cash Flow, Consistency, and Period-End Events (IAS 7, 8, & 10)

Beyond the foundational structure and specific asset valuation, the simulator must accurately model the flow of cash and maintain the integrity of its accounting principles over time. This section details the implementation of IAS 7 *Statement of Cash Flows*, IAS 8 *Accounting Policies, Changes in Accounting Estimates and Errors*, and IAS 10 *Events After the Reporting Period*.

8.5.1 IAS 7: Modelling the Statement of Cash Flows

The model's logic will ensure that all cash flows during a period are classified into one of three distinct categories, as mandated by IAS 7:

- **Operating Activities:** The principal revenue-producing activities of the entity.
- **Investing Activities:** The acquisition and disposal of long-term assets and other investments not included in cash equivalents.
- **Financing Activities:** Activities that result in changes in the size and composition of the contributed equity and borrowings of the entity.

The simulator will implement the **indirect method** for presenting cash flows from operating activities. The model's algorithm will derive the net cash from operating activities by adjusting the net profit or loss for the effects of:

1. Non-cash transactions (e.g., depreciation, amortisation, impairment losses, provisions).
2. Gains or losses associated with investing or financing cash flows (e.g., gain or loss on the disposal of PPE).
3. Changes during the period in inventories and operating receivables and payables (i.e., changes in working capital).

The final output will provide a clear reconciliation between the reported profit and the net cash flow generated from core operations.

8.5.2 IAS 8: Ensuring Consistency and Handling Changes

IAS 8 provides the crucial framework for ensuring the consistency and comparability of financial statements over time. It prescribes the criteria for selecting and changing accounting policies, and the accounting treatment for changes in accounting estimates and the correction of errors.

For the initial development phase of this simulator, a simplifying assumption will be made regarding accounting policies. The model will be built on the principle of consistency, as required by IAS 8. However, it will operate under the assumption that **no voluntary changes in accounting policies or corrections of prior period errors occur**. All entities within the simulation will apply their selected accounting policies consistently from one period to the next.

8.5.3 IAS 10: Events After the Reporting Period

To streamline the initial model, the simulator will operate under a further simplifying assumption. While acknowledging the importance of IAS 10, for the current scope of the project, we will assume that **no material adjusting events occur**. Therefore, the financial statements generated by the simulator for any given period will be considered final based on the information and conditions known to exist at the reporting date.

8.6 Modelling IAS 12 *Income Taxes*

The accounting for income taxes is a critical and complex area of financial reporting. The simulator must incorporate a robust module based on IAS 12 *Income Taxes* to handle the differences between accounting principles and tax regulations, primarily through the mechanism of deferred tax.

8.6.1 Rationale: Reconciling the "Two Worlds" of Profit

The fundamental need for IAS 12 arises from the divergence between two distinct frameworks for determining profit:

The Accounting World Governed by IFRS, the objective is to provide a fair presentation of an entity's financial position and performance. This results in the calculation of **Accounting Profit**.

The Tax World Governed by national tax legislation, the objective is to determine the amount of income tax payable to fiscal authorities. This results in the calculation of **Taxable Profit**.

Resulting Differences The rules in these two worlds are often inconsistent, leading to timing differences in the recognition of income and expenses. For example:

- **Depreciation:** Accounting depreciation (e.g., straight-line over 10 years) may be slower than the accelerated depreciation permitted by tax law.
- **Revenue Recognition:** Revenue may be recognised over time under IFRS 15, whereas tax law might require tax to be paid when cash is received.
- **Provisions:** A provision for future warranty costs is recognised as an expense under IAS 37, but tax law may only permit a deduction when the cost is actually paid.

This mismatch creates a significant accounting challenge. Simply recognising the current tax payable as the total income tax expense in the profit and loss statement would be misleading. An entity might report high profits due to low current tax payments (e.g., from accelerated tax depreciation), while ignoring the fact that these tax benefits will reverse in the future, leading to higher tax payments then. The purpose of the IAS 12 module in the simulator is to resolve this inter-period allocation problem through the calculation and recognition of **Deferred Tax**.

8.6.2 Key Modelling Requirements and Challenges

To accurately simulate the requirements of IAS 12, the model must address several specific complexities:

1. **Calculation of Deferred Tax Balances:** The simulator will calculate Deferred Tax Liabilities (DTL) and Deferred Tax Assets (DTA) based on the temporary differences between the carrying amount of assets and liabilities in the statement of financial position and their tax base. The core formulas to be implemented are:

$$\text{DTL} = \text{Taxable Temporary Differences (TTD)} \times \text{Future Tax Rate}$$

$$\text{DTA} = \text{Deductible Temporary Differences (DTD)} \times \text{Future Tax Rate}$$

A critical feature of the model will be its use of expected *future* tax rates (specifically, those that have been enacted or substantively enacted by the end of the reporting period), rather than the current period's tax rate, to measure these balances.

2. **Strict Recognition Criteria for DTA:** The simulator must model the asymmetric prudence required for DTA recognition. While DTLs are recognised for almost all TTDs, a DTA is only recognised to the extent that it is *probable* that future taxable profit will be available against which the DTA can be utilised. The model's logic will require that the forecasted future taxable profit, when multiplied by the tax rate, is sufficient to absorb the DTA being recognised. This necessitates a module that can assess future profitability forecasts as an input for the DTA recognition test.
3. **The Balance Sheet Approach to Tax Expense:** The total tax expense will be calculated as:

$$\text{Total Tax Expense} = \text{Current Tax Expense} + \text{Deferred Tax Expense}$$

Where,

$$\text{Current Tax Expense} = \text{Taxable Profit} \times \text{Current Tax Rate}$$

$$\text{Deferred Tax Expense} = (\text{DTL}_{\text{end}} - \text{DTL}_{\text{start}}) - (\text{DTA}_{\text{end}} - \text{DTA}_{\text{start}})$$

It is exhibited that traditionally the required closing DTL and DTA balances (the stocks) are calculated first, and then people derive the deferred tax expense (the flow) for the income statement from the change in these stocks. The simulator should explore different algorithms to estimate the flow variable first and then obtain the closing variable.

4. **Offsetting of DTA and DTL:** The simulator will incorporate the logic for offsetting DTA and DTL balances in the statement of financial position.
5. **Balance Sheet Classification:** In accordance with IAS 1, the model will classify all DTA and DTL balances as **non-current** assets and liabilities.
6. **Intra-period Tax Allocation:** The simulator's tax module will be designed to trace the origin of the temporary differences. If a transaction is recognised directly in Other Comprehensive Income (OCI) (e.g., a PPE revaluation under IAS 16) or equity, the related deferred tax effect will also be allocated directly to OCI or equity, respectively, rather than to the profit or loss statement.

8.7 Modelling IAS 16 *Property, Plant and Equipment*

Property, Plant and Equipment (PPE) often represent the most significant component of an entity's total assets. The simulator will feature a comprehensive module to account for the entire lifecycle of PPE, from acquisition to disposal, in accordance with IAS 16 *Property, Plant and Equipment*.

8.7.1 Initial Measurement of Cost

The simulator will recognise an item of PPE at its initial cost. The model's logic for determining this cost will be designed to include all components specified by IAS 16. The cost will comprise:

- (a). **Purchase price**, including import duties and non-refundable purchase taxes, after deducting trade discounts and rebates.
- (b). **Directly attributable costs** of bringing the asset to the location and condition necessary for it to be capable of operating in the manner intended by management. The model will include parameters for costs such as:
 - Costs of employee benefits arising directly from the construction or acquisition.
 - Costs of site preparation.

- Initial delivery and handling costs.
- Installation and assembly costs.
- Costs of testing whether the asset is functioning properly, net of the proceeds from selling any items produced while testing.
- Professional fees (e.g., architects, engineers).

(c). **The initial estimate of the costs of dismantling and removing the item** and restoring the site on which it is located, the obligation for which an entity incurs either when the item is acquired or as a consequence of having used the item during a particular period. This will be modelled in line with IAS 37 *Provisions, Contingent Liabilities and Contingent Assets*.

Conversely, the simulator will correctly identify costs that are not part of the cost of an item of PPE and will expense them in the P/L module as incurred. These include costs of opening a new facility, costs of introducing a new product or service (including advertising), costs of conducting business in a new location, employee training costs, and general administrative overheads.

For the initial implementation of the simulator, the range of directly attributable costs can be simplified to a core set (e.g., purchase price and installation costs), while ensuring the fundamental distinction between capitalisable costs and period expenses is maintained.

8.7.2 Subsequent Measurement

A key parameter for each entity class within the simulation will be its chosen accounting policy for the subsequent measurement of PPE. The simulator will support both models permitted by IAS 16 and will apply the chosen policy consistently to an entire class of PPE.

Cost Model The default and simplest model. The asset will be carried at its cost less any accumulated depreciation and any accumulated impairment losses.

$$\text{Carrying Amount} = \text{Cost} - \text{Accumulated Depreciation} - \text{Accumulated Impairment Losses}$$

Revaluation Model For entities modelled with this policy, an asset will be carried at a revalued amount, being its fair value at the date of the revaluation less any subsequent accumulated depreciation and impairment losses. The accounting for changes in value will be modelled as follows:

- **Revaluation Increase:** If an asset's carrying amount is increased as a result of a revaluation, the increase shall be recognised in **Other Comprehensive Income (OCI)** and accumulated in equity under the heading of **Revaluation Surplus**.
- **Revaluation Decrease:** If an asset's carrying amount is decreased, the decrease shall first reduce any credit balance existing in the revaluation surplus in respect of that same asset. Any excess decrease shall be recognised in **Profit or Loss (P/L)**.

The simulator will enforce the rule that if the revaluation model is chosen, all assets within that class of PPE must be revalued.

8.7.3 Depreciation Logic

The simulator's architecture will define a 'PPE' class, where each company will have a set of properties that determine its depreciation.

The model for each company will be built upon the three core elements of depreciation:

1. **Depreciable Amount:** Calculated as the cost of the asset less its residual value.
2. **Useful Life:** The period over which an asset is expected to be available for use by the entity. This will be a key input parameter, treated as an accounting estimate.
3. **Residual Value:** The estimated amount that the entity would currently obtain from disposal of the asset, after deducting the estimated costs of disposal. This will also be an input parameter treated as an estimate.

Furthermore, the simulator will support multiple depreciation methods to reflect different patterns of consumption of economic benefits, including:

- Straight-line method.
- Diminishing balance method.
- Units of production method.

8.7.4 Derecognition (Disposal)

The simulator will derecognise (remove from the statement of financial position) an item of PPE upon its disposal or when no future economic benefits are expected from its use or disposal. The carrying amount of the disposed asset will be set to zero.

The gain or loss arising from the derecognition will be calculated and recognised in P/L as per the standard. The model will compute this as:

$$\text{Gain or Loss on Disposal} = \text{Net Disposal Proceeds} - \text{Carrying Amount of the Asset}$$

This gain or loss will be included in the profit or loss for the period in which the item is derecognised.

8.8 Modelling IAS 19 *Employee Benefits*

Employee benefits represent a significant cost for many entities, and their accounting treatment varies widely depending on their nature. The accounting simulator will incorporate a module to handle the main categories of employee benefits as prescribed by IAS 19 *Employee Benefits*, with a particular focus on the complexities of post-employment benefit plans.

8.8.1 Short-term Employee Benefits and DC Plans

The model will handle two simpler forms of employee benefits by recognising them as an expense as the service is rendered by the employee.

Short-term Employee Benefits This category includes items such as wages, salaries, paid annual leave, and bonuses payable within twelve months. The simulator will recognise the cost of these benefits in **Profit or Loss (P/L)** in the period the employee service is received, on an undiscounted basis.

Defined Contribution (DC) Plans For post-employment benefits classified as DC plans, the entity's obligation is limited to the amount it agrees to contribute to the fund. The simulator will model this by recognising the contribution payable as an expense in **Profit or Loss (P/L)** in the period the employee renders the related service.

8.8.2 Defined Benefit (DB) Plans

The accounting for DB plans is significantly more complex as the entity retains the actuarial and investment risk. The simulator will feature a detailed sub-module to model the accounting for these plans, implementing the following core logic as specified by IAS 19.

Balance Sheet Recognition The core of the DB plan model will be the calculation of the net position to be recognised on the statement of financial position. This involves two key variables:

- **The Defined Benefit Obligation (DBO):** The present value of the future benefits that employees have earned in return for their service in the current and prior periods. This will be an input to the simulator, representing an actuarial valuation.
- **The Fair Value of Plan Assets:** The value of the assets held in a separate fund to meet the future obligations.

The simulator will then calculate the net balance sheet position as follows:

$$\text{Net Defined Benefit (Asset)/Liability} = \text{Fair Value of Plan Assets} - \text{DBO}$$

If the result is positive (a surplus), a *Net Defined Benefit Asset* will be recognised. If it is negative (a deficit), a *Net Defined Benefit Liability* will be recognised.

Recognition of Changes in P/L and OCI A critical feature of the simulator will be to correctly disaggregate the change in the Net Defined Benefit Asset/Liability during a period into its three components, with each being allocated to either P/L or OCI as required by the standard.

1. **Service Cost:** This includes the current service cost (the increase in the DBO resulting from employee service in the current period), past service cost, and any gain or loss on settlement. This component will be recognised in **Profit or Loss (P/L)**.
2. **Net Interest:** This is the change in the net defined benefit liability/asset due to the passage of time. It is calculated by applying a discount rate to the opening net defined benefit liability/asset. This component will also be recognised in **Profit or Loss (P/L)**.
3. **Remeasurements:** This component comprises all other changes, including actuarial gains and losses (arising from changes in actuarial assumptions) and the return on plan assets (excluding amounts included in net interest). The simulator will recognise all remeasurement components in **Other Comprehensive Income (OCI)**. A crucial part of the model's logic is that these OCI amounts are **never reclassified** ("recycled") to P/L in subsequent periods.

By implementing this logic, the simulator will accurately reflect the performance-related aspects of DB plans in P/L, while containing the volatile, non-operational valuation adjustments within OCI, fully complying with the presentation requirements of IAS 19.

8.9 Modelling IAS 20 *Accounting for Government Grants*

The simulator will incorporate the principles of IAS 20 *Accounting for Government Grants and Disclosure of Government Assistance* to model how entities account for resources received from government bodies. A key function of the model will be to classify grants based on their nature and to apply the entity's chosen accounting policy consistently. The core accounting treatment is based on the matching principle, recognising grant income in the same periods as the related costs they are intended to compensate.

8.9.1 Grants Related to Assets

For grants whose primary condition is the acquisition or construction of long-term assets, the simulator will support the two alternative presentation methods allowed by IAS 20. A key parameter for each entity within the simulation will be its selected policy for this class of grants, which will be applied consistently.

Deferred Income Method Described by the standard as the more common approach, this method will be modelled as follows:

1. The grant is initially recognised as a liability under *deferred income*.
2. This deferred income balance is then systematically recognised in profit or loss on a basis consistent with the depreciation of the related asset.

This ensures a direct matching of the grant income with the expense of consuming the asset's economic benefits.

Deduction from Asset Cost Method As an alternative, the simulator will also support the method of deducting the grant from the asset's initial cost.

- The logic will deduct the grant from the carrying amount of the asset upon initial recognition.
- The benefit of the grant is then recognised in profit or loss over the life of the asset via a *reduced depreciation charge*.

The model will demonstrate that while the presentation in the financial statements differs (i.e., no "other income" line item), the net impact on an entity's profit for the period is identical to the deferred income method.

8.9.2 Grants Related to Income

For grants that compensate an entity for expenses or losses incurred (i.e., all grants other than those related to assets), the simulator will also support the two alternative presentation methods.

Gross Presentation The model will have the option to present the grant income as a separate line item in the statement of profit or loss, for example under 'Other Income'.

Net Presentation Alternatively, the model will support deducting the grant directly from the related expense that it is intended to compensate. For instance, if a simulated entity incurs R&D expenses of \$1,000,000 and receives a related grant of \$300,000, the P/L module will be able to report a net 'R&D Expense' of \$700,000.

8.9.3 Repayment of Government Grants

IAS 20 provides specific guidance for the event that a grant becomes repayable. The standard requires such a repayment to be accounted for as a change in an accounting estimate in accordance with IAS 8. The accounting treatment depends on the original method used for the grant:

- If the grant was recognised as deferred income, the repayment would first reduce any unamortised deferred income balance. Any excess would be recognised as an immediate expense in profit or loss.

- If the grant was deducted from the cost of an asset, the repayment would be treated as an increase in the carrying amount of that asset.

For the initial phase of the simulator's development, a simplifying assumption will be made. The logic for the **repayment of grants will be excluded** from the model's scope. The focus will be on the correct initial classification and the subsequent income recognition based on the entity's chosen policies.

8.10 Modelling IAS 21 *The Effects of Changes in Foreign Exchange Rates* - (Out of Scope)

IAS 21 prescribes how to include foreign currency transactions and foreign operations in an entity's financial statements. A correct application of this standard requires a clear distinction between three different currency roles. The simulator's conceptual framework will be built on these definitions.

8.10.1 Core Currency Definitions

Functional Currency This is the most fundamental concept in IAS 21. It is the currency of the primary economic environment in which the entity operates. The determination of the functional currency is a matter of fact and is not a free choice for the entity. The simulator will require this to be defined for each entity based on the following factors:

- **Primary Factors:** The currency that mainly influences sales prices for goods and services, and the currency of the country whose competitive forces and regulations mainly determine the sales prices; the currency that mainly influences labour, material, and other costs of providing goods or services.
- **Secondary Factors:** The currency in which funds from financing activities (e.g., issuing debt and equity instruments) are generated, and the currency in which receipts from operating activities are usually retained.

For example, for a UK company whose sales, procurement, and financing activities are primarily conducted in pound sterling (GBP), the functional currency would be GBP.

Foreign Currency A foreign currency is defined simply as any currency other than the entity's functional currency. For the UK company with a GBP functional currency, currencies such as the US dollar (USD), Euro (EUR), or Japanese Yen (JPY) would all be considered foreign currencies.

Presentation Currency This is the currency in which the financial statements are presented. An entity has a free choice of presentation currency. For instance, the aforementioned UK company (with a GBP functional currency) might choose to present its financial statements in USD to cater to its American investors.

8.10.2 Simplifying Assumption for the Current Model

The accounting treatment for foreign currency transactions and the translation of foreign operations are complex, requiring the re-measurement of monetary items at the closing rate, the recognition of exchange differences in profit or loss or OCI, and the application of different rates for different financial statement elements.

Therefore, for the initial development phase of this accounting simulator, a significant simplifying assumption will be made. The model will operate on the basis that, for any given simulated entity, **the functional currency, foreign currency, and presentation currency are all the same.**

8.11 Modelling IAS 23 *Borrowing Costs* - (Out of Scope)

The core principle of IAS 23 *Borrowing Costs* provides a significant exception to the general rule of expensing finance costs as they are incurred. The standard mandates that under specific circumstances, borrowing costs should not be treated as a period expense but should instead be considered a necessary part of constructing a long-term asset. The simulator's design will acknowledge these principles, even where simplifying assumptions are made.

The general rule is that borrowing costs are recognised as an expense in profit or loss in the period in which they are incurred. However, IAS 23 requires that borrowing costs that are directly attributable to the acquisition, construction, or production of a **qualifying asset** must be capitalised as part of the cost of that asset.

8.11.1 Key Definitions

To apply the standard, the model's logic must first be based on two core definitions.

Borrowing Costs These are defined as interest and other costs that an entity incurs in connection with the borrowing of funds. The most significant component to be modelled is interest expense calculated using the effective interest method under IFRS 9.

Qualifying Asset This is the central condition for capitalization. A qualifying asset is an asset that necessarily takes a **substantial period of time** to get ready for its intended use or sale. While "substantial period" is not explicitly defined and requires judgement, it is generally considered to be a period of one year or more.

- **Examples of qualifying assets** include:
 - Property, plant, and equipment during its construction phase (e.g., a factory or power plant).
 - Investment property during its development phase.
 - Intangible assets during their development phase (e.g., a major software system).
 - Inventories that require a long period of time to bring to a saleable condition (e.g., wine or whiskey that requires ageing, or forests under cultivation).
- **Examples of non-qualifying assets** include:
 - Inventories that are manufactured in large quantities on a repetitive basis over a short period.
 - Assets that are ready for their intended use or sale when acquired.
 - Assets that are measured at fair value, such as biological assets.

8.11.2 The Capitalisation Period

IAS 23 provides clear rules for the capitalisation period. It specifies when to start, suspend, and stop capitalising borrowing costs.

Commencement Capitalisation begins on the commencement date. This is the date when all three of the following conditions are first met:

1. The entity incurs expenditures for the asset.
2. The entity incurs borrowing costs.
3. The entity undertakes activities to prepare the asset for its use or sale.

Suspension An entity must suspend capitalisation during extended periods of interruption. For example, if active development of the asset stops for a long time. Any borrowing costs incurred during this suspension period are recognised as an expense in profit or loss. Normal, minor delays are not considered interruptions.

Cessation Capitalisation must stop on the cessation date. This occurs when the asset is substantially complete and ready for its intended use or sale. At this point, no more borrowing costs can be added to the asset's cost. If an asset is built in parts, and each part can be used independently, capitalisation stops for each part as it is completed.

8.11.3 Simplifying Assumption for the Current Model

The identification of a qualifying asset and the precise determination of the capitalisation period involve a significant degree of professional and subjective judgement. For example, assessing whether a period of time is “substantial” or whether an interruption is “extended” requires context-specific analysis.

Given this subjectivity, and to maintain focus on the core stock-flow mechanics of the simulator in its initial phase, a simplifying assumption will be adopted: **no borrowing costs will be capitalised**. All borrowing costs incurred by a simulated entity will be recognised as an expense in profit or loss in the period in which they are incurred.

This approach aligns with the default treatment for borrowing costs. The implementation of a module to identify qualifying assets and apply the capitalisation logic of IAS 23 is designated as a potential area for future enhancement of the simulator.

8.12 IAS 26 *Accounting and Reporting by Retirement Benefit Plans* - (Out of Scope)

To fully understand the accounting for pensions, it is crucial to distinguish the requirements of IAS 19 from those of IAS 26 *Accounting and Reporting by Retirement Benefit Plans*. These two standards are often confused but govern two entirely separate reporting entities.

Distinction from IAS 19 Before outlining the scope of IAS 26, it is important to clarify this distinction.

- **IAS 19 *Employee Benefits*** dictates the accounting from the perspective of the **sponsoring employer**. It prescribes how a company should recognise the net liability or asset arising from the pension benefits promised to its employees within its own corporate financial statements. To use an analogy, this is the company stating in its annual report: “I owe, or have pre-funded, an amount for my employees’ pensions.”
- **IAS 26 *Retirement Benefit Plans*** dictates the accounting from the perspective of the **retirement benefit plan itself**. It prescribes how the pension fund or plan, as a separate reporting entity, should prepare and present its own financial statements for its members and trustees. To continue the analogy, this is the pension plan itself issuing a separate annual report stating: “This is how much money (assets) I am managing, this is how much I am obligated to pay out in the future (liabilities), and here is my overall financial health.”

Objective and Scope of IAS 26 The objective of IAS 26 is to specify the content of the financial reports prepared by retirement benefit plans. These reports help users, who are primarily the members and trustees of the plan, to assess the resources and investment performance of the plan. The standard applies to the financial statements of all retirement benefit plans, whether they are constituted as separate legal entities or have trustees.

Applicability to the Accounting Simulator Since IAS 26 applies only to the financial statements of the retirement benefit plan as a separate entity, and not to the sponsoring company's annual report, modelling the specific requirements of IAS 26 is considered **out of scope** for the current phase of this research project.

8.13 IAS 27 *Separate Financial Statements* - (Out of Scope)

The IFRS framework places primary importance on consolidated financial statements for entities that control subsidiaries. The requirements for consolidation are mandated by IFRS 10 *Consolidated Financial Statements*. However, a parent entity may also be required or may choose to present its own individual financial statements. IAS 27 *Separate Financial Statements* is the standard that prescribes the accounting and disclosure requirements for these specific reports.

The Role of Separate Financial Statements A parent entity must present consolidated financial statements. In addition to these, it may present separate financial statements. It is critical to note that where consolidated financial statements are required, the separate financial statements are **supplementary** and **cannot replace** the consolidated reports. They serve specific purposes, often for local legal or statutory requirements.

Core Accounting Requirement The central requirement of IAS 27 is to define how an entity should account for its investments in subsidiaries, joint ventures, and associates in its separate financial statements. The standard provides an accounting policy choice, allowing the entity to account for these investments either:

1. **At cost;**
2. **In accordance with IFRS 9 *Financial Instruments*** (which typically implies measurement at fair value); or
3. **Using the equity method** as described in IAS 28 *Investments in Associates and Joint Ventures*.

The chosen policy must be applied consistently for each category of investment.

Applicability to the Accounting Simulator Given that separate financial statements are supplementary in nature and not the primary report for a parent company, and to maintain a clear focus on the core consolidation and operational accounting logic, a simplifying assumption will be made for the current project. The simulator will, for now, **not consider the preparation of separate financial statements**.

8.14 IAS 28 *Investments in Associates and Joint Ventures* - (Out of Scope)

IAS 28 *Investments in Associates and Joint Ventures* is a critical component of the suite of standards on group accounting. It addresses the "middle-ground" of investment relationships, where an investor has more than passive influence over an investee, but less than full control.

- **Less than control:** If a control relationship exists, IFRS 10 *Consolidated Financial Statements* requires full consolidation.
- **More than passive influence:** If it is a simple equity investment without significant influence, IFRS 9 *Financial Instruments* requires measurement at fair value or cost.

IAS 28 prescribes that for this middle-ground of investments, the **equity method** must be applied.

8.14.1 Core Definitions: Associates and Joint Ventures

To apply the standard, it is first necessary to identify whether an investment qualifies as an associate or a joint venture.

Associate An associate is an entity over which the investor has **significant influence**. Significant influence is the power to *participate* in the financial and operating policy decisions of the investee but is not control or joint control over those policies. This is a key area of judgement.

- **Quantitative Indicator (Presumption):** If an investor holds, directly or indirectly, 20% or more of the voting power of the investee, it is presumed that the investor has significant influence, unless it can be clearly demonstrated that this is not the case.
- **Qualitative Indicators:** Even if the holding is less than 20%, significant influence may be evidenced by one or more of the following: representation on the board of directors; participation in policy-making processes; material transactions between the investor and the investee; interchange of managerial personnel; or provision of essential technical information.

Joint Venture A joint venture is a joint arrangement whereby the parties that have **joint control** of the arrangement have rights to the **net assets** of the arrangement. This definition aligns with IFRS 11 *Joint Arrangements*. Joint control implies that decisions about the relevant activities require the unanimous consent of the parties sharing control.

8.14.2 The Equity Method

Once an investment is identified as an associate or joint venture, the application of the equity method is mandatory. This method is sometimes referred to as a "one-line consolidation" because it presents the investment in the investor's statement of financial position as a single line item, which is adjusted for the investor's share of the investee's changing net assets.

Initial Measurement The investment is initially recognised at **cost**.

Subsequent Measurement After initial recognition, the carrying amount of the investment is adjusted to reflect the investor's share of the investee's post-acquisition performance and other equity changes. The change in the carrying amount is determined as follows:

$$\begin{aligned}\text{Closing Balance} &= \text{Opening Balance} \\ &+ \text{Investor's share of the investee's profit for the period} \\ &- \text{Investor's share of the investee's loss for the period} \\ &\pm \text{Investor's share of the investee's OCI for the period} \\ &- \text{Dividends received from the investee}\end{aligned}$$

Share of Profit/Loss and OCI: The investor recognises its share of the investee's profit or loss and OCI in its own statement of comprehensive income. These amounts also adjust the carrying amount of the investment.

Treatment of Dividends: Critically, under the equity method, dividends received from an associate or joint venture are **not** recognised as income. They are treated as a **return of the investment**, and therefore reduce the carrying amount of the investment on the statement of financial position. This is because the underlying profit that enabled the dividend has already been recognised by the investor through its share of the investee's profit.

8.14.3 Other Important Aspects of the Equity Method

The application of the equity method also involves considering impairment, losses, and intra-group transactions.

- **Impairment Testing:** The investment is tested for impairment as a single asset under IAS 36 *Impairment of Assets* whenever there is an indication that it may be impaired.
- **Losses in Excess of Investment:** The investor's share of losses is recognised until the carrying amount of the investment is reduced to zero. Further losses are not recognised unless the investor has incurred legal or constructive obligations on behalf of the investee.
- **Elimination of Unrealised Profits:** Unrealised profits on transactions between the investor and the investee (e.g., "upstream" or "downstream" sales of inventory) must be eliminated to the extent of the investor's interest in the entity.

8.14.4 Applicability to the Accounting Simulator

The position of IAS 28 within the IFRS framework can be summarised by the level of influence an investor holds:

Table 2: Spectrum of Influence and Applicable Standards

Level of Influence	Relationship	Applicable Standard	Core Method
No Significant Influence	Financial Asset	IFRS 9	Fair Value or Cost
Significant Influence	Associate	IAS 28	Equity Method
Joint Control	Joint Venture	IAS 28 & IFRS 11	Equity Method
Control	Subsidiary	IFRS 10	Consolidation

The equity method is thus a more simplified approach than full consolidation but provides a more faithful representation of the close economic link between an investor and its investee than the fair value method.

The primary focus of the simulator in its initial phase is to perfectly model the two ends of this spectrum: simple financial assets governed by IFRS 9 and full control relationships requiring consolidation under IFRS 10. Therefore, to manage the complexity of the initial model, the simulator will temporarily **not consider the cases of associates and joint ventures**. The implementation of the equity method as prescribed by IAS 28 is identified as a significant module for future development.

8.15 IAS 29 *Financial Reporting in Hyperinflationary Economies* - (Out of Scope)

IAS 29 is a highly specialized standard that applies only under specific, extreme economic conditions. Its principles are fundamentally different from the standard historical cost basis of accounting.

8.15.1 Rationale for IAS 29

In a normal economic environment, financial accounting is based on the historical cost principle, which operates under the stable measuring unit assumption (i.e., the purchasing power of the currency is relatively stable). However, in a **hyperinflationary economy**, the currency loses its purchasing power at such a rapid rate that this assumption completely fails. Comparing amounts from different periods becomes meaningless, and the financial statements lose their relevance. The objective of IAS 29 is to provide a methodology to restate the financial statements

so that all items are expressed in a common measuring unit, thereby restoring their informational value.

8.15.2 Indicators of Hyperinflation

The standard does not establish an absolute scenario for when hyperinflation is deemed to arise. Instead, it provides a series of characteristics that indicate such an environment. An entity should consider applying the standard when several of these characteristics are present.

Qualitative Characteristics:

- The general population prefers to keep its wealth in non-monetary assets or in a relatively stable foreign currency.
- The general population regards monetary amounts not in terms of the local currency but in terms of a relatively stable foreign currency. Prices may be quoted in that currency.
- Sales and purchases on credit take place at prices that compensate for the expected loss of purchasing power during the credit period, even if the period is short.
- Interest rates, wages and prices are linked to a price index.

Quantitative Characteristic (Indicative):

- The cumulative inflation rate over three years is approaching, or exceeds, 100%. This is the most commonly cited indicative threshold.

8.15.3 The Core Principle of Restatement

Once an economy is identified as hyperinflationary, an entity operating within it **must** restate its financial statements. The core principle is as follows:

The financial statements of an entity whose functional currency is the currency of a hyperinflationary economy shall be stated in terms of the **measuring unit current at the end of the reporting period** (e.g., December 31).

8.15.4 Summary and Applicability to the Simulator

IAS 29 represents a "special accounting model" designed for extreme economic circumstances. Its core logic is to abandon the historical cost principle and, by means of a general price index (GPI), adjust all non-monetary items and income statement items to the single point of purchasing power that exists at the end of the reporting period.

The specific accounting treatments for this restatement are complex and significantly alter the foundational logic of the simulator, which is built upon the stable measuring unit assumption. Given the specialized nature of these requirements and their rare application in most major economies, the simulator will, for the time being, **not consider hyperinflationary accounting**.

8.16 Modelling IAS 32 *Financial Instruments: Presentation*

The accounting for financial instruments is governed by a suite of standards. IAS 32 *Financial Instruments: Presentation* serves as the foundational standard, providing the principles for classifying financial instruments on the statement of financial position. Before determining how to measure (IFRS 9) or disclose (IFRS 7) a financial instrument, an entity must first use IAS 32

to answer a fundamental question: should the instrument be presented as a financial liability or as an equity instrument?

The core principle of IAS 32 is **substance over form**. The classification is not based on the legal name of an instrument (e.g., “preference share”) but on the economic substance of its contractual arrangement.

8.16.1 The Core Principle: Distinguishing Financial Liabilities from Equity

The cornerstone of IAS 32 is the distinction based on whether the issuer has an unavoidable contractual obligation to deliver cash or another financial asset.

Financial Liability An instrument is classified as a financial liability if it contains a contractual obligation:

1. To deliver cash or another financial asset to another entity. Common examples include bank loans, bonds payable, and trade payables, all of which contain a clear obligation to pay cash in the future.
2. To exchange financial assets or financial liabilities with another entity under conditions that are potentially unfavourable to the issuer. This typically covers certain derivative contracts.

Equity Instrument An equity instrument is defined as any contract that evidences a **residual interest** in the assets of an entity after deducting all of its liabilities. The core feature is the *absence* of a contractual obligation to deliver cash. The decision of whether to pay, when to pay, and how much to pay rests entirely at the discretion of the issuer. The classic example is an ordinary share, for which the entity has no legal obligation to pay dividends.

8.16.2 Application to Complex Financial Instruments

The strength of IAS 32’s principle-based approach is demonstrated in its application to complex instruments with mixed characteristics.

Preference Shares The classification of preference shares depends entirely on their contractual terms:

- If a preference share has a **mandatory redemption** clause (e.g., it must be redeemed for cash after 5 years), it creates an unavoidable obligation and is therefore substantively a **liability**.
- If a preference share has **mandatory dividends** (i.e., dividends must be paid and are not at the discretion of the issuer), this also creates a contractual obligation, leading to a **liability** classification.
- If both redemption and dividends are purely at the discretion of the issuer, no obligation exists, and the instrument is classified as **equity**.

Compound Instruments (e.g., Convertible Bonds) A typical convertible bond contains both a liability and an equity component from the issuer’s perspective:

- **Liability component:** The obligation to make scheduled payments of interest and principal.
- **Equity component:** The option granted to the holder to convert the bond into the issuer’s ordinary shares.

IAS 32 requires “split accounting” for such instruments upon initial recognition. The fair value of the liability component is determined first (e.g., by pricing a similar bond without a conversion option). The residual amount of the total proceeds is then allocated to the equity component.

8.16.3 Other Important Presentation Rules

- **Interest, Dividends, Losses and Gains:** The classification of the instrument dictates the treatment of its associated payments. Payments on an instrument classified as a liability (e.g., interest on a bond) are recognised as an expense in **profit or loss**. Distributions on an instrument classified as equity (e.g., dividends on ordinary shares) are recognised directly in **equity**.
- **Offsetting:** A financial asset and a financial liability can be offset only when an entity has a currently legally enforceable right to set off the recognised amounts, AND it intends either to settle on a net basis or to realise the asset and settle the liability simultaneously. These strict conditions make offsetting rare in practice.
- **Treasury Shares:** An entity’s own shares that it has reacquired are presented as a deduction from equity. No gain or loss is recognised in profit or loss on the purchase, sale, issue or cancellation of an entity’s own equity instruments.

8.16.4 Simulator Implementation Logic

The simulator’s architecture will be designed to ensure compliance with the principles of IAS 32.

- A core logic principle will be consistency. For standard financial instruments, the classification will default to be the same as in the previous period unless the instrument is derecognised or its terms are modified.
- For complex instruments, the object-oriented design will be leveraged. For an item like a ‘Preference Share’, the underlying class in the code will include attributes (e.g., is it mandatorily redeemable or are dividends mandatory). A classification function will then analyse these attributes to determine whether the instance of the class should be treated as a liability or as equity within the simulation.
- The model will also implement the “split accounting” logic for compound instruments at the point of their simulated issuance.

8.17 Modelling IAS 34 *Interim Financial Reporting*

The very objective of this research project—to create a simulator that generates reports on a quarterly basis—is conceptually aligned with the principles of IAS 34 *Interim Financial Reporting*. This standard governs the minimum content of an interim financial report and the principles for recognition and measurement in such a report.

The standard views an interim report as an update on the last full annual financial statements. While the simulator will generate a full set of statements for each period for completeness, the underlying measurement principles for interim periods must follow IAS 34’s logic to be compliant.

A critical implementation of these principles within the simulator will be the calculation of the interim income tax expense. IAS 34 requires that this measurement be made on a year-to-date basis to avoid distorting quarterly earnings. Therefore, a key feature of the simulator’s tax module will be its adherence to this rule. The model will not apply a discrete tax rate to the profit of the interim period (e.g., the quarter). Instead, it will calculate the tax expense

based on the **best estimate of the weighted-average annual income tax rate** expected for the full financial year. This ensures that the tax expense is allocated appropriately across the interim periods, reflecting the expected relationship between tax expense and profit for the year as a whole.

8.18 Modelling IAS 36 *Impairment of Assets*

A core principle of prudent accounting is that assets should not be carried at a value higher than their economic potential. The simulator will implement a robust impairment module based on IAS 36 *Impairment of Assets* to ensure that the carrying amounts of assets in the simulated balance sheet remain relevant and are not overstated.

8.18.1 Core Principle

The simulator's impairment logic will be built on the fundamental principle of IAS 36: an asset's **carrying amount** must not exceed its **recoverable amount**. If the carrying amount is higher, the asset is considered impaired, and an impairment loss must be recognised.

The *recoverable amount* is defined as the higher of an asset's:

- **Fair Value Less Costs of Disposal (FVLCD):** The price that would be received to sell an asset in an orderly transaction, less the incremental costs directly attributable to the disposal.
- **Value in Use (VIU):** The present value of the future cash flows expected to be derived from the asset.

8.18.2 Scope of the Impairment Module

The IAS 36 module will apply to almost all non-financial assets of a simulated entity. The scope will include:

- Property, Plant and Equipment (IAS 16)
- Intangible Assets (IAS 38)
- Goodwill acquired in a business combination (IFRS 3)
- Investment Property carried at cost (IAS 40)
- Right-of-use assets (IFRS 16)
- Investments in associates and joint ventures (IAS 28)

The module will not apply to assets whose impairment is governed by other standards, such as inventories (IAS 2), financial assets (IFRS 9), and deferred tax assets (IAS 12).

8.18.3 Impairment Testing Triggers

While the standard requires testing for most assets when there is an indication of impairment, certain assets are more susceptible to impairment and require more frequent testing. The simulator's logic will enforce the mandatory annual impairment test for the following assets, regardless of whether any indication of impairment exists:

- Goodwill.
- Intangible assets with an indefinite useful life.
- Intangible assets not yet available for use.

8.18.4 Recognition of Impairment Losses

When an impairment loss is identified (i.e., Carrying Amount > Recoverable Amount), the simulator will recognise the loss. The accounting treatment will depend on the asset's measurement basis:

- **For assets carried at cost:** The impairment loss will be recognised immediately in **Profit or Loss (P/L)**.
- **For assets carried under the revaluation model (IAS 16):** The impairment loss is treated as a revaluation decrease. It will first be used to reduce any existing **Revaluation Surplus** for that same asset in Other Comprehensive Income (OCI). Any remaining loss will then be recognised in **P/L**.

8.18.5 Reversal of Impairment Losses

The simulator will also model the logic for the potential reversal of impairment losses if the asset's recoverable amount subsequently increases.

Recognition of Reversal The increase in the asset's carrying amount due to a reversal will be recognised in P/L (or OCI for revalued assets).

Ceiling on Reversal The reversal is capped. The increased carrying amount of an asset due to a reversal of an impairment loss shall not exceed the carrying amount that would have been determined (net of depreciation or amortisation) had no impairment loss been recognised for the asset in prior years.

The Absolute Exception: Goodwill A critical and non-negotiable rule will be hard-coded into the simulator: **an impairment loss recognised for goodwill shall not be reversed in a subsequent period.**

8.19 Modelling IAS 37 *Provisions, Contingent Liabilities and Contingent Assets*

The simulator must be able to account for uncertainties regarding the timing and amount of liabilities and assets. The model will implement the core principles of IAS 37 *Provisions, Contingent Liabilities and Contingent Assets* to ensure that these uncertainties are treated with the appropriate level of prudence and recognition.

8.19.1 Recognition of Provisions and Contingent Liabilities

The model's logic will differentiate between a *Provision* (a liability that is recognised) and a *Contingent Liability* (which is only disclosed). The key decision rule, as specified by IAS 37, is based on probability and measurability.

A Provision will be recognised on the statement of financial position only when three conditions are met: (1) a present obligation exists from a past event; (2) the outflow of economic benefits is **probable** (interpreted as a probability of more than 50%); and (3) the amount can be **reliably estimated**.

The simulator's design will explicitly model this. For potential obligations arising in the simulation, a probability parameter will be assigned. If this probability of a liability existing and requiring settlement exceeds 50% and a reliable estimate is available, the model will recognise a Provision. If the probability is 50% or less, or a reliable estimate cannot be made, the item will be treated as a contingent liability, for disclosure purposes only, and not recognised on the balance sheet.

8.19.2 Treatment of Contingent Assets

In line with the principle of prudence, IAS 37 forbids the recognition of contingent assets. A potential asset is only recognised when its realisation is virtually certain, at which point it ceases to be contingent.

The simulator will strictly adhere to this rule. The model will be programmed to **not recognise contingent assets** on the statement of financial position. This ensures that potential, unrealised gains do not overstate an entity's financial position.

8.20 Modelling IAS 38 *Intangible Assets*

The simulator's accounting logic must extend beyond tangible assets to incorporate the principles of IAS 38 *Intangible Assets*. This standard can be seen as a "sister standard" to IAS 16, providing the framework for non-physical long-term operational assets such as patents, trademarks, software, and customer lists.

8.20.1 Core Definition of an Intangible Asset

For an item to be classified as an intangible asset within the simulator, it must meet all three of the following core characteristics:

Identifiability This is the key criterion that distinguishes intangible assets from Goodwill. An asset is identifiable if it is either:

- *Separable*, i.e., capable of being separated or divided from the entity and sold, transferred, licensed, rented, or exchanged.
- Arises from *contractual or other legal rights*, regardless of whether those rights are transferable or separable.

Control The entity must control the asset, meaning it has the power to obtain the future economic benefits flowing from it and can restrict the access of others to those benefits (e.g., through legal protection like a patent).

Future Economic Benefits The asset must be expected to generate future economic benefits for the entity, such as revenues or cost savings.

8.20.2 Recognition and Measurement in the Simulator

The simulator will model the recognition and measurement of intangible assets based on their mode of acquisition. For the purpose of the simulation, it will be assumed that intangible assets primarily exist on the opening statement of financial position and their state can be carried over from the previous period. The underlying principles for their initial measurement are as follows:

- **Separate Acquisition:** An intangible asset purchased separately is measured at cost, which comprises its purchase price and any directly attributable costs.
- **Acquired in a Business Combination:** An intangible asset acquired in a business combination is measured at its fair value at the acquisition date, in line with IFRS 3 *Business Combinations*.

- **Internally Generated Intangible Assets:** IAS 38 mandates a strict distinction between a research phase and a development phase. All expenditure during the *research phase* must be expensed as incurred. Expenditure during the *development phase* may only be capitalised under very strict criteria. A critical rule is that internally generated brands, mastheads, publishing titles, customer lists, and similar items **can never be recognised** as assets.

A key simplification for the initial model will be the treatment of internally generated assets. The complex logic for capitalising development expenditure will not be modelled in the initial phase.

8.20.3 Subsequent Measurement and Amortisation

The subsequent measurement of intangible assets within the simulator will follow a logic parallel to that for PPE under IAS 16 (i.e., a choice between the cost model and the revaluation model). The model will focus on the amortisation process, which is dependent on the asset's useful life.

A critical attribute for each intangible asset class within the simulator will be the classification of its useful life as either finite or indefinite, as this distinction dictates the entire subsequent accounting treatment.

Assets with a Finite Useful Life These assets will be amortised on a systematic basis over their useful life. For the purposes of the simulator, the following simplifying assumptions will be made, reflecting common practice:

- The **straight-line method** of amortisation will be the default method applied.
- The **residual value** will be assumed to be zero.

Assets with an Indefinite Useful Life For assets determined to have an indefinite useful life (e.g., a perpetually renewable brand), the simulator will apply the following rules:

- They will **not be amortised**.
- They will be subject to a mandatory **annual impairment test** in accordance with the IAS 36 module.

8.21 Modelling IAS 40 *Investment Property*

The simulator will include a module for a special class of assets: investment property. The accounting rules for these assets are prescribed by IAS 40 *Investment Property*. Understanding this standard is crucial as it requires a clear distinction between properties held for investment purposes versus those used for operational or sales activities.

8.21.1 Definition and Scope

The core of IAS 40 lies in its definition of investment property, which is based on the **purpose for which the property is held**. Investment property is defined as property (land or a building, or part of a building, or both) held to earn rentals or for capital appreciation or both.

This distinguishes it from two other categories of property:

- **Owner-occupied property (governed by IAS 16):** Property held for use in the production or supply of goods or services or for administrative purposes. Examples include a company's head office, its factories, or an owner-managed hotel.

- **Property held for sale (governed by IAS 2):** Property held for sale in the ordinary course of business. This would apply, for example, to a real estate developer.

In simple terms, the simulator's classification logic will be:

- A property used by the company itself → IAS 16 PPE.
- A property built or acquired to be sold → IAS 2 Inventory.
- A property built or acquired to be rented out or held for value increase → IAS 40 Investment Property.

8.21.2 Measurement Principles

The simulator will model the initial measurement of investment property at cost, including transaction costs, which is a principle consistent with IAS 16. The key complexity and a core feature of the simulator's IAS 40 module will be in the subsequent measurement, where an accounting policy choice is available.

Cost Model An entity may choose the cost model. If this policy is selected, the accounting treatment is identical to the cost model in IAS 16. The property is carried at cost less accumulated depreciation and accumulated impairment losses.

Fair Value Model Alternatively, an entity may choose the fair value model. The simulator will model the unique mechanics of this approach, which differ significantly from the revaluation model in IAS 16:

- The investment property is remeasured to fair value at the end of each reporting period.
- Any gain or loss arising from a change in the fair value is recognised in **Profit or Loss (P/L)** for the period in which it arises. This is a direct contrast to the IAS 16 revaluation model, where gains are recognised in OCI.
- Under the fair value model, the investment property is **not depreciated**.

8.21.3 Transfers and Derecognition

IAS 40 provides rules for when a property's use changes, requiring a transfer between Investment Property (IAS 40), PPE (IAS 16), and Inventory (IAS 2). However, for the initial development phase of the simulator, a simplifying assumption will be made: **the model will not consider transfers** between property categories.

The simulator will model the derecognition of investment property. An asset will be removed from the statement of financial position on disposal or when it is permanently withdrawn from use and no future economic benefits are expected. The gain or loss on disposal, calculated as the difference between the net disposal proceeds and the carrying amount of the asset, will be recognised in **Profit or Loss (P/L)**.

8.22 Modelling IAS 41 *Agriculture*

To accommodate entities involved in agricultural activities, the simulator must incorporate the principles of IAS 41 *Agriculture*. This standard is unique as it largely departs from the historical cost model in favour of a fair value approach to reflect the economic reality of biological transformation.

8.22.1 Scope and Interaction with IAS 2

The simulator's logic will correctly apply IAS 41 to the two categories of assets within its scope:

- **Biological Assets:** Living animals and plants managed by the entity.
- **Agricultural Produce at the point of harvest:** The harvested product from the entity's biological assets.

A critical feature of the model will be the interface between IAS 41 and IAS 2. The simulator will be programmed to apply IAS 41 up to the point of harvest. Immediately **after harvest**, the agricultural produce will be reclassified as inventory, and its subsequent accounting will be governed by the IAS 2 module.

8.22.2 Measurement Principle

The core measurement principle of IAS 41, which will be implemented in the simulator, is measurement at fair value less costs to sell.

- **Biological Assets** will be measured at 'Fair Value Less Costs to Sell', with any gains or losses arising from changes in this value being recognised directly in profit or loss.
- **Agricultural Produce** will be measured at 'Fair Value Less Costs to Sell' at the point of harvest.
- **Change in Fair Value** will go into **profit and loss**.

This establishes the crucial link to the inventory system. The simulator will use the carrying amount determined at the point of harvest as the '**deemed cost**' of the inventory when it is initially recognised under the rules of IAS 2.

8.23 IFRS 2 *Share-based Payment* - (Out of Scope)

IFRS 2 *Share-based Payment* prescribes the accounting treatment for transactions in which an entity receives goods or services as consideration for its own equity instruments (e.g., share options granted to employees). While a significant standard, modelling the complex valuation and vesting conditions of share-based payments is a specialized field. Therefore, for the initial development phase of the simulator, transactions governed by IFRS 2 will be considered **out of scope**.

8.24 Modelling IFRS 3 *Business Combinations*

A core module of the accounting simulator will be its ability to model mergers and acquisitions. The logic for this module will be built exclusively on the principles of IFRS 3 *Business Combinations*. The central tenet of this standard is that all business combinations must be accounted for using the **acquisition method**. The simulator will implement the key steps of this method as follows.

8.24.1 Recognition and Measurement of Acquired Assets and Liabilities

Upon a simulated acquisition, the model will identify all identifiable assets acquired and liabilities assumed from the acquiree. In line with IFRS 3, all of these items will be measured at their **fair value at the acquisition date**. A critical feature of the simulator's logic will be its ability to recognise assets that may not have been on the acquiree's own statement of financial position. This includes intangible assets such as brand names, customer relationships, or in-process research and development projects, provided they are identifiable and their fair value can be measured reliably.

8.24.2 Measurement of Non-controlling Interest (NCI)

For acquisitions of less than 100% of an entity, the simulator will account for the non-controlling interest (NCI). The model will incorporate the two measurement choices for NCI provided by IFRS 3, which will be a key parameter in any acquisition scenario:

1. **The Fair Value Method:** The NCI is measured at its fair value on the acquisition date (e.g., based on the market price of the shares held by the NCI).
2. **The Proportionate Share Method:** The NCI is measured at its proportionate share of the acquiree's identifiable net assets at fair value.

8.24.3 Calculation of Goodwill or Bargain Purchase Gain

The final step in the acquisition accounting module will be the calculation of the residual amount, which is either goodwill or a gain from a bargain purchase. The simulator will use the following formula:

$$\text{Goodwill} = (A + B) - C$$

Where:

- **A** is the *consideration transferred*: the fair value of the assets transferred by the acquirer (e.g., cash, equity instruments issued, contingent consideration).
- **B** is the *amount of the NCI*, measured using one of the two methods described above.
- **C** is the *fair value of the net identifiable assets acquired*.

Goodwill If the result of the formula $(A + B) - C$ is positive, the amount will be recognised as **Goodwill**. The simulator will model goodwill as an asset representing future economic benefits arising from assets that are not individually identified (e.g., synergies, skilled workforce, customer loyalty). In line with IFRS, this goodwill will *not* be amortised, but will be subject to an annual impairment test under the IAS 36 module.

Gain from a Bargain Purchase If the result of the formula is negative (a "bargain purchase"), this indicates that the acquirer has purchased the net assets for less than their fair value. The simulator will recognise this negative difference immediately as a gain in **profit or loss**.

8.25 Modelling IFRS 5 *Non-current Assets Held for Sale and Discontinued Operations*

The simulator will be designed to account for changes in the intended use of an entity's long-term assets, specifically when management decides to sell an asset rather than continue to use it in operations. This is governed by IFRS 5 *Non-current Assets Held for Sale and Discontinued Operations*.

8.25.1 Modelling Assets Held for Sale

When a non-current asset's carrying amount is expected to be recovered principally through a sale transaction rather than through continuing use, its accounting treatment changes significantly. The simulator will incorporate a specific module to handle this reclassification and subsequent measurement.

Classification Criteria The model’s logic will reclassify a non-current asset (or a disposal group) to ‘Held for Sale’ only when both of the following criteria are met:

1. The asset is **available for immediate sale in its present condition**.
2. Its sale is **highly probable**.

Measurement and Impairment Upon reclassification, the simulator will apply the specific measurement principle of IFRS 5. The asset will be measured at the **lower of** its:

- Carrying Amount, and
- Fair Value less Costs to Sell (FVLCTS).

If the FVLCTS is lower than the carrying amount, the simulator will recognise an impairment loss. The model will also support the subsequent reversal of an impairment loss if the FVLCTS increases, capped at the amount of the cumulative impairment loss that has been previously recognised.

Cessation of Depreciation A critical rule that will be implemented is the immediate cessation of depreciation and amortisation. Once an asset is classified as held for sale, the simulator will stop charging depreciation or amortisation expense for that asset.

Presentation To reflect the change in nature and liquidity, the simulator’s Statement of Financial Position will present assets and disposal groups classified as held for sale separately from other assets. A specific line item, ‘**Non-current Assets Held for Sale**’, will be created. Furthermore, if a disposal group is classified as held for sale, the assets and liabilities within that group will be presented separately on the face of the balance sheet and will not be offset against one another.

8.25.2 Discontinued Operations - (Out of Scope)

The second major part of IFRS 5 deals with the presentation and disclosure of *discontinued operations*, which is a component of an entity that either has been disposed of or is classified as held for sale, and represents a separate major line of business or geographical area of operations. This requires a significant re-presentation of the Statement of Comprehensive Income.

For the current phase of the simulator’s development, the complex presentation and disclosure requirements for discontinued operations will be considered **out of scope**. The model will focus on the accounting for individual assets and disposal groups held for sale.

8.26 IFRS 6 *Exploration for and Evaluation of Mineral Resources* - (Out of Scope)

IFRS 6 *Exploration for and Evaluation of Mineral Resources* is a unique standard within the IFRS framework due to its highly industry-specific nature and the significant flexibility it permits. It is designed exclusively for entities in the extractive industries (e.g., oil, natural gas, and mineral exploration) and applies only to the costs incurred during the ‘exploration and evaluation’ phase of their operations.

The standard is considered a temporary and transitional measure. Its most notable feature is that it allows entities to develop their own accounting policy for exploration and evaluation expenditures. This provides a great deal of flexibility, permitting entities to determine which costs are capitalised as assets and which are recognised as expenses, so long as the policy is applied consistently and meets certain minimum requirements. This contrasts sharply with most other standards, which prescribe a single, detailed accounting model.

Given that the accounting treatment under IFRS 6 is highly dependent on an entity’s pre-existing and industry-specific policies rather than a uniform IFRS model, and because the primary goal of this simulator is to model the application of more broadly applicable IFRS standards, the specific requirements of IFRS 6 will be considered **out of scope** for the current project.

8.27 Modelling IFRS 9 *Financial Instruments*

The content of IFRS 9 *Financial Instruments* is vast and complex, representing a significant shift from its predecessor, IAS 39. The simulator will focus on implementing the most important components of this standard, which are structured around three core pillars.

8.27.1 Pillar 1: Classification and Measurement

The simulator’s foundational logic for financial instruments will be based on the IFRS 9 classification model, which replaced the complex, intent-based rules of IAS 39.

Financial Asset Classification The model will classify financial assets based on the two key tests mandated by IFRS 9:

1. **The Business Model Test:** This assesses the entity’s strategy for managing the financial assets. The simulator will assign assets to one of three business models: (a) Held to Collect contractual cash flows; (b) Held to Collect contractual cash flows and Sell the asset; or (c) Other business models (e.g., holding for trading).
2. **The SPPI Test:** This assesses whether the contractual cash flows of the asset represent *Solely Payments of Principal and Interest* on the principal amount outstanding.

Based on the outcome of these two tests, the simulator will classify and measure financial assets as shown in Table 3.

Table 3: IFRS 9 Financial Asset Classification Logic

Business Model	SPPI Test	Classification & Measurement
1. Held to Collect	Passes SPPI	Amortised Cost (AC)
2. Held to Collect & Sell	Passes SPPI	Fair Value through OCI (FVOCI)
3. Other / Fails SPPI	Fails SPPI	Fair Value through P&L (FVTPL)

For **equity instruments**, which do not have contractual cash flows, the simulator will apply the default classification of FVTPL. However, the model will include a parameter to allow for the irrevocable election to designate non-trading equity investments at FVOCI, where fair value changes are recognised in OCI and are never reclassified to profit or loss.

Financial Liability Classification The classification for financial liabilities is simpler. The simulator will measure most financial liabilities at **Amortised Cost**. The main exception will be for liabilities held for trading, which will be measured at FVTPL. The model will also incorporate the specific requirement that for liabilities designated at FVTPL, changes in fair value attributable to the entity’s *own credit risk* are presented in OCI.

8.27.2 Pillar 2: Impairment - The Expected Credit Loss (ECL) Model

A revolutionary aspect of IFRS 9 is its forward-looking impairment model. The simulator will feature a sophisticated ECL module to replace the backward-looking “incurred loss” model

of IAS 39. This “expected credit loss” (ECL) model requires recognising loss provisions in anticipation of future defaults.

The Three-Stage General Model For financial assets like loans and bonds, the simulator will implement the three-stage model:

- **Stage 1:** For assets where credit risk has not increased significantly since initial recognition, the loss allowance will be measured at an amount equal to **12-month ECL**.
- **Stage 2:** For assets where credit risk has increased significantly, the loss allowance will be measured at an amount equal to **lifetime ECL**. The model will use parameters such as ‘>30 days past due’ as a trigger for this significant increase.
- **Stage 3:** For assets that are credit-impaired (e.g., ‘>90 days past due’), the loss allowance will continue to be measured at **lifetime ECL**. The interest revenue calculation for these assets will be based on the net carrying amount (amortised cost less the ECL provision).

The Simplified Approach For trade receivables, contract assets, and lease receivables, the simulator will apply the simplified approach permitted by IFRS 9. This approach does not require tracking the stage transfers; instead, the loss allowance will always be measured at an amount equal to **lifetime ECL**.

8.27.3 Pillar 3: Hedge Accounting

IFRS 9’s hedge accounting model is designed to align more closely with an entity’s risk management activities, offering more flexibility than the rigid rules of IAS 39. The key principles include relaxed effectiveness testing, the concept of ‘cost of hedging’, and ‘rebalancing’.

While the simulator’s framework will be designed to be compatible with these principles, the detailed implementation of the highly complex rules for hedge accounting will be considered for future development. The initial focus of the simulator will be on the correct implementation of the Classification & Measurement and Impairment pillars of IFRS 9.

8.28 Group Accounting and Fair Value Measurement (IFRS 10, 11, & 13)

This section outlines the simulator’s approach to standards governing complex control structures and the measurement of fair value. To maintain focus on the core single-entity stock-flow dynamics in the initial phase, simplifying assumptions will be adopted for these standards.

8.28.1 IFRS 10 & IFRS 11: Consolidation and Joint Arrangements - (Out of Scope)

The IFRS framework provides detailed guidance for accounting for interests in other entities.

IFRS 10 Consolidated Financial Statements The core contribution of IFRS 10 is the establishment of a single, uniform definition of ‘control’. This definition serves as the sole basis for determining whether an investor must consolidate an investee.

IFRS 11 Joint Arrangements The core contribution of IFRS 11 is to simplify and clarify the accounting for all joint arrangements. It mandates that entities must classify such arrangements as either a *Joint Operation* or a *Joint Venture* based on the entity’s contractual rights and obligations.

Applicability to the Simulator The accounting for subsidiaries (consolidation under IFRS 10) and joint arrangements (IFRS 11) introduces significant complexity, involving procedures for eliminating intra-group transactions and calculating goodwill or non-controlling interests. Therefore, for the initial phase of development, the simulator will focus on the accounting for single-entity operations. The complex mechanics of consolidation and joint arrangement accounting are considered **out of scope**.

8.28.2 IFRS 13: Fair Value Measurement - (Simplified Application)

IFRS 13 *Fair Value Measurement* provides a single framework and definition for measuring fair value where its use is required or permitted by other IFRS standards. It establishes a hierarchy of inputs for valuation techniques (Level 1, 2, and 3) to increase consistency and comparability.

Applicability to the Simulator While the IFRS framework frequently requires fair value measurements (e.g., for certain financial instruments under IFRS 9), the process of estimating fair value is complex. For the purpose of the current model, a simplifying assumption will be made. Where a standard requires a fair value measurement, it will be assumed that the **fair value is a known, exogenous input** to the simulator. The simulator will not contain a module to estimate fair values itself, but will be designed to correctly apply the required accounting treatment based on the provided fair value data.

8.29 Modelling IFRS 15 *Revenue from Contracts with Customers*

The simulator will follow the rules of IFRS 15 for all revenue. The main idea of this standard is to recognise revenue when a good or service is transferred to a customer. The amount recognised should be what the company expects to receive.

IFRS 15 uses a five-step model to achieve this. This section will not detail all five steps. Instead, it focuses on three key outcomes the simulator must model correctly.

8.29.1 Cash Received and Liabilities

The simulator will separate when cash is received from when revenue is earned.

- When a customer pays in advance, this is not yet revenue.
- The model will increase cash and also increase a liability. This liability is called a **Contract Liability**.
- The model will only recognise revenue at when the company delivers the good or provides the service.

8.29.2 Long Payment Terms and Interest

The model will check for long payment terms in contracts.

- If the time between payment and performance is more than one year, this is a financing deal.
- The simulator will then adjust the revenue amount for the time value of money.
- This creates interest. If the customer pays much later than the service is provided, the simulator will record **interest income**. If the customer pays far in advance, it will record **interest expense**.

8.29.3 Timing of Revenue Recognition

The simulator must know when to recognise revenue. There are two patterns, and the model will support both.

Recognition at a point in time. This happens all at once. For example, when a customer buys a product from a shop. The simulator will record all the revenue on that specific date.

Recognition over time. This happens over a period. For example, a one-year software subscription. The simulator will spread the revenue out over the contract period, often recognising it on a straight-line basis.

8.30 Modelling IFRS 16 *Leases*

IFRS 16 *Leases* represents one of the most significant recent changes in accounting, fundamentally altering how entities account for lease contracts. The simulator will incorporate a comprehensive module to reflect the standard's transformative impact, particularly for lessees.

8.30.1 Core Change: The Elimination of the Lessee's Operating Lease

Under the previous standard, IAS 17, lessees classified leases as either a finance lease or an operating lease. A finance lease was recognised on the balance sheet, whereas an operating lease was treated as a simple rental, with lease payments expensed to the income statement and no asset or liability recognised.

This created the problem of **off-balance-sheet financing**. A significant volume of lease obligations (e.g., for aircraft or retail stores) remained "hidden" from the balance sheet, making it difficult for investors to assess an entity's true level of leverage and assets.

IFRS 16 resolves this by introducing a **single lessee accounting model**, which eliminates the distinction between finance and operating leases. The core principle is:

At the commencement date, a lessee shall recognise a **right-of-use asset** and a **lease liability** on its statement of financial position for almost all lease contracts.

In simple terms, leases that were formerly "off-balance sheet" must now be brought "on-balance sheet".

8.30.2 The Lessee Accounting Model

The simulator's logic will be built around the new lessee accounting model.

Initial Measurement At the start of a lease, the model will recognise:

- **A Lease Liability**, measured at the present value of the future lease payments. The lease payments will include fixed payments, variable payments dependent on an index or rate, and amounts expected to be payable under purchase options or termination penalties. The discount rate used will be the interest rate implicit in the lease or the lessee's incremental borrowing rate.
- **A Right-of-use (ROU) Asset**, measured at cost. The cost will comprise the amount of the initial lease liability, any initial direct costs, any lease payments made at or before the commencement date (less any lease incentives received), and an estimate of dismantling and restoration costs.

Subsequent Measurement After commencement, the asset and liability are accounted for separately:

- **The Lease Liability** is accounted for similarly to a loan. The simulator will increase the carrying amount to reflect interest using the *effective interest method* and decrease it for lease payments made.
- **The ROU Asset** is typically accounted for under the cost model, similar to PPE under IAS 16. The simulator will charge *depreciation* over the useful life of the asset and test it for impairment under the IAS 36 module.

Impact on the Income Statement The single, straight-line ‘rent expense’ of an old operating lease is replaced by two separate expenses: (1) *depreciation expense* on the ROU asset (typically straight-line) and (2) *interest expense* on the lease liability (higher in the earlier years of the lease). This results in a front-loaded total expense profile and impacts key performance metrics such as EBITDA.

8.30.3 Recognition Exemptions

To simplify implementation, the simulator will also support the two recognition exemptions for lessees. A simulated entity can elect not to apply the on-balance sheet model for:

1. **Short-term leases** (leases with a term of 12 months or less).
2. **Leases of low-value assets** (e.g., tablets, office furniture).

8.30.4 Definition of a Lease

The model will adopt the updated IFRS 16 definition of a lease. A contract will be classified as a lease if it conveys the right to **control the use of an identified asset** for a period of time in exchange for consideration. This requires assessing if there is an *identified asset* and if the customer has the right to obtain substantially all the economic benefits from, and to *direct the use of*, that asset.

8.30.5 Summary of Impacts and Simulator Implementation

The introduction of IFRS 16 has a pervasive impact on a lessee’s financial statements:

- **Statement of Financial Position:** Assets and liabilities both increase, raising financial leverage ratios.
- **Statement of Comprehensive Income:** Expense recognition is front-loaded, and the nature of expenses changes from ‘rent’ to ‘depreciation and interest’, typically increasing EBITDA and EBIT.
- **Statement of Cash Flows:** The principal portion of a lease payment is classified under financing activities, while the interest portion can be in operating or financing activities. This generally increases cash from operations.

The accounting simulator will fully apply the IFRS 16 model to all relevant lease contracts entered into by a simulated entity, thereby providing a realistic depiction of this standard’s significant effects on modern financial reporting.

9 Application Example: Standards to Key Working Capital Items

The accounting treatment for specific line items is determined by the interplay of several standards. The simulator will be designed to correctly apply this multi-standard framework to key working capital accounts. This section summarises the governing principles for four such items.

9.1 Trade Receivable

An Trade Receivable is a financial asset representing an entity's right to consideration in exchange for goods or services. Its accounting lifecycle is governed by a combination of standards:

- **IFRS 15 *Revenue from Contracts with Customers*** governs its initial recognition and measurement. The receivable is recognised when the entity's right to consideration becomes unconditional, and its amount is determined by the transaction price allocation.
- **IFRS 9 *Financial Instruments*** governs its subsequent measurement. It is classified as a financial asset measured at **amortised cost**. Critically, IFRS 9 also mandates that the receivable is subject to **impairment** testing under the Expected Credit Loss (ECL) model.
- **IAS 1 *Presentation of Financial Statements*** guides its presentation, classifying it as a **current asset** as it is expected to be realised within the entity's normal operating cycle.
- **IFRS 7 *Financial Instruments: Disclosures*** specifies the detailed disclosures required in the notes regarding the receivable's credit risk and the ECL calculation.

9.2 Trade Payable

An Trade Payable represents an entity's obligation to pay for goods or services received from a supplier. It is a financial liability.

- **IAS 32 *Financial Instruments: Presentation*** provides the fundamental definition of a **financial liability**, which includes the contractual obligation to deliver cash, thereby classifying accounts payable.
- **IFRS 9 *Financial Instruments*** governs its subsequent measurement at **amortised cost**. Unlike financial assets, financial liabilities are **not subject to impairment** testing under the ECL model.
- **IAS 1 *Presentation of Financial Statements*** guides its presentation, classifying it as a **current liability** as it is expected to be settled within the entity's normal operating cycle.

9.3 Prepayment

A Prepaid Expense is a non-financial asset representing a right to receive goods or services in the future for which payment has already been made.

- **IAS 1 *Presentation of Financial Statements*** guides its presentation as a **current asset**. The recognition of a prepaid expense is a direct result of the accrual basis of accounting, also prescribed by IAS 1.
- The subsequent treatment involves the asset being amortised or expensed to profit or loss as the entity consumes the related economic benefits (i.e., receives the goods or services). This process ensures costs are matched with the period in which they are incurred.

9.4 Unearned Revenue (Contract Liability)

Contract Liability is a non-financial liability representing an entity's obligation to provide goods or services in the future for which payment has already been received.

- **IFRS 15 *Revenue from Contracts with Customers*** provides the specific terminology and governing principles for this item, defining it as a **Contract Liability**. It is recognised when a customer pays consideration before the entity transfers the related good or service.
- The subsequent treatment involves the liability being derecognised and **recognised as revenue** in profit or loss as the entity satisfies its performance obligations.
- **IAS 1 *Presentation of Financial Statements*** guides its presentation, classifying it as a **current liability** as the obligation is expected to be settled within the entity's normal operating cycle.

10 Complete Consolidated statement of Profit and Loss, and Statement of Cash Flow

As previously discussed, no single standard provides a complete chart of accounts for the financial statements. Instead, IAS 1 *Presentation of Financial Statements* provides the framework and minimum requirements, while the detailed rules for the composition and calculation of specific line items are derived from various topical IFRS and IAS standards. This section provides a comprehensive guide to the sources of common line items for the Statement of Comprehensive Income and the Statement of Cash Flows.

10.1 Statement of Comprehensive Income

This statement is composed of two sections: the Profit or Loss section and the Other Comprehensive Income (OCI) section.

10.1.1 Part 1: Profit or Loss (P/L)

The P/L section reflects the entity's financial performance over a period. The following table maps key line items to their governing standards.

Table 4: Source of Key Profit or Loss Line Items

P/L Line Item	Governing Standard	What the Standard Prescribes
Revenue	IFRS 15	The five-step model for revenue recognition: when to recognise and how much to measure.
Cost of Sales (COGS)	IAS 2	The composition of inventory costs, costing methods (FIFO/WAC), and when to expense inventory.
<i>Gross Profit</i>	<i>IAS 1</i>	<i>An important subtotal required by IAS 1 when expenses are classified by function.</i>
Other Income	Various (e.g., IAS 16, 20, IFRS 9)	Rules for specific income, e.g., gain on disposal of PPE (IAS 16), government grant income (IAS 20).
Distribution / Selling Costs	IAS 1	A functional classification. The specific components (e.g., salaries, depreciation) are governed by other standards like IAS 19 and IAS 16.
Administrative Expenses	IAS 1	A functional classification, with components governed by other standards.
Employee Benefits Expense	IAS 19	The recognition and measurement of salaries, bonuses, pensions (service cost and net interest for DB plans), etc.
Depreciation & Amortisation	IAS 16, IAS 38, IFRS 16	Rules for the method, useful life, and residual value for PPE, intangible assets, and right-of-use assets, respectively.
Impairment Losses	IAS 36, IFRS 9	The impairment testing and loss recognition for non-financial assets (e.g., PPE, goodwill) and financial assets (ECL model), respectively.
Finance Costs	IFRS 9, IAS 23	The calculation of interest on financial liabilities (effective interest method under IFRS 9) and the rules for capitalising borrowing costs (IAS 23).
Share of Profit of Associates/JVs	IAS 28	The application of the equity method to recognise the investor's share of profit or loss from its associates and joint ventures.
<i>Profit before Tax</i>	<i>IAS 1</i>	<i>A mandatory subtotal.</i>
Income Tax Expense	IAS 12	The calculation and recognition of current tax and deferred tax.
Profit for the year	IAS 1	The final result of the Profit or Loss section.

10.1.2 Part 2: Other Comprehensive Income (OCI)

The OCI section includes items of income and expense that are not recognised in profit or loss as required or permitted by other IFRSs.

Table 5: Source of Key Other Comprehensive Income Line Items

OCI Category & Line Item	Governing Standard
Items that will <i>not</i> be reclassified to P/L	
Changes in Revaluation Surplus	IAS 16
Remeasurements of a Defined Benefit Plan	IAS 19
Gains and Losses on FVOCI Equity Instruments	IFRS 9
Items that <i>may</i> be reclassified to P/L	
Gains and Losses on Cash Flow Hedges	IFRS 9
Gains and Losses on FVOCI Debt Instruments	IFRS 9
Exchange Differences on Translating Foreign Operations	IAS 21

10.2 Statement of Cash Flows

The primary governing standard for this statement is IAS 7 *Statement of Cash Flows*, which provides the structure and classification rules. The specific line items are a result of the entity's actual cash transactions during the period.

Table 6: Source and Classification of Key Cash Flow Items

Cash Flow Category	Example Line Item	Related Standard(s)
Operating Activities	Cash receipts from customers	IFRS 15, IFRS 9
	Cash paid to suppliers and employees	IAS 2, IAS 19
	Income taxes paid	IAS 12
	Interest and dividends received / paid	IAS 7 (provides classification choices)
Investing Activities	Purchase / sale of PPE & Intangibles	IAS 16, IAS 38
	Purchase / sale of equity/debt instruments	IFRS 9
	Loans made to other parties	IFRS 9
Financing Activities	Proceeds from issuing shares	IAS 32
	Proceeds from issuing debt / borrowings	IFRS 9
	Repayment of borrowings (principal)	IFRS 9
	Principal payments on lease liabilities	IFRS 16
	Dividends paid	IAS 1, IAS 32

10.3 Balance Sheets

Table 7: Statement of Financial Position - Assets

Account	Governing Standard	Explanation
Non-current assets		
Property, plant and equipment	IAS 16	Rules for recognizing, measuring, and depreciating tangible fixed assets.
Right-of-use assets	IFRS 16	Recognition of leased assets on the balance sheet for lessees, representing the right to use an underlying asset.
Intangible assets and goodwill	IAS 38, IFRS 3	Rules for non-physical assets (IAS 38) and goodwill from acquisitions (IFRS 3).
Investment property	IAS 40	Property held for rental income or capital appreciation, measured at cost or fair value.
Financial assets	IFRS 9	Classification and measurement rules for financial investments (e.g., shares, bonds) not classified elsewhere.
Biological assets	IAS 41	Accounting for agricultural assets, measured at fair value less costs to sell.
Employee benefits	IAS 19	Asset arising when a defined benefit pension plan is overfunded (plan assets exceed obligation).
Deferred tax assets	IAS 12	Assets arising from future tax deductions or tax loss carryforwards.
Total non-current assets		
Current assets		
Inventories	IAS 2	Valuation at the lower of cost and net realisable value (NRV).
Trade and other receivables	IFRS 9	Measurement of receivables, including an allowance for expected credit losses (ECL).
Prepayments	IAS 1	Payments made in advance for goods or services to be received in the future.
Financial assets	IFRS 9	Classification and measurement rules for financial investments (e.g., shares, bonds) not classified elsewhere.
Biological assets	IAS 41	Agricultural assets expected to be harvested or sold within one operating cycle.
Current tax assets	IAS 12	Income tax recoverable from tax authorities for the current or prior periods.
Assets held for sale	IFRS 5	Non-current assets intended for sale, measured at lower of carrying amount and fair value.
Cash and cash equivalents	IAS 7	Defines cash on hand and highly liquid short-term investments.
Total current assets		
Total assets		

Table 8: Statement of Financial Position - Liabilities

Account	Governing Standard	Explanation
Non-current liabilities		
Financial liabilities	IFRS 9	Non-derivative financial liabilities not designated at FVTPL, measured using the effective interest method. Due after 12 months.
Other financial liabilities	IFRS 9	Liabilities held for trading or designated at fair value through profit or loss (FVTPL). Due after 12 months.
Lease liabilities	IFRS 16	Portion of lease payment obligations due for settlement after more than 12 months.
Provisions	IAS 37	A liability of uncertain timing or amount, expected to be settled after 12 months.
Contract liabilities	IFRS 15	Payment received for services/goods to be delivered after 12 months.
Employee benefits	IAS 19	Long-term liabilities for employee benefits, such as defined benefit pension plans.
Trade and other payables	IFRS 9	In rare cases, payables with settlement terms longer than 12 months.
Deferred tax liabilities	IAS 12	Future income tax payable resulting from temporary differences.
Total non-current liabilities		
Current liabilities		
Financial liabilities	IFRS 9	The portion of financial liabilities at amortised cost due to be settled within 12 months.
Other financial liabilities	IFRS 9	The portion of financial liabilities at FVTPL due to be settled within 12 months.
Lease liabilities	IFRS 16	The portion of the lease liability due within 12 months.
Trade and other payables	IFRS 9	Obligations to pay for goods or services received from suppliers, due within 12 months.
Contract liabilities	IFRS 15	Obligation to provide goods/services in the future for which payment has been received.
Provisions	IAS 37	The portion of provisions expected to be settled within 12 months.
Employee benefits	IAS 19	Short-term obligations like accrued salaries, bonuses, and holiday pay.
Current tax payable	IAS 12	Unpaid income tax for the current and prior periods.
Liabilities assoc. with assets held for sale	IFRS 5	Liabilities directly linked to a disposal group classified as held for sale.
Total current liabilities		
Total liabilities		

Table 9: Statement of Financial Position - Equity

Account	Governing Standard	Explanation
Equity		
Share capital and premium	IAS 1	Funds contributed directly by shareholders in exchange for shares.
Reserves	Various	Other equity components, such as revaluation surplus or foreign currency translation reserves.
Retained earnings	IAS 1	Cumulative net profits of the company not paid out as dividends.
Equity attributable to owners of the parent		
Non-controlling interests	IFRS 10	The portion of a subsidiary's equity not owned by the parent company.
Total equity		

11 Asset

11.1 Cash and Cash Equivalents

$$\begin{aligned}
 \text{Cash and Cash Equivalents}_t &= \text{Cash and Cash Equivalents}_{t-1} \\
 &\quad + \text{CFO}_t \\
 &\quad + \text{CFI}_t \\
 &\quad + \text{CFF}_t \\
 &\quad + \text{FXRecon}_t
 \end{aligned}$$

Table 10: Explanation of Flow Variables for Cash and Cash Equivalents

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
FXRecon _t	IAS 7	FX effect and reclassifications between cash & cash equivalents	SCF	×

11.2 Trade and Other Receivables

This section details the dynamics of trade receivables, broken down into two components: the gross amount owed by customers, and the allowance for expected credit losses (ECL). The net amount is presented on the Statement of Financial Position.

11.2.1 Gross Trade Receivables

This represents the total invoiced amount owed by customers from credit sales, before any reduction for expected credit losses.

$$\begin{aligned}
 \text{Gross TR}_t &= \text{Gross TR}_{t-1} \\
 &\quad + \text{New receivables recognised (Credit Sales)}_t \\
 &\quad - \text{Cash collections}_t \\
 &\quad - \text{Write-offs}_t
 \end{aligned}$$

Table 11: Explanation of Flow Variables for Gross Trade Receivables

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New receivables recognised _t	IFRS 15	The value of credit sales made during the period. This flow increases revenue.	P&L	✓
Cash collections _t	IAS 7, IFRS 9	Cash received from customers, which settles the receivable.	SCF	✓
Write-offs _t	IFRS 9	Derecognition of a specific receivable when it is deemed uncollectible. This is a non-cash event that also reduces the allowance account.	SFP (reclass)	×

11.2.2 Allowance for Expected Credit Losses (ECL)

This is a contra-asset account that reduces the gross trade receivables to their net realisable value. Its movement is driven by the P&L charge for bad debts.

Dynamic Model of ECL Allowance

$$\begin{aligned} \text{ECL Allowance}_t &= \text{ECL Allowance}_{t-1} \\ &\quad + \text{ECL expense recognised in P\&L}_t \\ &\quad - \text{Write-offs}_t \end{aligned}$$

Calculation of ECL Expense The ECL expense for the period is the amount needed to adjust the opening allowance (net of write-offs) to the required closing target allowance.

$$\text{Target Allowance}_t = \text{Gross TR}_t \times \text{Expected Loss Rate}_t$$

$$\text{ECL expense}_t = \text{Target Allowance}_t - (\text{ECL Allowance}_{t-1} - \text{Write-offs}_t)$$

Table 12: Explanation of Flow Variables for ECL Allowance

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
ECL expense recognised in P&L _t	IFRS 9	The charge for the period representing the change in lifetime expected credit losses. Also known as bad debt expense.	P&L	✓
Write-offs _t	IFRS 9	The utilisation of the allowance to derecognise a specific uncollectible receivable. This links the two dynamic models.	SFP (reclass)	×

Calculation of Net Trade Receivables The final amount presented on the Statement of Financial Position is the net of the gross receivables and the allowance.

$$\text{Net TR}_t = \text{Gross TR}_t - \text{ECL Allowance}_t$$

11.3 Return Asset

This asset arises from sales with a right of return and is the counterpart to the refund liability. It represents the entity's right to recover goods from customers on settling the refund liability. It is measured at the former carrying amount of the goods (usually cost).

$$\begin{aligned} \text{Return Asset}_t &= \text{Return Asset}_{t-1} \\ &\quad + \text{New return assets recognised}_t \\ &\quad - \text{Cost of goods actually returned}_t \\ &\quad \pm \text{Changes in estimate (adjustments)}_t \end{aligned}$$

Initial Recognition When a sale is made, the cost of goods expected to be returned is not recognised in Cost of Sales (COGS). Instead, it is recognised as a return asset.

$$\text{New return asset}_t = \text{Cost of goods expected to be returned from current sales}$$

Table 13: Explanation of Flow Variables for Return Asset

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New return assets recognised _t	IFRS 15, IAS 2	The asset recognised for the right to recover goods, measured at their cost. This is paired with a reduction in recognised Cost of Sales.	P&L	×
Cost of goods actually returned _t	IFRS 15, IAS 2	When goods are physically returned by customers, the Return Asset is derecognised and the goods are recognised back into Inventory.	SFP (reclass)	×
Changes in estimate (adjustments) _t	IFRS 15, IAS 2	Adjustments due to changes in expected returns. A decrease in expected returns reduces the Return Asset and increases COGS. Also includes any write-downs if the asset's expected value is impaired.	P&L	×

11.4 Inventory

Inventory is measured at the lower of cost and net realisable value (NRV). Its dynamic model captures purchases, consumption, and various reclassifications.

$$\begin{aligned}
\text{Inventory}_t = & \text{Inventory}_{t-1} + \text{Purchases on credit}_t + \text{Cash purchases}_t \\
& + \text{Capitalised Borrowing Costs}_t + \text{Goods Returned to Stock}_t \\
& + \text{Increases from harvest}_t + \text{Reclass from IP to Inventory}_t \\
& - \text{Cost of Goods Sold (COGS)}_t - \text{Used in Self-Construction}_t \\
& - \text{Physical Loss Expense}_t - \text{NRV Write-down}_t \\
& + \text{NRV Reversal}_t - \text{Reclass from Inventory to IP}_t
\end{aligned}$$

Table 14: Explanation of Flow Variables for Inventory

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Purchases on credit _t	IAS 2	Cost of inventory purchased on credit, increasing Trade Payables.	SFP	✓
Cash purchases _t	IAS 2, IAS 7	Cost of inventory purchased with cash.	SCF	✓
Capitalised Borrowing Costs _t	IAS 23	Borrowing costs capitalized for qualifying inventory.	P&L (Reduces Int. Exp.)	×
Goods Returned to Stock _t	IFRS 15, IAS 2	Inventory increase from physical customer returns (from Return Asset).	SFP (reclass)	×
Increases from harvest _t	IAS 41, IAS 2	Reclassification from biological assets to inventory upon harvest.	SFP (reclass)	✓
Cost of Goods Sold (COGS) _t	IAS 2	Cost of inventory sold to customers.	P&L	✓
Used in Self-Construction _t	IAS 16	Cost of inventory reclassified to another asset (e.g., PPE) when used internally.	SFP (reclass)	✓
Physical Loss / NRV Write-down _t	IAS 2	Write-off for abnormal loss or write-down to NRV.	P&L	✓
NRV Reversal _t	IAS 2	Reversal of a prior NRV write-down.	P&L (Reduces COGS)	✓
Reclass to/from Inv. Property _t	IAS 40, IAS 2	Reclassification between Inventory and Investment Property.	SFP (reclass)	×

11.5 Property, Plant, and Equipment (PPE)

Measurement basis note: PPE is accounted for using the **cost model** (IAS 16.30). The carrying amount is cost less accumulated depreciation and accumulated impairment losses.

Dynamic Model of Net PPE

$$\begin{aligned}
 \text{PPE Net}_t &= \text{PPE Net}_{t-1} + \text{Additions}_t - \text{Depreciation Expense}_t \\
 &\quad - \text{Impairment Loss}_t + \text{Impairment Reversal}_t \\
 &\quad \pm \text{Net Reclassifications}_t - \text{Carrying Amount of Disposals}_t \\
 &= \text{PPE Net}_{t-1} + \text{Additions}_t - \text{Depreciation Expense}_t \\
 &\quad - \text{Impairment Loss}_t + \text{Impairment Reversal}_t \pm \text{Net Reclassifications}_t \\
 &\quad + \text{Gain/Loss on Disposal}_t - \text{Proceeds from sale}_t
 \end{aligned}$$

Table 15: Explanation of Main Flow Variables for PP&E (Net)

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Additions _t	IAS 16, 23, 37	Total cost of new PPE. See further decomposition below.	SCF / SFP / P&L	✓*
Depreciation Expense _t	IAS 16	The systematic allocation of the depreciable amount over the useful life.	P&L	✓
Impairment Loss _t	IAS 36	A write-down of the asset's carrying amount to its recoverable amount.	P&L	×
Carrying Amount of Disposals _t	IAS 16	The net book value of assets sold. See further decomposition below.	P&L / SCF	✓*
Net Reclassifications _t	IAS 40, IAS 16	The net carrying amount transferred between PPE and Investment Property.	SFP (reclass)	✓

*See further decomposition table for details.

Further Decomposition of Main Flows This table details the components of the main flow variables.

Table 16: Further Decomposition of PP&E Flows

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Components of Additions_t				
Cash Purchases _t	IAS 16, IAS 7	Cash paid to acquire PPE.	SCF (Investing)	✓
Capitalised Borrowing Costs _t	IAS 23	Borrowing costs directly attributable to a qualifying asset.	P&L (Reduces Int. Exp.)	×
Inventory Used in Self-Construction _t	IAS 2, IAS 16	Reclassification of inventory consumed to build PPE.	SFP (reclass)	✓
ARO Recognition _t	IAS 37, IAS 16	Initial recognition of an ARO, added to the cost of PPE.	SFP (non-cash)	×
Non-monetary Acquisitions _t	IAS 16	PPE acquired via non-cash consideration.	SFP (non-cash)	×
Components of Disposals_t				
Proceeds from sale of PPE _t	IAS 7, IAS 16	Cash received from the disposal of the asset.	SCF (Investing)	✓
Gain/Loss on Disposal _t	IAS 16	Calculated as: 'Proceeds - Carrying Amount of Disposals'.	P&L	✓

11.6 Assets Held for Sale (AHFS)

This is a temporary classification for non-current assets (or a disposal group) whose value will be recovered principally through a sale transaction rather than through continuing use.

$$\begin{aligned}
 \text{AHFS}_t &= \text{AHFS}_{t-1} \\
 &\quad + \text{Assets reclassified as HFS}_t \\
 &\quad - \text{Impairment loss post-reclassification}_t \\
 &\quad - \text{Assets derecognised on disposal}_t \\
 &\quad - \text{Assets reclassified back}_t
 \end{aligned}$$

Measurement Changes Post-Reclassification A key rule under IFRS 5 is that once an asset is classified as held for sale, **depreciation and amortisation must cease**. The primary subsequent measurement is an impairment test. Any write-down to fair value less costs to sell (FVLCTS) is recognised as an impairment loss.

$$\text{Impairment Loss}_t = \max(0, \text{Carrying Amount}_t - \text{FVLCTS}_t)$$

Table 17: Explanation of Flow Variables for AHFS

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Assets reclassified as HFS _t	IFRS 5	Reclassification from original non-current asset categories (e.g., PPE, Intangibles) to AHFS upon meeting the criteria.	SFP (reclass)	×
Impairment loss post-reclass _t	IFRS 5	Unlike LAHFS, depreciation ceases. The asset is written down to its FVLCTS if lower than the carrying amount. The loss is recognised in P&L.	P&L	×
Assets derecognised on disposal _t	IFRS 5	Removal from the SFP when the disposal group is sold. Cash proceeds are part of investing activities.	P&L / SFP / SCF	×
Assets reclassified back _t	IFRS 5	Reclassification back to original categories if the held-for-sale criteria are no longer met. The asset is measured at the lower of its recoverable amount and its pre-HFS carrying amount adjusted for depreciation that would have been recognised.	SFP (reclass)	×

11.7 Current Tax Asset (CTA)

This represents the amount of income taxes recoverable from taxation authorities, typically arising from overpayment of tax instalments. It is distinct from Deferred Tax Asset and is classified as current.

$$\begin{aligned}
 \text{CTA}_t &= \text{CTA}_{t-1} \\
 &\quad - \text{Current tax expense for the period}_t \\
 &\quad + \text{Cash tax paid during the period}_t
 \end{aligned}$$

Nature of Current Tax Asset A CTA typically arises when the cash tax paid in instalments during the year exceeds the final current tax expense calculated for that year. The excess represents an amount receivable from the tax authority.

Table 18: Explanation of Flow Variables for Current Tax Asset

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Current tax expense _t	IAS 12	The tax charge for the period. This utilises or reduces the asset created by tax prepayments.	P&L	×*
Cash tax paid _t	IAS 7	Represents cash paid to tax authorities, often as prepayments, which creates the CTA.	SCF	×*

*Not applicable in the simulator, as it assumes tax paid exactly equals tax expense, meaning no CTP or CTA arises.

11.8 Prepayments (PP)

This represents the right to receive goods or services in the future for which the entity has been paid. It is the mirror image of a Contract Liability.

$$\begin{aligned}
 PP_t = & PP_{t-1} \\
 & + \text{Cash outflow for advance payments}_t \\
 & + \text{Interest income (financing component)}_t \\
 & - \text{Expense recognised}_t
 \end{aligned}$$

Initial Recognition of Cash Outflow The asset is initially recognised at the nominal cash value paid. For long-term prepayments, this consideration is conceptually split into the asset/expense itself (at present value) and a financing component.

Calculation of Interest Income This applies only to long-term prepayments with a significant financing component.

$$\text{Interest income}_t = \begin{cases} 0 & \text{if short-term (current)} \\ \text{Carrying Amount}_{t-1} \times \text{Discount Rate} & \text{if long-term (non-current)} \end{cases}$$

Table 19: Explanation of Flow Variables for Prepayments

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Cash outflow (advance payments) _t	IFRS 15, IAS 7	Cash paid to suppliers before the goods or services are received.	SCF	✓
Interest income _t	IFRS 15	Arises from a significant financing component in long-term prepayments. This is zero for short-term prepayments.	P&L	✓
Expense recognised _t	IAS 1, IAS 2	When the goods/services are received, the prepayment is reduced and an expense (or another asset like inventory) is recognised.	P&L	✓

11.9 Right-of-Use Asset (ROU Asset)

Under IFRS 16, a ROU asset is recognised for most leases, representing the lessee's right to use an underlying asset for the lease term. It is typically measured at cost and subsequently depreciated.

$$\begin{aligned}
 \text{ROU}_t &= \text{ROU}_{t-1} \\
 &\quad + \text{New ROU assets recognised}_t \\
 &\quad - \text{ROU Depreciation charge}_t \\
 &\quad - \text{Impairment loss}_t \\
 &\quad \pm \text{Adjustments from lease reassessments}_t
 \end{aligned}$$

Initial Recognition of ROU Asset The initial cost of the ROU asset is based on the initial measurement of the lease liability, plus other direct costs.

$$\begin{aligned}
 \text{New ROU Asset}_t &= \text{Initial Lease Liability}_t + \text{Initial direct costs}_t \\
 &\quad + \text{Prepayments}_t - \text{Lease incentives received}_t
 \end{aligned}$$

Table 20: Explanation of Flow Variables for Right-of-Use Asset

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New ROU assets recognised _t	IFRS 16	Initial recognition of the asset, corresponding to the new lease liability. This is a non-cash transaction.	SFP (non-cash)	✓
Depreciation charge _t	IFRS 16, IAS 16	The systematic allocation of the ROU asset's cost over its useful life or the lease term.	P&L	✓
Impairment loss _t	IAS 36	Reduction in the carrying amount if the asset is impaired.	P&L	×
Adjustments from reassessments _t	IFRS 16	Corresponds to adjustments made to the lease liability (e.g., from a change in the lease term).	SFP (non-cash)	×

11.10 Intangible Assets and Goodwill

Measurement basis note: Finite-life intangible assets are accounted for using the **cost model** (IAS 38). Goodwill and indefinite-life intangibles are not amortised but tested for impairment annually (IFRS 3, IAS 36).

11.10.1 Intangible Assets (Finite Life)

Dynamic Model of Net Intangibles

$$\begin{aligned}
 \text{Intangible Net}_t &= \text{Intangible Net}_{t-1} + \text{Additions}_t - \text{Amortisation Expense}_t \\
 &\quad - \text{Impairment Loss}_t + \text{Impairment Reversal}_t \\
 &\quad - \text{Carrying Amount of Disposals}_t \\
 &= \text{Intangible Net}_{t-1} + \text{Additions}_t - \text{Amortisation Expense}_t \\
 &\quad - \text{Impairment Loss}_t + \text{Impairment Reversal}_t \\
 &\quad + \text{Gain/Loss on Disposal}_t - \text{Proceeds from sale}_t
 \end{aligned}$$

Table 21: Explanation of Main Flow Variables for Intangible Assets

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Additions _t	IAS 38, IAS 23	Total cost of new intangibles. See decomposition below.	SCF / SFP / P&L	✓*
Amortisation Expense _t	IAS 38	Systematic allocation of the asset's cost over its useful life.	P&L	✓
Impairment Loss _t	IAS 36	A write-down to recoverable amount. Reversals are permitted.	P&L	✓
Carrying Amount of Disposals _t	IAS 38	The net book value of intangible assets sold or retired.	P&L / SCF	✓*

*See further decomposition table for details.

Table 22: Further Decomposition of Intangible Asset Flows

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Components of Additions_t				
Cash Purchases _t	IAS 38, IAS 7	Cash paid to acquire or internally develop qualifying intangibles.	SCF (Investing)	✓
Capitalised Borrowing Costs _t	IAS 23, IAS 38	Borrowing costs capitalised to qualifying intangible asset projects.	P&L (Reduces Int. Exp.)	×
Non-monetary Acquisitions _t	IAS 38	Intangibles acquired via non-cash consideration.	SFP (non-cash)	×
Components of Disposals_t				
Proceeds from sale of Intangibles _t	IAS 7, IAS 38	Cash received from the disposal of the asset.	SCF (Investing)	✓
Gain/Loss on Disposal _t	IAS 38	Calculated as: 'Proceeds - Carrying Amount of Disposals'.	P&L	✓

Further Decomposition of Main Flows for Intangibles

11.10.2 Goodwill

Dynamic Model of Goodwill

$$\begin{aligned}\text{Goodwill}_t &= \text{Goodwill}_{t-1} + \text{Goodwill on Acquisition}_t \\ &\quad - \text{Impairment Loss}_t - \text{Goodwill Derecognised on Disposal}_t\end{aligned}$$

Table 23: Explanation of Flow Variables for Goodwill

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Goodwill on Acquisition _t	IFRS 3	Goodwill recognised on a new business combination. This is a non-cash item.	SFP (non-cash)	×
Impairment Loss _t	IAS 36	Write-down of goodwill when the recoverable amount of the CGU is less than its carrying amount. Impairment losses on goodwill cannot be reversed.	P&L	×
Goodwill Derecognised _t	IFRS 10, IFRS 3	The carrying amount of goodwill allocated to a subsidiary or CGU that is disposed of.	P&L / SFP	×

11.11 Investment Property (IP)

Under IAS 40, entities can choose between the cost model or the fair value model.

11.11.1 Investment Property (Cost Model)

Measurement basis note: The carrying amount is cost less accumulated depreciation and accumulated impairment losses. This model is identical in mechanics to PPE.

Dynamic Model of Net IP

$$\begin{aligned}\text{IP Net}_t &= \text{IP Net}_{t-1} + \text{Additions}_t - \text{Depreciation Expense}_t \\ &\quad - \text{Impairment Loss}_t + \text{Impairment Reversal}_t \\ &\quad - \text{Carrying Amount of Disposals}_t \pm \text{Net Reclassifications}_t \\ &= \text{IP Net}_{t-1} + \text{Additions}_t - \text{Depreciation Expense}_t \\ &\quad - \text{Impairment Loss}_t + \text{Impairment Reversal}_t \pm \text{Net Reclassifications}_t \\ &\quad + \text{Gain/Loss on Disposal}_t - \text{Proceeds from sale}_t\end{aligned}$$

Table 24: Explanation of Main Flow Variables for IP (Cost Model)

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Additions _t	IAS 40, 23	Total cost of new IP from cash purchases or subsequent capex. See decomposition below.	SCF / P&L	✓*
Depreciation Expense _t	IAS 40, 16	The systematic allocation of the depreciable amount over the useful life.	P&L	✓
Impairment Loss/Reversal _t	IAS 36	Write-down to recoverable amount, and reversal of prior impairments.	P&L	✓
Carrying Amount of Disposals _t	IAS 40	The net book value of IP sold or retired. See decomposition below.	P&L / SCF	✓*
Net Reclassifications _t	IAS 40, 16, 2	The net carrying amount transferred between IP, PPE, and Inventory due to a change in use.	SFP (reclass)	✓

*See further decomposition table for details.

Further Decomposition of Main Flows This table details the components of the main flow variables for the cost model.

Table 25: Further Decomposition of IP Flows (Cost Model)

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Components of Additions_t				
Cash Purchases & Capex _t	IAS 40, IAS 7	Cash paid to acquire or subsequently improve IP.	SCF (Investing)	✓
Capitalised Borrowing Costs _t	IAS 23	Borrowing costs capitalised to qualifying IP projects.	P&L (Reduces Int. Exp.)	×
Components of Disposals_t				
Proceeds from sale of IP _t	IAS 7, IAS 40	Cash received from the disposal of the asset.	SCF (Investing)	✓
Gain/Loss on Disposal _t	IAS 40	Calculated as: 'Proceeds - Carrying Amount of Disposals'.	P&L	✓

11.11.2 Investment Property (Fair Value Model)

Measurement basis note: The carrying amount is the fair value at the measurement date. No depreciation is charged.

Dynamic Model of Net IP (Fair Value)

$$\begin{aligned} \text{IP Net}_t &= \text{IP Net}_{t-1} + \text{Additions}_t - \text{Carrying Amount of Disposals}_t \\ &\quad \pm \text{Fair Value Gains or Losses}_t \pm \text{Net Reclassifications}_t \end{aligned}$$

Table 26: Explanation of Main Flow Variables for IP (Fair Value Model)

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Additions _t	IAS 40, IAS 7	Total cost of new IP from cash purchases or subsequent capex.	SCF	×
Fair Value Gains or Losses _t	IAS 40	The change in the fair value of the investment property between measurement dates. This goes directly to P&L.	P&L	×
Carrying Amount of Disposals _t	IAS 40	The net book value (last fair value) of IP sold or retired.	P&L / SCF	×
Net Reclassifications _t	IAS 40, 16, 2	The fair value of assets transferred between IP, PPE, and Inventory. Transfers can trigger gains/losses.	SFP / P&L / OCI	×

11.12 Biological Assets

This covers living animals or plants managed for agricultural activity. They are measured at fair value less costs to sell under IAS 41. The accounting treatment can differ based on whether the asset itself is consumed upon harvest.

11.12.1 Bearer Biological Assets (e.g., Livestock for production)

Bearer assets are not themselves harvested but are used to produce other agricultural produce (e.g., dairy cows for milk, sheep for wool). The asset remains on the SFP after the produce is harvested.

$$\begin{aligned} \text{BA}_{\text{bearer},t} &= \text{BA}_{\text{bearer},t-1} + \text{Purchases}_t \pm \text{Fair value gains or losses}_t \\ &\quad - \text{Carrying Amount of Disposals}_t \\ &= \text{BA}_{\text{bearer},t-1} + \text{Purchases}_t \pm \text{Fair value gains or losses}_t \\ &\quad + \text{Gain/Loss on Disposal}_t - \text{Proceeds from sale}_t \end{aligned}$$

Table 27: Explanation of Main Flow Variables for Bearer Biological Assets

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Purchases _t	IAS 41	The acquisition of new bearer assets during the period.	SCF	✓
Fair value gains or losses _t	IAS 41	Gains/losses from changes in FV less costs to sell, arising from both physical changes (growth, ageing) and price changes.	P&L	✓
Carrying Amount of Disposals _t	IAS 41	The net book value (FVLCTS) of assets sold or that have died. The flow is decomposed to show the P&L and SCF impact separately.	P&L / SCF	✓*

*See further decomposition table for details.

Table 28: Further Decomposition of Bearer Biological Asset Flows

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Components of Disposals_t				
Proceeds from sale of Bearer Assets _t	IAS 7, IAS 41	Cash received from the disposal of the asset.	SCF (Investing)	✓
Gain/Loss on Disposal _t	IAS 41	Calculated as: 'Proceeds - Carrying Amount of Disposals'.	P&L	✓

Further Decomposition of Main Flows

11.12.2 Consumable Biological Assets (e.g., Crops for harvest)

Consumable assets are those that are to be harvested as agricultural produce, at which point the asset is reclassified to inventory (e.g., wheat to be harvested, trees grown for timber).

$$\begin{aligned}
 BA_{\text{consumable},t} &= BA_{\text{consumable},t-1} \\
 &\quad + \text{Purchases}_t \\
 &\quad \pm \text{Fair value gains or losses}_t \\
 &\quad - \text{Decreases due to harvest}_t
 \end{aligned}$$

Table 29: Explanation of Flow Variables for Consumable Biological Assets

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Purchases _t	IAS 41	The acquisition of new consumable assets during the period.	SCF	✓
Fair value gains or losses _t	IAS 41	Gains/losses from changes in FV less costs to sell, from both physical changes (growth) and price changes.	P&L	✓
Decreases due to harvest _t	IAS 41, IAS 2	Reclassification from biological assets to inventory upon harvest. The value is the FV less costs to sell at the point of harvest.	SFP (reclass)	✓

11.13 Employee Benefits

This asset arises when a Defined Benefit (DB) pension plan is in a surplus position (overfunded).

11.13.1 Defined Benefit (DB) Asset

The asset recognised in the statement of financial position is the net total of the Fair Value of Plan Assets (FVPA) minus the Defined Benefit Obligation (DBO).

$$\text{Net DB Asset}_t = \text{FVPA}_t - \text{DBO}_t$$

Its dynamic is highly complex and involves actuarial assumptions.

$$\begin{aligned} \text{Net DB Asset}_t = & \text{Net DB Asset}_{t-1} - \text{Service Cost}_t \\ & + \text{Net Interest on Net DB Asset}_t \\ & \pm \text{Remeasurements}_t + \text{Contributions Paid}_t \end{aligned}$$

Table 30: Explanation of Flow Variables for Defined Benefit Asset

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Service Cost _t	IAS 19	The increase in the DBO from employee service. This reduces the net asset.	P&L	×
Net Interest on Net DB Asset _t	IAS 19	The change in the net asset due to the passage of time (interest income on the net asset).	P&L	×
Remeasurements _t	IAS 19	Comprises actuarial gains/losses and the actual return on plan assets less amounts in net interest.	OCI	×
Contributions Paid _t	IAS 19	Cash contributions paid by the employer into the plan fund, increasing the net asset.	SCF	×

11.14 Financial Assets

This section details the dynamic models for financial assets, which are classified into three main categories under IFRS 9: Amortised Cost, Fair Value Through Other Comprehensive Income (FVOCI), and Fair Value Through Profit or Loss (FVTPL).

11.14.1 Financial Assets at Amortised Cost

This category applies to debt instruments where the business model is to hold the asset to collect contractual cash flows, and the cash flows represent solely payments of principal and interest (SPPI test).

$$\begin{aligned}
 AC_t = & AC_{t-1} \\
 & + \text{New assets purchased}_t \\
 & - \text{Principal repayments received}_t \\
 & + \text{Interest income}_t \\
 & - \text{Impairment loss (ECL)}_t
 \end{aligned}$$

Calculation of Interest Income The interest income is calculated using the effective interest method on the asset's gross carrying amount.

$$\text{Interest income}_t = \text{Gross Carrying Amount}_{t-1} \times \text{EIR}$$

Table 31: Explanation of Flow Variables for Financial Assets at Amortised Cost

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New assets purchased _t	IFRS 9	Initial recognition at fair value plus directly attributable transaction costs.	SCF	✓
Principal repayments received _t	IFRS 9	The portion of cash received from the debtor that represents a repayment of the principal amount.	SCF	✓
Interest income _t	IFRS 9	Interest revenue calculated using the effective interest rate (EIR) on the gross carrying amount.	P&L	✓
Impairment loss (ECL) _t	IFRS 9	Recognition of expected credit losses under the ECL model. This creates or increases a loss allowance, reducing the net carrying amount.	P&L	✓

11.14.2 Financial Assets at Fair Value Through Profit or Loss (FVTPL)

This is the default category for equity instruments and for debt instruments that fail the SPPI test or are held in a trading business model.

$$\begin{aligned}
 FV_t = & FV_{t-1} \\
 & + \text{New assets purchased}_t \\
 & - \text{Carrying amount of assets sold}_t \\
 & \pm \text{Fair value gains or losses}_t
 \end{aligned}$$

Table 32: Explanation of Flow Variables for Financial Assets at FVTPL

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New assets purchased _t	IFRS 9	Initial recognition at fair value. Transaction costs are expensed immediately in the P&L.	SCF	✓
Carrying amount of assets sold _t	IFRS 9	Derecognition of the asset's carrying value upon sale. The cash proceeds are reported in the SCF.	SCF	✓
Fair value gains or losses _t	IFRS 9, IFRS 13	All changes in fair value, including those arising from market price fluctuations and credit risk, are recognised in the P&L.	P&L	✓

11.14.3 Financial Assets at Fair Value Through Other Comprehensive Income (FVOCI)

This category applies to debt instruments held in a 'hold to collect and sell' business model, and to non-trading equity instruments where an irrevocable election is made. This section focuses on debt instruments.

Dynamic Model of FVOCI Assets

$$\begin{aligned} \text{FV}_t = & \text{FV}_{t-1} + \text{New assets purchased}_t - \text{Principal repayments received}_t \\ & - \text{Carrying amount of assets sold}_t + \text{Interest income}_t \\ & - \text{Impairment loss (ECL)}_t + \text{Gains/losses in OCI}_t \end{aligned}$$

Shadow Accounting: Amortised Cost Track For FVOCI debt instruments, a separate amortised cost calculation must be maintained. This "shadow" value is not reported on the SFP but is crucial for calculating the interest income to be recognised in P&L.

$$\begin{aligned} \text{AC}_t = & \text{AC}_{t-1} \\ & + \text{Interest income}_t \\ & - \text{Cash received (principal and interest)}_t \end{aligned}$$

Calculation of Interest Income The interest income recognised in P&L is based on the shadow amortised cost, not the fair value.

$$\text{Interest income}_t = \text{Amortised Cost}_{t-1} \times \text{EIR}$$

Reporting Convention: Clean Price Assumption For simplicity in the simulator, we assume coupon payments are continuous. Therefore, the reported Fair Value (FV) on the balance sheet is always equal to the Clean Price, and we do not need to model the separate calculation of Accrued Interest and Dirty Price.

Table 33: Explanation of Main Flow Variables for Financial Assets at FVOCI (Debt)

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New assets purchased _t	IFRS 9	Initial recognition at fair value plus directly attributable transaction costs.	SCF	✓
Principal repayments received _t	IFRS 9	The cash inflow representing repayment of the principal amount.	SCF	✓
Carrying amount of assets sold _t	IFRS 9	The net book value (fair value) of assets sold. See decomposition below.	P&L / SCF	✓*
Interest income _t	IFRS 9	Calculated on the 'shadow' amortised cost basis using the EIR.	P&L	✓
Impairment loss (ECL) _t	IFRS 9	Expected credit losses are recognised in the P&L, with a corresponding adjustment made in OCI.	P&L / OCI	✓
Gains/losses in OCI _t	IFRS 9	The residual change in fair value after accounting for P&L items. These are unrealised gains/losses.	OCI	✓

*See further decomposition table for details.

Further Decomposition of Disposals When an FVOCI debt instrument is sold, the cumulative gain or loss previously recognised in OCI is "recycled" to the P&L.

Table 34: Further Decomposition of FVOCI Disposal Flows

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Components of Disposals_t				
Proceeds from sale of FVOCI assets _t	IAS 7, IFRS 9	Cash received from the disposal of the asset.	SCF (Investing)	✓
Recycled gain/loss on disposal _t	IFRS 9	The reclassification of the cumulative gain or loss from OCI to P&L upon sale.	P&L / OCI	✓

11.15 Deferred Tax Asset (DTA)

This asset arises from deductible temporary differences and the carryforward of unused tax losses and tax credits. It represents a future reduction in tax payments.

$$\begin{aligned}
 \text{DTA}_t &= \text{DTA}_{t-1} \\
 &\quad \pm \text{Deferred tax income/(expense) recognised in P\&L}_t \\
 &\quad \pm \text{Deferred tax recognised directly in OCI/Equity}_t
 \end{aligned}$$

Utilisation of DTA A DTA is not received in cash. It is "utilised" by reducing future current tax payments. This happens when the underlying temporary difference reverses, making the taxable profit lower than the accounting profit.

Table 35: Explanation of Flow Variables for Deferred Tax Asset

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Deferred tax in P&L _t	IAS 12	Arises from deductible temporary differences originating from P&L items (e.g., provisions for warranties expensed in P&L but not yet tax-deductible).	P&L	×*
Deferred tax in OCI/Equity _t	IAS 12	Arises from deductible temporary differences originating from OCI/Equity items (e.g., losses on FVTOCI assets).	OCI	✓

*Not applicable in the simulator, as it assumes Taxable Profit equals Profit Before Tax, implying no temporary differences.

12 Equity

This section details the dynamic models for key equity accounts, incorporating the principle of capital maintenance where losses first reduce retained earnings, then share premium, and finally share capital.

12.1 Retained Earnings (RE)

The dynamic model for Retained Earnings, updated to ensure its balance does not fall below zero. Any excess loss is transferred to reduce other capital reserves.

$$\text{Intermediate RE}_t = \text{RE}_{t-1} + \text{P\&L attributable to parent}_t - \text{Dividends}_t \\ \pm \text{Transfers from/to reserves}_t \pm \text{Other direct adjustments}_t$$

$$\text{Loss absorbed by capital}_t = |\min(0, \text{Intermediate RE}_t)| \\ \text{RE}_t = \text{Intermediate RE}_t + \text{Loss absorbed by capital}_t$$

Table 36: Explanation of Flow Variables for Retained Earnings

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
P&L attributable to parent _t	IFRS 10, IAS 1	The portion of the period's profit or loss from the consolidated P&L that belongs to the parent's owners.	P&L	✓
Dividends _t	IAS 1, IAS 32	Dividends declared by the parent company to its shareholders.	SCF	✓
Transfers from/to reserves _t	IAS 16, IFRS 9	Represents reclassifications within equity, e.g., transferring a revaluation surplus to RE upon asset disposal.	Within Equity	×
Other direct adjustments _t	IFRS 10, IAS 32	Direct equity adjustments not routed through P&L or OCI, e.g., from transactions with NCI without a loss of control.	Within Equity	×
Intermediate Variable				
Loss absorbed by capital _t	Company Law, IAS 1	The absolute value of the negative part of intermediate retained earnings, i.e., $ \min(0, \text{Intermediate RE}_t) $. This amount is used to reduce Share Premium and Share Capital.	Within Equity	✓

12.2 Share Capital (SC) and Share Premium (SP)

The dynamic models for Share Capital and Share Premium, now updated to absorb losses when Retained Earnings are depleted. Losses first reduce the Share Premium balance, and any remaining loss then reduces Share Capital.

Share Premium (SP) Dynamics

$$\begin{aligned}
 \text{Loss absorbed by SP}_t &= \min(\text{SP}_{t-1}, \text{Loss absorbed by capital}_t) \\
 \text{SP}_t &= \text{SP}_{t-1} + \text{Premium on new shares issued}_t \\
 &\quad - \text{Loss absorbed by SP}_t
 \end{aligned}$$

Share Capital (SC) Dynamics

$$\text{Loss absorbed by SC}_t = \text{Loss absorbed by capital}_t - \text{Loss absorbed by SP}_t$$

$$\begin{aligned}
 \text{SC}_t &= \text{SC}_{t-1} + \text{Nominal value of shares issued}_t \\
 &\quad - \text{Nominal value of shares cancelled}_t \\
 &\quad - \text{Loss absorbed by SC}_t
 \end{aligned}$$

Table 37: Explanation of Flow Variables for Share Capital & Premium

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Share Premium Flows				
Premium on new shares issued _t	IAS 32	The excess of cash received over the nominal value of new shares issued. The calculation is: (Issue Price – Nominal Value) × No. of Shares Issued.	SCF	✓
Loss absorbed by SP _t	Company Law, IAS 1	The amount of loss absorbed by share premium, i.e., min(SP _{t-1} , Loss absorbed by capital _t).	Within Equity	✓
Share Capital Flows				
Nominal value of shares issued _t	IAS 32, IFRS 2	The nominal (par) value of new shares issued through events like IPOs, employee share schemes, or convertible bond conversions.	SCF	✓
Nominal value of shares cancelled _t	IAS 32	The nominal value of shares that have been legally cancelled, typically following a share buyback.	SCF	×
Loss absorbed by SC _t	Company Law, IAS 1	The amount of loss absorbed by share capital, i.e., Loss absorbed by capital _t – Loss absorbed by SP _t .	Within Equity	✓

12.3 Non-controlling Interests (NCI)

The dynamic model for NCI is only applicable in consolidated financial statements. It represents the portion of a subsidiary's equity not attributable to the parent.

$$\begin{aligned}
 \text{NCI}_t = & \text{NCI}_{t-1} \\
 & + \text{NCI share of P\&L}_t \\
 & + \text{NCI share of OCI}_t \\
 & - \text{Dividends paid to NCI}_t \\
 & \pm \text{Transactions with parent}_t \\
 & \pm \text{NCI on new business combination}_t
 \end{aligned}$$

Table 38: Explanation of Flow Variables for Non-controlling Interests

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
NCI share of P&L _t	IFRS 10	The portion of the subsidiary's profit for the year allocated to NCI.	P&L	✓
NCI share of OCI _t	IFRS 10	The portion of the subsidiary's Other Comprehensive Income (OCI) allocated to NCI.	OCI	×
Dividends paid to NCI _t	IFRS 10	Dividends declared and paid by the subsidiary to its non-controlling shareholders. This reduces the NCI balance.	SCF	✓
Transactions with parent _t	IFRS 10	Changes in NCI resulting from the parent buying more shares from, or selling shares to, NCI without a loss of control.	Within Equity	×
NCI on new business combination _t	IFRS 3, IFRS 10	The recognition of NCI at fair value upon the acquisition of a new non-wholly-owned subsidiary.	SFP (non-cash)	×

12.4 Other Reserves

This section covers equity reserves other than share capital, share premium, and retained earnings. Each reserve follows a unique dynamic model based on its governing standard.

12.4.1 Revaluation Reserve (RR)

The dynamic model for the Revaluation Reserve, used for Property, Plant and Equipment (PPE) and Intangible Assets accounted for under the revaluation model.

$$\begin{aligned}
 RR_t = & RR_{t-1} \\
 & + \text{Revaluation gains in OCI}_t \\
 & - \text{Reversal of revaluation gains}_t \\
 & - \text{Transfer to RE (excess depreciation)}_t \\
 & - \text{Transfer to RE (asset disposal)}_t \\
 & \pm \text{Related deferred tax effect}_t
 \end{aligned}$$

Calculation of Excess Depreciation The excess depreciation is the difference between the depreciation based on the revalued amount and the depreciation that would have been charged based on the asset's historical cost.

$$\begin{aligned}
 \text{Excess Depr.}_t = & \text{Depr. on Revalued Amount}_t \\
 & - \text{Depr. on Historical Cost}_t
 \end{aligned}$$

Calculation of Deferred Tax Effect A revaluation surplus creates a taxable temporary difference (TTD), which gives rise to a Deferred Tax Liability (DTL).

$$\text{TTD}_t = \text{Carrying Amount}_t - \text{Tax Base}_t$$

$$\text{DTL}_t = \text{TTD}_t \times \text{Tax Rate}$$

Table 39: Explanation of Main Flow Variables for Revaluation Reserve

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Revaluation gains / Reversals _t	IAS 16, IAS 38	The net impact of upward or downward revaluations. See decomposition below.	OCI / P&L	×*
Transfer to RE (excess depr.) _t	IAS 16	An optional annual transfer from RR to RE for the higher depreciation charge in P&L.	Within Equity	×
Transfer to RE (disposal) _t	IAS 16	Upon disposal, the remaining RR balance is transferred directly to Retained Earnings.	Within Equity	×
Related deferred tax effect _t	IAS 12	The deferred tax impact arising from the revaluation is recognised directly in OCI.	OCI	×

*See further decomposition table for details.

Further Decomposition of Revaluation Changes The accounting for revaluation changes depends on whether it is a gain or a loss, and whether a previous surplus exists.

Table 40: Further Decomposition of Revaluation Gains and Reversals

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Components of Revaluation gains / Reversals_t				
Revaluation Gain _t	IAS 16, IAS 38	An upward revaluation of an asset's carrying amount to its fair value.	OCI	×
Revaluation Reversal (from OCI) _t	IAS 16, IAS 38	A downward revaluation that reverses a previous gain for the same asset. This reduces the RR balance.	OCI	×
Revaluation Loss (to P&L) _t	IAS 16, IAS 36	A downward revaluation that exceeds the existing revaluation surplus for that asset. This excess is treated as an impairment loss.	P&L	×

12.4.2 FVTOCI Reserve (for Financial Instruments)

This reserve accumulates gains and losses on financial assets designated as Fair Value Through Other Comprehensive Income (FVTOCI). The treatment of losses differs for impairment on

debt instruments.

$$\begin{aligned}
\text{FVTOCI } R_t &= \text{FVTOCI } R_{t-1} \\
&\pm \text{Fair value changes in OCI}_t \\
&- \text{Impairment loss reclassified to P\&L (debt)}_t \\
&\mp \text{Reclassification to P\&L on disposal (debt)}_t \\
&- \text{Transfer to RE on disposal (equity)}_t
\end{aligned}$$

Reclassification and Transfer on Disposal Upon disposal, the cumulative balance in the reserve for the specific asset is cleared. For debt instruments, it is recycled to P&L. For equity instruments, it is transferred directly to Retained Earnings.

$\text{Reclassification}_t = \text{Cumulative gain/loss in FVTOCI-R for the disposed debt asset}$

$\text{Transfer}_t = \text{Cumulative gain/loss in FVTOCI-R for the disposed equity asset}$

Table 41: Explanation of Flow Variables for FVTOCI Reserve

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Fair value changes in OCI _t	IFRS 9	Represents total unrealised gains/losses from FV changes recognised in OCI. Reserve can be negative.	OCI	✓
Impairment loss (debt) to P&L _t	IFRS 9	For debt instruments, expected credit losses (ECL) are reclassified from OCI to P&L.	OCI	✓*
Reclassification to P&L (debt disposal) _t	IFRS 9	Upon disposal of a debt instrument, the cumulative gain/loss is 'recycled' from the reserve to P&L.	P&L	✓
Transfer to RE (equity disposal) _t	IFRS 9	Upon disposal of an equity instrument, the cumulative gain/loss is transferred directly to Retained Earnings.	Within Equity	✓

*In the simulator, it is a final residual term.

12.4.3 Foreign Currency Translation Reserve (FCTR)

The dynamic model for the FCTR, which arises from the consolidation of foreign operations with a different functional currency.

$$\begin{aligned}
\text{FCTR}_t &= \text{FCTR}_{t-1} \\
&\pm \text{Exchange differences on translation}_t \\
&\mp \text{Reclassification to P\&L on disposal}_t
\end{aligned}$$

Table 42: Explanation of Flow Variables for FCTR

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Exchange differences on translation _t	IAS 21	Gains or losses arising from translating the financial statements of a foreign operation (e.g., subsidiary) from its functional currency to the parent's presentation currency.	OCI	×
Reclassification to P&L on disposal _t	IAS 21	Upon disposal of the foreign operation, the entire cumulative balance in the FCTR for that operation is reclassified ("recycled") from equity to P&L.	P&L	×

13 Liabilities

This section details the dynamic models for key liability accounts under IFRS.

13.1 Financial Liabilities (at Amortised Cost)

This category covers financial liabilities measured at amortised cost, such as loans and borrowings. These can be classified as either current or non-current depending on their maturity. The dynamic model is based on the amortised cost method.

$$\begin{aligned}
 FL_t = & FL_{t-1} \\
 & + \text{New financial liabilities issued}_t \\
 & + \text{Interest expense}_t \\
 & - \text{Cash payments}_t \\
 & - \text{Derecognition}_t
 \end{aligned}$$

Calculation of Interest Expense The interest expense for the period is calculated using the effective interest method, based on the carrying amount at the beginning of the period.

$$\text{Interest expense}_t = \text{Carrying Amount}_{t-1} \times \text{EIR}$$

Table 43: Explanation of Flow Variables for Financial Liabilities

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New financial liabilities issued _t	IFRS 9	Initially recognised at present value (net proceeds received), discounted using the Effective Interest Rate (EIR).	SCF	✓
Interest expense _t	IFRS 9	The EIR is determined at inception and remains fixed. This is a P&L charge.	P&L	✓
Cash payments _t	IFRS 7, IFRS 9	Represents the total cash outflow for the period, including both principal and interest (coupon) payments.	SCF	✓
Derecognition _t	IFRS 9	Occurs on extinguishment (e.g., discharge, cancellation, expiry). Any difference between the carrying amount and consideration paid is recognised in P&L as a gain/loss.	P&L / SCF	×

13.2 Other Financial Liabilities (at FVTPL)

This category primarily includes financial liabilities required to be measured at Fair Value Through Profit or Loss (FVTPL), such as derivative liabilities or contingent consideration from a business combination.

$$\begin{aligned}
 \text{OFL}_t &= \text{OFL}_{t-1} \\
 &\quad + \text{New liability recognised at fair value}_t \\
 &\quad \pm \text{Fair value gains or losses}_t \\
 &\quad - \text{Cash payments or settlements}_t \\
 &\quad - \text{Derecognition}_t
 \end{aligned}$$

Table 44: Explanation of Flow Variables for Other Financial Liabilities

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New liability recognised at FV_t	IFRS 9, IFRS 3	Initial recognition of liabilities like derivatives or contingent consideration at their fair value on that date.	SFP (non-cash)	✓
Fair value gains or losses _t	IFRS 9	Changes in the fair value of the liability are recognised in P&L. An increase in liability FV results in a loss in P&L, and vice versa.	P&L	✓
Cash payments or settlements _t	IFRS 7, IAS 7	Represents the cash outflow to settle the liability, e.g., payment of contingent consideration or settlement of a derivative.	SCF	✓
Derecognition _t	IFRS 9	Occurs when the obligation is discharged or cancelled. The final settlement is made.	P&L / SCF	×

13.3 Trade and Other Payables

This section is broken down into two distinct components: traditional trade payables owed to suppliers, and consideration payable to customers, which is governed by revenue recognition standards.

13.3.1 Trade Payables (to Suppliers)

This represents liabilities to pay for goods or services that have been received from suppliers but have not yet been paid for. It is a core operational liability.

Dynamic Model of Trade Payables

$$\begin{aligned}
 TP_t &= TP_{t-1} \\
 &\quad + \text{New purchases on credit}_t \\
 &\quad - \text{Cash payments to suppliers}_t
 \end{aligned}$$

Table 45: Explanation of Main Flow Variables for Trade Payables

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New purchases on credit _t	IAS 2, IAS 1	Total value of goods and services purchased on credit during the period. See decomposition below.	SFP / P&L	✓*
Cash payments to suppliers _t	IAS 7	The cash outflow to settle liabilities with suppliers. This is a core component of operating cash flow.	SCF	✓

*See further decomposition table for details.

Further Decomposition of New Purchases on Credit The accounting impact of a new credit purchase depends on whether an asset or an expense is acquired.

Table 46: Further Decomposition of New Purchases on Credit

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Components of New purchases on credit_t				
Purchases of Goods (Inventory) _t	IAS 2	The cost of inventory purchased on credit. This increases the Inventory asset account.	SFP (reclass)	✓
Purchases of Services (Expense) _t	IAS 1	The cost of services consumed during the period, purchased on credit (e.g., consulting, advertising).	P&L	✓

13.3.2 Consideration Payable to Customers

This represents amounts that an entity expects to pay to its customers, such as volume rebates or other bonuses. Under IFRS 15, this is treated as variable consideration that reduces the transaction price.

$$\begin{aligned}
 CP_t &= CP_{t-1} \\
 &\quad + \text{New consideration payable recognised}_t \\
 &\quad - \text{Cash paid to customers}_t \\
 &\quad \pm \text{Changes in estimate (reversals)}_t
 \end{aligned}$$

Table 47: Explanation of Flow Variables for Consideration Payable to Customers

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New consideration payable recognised _t	IFRS 15	The liability recognised for expected payments to customers (e.g., rebates). This is paired with a reduction in recognised revenue.	P&L	×
Cash paid to customers _t	IFRS 15, IAS 7	The cash outflow when the consideration (e.g., rebate) is paid to the customer, which settles the liability.	SCF	×
Changes in estimate (reversals) _t	IFRS 15	Adjustments to the liability due to changes in expectations (e.g., customer not meeting volume targets). A reversal increases revenue.	P&L	×

13.4 Current Tax Payable (CTP)

This represents the amount of income taxes payable to taxation authorities for the current and prior periods. It is distinct from Deferred Tax Liability and is classified as current.

$$\begin{aligned}
 \text{CTP}_t &= \text{CTP}_{t-1} \\
 &\quad + \text{Current tax expense for the period}_t \\
 &\quad - \text{Cash tax paid during the period}_t
 \end{aligned}$$

Calculation of Current Tax Expense The current tax expense is based on the taxable profit for the period, which often differs from the accounting profit (PBT).

$$\begin{aligned}
 \text{Taxable Profit}_t &= \text{Profit Before Tax (PBT)}_t \\
 &\quad + \text{Non-deductible expenses}_t \\
 &\quad - \text{Non-taxable income}_t \\
 &\quad \pm \text{Adjustments for temporary differences}_t
 \end{aligned}$$

$$\text{Current Tax Expense}_t = \text{Taxable Profit}_t \times \text{Applicable Tax Rate}$$

Tax are assumed paid immediately at the exact same amount, so current tax payable is not considered in the simulator.

Table 48: Explanation of Flow Variables for Current Tax Payable

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Current tax expense _t	IAS 12	The tax charge for the period based on taxable profit. Non-deductible/non-taxable items are permanent differences. Temporary differences (e.g., from depreciation after revaluation) reverse over time.	P&L	×
Cash tax paid _t	IAS 7	Represents the actual cash paid to tax authorities during the period, which may relate to the previous year's liability or current year's instalments.	SCF	×

13.5 Provisions

This represents liabilities of uncertain timing or amount, such as provisions for warranties, restructuring, or litigation, governed by IAS 37.

Dynamic Model of Provisions

$$\begin{aligned}
 \text{Prov}_t &= \text{Prov}_{t-1} \\
 &\quad + \text{Additional provisions recognised}_t \\
 &\quad + \text{Unwinding of discount}_t \\
 &\quad - \text{Provisions utilised}_t \\
 &\quad - \text{Reversal of unused provisions}_t
 \end{aligned}$$

Table 49: Explanation of Main Flow Variables for Provisions

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Additions recognised _t	IAS 37	Total new provisions or increases to existing ones. See decomposition below.	P&L / SFP	✓*
Unwinding of discount _t	IAS 37	The increase in a long-term provision's PV due to the passage of time. Recognised as a finance cost in P&L.	P&L	✓*
Provisions utilised _t	IAS 37, IAS 7	The cash outflow to settle the obligation for which the provision was made.	SCF	✓*
Reversal of unused provisions _t	IAS 37	Reversal of a provision if no longer required; results in a credit to the P&L.	P&L	✓

*See further decomposition table for details.

Further Decomposition of Main Flows An important component of provisions can be an Asset Retirement Obligation (ARO), which is a legal obligation associated with the retirement of a tangible long-lived asset (e.g., decommissioning an oil rig). Its accounting treatment differs from other provisions.

Table 50: Further Decomposition of Provision Flows (ARO)

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Components of Additions_t				
ARO Recognition _t	IAS 37, IAS 16	Initial recognition of an ARO provision, capitalised into the cost of the related PPE.	SFP (non-cash)	×
Other New Provisions _t	IAS 37	All other new provisions (e.g., for warranties, restructuring), which are expensed.	P&L	✓
Components of Unwinding_t				
ARO Discount Unwind _t	IAS 37	The unwinding of the discount specifically for the ARO provision.	P&L	×
Other Discount Unwind _t	IAS 37	The unwinding of discount for all other long-term provisions.	P&L	✓
Components of Utilised_t				
ARO Settlement Cash _t	IAS 37, IAS 7	Cash paid to settle the ARO obligation at the end of the asset's life.	SCF	×
Other Provisions Utilised _t	IAS 37, IAS 7	Cash paid to settle all other provisions (e.g., warranties).	SCF	✓

13.6 Contract Liability (CL)

This represents the obligation to transfer goods or services to a customer for which the entity has received consideration from the customer.

$$\begin{aligned}
 CL_t = & CL_{t-1} \\
 & + \text{Cash inflow from advance payments}_t \\
 & + \text{Interest expense (financing component)}_t \\
 & - \text{Revenue recognised}_t
 \end{aligned}$$

Initial Recognition of Cash Inflow The liability is initially recognised at the nominal cash value received. For long-term contracts, this consideration is conceptually split into the performance obligation (at present value) and a financing component.

Calculation of Interest Expense This applies only to long-term contracts with a significant financing component.

$$\text{interest expense}_t = \begin{cases} 0 & \text{if short-term (current)} \\ \text{Carrying Amount}_{t-1} \times \text{Discount Rate} & \text{if long-term (non-current)} \end{cases}$$

Table 51: Explanation of Flow Variables for Contract Liability

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Cash inflow (advance payments) _t	IFRS 15, IAS 7	Cash received from customers before the goods or services are delivered.	SCF	✓
Interest expense _t	IFRS 15	Arises from a significant financing component in long-term contracts. This is zero for short-term contracts.	P&L	✓
Revenue recognised _t	IFRS 15	When the performance obligation is satisfied (goods/services delivered), the liability is reduced and revenue is recognised.	P&L	✓

13.7 Deferred Tax Liability (DTL)

This liability arises from taxable temporary differences, representing income tax payable in future periods in respect of taxable temporary differences. Its settlement occurs through higher future current tax payments, not direct cash payments.

$$\begin{aligned}
 \text{DTL}_t &= \text{DTL}_{t-1} \\
 &\quad \pm \text{Deferred tax expense/(income) recognised in P\&L}_t \\
 &\quad \pm \text{Deferred tax recognised directly in OCI/Equity}_t
 \end{aligned}$$

Reversal of DTL (Settlement) A DTL is not paid directly. It reverses over time as the underlying temporary difference reverses. For instance, when accounting depreciation (which was initially lower) exceeds tax depreciation in later years, the taxable profit becomes higher than the accounting profit, leading to higher current tax payments. This higher payment effectively "settles" the DTL over time.

Table 52: Explanation of Flow Variables for Deferred Tax Liability

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Deferred tax in P&L _t	IAS 12	Arises from temporary differences originating from P&L items (e.g., depreciation differences on assets at cost).	P&L	×
Deferred tax in OCI/Equity _t	IAS 12	Arises from temporary differences originating from OCI/Equity items (e.g., asset revaluations, FVTOCI instruments).	OCI	✓

13.8 Employee Benefits

This covers all forms of consideration given by an entity in exchange for service rendered by employees. It is governed by IAS 19.

13.8.1 Short-term Benefits and Defined Contribution (DC) Plans

Defined Contribution (DC) Plans Under DC plans, the entity pays fixed contributions into a separate fund and has no further legal or constructive obligation. The accounting is simple: the contribution payable for a period is recognised as an expense in P&L. No significant liability arises.

Accrued Salaries This represents unpaid salaries at the end of a period. Its dynamic is a simple accrual and payment cycle.

$$\text{Accrued Salaries}_t = \text{Accrued Salaries}_{t-1} + \text{DC Plan Contribution}_t + \text{Salary Expense}_t - \text{Cash Paid}_t$$

Table 53: Explanation of Flow Variables for Simple Employee Benefits

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
DC Plan Contribution _t	IAS 19	The required contribution for the period.	P&L	✓
Salary Expense _t	IAS 19	The salary cost incurred for employee services during the period.	P&L	✓
Cash Paid _t	IAS 7	The cash outflow for salaries and contributions.	SCF	✓

13.8.2 Defined Benefit (DB) Plans

The liability recognised in the statement of financial position for a DB plan is the net total of the DBO minus the FVPA.

$$\text{Net DB Liability}_t = \text{DBO}_t - \text{FVPA}_t$$

Its dynamic is highly complex and involves actuarial assumptions.

$$\begin{aligned} \text{Net DB Liab.}_t = & \text{Net DB Liab.}_{t-1} + \text{Service Cost}_t \\ & + \text{Net Interest on Net DB Liab.}_t \\ & \pm \text{Remeasurements}_t - \text{Contributions Paid}_t \end{aligned}$$

Table 54: Explanation of Flow Variables for Defined Benefit Plans

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Service Cost _t	IAS 19	The increase in the DBO resulting from employee service in the current period.	P&L	×
Net Interest on Net DB Liab. _t	IAS 19	The change in the net DB liability due to the passage of time.	P&L	×
Remeasurements _t	IAS 19	Comprises actuarial gains/losses and the return on plan assets (excluding amounts in net interest).	OCI	×
Contributions Paid _t	IAS 19	Cash contributions paid by the employer into the plan fund.	SCF	×

13.9 Lease Liability (LL)

Under IFRS 16, a liability is recognised for most leases, representing the obligation to make lease payments. Its subsequent measurement follows the amortised cost method.

$$\begin{aligned}
 LL_t = & LL_{t-1} \\
 & + \text{New lease liabilities recognised}_t \\
 & + \text{Interest expense on lease liability}_t \\
 & - \text{Lease payments made}_t \\
 & \pm \text{Adjustments from lease reassessments}_t
 \end{aligned}$$

Initial Recognition The liability is initially recognised at the present value (PV) of future lease payments.

$$\text{New lease liability}_t = \text{PV}(\text{Future Lease Payments})$$

Discount Rate The discount rate (r) is the *interest rate implicit in the lease*, or the lessee's *incremental borrowing rate* if the former is not readily determinable.

Definition of Interest Rate Implicit in the Lease (r) This rate equates the lessor's investment in the lease with the PV of the inflows the lessor expects.

$$\begin{aligned}
 & \text{PV}(\text{Lease Payments}) + \text{PV}(\text{Unguaranteed Residual Value}) \\
 & = \text{Fair Value of Asset} + \text{Lessor's Initial Direct Costs}
 \end{aligned}$$

Calculation of Interest Expense The interest expense for the period is calculated using the same discount rate determined at inception.

$$\text{Interest expense}_t = \text{Carrying Amount}_{t-1} \times \text{Discount Rate}$$

Table 55: Explanation of Flow Variables for Lease Liability

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New lease liabilities recognised _t	IFRS 16	The present value of future lease payments. This is a non-cash transaction that also creates a Right-of-Use asset.	SFP (non-cash)	✓
Interest expense _t	IFRS 16	The interest accrued on the outstanding liability balance during the period.	P&L	✓
Lease payments made _t	IFRS 16, IAS 7	The cash outflow for the lease. Can be classified entirely as financing, or split into principal (financing) and interest (operating/financing).	SCF	✓*
Adjustments from reassessments _t	IFRS 16	Changes to the liability due to modifications (e.g., change in lease term or scope). This adjustment typically also affects the Right-of-Use asset.	SFP (non-cash)	×

*In the simulator, the entire lease payment is classified as a financing activity.

13.10 Liabilities Associated with Assets Held for Sale (LAHFS)

This is a temporary classification for liabilities of a disposal group that is expected to be sold within one year.

$$\begin{aligned}
 \text{LAHFS}_t &= \text{LAHFS}_{t-1} \\
 &\quad + \text{Liabilities reclassified as HFS}_t \\
 &\quad \pm \text{Measurement changes post-reclassification}_t \\
 &\quad - \text{Liabilities derecognised on disposal}_t \\
 &\quad - \text{Liabilities reclassified back}_t
 \end{aligned}$$

Calculation of Measurement Changes After reclassification, liabilities within the disposal group continue to be measured under their original standards. For example:

$$\text{Measurement changes}_t = \begin{cases} \text{Interest Expense}_t & \text{for amortised cost liabilities} \\ \text{Unwinding of Discount}_t & \text{for provisions} \end{cases}$$

Table 56: Explanation of Flow Variables for LAHFS

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
Liabilities reclassified as HFS _t	IFRS 5	Reclassification from original liability categories to LAHFS upon meeting the held-for-sale criteria.	SFP (reclass)	×
Measurement changes post-reclass _t	IFRS 5, IFRS 9, IAS 37	Liabilities continue to be measured under their original standards. E.g., interest expense on a loan or unwinding of discount on a provision is recognised.	P&L	×
Liabilities derecognised on disposal _t	IFRS 5	Removal from the SFP when the disposal group is sold. Cash proceeds are part of investing activities.	P&L / SFP / SCF	×
Liabilities reclassified back _t	IFRS 5	Reclassification back to original categories if the held-for-sale criteria are no longer met.	SFP (reclass)	×

13.11 Refund Liability

This liability arises from sales with a right of return. Under IFRS 15, an entity recognises a refund liability for the consideration it expects to refund to customers. It is a component of variable consideration.

$$\begin{aligned}
 \text{Refund Liability}_t &= \text{Refund Liability}_{t-1} \\
 &\quad + \text{New refund liabilities recognised}_t \\
 &\quad - \text{Cash refunds paid}_t \\
 &\quad \pm \text{Changes in estimate (reversals)}_t
 \end{aligned}$$

Initial Recognition When a sale is made, the portion of the transaction price that the entity expects to refund is not recognised as revenue. Instead, it is recognised as a refund liability.

$$\text{New refund liability}_t = \text{Value of expected returns from current period sales}$$

Table 57: Explanation of Flow Variables for Refund Liability

Flow Variable	Standard(s)	Explanation	Statement Impact	In Sim?
New refund liabilities recognised _t	IFRS 15	The liability recognised for expected returns from sales made in the current period. This is paired with a reduction in recognised revenue.	P&L	×
Cash refunds paid _t	IFRS 15, IAS 7	The cash paid to customers who exercise their right of return. This settles the liability.	SCF	×
Changes in estimate (reversals) _t	IFRS 15	Adjustments to the liability due to changes in the expectation of returns for past sales. A decrease in expected returns increases revenue (reversal).	P&L	×

14 Financial Statement with Flow variables

This section presents the dynamic models for the primary financial statements. The construction of these statements is based on the flow variables previously identified as applicable 'In Simulator' within the balance sheet account analyses. This ensures a direct and consistent link between the stock changes on the balance sheet and the flows reported in the income statement, statement of cash flows, and statement of changes in equity.

14.1 Comprehensive Income Statement

The following table details the line-by-line composition of the Profit or Loss statement, linking each flow variable to its corresponding source balance sheet account.

Table 58: Income Statement Structure (Part 1)

Item	Source Report / Item
Revenue	
Sales received in cash	Operating Cash Flow
New Receivables recognised	Trade Receivable
Revenue recognized	Contract Liability
Cost of sales	
COGS	Inventory
Gross Result = Revenue - Cost of sales	
Other operating expenses	
Cost on services	Operating Cash Flow
ECL expense	Trade Receivable
Physical Loss/NRV write-down	Inventory
NRV reversal	Inventory
Depreciation Expenses	PP&E
Expense recognized	Prepayments
Depreciation Expenses	ROU Asset
Amortized Expense	Intangible Assets
Depreciation Expenses	Investment Property
Purchase of services	Trade Payable
Other new provisions	Provisions
DC plan	Employee Benefits
Salary expense	Accrued Salaries
EBIT = Gross Result - Other operating expenses	

Table 59: Income Statement Structure (Part 2)

Item	Source Report / Item
Other Income	
Reversal of unused provisions	Provisions
Gain on Disposal	PP&E
Impairment reversal	PP&E
Gain on Disposal	Intangible Assets
Impairment reversal	Intangible Assets
Gain on Disposal	Investment Property
Impairment reversal	Investment Property
Fair Value Gain (bearer)	Biological Assets
Gain on Disposal (bearer)	Biological Assets
Fair Value Gain (consumable)	Biological Assets
Other expense	
Impairment loss	PP&E
Loss on Disposal	PP&E
Impairment loss	Intangible Assets
Loss on Disposal	Intangible Assets
Impairment loss	Investment Property
Loss on Disposal	Investment Property
Fair Value Loss (bearer)	Biological Assets
Loss on Disposal (bearer)	Biological Assets
Fair Value Loss (consumable)	Biological Assets
Finance Income	
Interest Income	Prepayments
Interest Income	Financial Assets (AC)
Interest Income	Financial Assets (FVOCI)
Fair value gains	Financial Assets (FVTPL)
Fair value gains	Other Financial Liabilities
Recycled gains on disposal	Financial Assets (FVOCI)
Finance expense	
Impairment loss (ECL)	Financial Assets (AC)
Interest expense	Financial Liability (AC)
Fair value loss	Financial Assets (FVTPL)
Fair value loss	Other Financial Liabilities
Other discount unwind	Provisions
Interest expense	Contract Liability
Interest expense	Lease Liability
Impairment loss (ECL)	Financial Assets (FVOCI)
Recycled losses on disposal	Financial Assets (FVOCI)
EBT = EBIT + Other Income - Other expense + Finance Income - Finance expense	
Tax	
tax paid	operating cash flow
Net Income = EBT - Tax	
Net Income attributable to:	
P&L attributable to parent	Retained Earnings
NCI share of P&L	Non-controlling Interests

Table 60: Comprehensive Income Statement

Item	From	To
Net Income for the period	P&L	
Other Comprehensive Income (OCI)		
Gain in OCI - Including Impairment Loss Reclassification (net of tax)	Financial Assets (FVOCI)	FVTOCI Reserve
(Losses in OCI - Including Impairment Loss Reclassification) (net of tax)	Financial Assets (FVOCI)	FVTOCI Reserve
(Recycled gain on disposal) (net of tax)	FVTOCI Reserve	P&L
Recycle loss on disposal (net of tax)	FVTOCI Reserve	P&L
Total Other Comprehensive Income		
Total Comprehensive Income for the period		

Tax Effects of Other Comprehensive Income

The presentation of Other Comprehensive Income (OCI) items "net of tax" reflects the accounting principle that the tax effects of items should be recognized in the same component of the financial statements as the items themselves. When an OCI item creates a temporary difference between its accounting carrying amount and its tax base, a deferred tax asset (DTA) or deferred tax liability (DTL) is recognized. The corresponding deferred tax expense or benefit is recorded directly within OCI.

The specific changes in DTA and DTL related to OCI items are calculated as follows:

- **For gains recognized in OCI (e.g., unrealized gains on FVOCI assets):** A deferred tax liability is created or increased.

$$\text{Increase in DTL} = \text{Taxable Gain in OCI (pre-tax)} \times \text{Tax Rate}$$

- **For losses recognized in OCI (e.g., unrealized losses on FVOCI assets):** A deferred tax asset is created or increased.

$$\text{Increase in DTA} = \text{Deductible Loss in OCI (pre-tax)} \times \text{Tax Rate}$$

- **For gains recycled from OCI to P&L on disposal:** The previously recognized DTL is reversed.

$$\text{Decrease in DTL} = \text{Recycled Gain (pre-tax)} \times \text{Tax Rate}$$

- **For losses recycled from OCI to P&L on disposal:** The previously recognized DTA is reversed.

$$\text{Decrease in DTA} = \text{Recycled Loss (pre-tax)} \times \text{Tax Rate}$$

14.2 Statement of Cash Flow (SCF)

The following table details the line-by-line composition of the statement of cash flow, linking each flow variable to its corresponding source balance sheet account.

Table 61: Statement of Cash Flows (Operating Activities)

Cash flows from operating activities (Direct Method)	
Cash collection from customers	Revenue / Trade Receivable
Cash from advance payments	Contract Liability
(Cash spent on services)	other operating expense
(Cash purchase of inventory)	Inventory
(Cash purchase of consumable assets)	Biological Assets
(Cash paid to suppliers)	Trade Payable
(Cash paid for other prepayments)	Prepayments
(Cash paid for salaries)	Accrued Salaries
(Cash paid for DC plan)	Employee Benefits
(Other provisions utilised)	Provisions
(Tax paid)	P&L (Tax assumed paid in cash)
Net cash from operating activities	
Cash flows from operating activities (Indirect Method)	
Net Income	P&L
<i>Adjustments to reconcile Net Income:</i>	
Depreciation and Amortization	PP&E, ROU, Intangible, Inv Prop
(Gains)/Losses on Disposal of Assets	P&L
Impairment Losses/(Reversals)	P&L
Fair Value (Gains)/Losses	P&L
Finance Income and Expense	P&L
<i>Changes in working capital:</i>	
(Increase)/Decrease in Trade Receivables	Trade Receivable
(Increase)/Decrease in Inventories	Inventory
(Increase)/Decrease in Prepayments	Prepayments
Increase/(Decrease) in Trade Payables	Trade Payable
Increase/(Decrease) in Contract Liabilities	Contract Liability
Increase/(Decrease) in Provisions	Provisions
Increase/(Decrease) in Accrued Salaries	Accrued Salaries
(Tax paid)	P&L (Tax assumed paid in cash)
Net cash from operating activities	

Table 62: Statement of Cash Flows (Investing & Financing Activities)

Cash flows from investing activities	Source Report / Item
(Purchase of PP&E)	PP&E
Proceeds from sale of PP&E	PP&E
(Purchase of Intangible Assets)	Intangible Assets
Proceeds from sale of Intangibles	Intangible Assets
(Purchase of Investment Property)	Investment Property
Proceeds from sale of Investment Property	Investment Property
(Purchase of Biological Assets)	Biological Assests
Proceeds from sales of bearer assets	Biological Assests
(Purchase of Financial Assets - AC)	Financial Assets (AC)
Principal repayments from Financial Assets (AC)	Financial Assets (AC)
(Purchase of Financial Assets - FVTPL)	Financial Assets (FVTPL)
Proceeds from sale of Financial Assets (FVTPL)	Financial Assets (FVTPL)
(Purchase of Financial Assets - FVOCI)	Financial Assets (FVOCI)
Principal repayments from Financial Assets (FVOCI)	Financial Assets (FVOCI)
Proceeds from sale of Financial Assets (FVOCI)	Financial Assets (FVOCI)
Net cash used in investing activities	
Cash flows from financing activities	
Proceeds from issuance of shares	Share capital / Share premium
Proceeds from new financial liabilities issued	Financial Liabilities (AC)
(Repayment of financial liabilities)	Financial Liabilities (AC)
(Repayment of other financial liabilities)	Other Financial Liabilities
(Lease payments made)	Lease Liability
(Dividends paid)	Retained Earnings
Net cash from financing activities	
Net increase/(decrease) in cash and cash equivalents	
Cash and cash equivalents at beginning of period	
Cash and cash equivalents at end of period	

14.3 Analysis of Within Balance Sheet Reclassifications

Please note: This table is not a standard requirement under IFRS. Its purpose is to explicitly map the reclassifications within the balance sheet to ensure that the principles of double-entry bookkeeping are correctly applied and represented.

Table 63: Summary of Within Balance Sheet Reclassifications

Item	Reclassification Event	Counterparty Account
Inventory	Purchase on credit	Trade Payable
Inventory	Harvest	Biological assets
Inventory	Used in Self-construction	PP&E
PP&E	Used in Self-construction	Inventory
PP&E	Net Reclassification	Investment Property
Right-of-Use Asset	New ROU assets recognized	Lease Liability
Investment Property	Net Reclassification	PP&E
Biological assets	Harvest	Inventory
Other Financial Liabilities	New liability recognised at FV	FVTPL
Trade Payable	Purchase on credit	Inventory
Lease Liability	New Lease liabilities recognized	Right-of-Use Asset
FVTPL	New asset purchased	Other Financial Liabilities

14.4 Statement of Changes in Equity

Table 64: Statement of Changes in Equity

	Share Capital	Share Premium	FVTOCI Reserve	Retained Earnings	Non-controlling Interests
Balance at beginning of period					
Total Comprehensive Income:					
Net income for the period				✓	✓
Other comprehensive income			✓		
Transactions with owners:					
Issue of share capital	✓	✓			
Dividends paid				(✓)	
Balance at end of period					

15 General Mathematical Form for Flow Variables under SSM

This section aims to develop a general mathematical framework for modeling flow and stock dynamics using the State-Space Model (SSM). The primary objective is to establish a form that guarantees the theoretical validity of specific model settings for flow variables. By casting the system dynamics into the canonical state-space representation, we can first determine whether a proposed model structure is coherent and feasible.

15.1 The Simplest Case: Decoupled Stock and Flow Dynamics

In the most basic formulation, the stock variable is treated as a pure consequence of the flow, without any feedback mechanism influencing the flow's generation. In this setup, the observation is solely the flow variable, which depends on a latent state, while the stock variable follows an exogenous, deterministic process driven by the history of observations.

The system comprises three distinct types of variables: a stock variable that accumulates over time, a flow variable that modifies the stock, and a set of intermediary variables that influence the flow.

Their formal definitions are as follows:

- **Stock Variable:** $\mathbf{u}_t^{\text{stock}} \in \mathbb{R}^N$, representing the accumulated stock at the end of period t .
- **Flow Variable:** $\mathbf{y}_t^{\text{flow}} \in \mathbb{R}^M$, representing the flow that occurs during period t .
- **Intermediary Variables:** $\mathbf{x}_t^{\text{others}} \in \mathbb{R}^K$, representing other variables that affect the flow generation process at period t .

The system is defined by:

$$\mathbf{u}_t^{\text{stock}} = \mathbf{u}_{t-1}^{\text{stock}} + h(\mathbf{y}_t^{\text{flow}}) \quad (52)$$

$$\mathbf{y}_t^{\text{flow}} = f(\mathbf{x}_t^{\text{others}}, \boldsymbol{\nu}_t) \quad (53)$$

$$\mathbf{x}_t^{\text{others}} = g(\mathbf{x}_{t-1}^{\text{others}}, \boldsymbol{\omega}_t) \quad (54)$$

Here, the state-space model is trivially defined. The latent state is simply $\mathbf{X}_t = \mathbf{x}_t^{\text{others}}$, and the observation is $\mathbf{Y}_t = \mathbf{y}_t^{\text{flow}}$. The system directly satisfies the core SSM properties:

- **Markov Property:** The state equation $\mathbf{X}_t = g(\mathbf{X}_{t-1}, \boldsymbol{\omega}_t)$ is already in the required first-order Markov form.
- **Conditional Independence:** The observation equation $\mathbf{Y}_t = f(\mathbf{X}_t, \boldsymbol{\nu}_t)$ ensures that the observation depends only on the current state.

The stock variable $\mathbf{u}_t^{\text{stock}}$ is not part of the latent state in the filtering process; it can be computed ex-post by accumulating the observed flows. However, this model's assumption that the stock level does not influence the flow is often too restrictive for most economic and physical systems. In the subsequent sections, we will progressively relax this assumption to build more realistic models.

15.2 A Foundational Model Structure with Feedback Loop

Now, we begin with the modeling structure that includes feedback.

The logical and temporal relationships between these variables are governed by the following system of equations:

$$\mathbf{u}_t^{\text{stock}} = \mathbf{u}_{t-1}^{\text{stock}} + h(\mathbf{y}_t^{\text{flow}}) \quad (55)$$

$$\mathbf{y}_t^{\text{flow}} = f(\mathbf{u}_{t-1}^{\text{stock}}, \mathbf{x}_t^{\text{others}}, \boldsymbol{\nu}_t) \quad (56)$$

$$\mathbf{x}_t^{\text{others}} = g(\mathbf{x}_{t-1}^{\text{others}}, \boldsymbol{\omega}_t) \quad (57)$$

where $\boldsymbol{\nu}_t$ and $\boldsymbol{\omega}_t$ are stochastic noise terms.

The economic intuition behind this system is straightforward. Equation (55) is an accounting identity stating that the stock is a deterministic process given the flow; its current value is simply the previous period's value plus the inflow. Equation (56) posits that the current flow is a function of the stock from the previous period, the value of other intermediary variables in the current period, and a random shock. Finally, Equation (57) states that the intermediary variables follow their own first-order Markov process, independent of the other parts of the system.

15.2.1 State-Space Representation and Validation

To validate this structure, we must cast it into the standard state-space form, which consists of an observation equation and a state transition equation. The key technique to achieve this is **state augmentation**.

Let us define the observation vector $\mathbf{Y}_t$ and the latent state vector $\mathbf{X}_t$.

- **Observation Vector:** The flow variable is what we observe or model, so we define $\mathbf{Y}_t = \mathbf{y}_t^{\text{flow}}$.
- **Augmented State Vector:** For the system to be Markovian, the state $\mathbf{X}_t$ must contain all necessary information to generate the observation $\mathbf{Y}_t$ and the next state $\mathbf{X}_{t+1}$. From the flow generation equation (56), we see that $\mathbf{y}_t^{\text{flow}}$ depends on $\mathbf{u}_{t-1}^{\text{stock}}$ and $\mathbf{x}_t^{\text{others}}$. We therefore define the augmented state vector as the concatenation of these two variables:

$$\mathbf{X}_t = \begin{bmatrix} \mathbf{x}_t^{\text{others}} \\ \mathbf{u}_{t-1}^{\text{stock}} \end{bmatrix} \quad (58)$$

With these definitions, we can now prove that the two core properties of the SSM are satisfied.

1. Conditional Independence of the Observation. The observation equation in an SSM requires the observation $\mathbf{Y}_t$ to be conditionally independent of all past information given the current state $\mathbf{X}_t$. Let's formulate our observation equation:

$$\mathbf{Y}_t = \mathbf{y}_t^{\text{flow}} = f(\mathbf{u}_{t-1}^{\text{stock}}, \mathbf{x}_t^{\text{others}}, \boldsymbol{\nu}_t) \quad (59)$$

By our definition of the state vector $\mathbf{X}_t$, its components are precisely $\mathbf{x}_t^{\text{others}}$ and $\mathbf{u}_{t-1}^{\text{stock}}$. Therefore, we can rewrite the equation as a function H of the state $\mathbf{X}_t$ and the observation noise $\boldsymbol{\nu}_t$:

$$\mathbf{Y}_t = H(\mathbf{X}_t, \boldsymbol{\nu}_t) \quad (60)$$

This confirms that the conditional independence property $p(\mathbf{Y}_t | \mathbf{X}_t, \mathbf{X}_{t-1}, \dots) = p(\mathbf{Y}_t | \mathbf{X}_t)$ holds.

2. Markov Property of the Latent State. The state transition equation requires the state vector $\mathbf{X}_t$ to follow a first-order Markov process, meaning its distribution depends only on the previous state $\mathbf{X}_{t-1}$. We must show that $\mathbf{X}_t$ can be expressed as a function of $\mathbf{X}_{t-1}$ and system noises. Let's examine each component of $\mathbf{X}_t$:

- **First component of $\mathbf{X}_t$ ($\mathbf{x}_t^{\text{others}}$):** From Equation (57), we have $\mathbf{x}_t^{\text{others}} = g(\mathbf{x}_{t-1}^{\text{others}}, \boldsymbol{\omega}_t)$. The term $\mathbf{x}_{t-1}^{\text{others}}$ is the first component of the state vector at time $t-1$, i.e., $\mathbf{X}_{t-1}$. This part is satisfied.
- **Second component of $\mathbf{X}_t$ ($\mathbf{u}_{t-1}^{\text{stock}}$):** Using the stock update rule (55) at time $t-1$, we have $\mathbf{u}_{t-1}^{\text{stock}} = \mathbf{u}_{t-2}^{\text{stock}} + \mathbf{y}_{t-1}^{\text{flow}}$. Now we express each term on the right-hand side using components of $\mathbf{X}_{t-1} = \begin{bmatrix} \mathbf{x}_{t-1}^{\text{others}} \\ \mathbf{u}_{t-2}^{\text{stock}} \end{bmatrix}$:

- The term $\mathbf{u}_{t-2}^{\text{stock}}$ is the second component of $\mathbf{X}_{t-1}$.
- The term $\mathbf{y}_{t-1}^{\text{flow}}$ is given by the flow equation at $t-1$:

$$\mathbf{y}_{t-1}^{\text{flow}} = f(\mathbf{u}_{t-2}^{\text{stock}}, \mathbf{x}_{t-1}^{\text{others}}, \boldsymbol{\nu}_{t-1})$$

The arguments of this function, $\mathbf{u}_{t-2}^{\text{stock}}$ and $\mathbf{x}_{t-1}^{\text{others}}$, are exactly the components of $\mathbf{X}_{t-1}$.

Thus, we can write $\mathbf{u}_{t-1}^{\text{stock}}$, the second element of $\mathbf{X}_t$, as a function of $\mathbf{X}_{t-1}$ and the noise term $\boldsymbol{\nu}_{t-1}$.

Since both components of $\mathbf{X}_t$ can be written as a function of $\mathbf{X}_{t-1}$ and the collective system noises $(\boldsymbol{\omega}_t, \boldsymbol{\nu}_{t-1})$, we can formulate a valid state transition equation:

$$\mathbf{X}_t = F(\mathbf{X}_{t-1}, \text{noises}_{t-1,t}) \quad (61)$$

This proves that the augmented state vector $\mathbf{X}_t$ follows a first-order Markov process. The proposed model structure is therefore valid and coherent within the SSM framework.

15.3 Model Extension with Flow Persistence

We now extend the foundational model to incorporate persistence, or auto-correlation, in the flow variable itself. This is a common empirical feature where the flow in one period is influenced by the flow in the preceding period, representing phenomena such as inertia, momentum, or partial adjustment dynamics.

The system of equations is modified as follows, with the change highlighted in the flow generation equation:

$$\mathbf{u}_t^{\text{stock}} = \mathbf{u}_{t-1}^{\text{stock}} + h(\mathbf{y}_t^{\text{flow}}) \quad (62)$$

$$\mathbf{y}_t^{\text{flow}} = f(\mathbf{u}_{t-1}^{\text{stock}}, \mathbf{y}_{t-1}^{\text{flow}}, \mathbf{x}_t^{\text{others}}, \boldsymbol{\nu}_t) \quad (63)$$

$$\mathbf{x}_t^{\text{others}} = g(\mathbf{x}_{t-1}^{\text{others}}, \boldsymbol{\omega}_t) \quad (64)$$

The key modification in Equation (63) is the inclusion of the lagged flow, $\mathbf{y}_{t-1}^{\text{flow}}$, as an explanatory variable for the current flow, $\mathbf{y}_t^{\text{flow}}$. While this adds a layer of dynamic complexity, we can show that the model remains valid within the SSM framework through a further augmentation of the state vector.

15.3.1 State-Space Representation for the Extended Model

The introduction of a lagged dependent variable in the observation equation requires us to expand our definition of the system's state.

- **Observation Vector:** This remains unchanged, $\mathbf{Y}_t = \mathbf{y}_t^{\text{flow}}$.
- **Augmented State Vector:** To compute the observation $\mathbf{Y}_t$ via Equation (63), we now need access to three variables: $\mathbf{u}_{t-1}^{\text{stock}}$, $\mathbf{y}_{t-1}^{\text{flow}}$, and $\mathbf{x}_t^{\text{others}}$. Consequently, the state vector $\mathbf{X}_t$ must be expanded to include all three pieces of information. We define the new, higher-dimensional state vector as:

$$\mathbf{X}_t = \begin{bmatrix} \mathbf{x}_t^{\text{others}} \\ \mathbf{u}_{t-1}^{\text{stock}} \\ \mathbf{y}_{t-1}^{\text{flow}} \end{bmatrix} \quad (65)$$

This redefinition allows us to once again validate the model against the core SSM properties.

1. Conditional Independence of the Observation. The observation equation is given by Equation (63). All its arguments— $\mathbf{u}_{t-1}^{\text{stock}}$, $\mathbf{y}_{t-1}^{\text{flow}}$, and $\mathbf{x}_t^{\text{others}}$ —are, by construction, components of our new state vector $\mathbf{X}_t$. We can therefore write a valid observation equation as a function H of only the current state $\mathbf{X}_t$ and noise:

$$\mathbf{Y}_t = H(\mathbf{X}_t, \boldsymbol{\nu}_t) \quad (66)$$

The conditional independence property continues to hold.

2. Markov Property of the Latent State. We must demonstrate that the new 3-dimensional state vector $\mathbf{X}_t$ can be fully determined by its lagged counterpart $\mathbf{X}_{t-1}$ and system noises. Let's

analyze each component of $\mathbf{X}_t$ in terms of $\mathbf{X}_{t-1} = \begin{bmatrix} \mathbf{x}_{t-1}^{\text{others}} \\ \mathbf{u}_{t-2}^{\text{stock}} \\ \mathbf{y}_{t-2}^{\text{flow}} \end{bmatrix}$:

- **First component of $\mathbf{X}_t$ ($\mathbf{x}_t^{\text{others}}$):** This transition is unchanged and depends only on the first component of $\mathbf{X}_{t-1}$:

$$\mathbf{x}_t^{\text{others}} = g(\mathbf{x}_{t-1}^{\text{others}}, \boldsymbol{\omega}_t)$$

- **Second component of $\mathbf{X}_t$ ($\mathbf{u}_{t-1}^{\text{stock}}$):** We use the stock update rule at $t - 1$: $\mathbf{u}_{t-1}^{\text{stock}} = \mathbf{u}_{t-2}^{\text{stock}} + \mathbf{y}_{t-1}^{\text{flow}}$.

- The term $\mathbf{u}_{t-2}^{\text{stock}}$ is the second component of $\mathbf{X}_{t-1}$.
- The term $\mathbf{y}_{t-1}^{\text{flow}}$ is given by the flow equation at $t - 1$:

$$\mathbf{y}_{t-1}^{\text{flow}} = f(\mathbf{u}_{t-2}^{\text{stock}}, \mathbf{y}_{t-2}^{\text{flow}}, \mathbf{x}_{t-1}^{\text{others}}, \boldsymbol{\nu}_{t-1})$$

All arguments of this function ($\mathbf{u}_{t-2}^{\text{stock}}$, $\mathbf{y}_{t-2}^{\text{flow}}$, and $\mathbf{x}_{t-1}^{\text{others}}$) are components of $\mathbf{X}_{t-1}$.

Therefore, $\mathbf{u}_{t-1}^{\text{stock}}$ can be expressed as a function of $\mathbf{X}_{t-1}$ and noise.

- **Third component of $\mathbf{X}_t$ ($\mathbf{y}_{t-1}^{\text{flow}}$):** This is the new component in our state definition. As shown immediately above, $\mathbf{y}_{t-1}^{\text{flow}}$ can be generated using only the components of $\mathbf{X}_{t-1}$ and the noise term $\boldsymbol{\nu}_{t-1}$.

Since all three components of $\mathbf{X}_t$ are functions of $\mathbf{X}_{t-1}$ and system noises, a valid state transition function F exists:

$$\mathbf{X}_t = F(\mathbf{X}_{t-1}, \text{noises}_{t-1,t}) \quad (67)$$

This confirms that even with the added complexity of flow persistence, the model structure is coherent and can be correctly represented within the SSM framework by increasing the dimensionality of the state vector. This demonstrates the flexibility and power of state augmentation for handling richer dynamic dependencies.

15.4 Model with Contemporaneous Flow Coupling

A further, more complex extension involves modeling the contemporaneous interdependence between different elements of the flow vector. This scenario arises when the elements of $\mathbf{y}_t^{\text{flow}}$ are determined jointly, forming a system of simultaneous equations.

The flow generation equation now includes the current flow vector $\mathbf{y}_t^{\text{flow}}$ as an argument, representing this coupling:

$$\mathbf{u}_t^{\text{stock}} = \mathbf{u}_{t-1}^{\text{stock}} + h(\mathbf{y}_t^{\text{flow}}) \quad (68)$$

$$\mathbf{y}_t^{\text{flow}} = f(\mathbf{u}_{t-1}^{\text{stock}}, \mathbf{y}_{t-1}^{\text{flow}}, \mathbf{y}_t^{\text{flow}}, \mathbf{x}_t^{\text{others}}, \boldsymbol{\nu}_t) \quad (69)$$

$$\mathbf{x}_t^{\text{others}} = g(\mathbf{x}_{t-1}^{\text{others}}, \boldsymbol{\omega}_t) \quad (70)$$

Equation (69) is no longer an explicit formula for $\mathbf{y}_t^{\text{flow}}$ but an **implicit function** that defines an equilibrium or consistency condition that $\mathbf{y}_t^{\text{flow}}$ must satisfy.

15.4.1 Validation via Explicit Formulation

To fit this model into the standard SSM framework, we must first convert the implicit observation equation into an explicit one. This requires that Equation (69) can be algebraically solved for $\mathbf{y}_t^{\text{flow}}$.

Let us define a vector $\mathbf{z}_t$ that consolidates all predetermined variables and noise:

$$\mathbf{z}_t = \text{pack}(\mathbf{u}_{t-1}^{\text{stock}}, \mathbf{y}_{t-1}^{\text{flow}}, \mathbf{x}_t^{\text{others}}, \boldsymbol{\nu}_t)$$

The implicit equation can then be viewed abstractly as $\mathbf{y}_t^{\text{flow}} = f(\mathbf{z}_t, \mathbf{y}_t^{\text{flow}})$. If this equation has a unique solution for $\mathbf{y}_t^{\text{flow}}$, we can define a new function f' such that:

$$\mathbf{y}_t^{\text{flow}} = f'(\mathbf{u}_{t-1}^{\text{stock}}, \mathbf{y}_{t-1}^{\text{flow}}, \mathbf{x}_t^{\text{others}}, \boldsymbol{\nu}_t) \quad (71)$$

For example, in a linear system where $\mathbf{y}_t^{\text{flow}} = A\mathbf{y}_t^{\text{flow}} + \text{terms not involving } \mathbf{y}_t^{\text{flow}}$, we can solve to get $\mathbf{y}_t^{\text{flow}} = (I - A)^{-1}(\dots)$, provided $(I - A)$ is invertible.

Validity Condition. A model with contemporaneous coupling is valid within the standard SSM framework *if and only if* an explicit form like Equation (71) can be derived.

If this condition is met, the problem reduces to the case discussed in the previous subsection. The state vector definition remains the same, as the set of *predetermined* variables required to compute the flow has not changed.

- **Augmented State Vector:** The definition is identical to the flow persistence model:

$$\mathbf{X}_t = \begin{bmatrix} \mathbf{x}_t^{\text{others}} \\ \mathbf{u}_{t-1}^{\text{stock}} \\ \mathbf{y}_{t-1}^{\text{flow}} \end{bmatrix} \quad (72)$$

- **Observation Equation:** The new, explicit function f' becomes the observation function H :

$$\mathbf{Y}_t = f'(\dots) = H(\mathbf{X}_t, \boldsymbol{\nu}_t)$$

Note that this transformation may also alter the distribution of the effective observation noise.

Since the state vector definition and its transition logic are unchanged from the previous model, the Markov property holds. The crucial step is the algebraic pre-processing to establish an explicit observation equation. If an explicit solution cannot be found, the subsequent section outlines the necessary alternative approaches.

15.4.2 Handling Models without an Explicit Solution

When an algebraic, closed-form solution for Equation (71) is intractable (e.g., in cases of high non-linearity), standard filtering algorithms like the Kalman Filter or its direct extensions (EKF, UKF) are no longer applicable. These situations require computational methods that do not rely on an explicit observation likelihood function $p(\mathbf{Y}_t|\mathbf{X}_t)$.

Simulation-Based Filtering. The most powerful and versatile approach in this scenario is the **Particle Filter (PF)**, a simulation-based, sequential Monte Carlo method. The PF approximates the posterior distribution of the state vector using a large set of weighted samples, or "particles". Its key advantage is in the weighting step, which can be adapted for implicit models.

For a given particle i with a hypothesized state $\mathbf{X}_t^{(i)}$, its weight is proportional to the likelihood of the actual observation $\mathbf{Y}_t^{\text{obs}}$ given that state, $p(\mathbf{Y}_t^{\text{obs}}|\mathbf{X}_t^{(i)})$. Without an explicit function, we compute this likelihood by inverting the logic:

1. Given the particle's state $\mathbf{X}_t^{(i)}$ and the true observation $\mathbf{Y}_t^{\text{obs}}$, we use the implicit equation (69) to solve for the value of the noise term $\boldsymbol{\nu}_t$ that would be required to make the equation hold. Let's call this the "implied noise", $\boldsymbol{\nu}_{\text{implied}}^{(i)}$.
2. We then evaluate the probability density of this implied noise under its assumed prior distribution (e.g., a multivariate normal distribution $\mathcal{N}(\mathbf{0}, R)$).

$$w_t^{(i)} \propto p(\boldsymbol{\nu}_t = \boldsymbol{\nu}_{\text{implied}}^{(i)}) \quad (73)$$

Particles whose state hypotheses require a low-probability noise to explain the observation are assigned low weights, and vice versa. While computationally intensive, this method is extremely general.

Other Strategies. Two other strategies are worth noting:

- **Embedded Numerical Solvers:** One can retain the structure of an EKF or UKF but, at each step, use a numerical root-finding algorithm (e.g., Newton-Raphson) to solve the implicit equation for the predicted observation. This is often complex to implement and can be numerically unstable if the solver fails to converge.
- **Model Approximation:** A pragmatic first step is often to simplify the model. For instance, one could approximate the contemporaneous coupling $\mathbf{y}_t^{\text{flow}}$ with its lagged value $\mathbf{y}_{t-1}^{\text{flow}}$. This transforms the model back into an explicit form at the cost of introducing a potential specification error.

15.5 Conclusion and Recommended Framework

This section has systematically built up a general framework for modeling stock and flow dynamics, demonstrating that progressively more complex and realistic assumptions can be accommodated within the State-Space Model paradigm through the technique of state augmentation.

Among the models discussed, the **Model Extension with Flow Persistence** provides a powerful and practical balance. It is computationally tractable while being comprehensive enough to capture many crucial empirical features, such as feedback from stock levels and auto-correlation (momentum) in the flow variables.

Therefore, for the specific modeling of flow variables, we are adopting this framework, as its mathematical validity has been formally established. Specifically, if one posits a stochastic process for a flow variable of the form:

$$\mathbf{y}_t^{\text{flow}} = \rho \mathbf{y}_{t-1}^{\text{flow}} + f(\mathbf{x}_t^{\text{others}}, \mathbf{u}_{t-1}^{\text{stock}}) + \boldsymbol{\nu}_t \quad (74)$$

where ρ captures the persistence, this model can be directly and correctly handled within the SSM framework. By defining the state vector as $\mathbf{X}_t = [\mathbf{x}_t^{\text{others}}, \mathbf{u}_{t-1}^{\text{stock}}, \mathbf{y}_{t-1}^{\text{flow}}]'$, we can successfully derive the required conditional distribution $p(\mathbf{Y}_t | \mathbf{X}_t)$, making the system amenable to standard filtering and estimation techniques. This provides a robust foundation for further empirical work.

16 A Toy Model for Drivers and Flow Variables

This section defines the selected flow variable and the model used to forecast it. While recent work often prioritizes predictive accuracy via machine learning, our emphasis is interpretability. Rather than a black-box approach, we ground the framework in established concepts from economics, finance, and accounting. This transparency is essential for isolating underlying drivers and enabling disciplined what-if analysis.

16.1 Sales (Revenue)

The composition of revenue depends on a firm's industry and business model. For example, a consulting firm recognizes service revenue, whereas a streaming platform generates subscription revenue. Here we focus on the most common case: sales revenue from physical products. Sales outcomes reflect both internal and external influences (Table 65), shaping sales volume, growth, and overall performance (Khajehzadeh et al., 2022; Azad et al., 2013; Pekkaya et al., 2024; Warsame, 2023; Cheong and Hoang, 2021; Alamsyah and Hasan, 2023). We begin with the two core components—price and quantity. If we can project the future price and units sold for each product, we can estimate revenue. Because price and quantity are jointly determined by supply and demand, a practical strategy is to condition on one and forecast the other.

Table 65: Internal and External Factors Affect Sales

Internal Factors	External Factors
Product quality and features	Market demand
Price	Economic environment
Marketing and promotion	Competition
Sales force effectiveness	Social/cultural trends
Distribution channels	Regulations/legal environment
Innovation	Political/geopolitical events

Baseline. Assume p major products. Let the price and quantity dynamics for product p follow

$$\begin{aligned}\log\text{Price}_t^p &= c + \phi_1^{PP} \cdot \log\text{Price}_{t-1}^p + \epsilon_t^P, \\ \log\text{Quantity}_t^p &= c + \phi_1^{PQ} \cdot \log\text{Price}_{t-1}^p + \phi_2^{QQ} \cdot \log\text{Quantity}_{t-1}^p + \epsilon_t^Q.\end{aligned}$$

Total sales are then

$$\text{Sales}_t = \sum_p \text{Price}_t^p \cdot \text{Quantity}_t^p.$$

This baseline is straightforward to estimate and explain, as each product is modeled independently. Its limitations stem from its assumptions: it does not capture cross-price interactions (e.g., substitution or cannibalization), it implies demand reacts to lagged rather than contemporaneous prices, and it omits exogenous drivers such as seasonality, promotion, and macroeconomic conditions.

Theory-driven extension. To introduce economically consistent substitution patterns and support rigorous what-if analysis, we consider the *Almost Ideal Demand System (AIDS)* (Deaton and Muellbauer, 1980):

$$w_i = a_i + \sum_j \gamma_{i,j} \log(p_j) + \beta_i \log(x/P),$$

where:

- w_i is the budget share of good i .
- p_j are the prices of the various goods.
- x is the total expenditure of the consumer.
- P is an overall price index.

- $a_i, \gamma_{i,j}, \beta_i$ are the parameters to be estimated. These parameters tell us how demand responds changes in price and income.

In practice, we define a relevant product category (e.g., smartphones), forecast own and competitor prices and category expenditure, and use the estimated shares to infer volumes. Revenue then follows as predicted market size times predicted share. The framework is well-suited to scenario analysis (price moves, expenditure shocks) but requires granular inputs—own and rival prices and a measure of category spending, and unless explicitly modeled, does not incorporate factors like marketing, discounts, or brand loyalty. There also exist some related work on market model (Berry et al., 1993; Hendel and Nevo, 2006; Hanssens et al., 2001), which we may consider later.

16.2 Accounts Receivable and Cash Collections

Forecasting accounts receivable and the resulting cash collections is essential for linking a firm’s operational sales to its balance sheet and cash flow statement. We explore two primary methods: a direct forecast of cash collection patterns or an indirect approach based on a key efficiency ratio. The direct method, while granular, requires estimating unobservable collection rates, reducing its transparency for an external analyst. Therefore, we adopt the more standard indirect approach, which models the Days Sales Outstanding (DSO) ratio.

$$\text{DSO}_t = \text{Days} \times \frac{\text{Average TR}_t}{\text{Net Credit Sales}_t}, \quad \text{Average TR}_t = \frac{TR_{t-1} + TR_t}{2}$$

This method is fundamentally grounded in public data, as DSO can be directly calculated, analyzed, and benchmarked against industry peers, making it a more transparent and defensible driver for a third-party forecast.

Credit Sales. Credit sales represent the portion of revenue that generates a trade receivable. Given the data limitation that public filings rarely separate cash and credit revenue, we must make a simplifying assumption. While DSO is formally defined with net credit sales, we will follow the standard practice of using total revenue as a proxy in its calculation. To maintain the integrity of the accounting identities, we still define credit sales as a percentage of total revenue.

$$\text{Credit Sales}_t = \text{Revenue}_t \times r_{\text{credit}}$$

where:

- r_{credit} is the assumed ratio of sales made on credit, treated as an exogenous variable based on industry norms and business model analysis.
- r_{credit} is should be within $(0, 1)$. The modeling of r_{credit} is in Appendix C to guarantee this feature obtained from a linear model with Gaussian Noise.

Therefore, Sales Received in Cash is calculated as

$$\text{Sales Received in Cash}_t = \text{Revenue}_t \times (1 - r_{\text{credit}})$$

In general, suppliers offer trade credit when they possess a financing advantage over traditional lenders, stemming from superior information, better control over the buyer, or a greater ability to salvage assets (Petersen and Rajan, 1997). Given that these advantages are most pronounced in business-to-business (B2B) settings, we may set this ratio high for B2B firms and low for business-to-consumer (B2C) firms, where such transactional advantages are less relevant.

Forecasting Trade Receivables. The cornerstone of our approach is to first forecast the DSO ratio and then use it to determine the ending trade receivables balance. We model DSO using a time-series approach, which can be augmented with other explanatory variables.

$$\log \text{DSO}_t = c + \phi_1 \cdot \log \text{DSO}_{t-1} + \beta' \mathbf{X}_t + \varepsilon_t$$

where:

- c is the intercept or constant term.
- ϕ_1 is the autoregressive coefficient capturing persistence in the DSO series.
- $\mathbf{X}_t$ is a vector of external variables reflecting company policy, seasonality, economic health, or firm-specific characteristics like size, leverage, and growth (Deloof, 2003; Knauer and Wöhrmann, 2013).

After obtaining the forecast for DSO_t , we determine the ending trade receivables balance. For consistency, we use total revenue in this formula, mirroring its use in the historical DSO calculation.

$$\text{Trade Receivable}_t = \left(\frac{\text{Revenue}_t}{N} \right) \times \text{DSO}_t$$

where:

- N is the number of days in the period (e.g., 365 for a year).

The main advantage of this approach is its direct link to an observable business driver. The primary limitation is that it assumes the historical relationship between sales, receivables, and DSO will continue into the future.

Cash Collected. Finally, with forecasts for credit sales and the beginning and ending trade receivables balances, we derive the cash collected from customers. This value is not forecasted directly but is the result of the accounting identity that links the balance sheet flows, ensuring model integrity.

$$\text{Cash Collected}_t = \text{Trade Receivable}_{t-1} + \text{Credit Sales}_t - \text{Trade Receivable}_t$$

This calculation ensures that the projected change in the trade receivables balance on the balance sheet is consistent with the net cash flow impact required for the cash flow statement.

16.3 Revenue Recognized from Contract Liability

The recognition of revenue from Contract Liability (specifically, the ‘Revenue recognised’ flow that reduces the ‘Contract Liability’ liability balance) is modelled as a proportion of the opening Contract Liability balance.

This relationship is defined by the following formula:

$$\text{Revenue recognised from CL}_t = r_{\text{CL},t} \times \text{CL}_{t-1}$$

In this model, $r_{\text{CL},t}$ is a stochastic variable, representing the portion of the opening prepayment balance that is consumed and recognised as an expense (or revenue, from the customer’s perspective in the case of a Contract Liability) during the period. The specific stochastic modelling of this ratio, which is bound between 0 and 1, is detailed in Appendix C.

16.4 Cost of Goods Sold and Working Capital (Tentative)

This section addresses the primary costs of production and the management of the firm’s core operating working capital. We first forecast the *Cost of Goods Sold (COGS)*, the direct cost of the products the firm sells. We then link this to the balance sheet by modeling *Inventory* and *Accounts Payable*. This approach ensures that our income statement forecast is directly and consistently tied to the operational assets and liabilities that support it.

Cost of Goods Sold. The most direct approach to forecasting COGS is to model the *Gross Margin*, which isolates the firm’s core profitability from its products. We model the gross margin using a time-series approach:

$$\text{Gross Margin}_t = c + \phi_1 \cdot \text{Gross Margin}_{t-1} + \beta' \mathbf{X}_t + \varepsilon_t$$

where $\mathbf{X}_t$ can include external variables such as commodity prices or inflation. Total COGS is then determined as a function of revenue:

$$\text{COGS}_t = \text{Revenue}_t \times (1 - \text{Gross Margin}_t)$$

The primary advantage of this approach is its direct link to a high-level, observable metric. The limitation, however, is that it treats the gross margin as an aggregate outcome, simplifying the complex underlying cost structure detailed in managerial accounting. For example, it does not explicitly separate costs into their *fixed* and *variable components*. In reality, a firm with high fixed manufacturing costs (e.g., a semiconductor fab) will see its margin expand significantly as sales volume increases due to *economies of scale*, an effect not captured by a simple margin forecast. Similarly, this approach does not distinguish between different cost systems; it is more reliable for a firm using *process costing* (homogenous products) than one using *job-order costing* (unique projects with variable profitability) (Warren and Tayler, 2022).

Inventory. We connect COGS to the balance sheet through the *Days Inventory Outstanding (DIO)* ratio. This driver measures the average number of days the company holds its inventory before selling it. We forecast DIO using an autoregressive model:

$$\text{DIO}_t = c + \phi_1 \cdot \text{DIO}_{t-1} + \beta' \mathbf{X}_t + \varepsilon_t$$

After forecasting DIO, we use it to determine the ending inventory balance.

$$\text{Inventory}_t = \left(\frac{\text{COGS}_t}{N} \right) \times \text{DIO}_t$$

This method is grounded in a standard measure of operational efficiency, but its main assumption is that the historical relationship between COGS and inventory will persist. This simplification acknowledges, but does not explicitly model, the strategic trade-offs in corporate *working capital management*. A firm’s chosen DIO reflects a deliberate balance between efficiency (a “Just-in-Time” strategy to minimize cash tied up in inventory) and resilience (holding “safety stock” to protect against *supply chain disruptions*) (Finance, 2007). Our forecast represents an assumption about the continuation of this strategic posture.

Accounts Payable. Finally, we model the trade credit the firm uses to finance its inventory purchases through the *Days Payable Outstanding (DPO)* ratio. DPO measures the average number of days the firm takes to pay its suppliers. We model its dynamics as:

$$\text{DPO}_t = c + \phi_1 \cdot \text{DPO}_{t-1} + \beta' \mathbf{X}_t + \varepsilon_t$$

This allows us to forecast the ending accounts payable balance based on the level of production activity (proxied by COGS).

$$\text{Accounts Payable}_t = \left(\frac{\text{COGS}_t}{N} \right) \times \text{DPO}_t$$

This approach treats payables management as a stable operational policy. It acknowledges that DPO is the outcome of the firm's *negotiating power* with its suppliers and its corporate finance strategy. A high DPO provides a valuable source of interest-free financing but can strain supplier relationships. Conversely, a lower DPO may be chosen to capture *early payment discounts*, which would boost the gross margin, an interaction our baseline model does not endogenously capture (Finance, 2007).

Cash Paid to Suppliers. As with cash collections, the cash paid to suppliers is not forecasted directly but is derived from the accounting identity that links the income statement and the changes in the relevant balance sheet accounts.

$$\text{Cash Paid to Suppliers}_t = \text{COGS}_t + \Delta \text{Inventory}_t - \Delta \text{Accounts Payable}_t$$

This calculation ensures the integrity of the integrated financial statements, where the cash flow statement correctly reflects the cash impact of the firm's production and working capital management.

16.5 Other Operating Expenses

This section focuses on modeling operating expenses other than the Cost of Goods Sold (COGS). We will break down this category into its key components and model each one individually before potentially aggregating them.

16.5.1 Cost on Services

We begin by modeling the 'Cost on Services'. Our approach is guided by two primary assumptions:

1. The cost is primarily driven by the scale of the business, which we proxy by 'Revenue'. We will therefore model the ratio of 'Cost on Services' to 'Revenue'.
2. This expense is paid fully in cash, implying a real-world constraint where the cost in any period cannot exceed the cash available at the beginning of that period.

Modeling Strategy: From Absolute Value to Ratio Directly modeling the absolute cost is challenging due to its non-negative nature and complex dynamics. A more stable and insightful approach is to model its ratio relative to revenue. We define the core variable of interest as:

$$r_t = \frac{\text{Cost on Services}_t}{\text{Revenue}_t} \tag{75}$$

This ratio r_t represents the proportion of revenue allocated to service costs. While often less than 1, this ratio can exceed 1 during periods of high investment or low revenue.

Distributional Assumption: The Log-Normal Distribution To accommodate the characteristics of this ratio ($r_t > 0$ and potentially $r_t > 1$), the **Log-Normal distribution** is an excellent choice. This assumption is equivalent to modeling the logarithm of the ratio, $\log(r_t)$, as a Gaussian (Normal) process. This approach has two major advantages:

- It naturally ensures that the ratio r_t is always positive.
- It transforms the model into a linear and Gaussian framework (on the log scale), making it perfectly compatible with the standard Kalman filter.

State-Space Formulation We can now define a Linear Gaussian State-Space Model (LGSSM) for the log-ratio.

1. **Latent State Definition:** The unobserved state, x_t , is defined as the logarithm of the true, underlying ratio.

$$x_t = \log(r_t) \quad (76)$$

2. **State Transition Equation:** We assume that the log-ratio follows a standard **AR(1) process**.

$$x_t = c + \phi x_{t-1} + w_t, \quad \text{where } w_t \sim \mathcal{N}(0, Q) \quad (77)$$

Here, c is a constant intercept, ϕ is the autoregressive coefficient, and w_t is the Gaussian process noise. This form is mathematically equivalent to a mean-reverting process. If the process is stationary ($|\phi| < 1$), the long-run mean of the log-ratio, μ , can be derived from the parameters as $\mu = c/(1 - \phi)$. This equation is linear in its states.

3. **Observation Equation:** We observe the actual cost and revenue, from which we can compute an observed ratio. The observation equation links the logarithm of this observed ratio to the latent state.

$$\log(r_t^{\text{obs}}) = x_t + v_t, \quad \text{where } v_t \sim \mathcal{N}(0, R) \quad (78)$$

Here, v_t is the Gaussian measurement noise.

This formulation constitutes a standard LGSSM, which can be optimally solved using the Kalman filter to estimate and forecast the latent log-ratio, x_t .

16.5.2 Purchase of Services on Trade Payables

This variable represents the portion of the total ‘Cost on Services’ for the period that is not paid in cash immediately but is instead accrued as a trade payable. We model this as a ratio.

Modeling Strategy: Ratio of Accrual We define the core variable to be modeled as the accrual ratio, $r_{TP,t}$, which is strictly bounded between 0 and 1:

$$r_{TP,t} = \frac{\text{Purchase of Services on Trade Payables}_t}{\text{Cost on Services}_t} \quad (79)$$

Here, $r_{TP,t} \in (0, 1)$ is the latent ratio representing the proportion of service costs that are put on credit.

Since the core latent variable to be modeled, $r_{TP,t}$, is a ratio bounded in $(0, 1)$, its modeling follows the methodology for such variables detailed in Appendix C. The approach involves applying a logit transformation to the ratio and modeling the transformed value using a linear Gaussian AR(1) process. This allows the system to be formulated as a standard **Linear Gaussian State-Space Model (LGSSM)** and solved using the **standard Kalman filter**.

16.5.3 Expected Credit Loss (ECL) of Trade Receivables

The modeling of ‘Expected Credit Loss (ECL)’ is based on tracking a latent **Expected Loss Rate**, denoted $r_{ECL,t}$, which is a ratio strictly bounded between 0 and 1. The final ‘ECL expense’ is then calculated as a function of this rate and other balance sheet items.

Modeling Strategy: Latent Ratio and Ex-Post Calculation The core of this model is the latent bad debt ratio, $r_{ECL,t} \in (0, 1)$. We assume that we can observe a noisy version of this ratio, $r_{ECL,t}^{\text{obs}}$, which can be derived from historical financial statements. The state-space model is built to filter and forecast the true underlying ratio. The final ‘ECL expense’ is not the direct observation for the filter, but rather an amount calculated *after* the filtering process using the estimated ratio.

The accounting identities are as follows:

$$\text{Target Allowance}_t = \text{Gross TR}_t \times r_{ECL,t} \quad (80)$$

$$\text{ECL expense}_t = \text{Target Allowance}_t - \text{ECL Allowance}_{t-1} \quad (81)$$

(Assuming write-offs are zero for simplicity).

Since the core latent variable to be modeled, $r_{ECL,t}$, is a ratio bounded in $(0, 1)$, its modeling follows the methodology for such variables detailed in Appendix C.

The final ‘ECL expense’ is calculated in a post-processing step, using the filtered or forecasted estimate of the state $\hat{x}_t$, which is first converted back to the ratio $\hat{r}_{ECL,t}$ via the inverse-logit (logistic) function, and then plugged into the accounting identities above. The non-linearities involved in this final calculation step are external to the filtering algorithm and thus do not affect the choice of the filter.

16.5.4 Expense Recognised in Prepayments

This expense represents the portion of prepaid assets that is recognised as an expense in the current period. The recognition of this expense simultaneously reduces the ‘Prepayments’ asset on the balance sheet. Following a similar logic to ‘Cost on Services’, we model this expense as a ratio of ‘Revenue’.

Modeling Strategy: Ratio to Revenue We define the core variable to be modeled as the ratio of the prepayment expense to carrying amount of prepayment:

$$r_t = \frac{\text{Expense Recognised in Prepayments}_t}{\text{prepayment}_{t-1}} \quad (82)$$

This ratio, r_t , represents the rate at which prepaid assets are expensed relative to the company’s top-line performance. As an expense ratio, it is expected to be positive ($r_t > 0$).

Distributional Assumption and State-Space Formulation Given that the ratio r_t must be positive, the **Log-Normal distribution** is an appropriate choice, which is equivalent to modeling its logarithm as a Gaussian process. This allows the model to be cast as a standard LGSSM, making it compatible with the Kalman filter.

The formulation directly mirrors that of the ‘Cost on Services’:

1. **Latent State:** The state x_t is the logarithm of the true, underlying ratio: $x_t = \log(r_t)$.
2. **State Transition:** The state is assumed to follow a linear AR(1) process:

$$x_t = c + \phi x_{t-1} + w_t, \quad \text{where } w_t \sim \mathcal{N}(0, Q) \quad (83)$$

3. **Observation:** The observation is the logarithm of the observed ratio, calculated from financial statements:

$$\log(r_t^{\text{obs}}) = x_t + v_t, \quad \text{where } v_t \sim \mathcal{N}(0, R) \quad (84)$$

This linear and Gaussian structure allows for the efficient estimation of the latent log-ratio x_t using the **standard Kalman filter**. The final expense amount is then calculated in a post-processing step by taking the exponent of the filtered state estimate and multiplying it by the revenue forecast.

16.5.5 DC Plan and Salary Expense

The **Defined Contribution (DC) Plan Expense** and **Salary Expense** are both key components of employee-related costs. Instead of modeling them as ratios to other variables, we model their absolute values directly, assuming they exhibit inherent autoregressive behavior.

Modeling Strategy and Formulation We assume that both expenses are positive and follow a **Log-Normal distribution**. This is equivalent to modeling the logarithm of each expense as a linear Gaussian AR(1) process.

For example, the dynamic for **Salary Expense** is specified as:

$$\log(\text{Salary Expense}_t) = c_{sal} + \phi_{sal} \log(\text{Salary Expense}_{t-1}) + w_{sal,t}, \quad w_{sal,t} \sim \mathcal{N}(0, Q_{sal}) \quad (85)$$

An analogous process is defined for **DC Plan Expense**.

This formulation ensures that the expenses are always positive. Since the model for the log-transformed expenses constitutes a standard LGSSM (with the state being the log-expense and the observation being the observed log-expense), the latent log-expenses can be estimated and forecasted efficiently using the **standard Kalman filter**. The final expense amounts are then obtained by taking the exponent of the filtered or forecasted log-values.

16.6 Other Income and Expenses

This section discusses income and expenses arising from non-core operating assets, specifically Property, Plant & Equipment (PP&E), intangible assets, investment property, and biological assets.

16.6.1 Gain/Loss on Disposal of Assets

This item captures the accounting gain or loss from selling a portion of an asset class. The modeling approach is consistent across PP&E, intangible assets, investment property, and biological assets. The gain or loss is determined by the proceeds from the sale relative to the carrying value of the asset portion sold.

Modeling Strategy: Decomposing the Gain/Loss We first derive the formula for the gain or loss on disposal. The starting point is the carrying value of the asset after accounting for depreciation and any impairment adjustments. First, we calculate the value after depreciation:

$$\text{Carrying Value after Depr.}_t = \text{Carrying Value}_{t-1} - \text{Depreciation}_t \quad (86)$$

Next, we adjust for any impairment losses or reversals recognized during the period:

$$\begin{aligned} \text{Carrying Value after Impairment Loss/Reversal}_t &= \text{Carrying Value after Depr.}_t \\ &\quad - \text{Impairment Loss}_t + \text{Impairment Reversal}_t \end{aligned} \quad (87)$$

The gain or loss is the difference between the sale proceeds and the book value of the specific portion of the asset that was sold. We can decompose this as follows:

$$\begin{aligned} \text{Gain/Loss}_t &= \text{Proceeds}_t - \text{Carrying Value Sold}_t \\ &= \left(\frac{\text{Market Price}}{\text{Unit Book Value}} \right)_t \times \text{Carrying Value Sold}_t - \text{Carrying Value Sold}_t \\ &= \text{Carrying Value Sold}_t \times \left[\left(\frac{\text{Market Price}}{\text{Unit Book Value}} \right)_t - 1 \right] \end{aligned} \quad (88)$$

We further express the ‘Carrying Value Sold’ as a fraction of the total carrying value after impairment adjustments:

$$\begin{aligned} \text{Carrying Value Sold}_t &= \text{Carrying Value after Impairment Loss/Reversal}_t \\ &\times \left(\frac{\text{Sold Amount}}{\text{Total Carrying Value}} \right)_t \end{aligned} \quad (89)$$

The core stochastic variable we aim to model is the disposal ratio, $r_{\text{dispose},t}$, which represents the fraction of the total asset base sold in a given period:

$$r_{\text{dispose},t} = \frac{\text{Sold Amount}_t}{\frac{\text{Carrying Value after Impairment Loss/Reversal}_t}{\text{Impairment Loss/Reversal}_t}} \quad (90)$$

This ratio is strictly bounded between 0 and 1, as the amount sold cannot exceed the total available amount.

Since the core latent variable to be modeled, $r_{\text{dispose},t}$, is a ratio bounded in $(0,1)$, its modeling follows the methodology for such variables detailed in Appendix C. The approach involves applying a logit transformation to the ratio and modeling the transformed value using a linear Gaussian AR(1) process. This allows the system to be formulated as a standard **Linear Gaussian State-Space Model (LGSSM)** and solved using the **standard Kalman filter**. The final ‘Gain/Loss on Disposal’ amount is then calculated in a post-processing step using the filtered estimate of this ratio and assumptions about the market-to-book ratio.

16.6.2 Impairment Loss/Reversal of Assets

The ‘Impairment Loss’ or ‘Impairment Reversal’ is an accounting adjustment to align an asset’s carrying value with its recoverable amount (e.g., Net Realisable Value or NRV). This item is calculated based on accounting rules rather than being modeled as a primary stochastic process itself.

Modeling Strategy: Ex-Post Calculation Based on Accounting Rules The calculation of impairment is triggered by comparing the asset’s NRV to its carrying value after depreciation. We also need to track the ‘Accumulated Previous Impairment Loss’ for each asset class to correctly handle reversals.

1. **Impairment Loss:** Occurs if the NRV is lower than the carrying value.

$$\text{Impairment Loss}_t = \max(0, \text{Carrying Value after Depr.}_t - \text{NRV}_t) \quad (91)$$

2. **Impairment Reversal:** Occurs if the NRV is higher than the carrying value, but the reversal is capped by the accumulated previous losses.

$$\text{Potential Reversal}_t = \max(0, \text{NRV}_t - \text{Carrying Value after Depr.}_t) \quad (92)$$

$$\text{Impairment Reversal}_t = \min(\text{Potential Reversal}_t, \text{Accumulated Previous Loss}_{t-1}) \quad (93)$$

3. **Accumulated Loss Dynamics:** The opening balance of accumulated losses is updated by the period’s impairment activities and adjusted for any asset disposals.

$$\begin{aligned} \text{Accumulated Previous Loss}_t &= \left(\text{Accumulated Previous Loss}_{t-1} \right. \\ &\quad \left. + \text{Impairment Loss}_t - \text{Impairment Reversal}_t \right) \times (1 - r_{\text{dispose},t}) \end{aligned} \quad (94)$$

where $r_{\text{dispose},t}$ is the disposal ratio defined in the previous subsection.

Theoretical Limitations of Ex-Post Truncation The use of ‘max’ and ‘min’ functions to enforce accounting rules represents a form of "hard-coding" or ex-post truncation. This approach poses significant theoretical challenges if one attempts to incorporate the ‘Impairment Loss/Reversal’ directly into an SSM estimation process, either as an observation variable (y_t) or as part of the state vector (x_t). Any process subjected to such truncation will no longer follow a Gaussian distribution. For instance, applying a ‘min’ function to a Gaussian forecast for a potential reversal will result in a truncated, non-Gaussian distribution. This fundamentally violates the core assumptions of the standard Kalman filter. A theoretically superior approach would involve modeling the reversal with a distribution whose parameters are functions of the constraints (e.g., a Truncated Normal distribution), but this would require more advanced non-Gaussian filters like **Particle Filters**.

Considerations for an Integrated State-Space Model If the asset’s ‘Carrying Value’ were to be included as a state variable within the SSM, a fully integrated model would require the ‘Impairment Loss/Reversal’ to be modeled endogenously as part of the state transition logic. The state vector would need to be augmented to include, at a minimum, the ‘Carrying Value’ and the ‘Accumulated Previous Loss’. The state transition equation for the carrying value would then become a complex, stochastic, and non-linear function due to the ‘max’ and ‘min’ operations. This would render the state transition function ‘F’ both non-linear and non-Gaussian, necessitating the use of advanced non-linear filtering methods and precluding the use of the standard Kalman filter.

Simplification for the Current Framework Given the aforementioned complexities, we adopt a simplifying assumption for the current framework. Under the assumption that the asset’s carrying value is not itself a state variable within the core SSM, the ‘Impairment Loss/Reversal’ is treated as an **ex-post (separately calculated) adjustment**. In this simplified approach, the only stochastic component requiring estimation is the Net Realisable Value (NRV). We assume the NRV, being a positive value, follows a Log-Normal process, which is modeled by specifying a linear AR(1) process for its logarithm:

$$\log(\text{NRV}_t) = c_{nrv} + \phi_{nrv} \log(\text{NRV}_{t-1}) + w_{nrv,t}, \quad w_{nrv,t} \sim \mathcal{N}(0, Q_{nrv}) \quad (95)$$

This process for NRV can be estimated separately. The resulting forecast for NRV_t is then used as an external input for the ex-post calculation of the impairment charge. This pragmatic approach maintains the linearity and Gaussianity of the other core models, ensuring overall compatibility with the Kalman filter.

16.6.3 Fair Value Gain and Loss

This item applies specifically to biological assets, which are measured at fair value less costs to sell under accounting standards. This includes both bearer and consumable biological assets.

Modeling Strategy: Ex-Post Calculation The fair value gain or loss is calculated as the change in fair value between the start and end of the reporting period, adjusted for any purchases or sales. The simplified calculation is:

$$\text{Fair Value Gain/Loss}_t = \text{Fair Value}_t - \text{Fair Value}_{t-1} \quad (96)$$

Similar to the impairment charge, the fair value itself might be driven by a complex stochastic process (e.g., commodity prices, biological growth models). For the purpose of the current framework, which prioritizes the use of the standard Kalman filter, the ‘Fair Value Gain/Loss’ is treated as an **ex-post calculation**. It is determined outside the core SSM estimation loop, based on a separate model or assumption for the fair value process, and is not directly estimated by the SSM.

16.7 Financial Assets

This section details the modeling approach for financial assets. To simplify the initial framework, we make several key assumptions regarding the types of assets held. We assume the entity does not hold any financial assets classified as Fair Value Through Other Comprehensive Income (FVOCI). Consequently, Other Comprehensive Income (OCI) and its associated tax effects on Deferred Tax Assets (DTA) or Deferred Tax Liabilities (DTL) are not considered in the current model. The company's financial asset portfolio is assumed to consist solely of assets measured at Fair Value Through Profit & Loss (FVTPL) and Amortised Cost (AC). The modeling approach for FVTPL assets is stochastic, reflecting their market-driven nature, while the approach for Amortised Cost assets is deterministic, based on contractual cash flows.

16.7.1 Flows from Financial Assets at Amortised Cost

Under our assumptions, financial assets at amortised cost are held to maturity, and their cash flow patterns are deterministic.

Interest Income Interest income from these assets is calculated using the Effective Interest Rate (EIR) method. We assume the EIR is constant throughout the life of the asset. The formula is:

$$\text{Interest Income}_{AC,t} = \text{EIR}_{AC} \times \text{Carrying Amount}_{AC,t-1} \quad (97)$$

Simplifying Assumption for Carrying Amount To further simplify, we assume the financial asset at amortised cost is a long-term bond for which the coupon rate is equal to the EIR at inception. This results in the bond's carrying amount remaining constant and equal to its par value throughout its life (before maturity).

$$\text{Carrying Amount}_{AC,t} = \text{Par Value} \quad (98)$$

As a consequence of this simplification, the cash coupon received in each period is exactly equal to the interest income recognised.

$$\text{Coupon Received}_t = \text{Interest Income}_{AC,t} \quad (99)$$

16.7.2 Flows from Financial Assets at FVTPL

This category includes gains and losses arising from changes in the fair value of FVTPL assets.

Fair Value Gains/Loss on FVTPL Assets Fair value gains and losses are recognised for financial assets classified as FVTPL. This is calculated as the change in the asset's market fair value over the period.

$$\text{Fair Value Gain/Loss}_t = \text{Market Fair Value}_t - \text{Market Fair Value}_{t-1} \quad (100)$$

We assume that the market fair value follows a stochastic process. To ensure positivity and compatibility with the LGSSM framework, we model its logarithm as a linear AR(1) process:

$$\log(\text{Market Fair Value}_t) = c_{mv} + \phi_{mv} \log(\text{Market Fair Value}_{t-1}) + w_{mv,t}, \quad w_{mv,t} \sim \mathcal{N}(0, Q_{mv}) \quad (101)$$

This allows the 'Market Fair Value' to be forecasted within an SSM, and the gain/loss is then calculated ex-post based on the model's output.

16.8 Financial Liabilities

This section details the modeling approach for financial liabilities, which presents a higher level of complexity due to the need to track a term structure of debt. We adopt the following simplifying assumptions:

1. The entity's financial liabilities consist solely of instruments measured at amortised cost.
2. All new financial liabilities are issued with a fixed maturity of one year.
3. These liabilities take the form of risk-free bonds that pay coupons quarterly.
4. The Effective Interest Rate (EIR) for a specific bond is fixed at its issuance, but the EIR for new issuances can vary from one period to the next.

Assumption 2 implies that if we model the financial statements on a quarterly basis, we must track the term structure of the liability portfolio. At any given time t , the company's liabilities will consist of cohorts with different remaining maturities: {Matured, 3 Months, 6 Months, 9 Months, 1 Year}. These correspond to bonds that are maturing, and bonds with 3, 6, 9 months remaining maturity, as well as newly issued 1-year bonds.

Amortised Cost Dynamics This set of equations describes the roll-forward of the amortised cost (AC) for each liability cohort from one quarter to the next.

$$\begin{aligned}
 AC1Y_t &= PV(\text{New Par Value Issued}_t + \text{Future Coupons}_t) \\
 AC9M_t &= AC1Y_{t-1} - \text{Coupon}1Y_t + \text{Interest Expense } 1Y_t \\
 AC6M_t &= AC9M_{t-1} - \text{Coupon}9M_t + \text{Interest Expense } 9M_t \\
 AC3M_t &= AC6M_{t-1} - \text{Coupon}6M_t + \text{Interest Expense } 6M_t \\
 \text{Principal Paid}_t &= AC3M_{t-1} - \text{Coupon}3M_t + \text{Interest Expense } 3M_t
 \end{aligned}$$

In this system, the first equation for new issuances is stochastic, as the amount and terms of new debt are uncertain. The subsequent four equations are deterministic, as they depend entirely on the known characteristics of previously issued debt.

Coupon Payments The coupon payments for each cohort are determined by the par value and coupon rate set at the time of issuance.

$$\begin{aligned}
 \text{Coupon}1Y_t &= \text{New Coupon Rate}_{t-1} \times \text{New Par Value}_{t-1} \\
 \text{Coupon}9M_t &= \text{Coupon}1Y_{t-1} \\
 \text{Coupon}6M_t &= \text{Coupon}9M_{t-1} \\
 \text{Coupon}3M_t &= \text{Coupon}6M_{t-1}
 \end{aligned}$$

Although the first equation involves "new" rates and values, these are determined in the prior period ($t - 1$), making all coupon payments in period t deterministic.

Interest Expense The interest expense for each cohort is calculated using the effective interest rate method.

$$\begin{aligned}
 IE1Y_t &= EIR1Y_t \times AC1Y_{t-1} \\
 IE9M_t &= EIR9M_t \times AC9M_{t-1} \\
 IE6M_t &= EIR6M_t \times AC6M_{t-1} \\
 IE3M_t &= EIR3M_t \times AC3M_{t-1}
 \end{aligned}$$

Effective Interest Rate (EIR) Dynamics The EIR for each bond cohort is fixed at issuance and travels with the cohort as it matures.

$$\begin{aligned}\text{EIR1Y}_t &= \text{New EIR}_{t-1} \\ \text{EIR9M}_t &= \text{EIR1Y}_{t-1} \\ \text{EIR6M}_t &= \text{EIR9M}_{t-1} \\ \text{EIR3M}_t &= \text{EIR6M}_{t-1}\end{aligned}$$

Similar to coupon payments, all EIRs for existing debt in period t are known from prior periods, making these dynamics deterministic.

16.8.1 Stochastic Processes for New Issuances

New Effective Interest Rate (EIR) The EIR for new issuances, New EIR_t , is a key stochastic driver. As an interest rate, it is naturally modeled as an autoregressive process. We assume it follows a standard AR(1) process:

$$\text{New EIR}_t = c_{eir} + \phi_{eir}\text{New EIR}_{t-1} + w_{eir,t}, \quad w_{eir,t} \sim \mathcal{N}(0, Q_{eir}) \quad (102)$$

New Coupon Rate To simplify the model, we assume that the coupon rate for new issuances is set to be equal to the prevailing effective interest rate for that issuance. This is a common simplification reflecting market practice for issuing debt at par.

$$\text{New Coupon Rate}_t = \text{New EIR}_t \quad (103)$$

A direct consequence of this assumption is that the amortised cost of a newly issued liability will be equal to its par value, as there are no premiums or discounts at inception. We can test this assumption in subsequent analyses.

New Par Value and Issuance Proceeds The amount of new debt issued, represented by New Par Value_t , is a strategic decision for the company. The cash received from this issuance, AC1Y_t , is a one-to-one mapping of this par value.

The stochastic process driving the amount of new debt issuance involves complex factors related to the company's financing needs, investment opportunities, and capital structure policy. The detailed modeling of this decision process will be discussed in a subsequent subsection.

16.9 Cash Flow Components

This section details the modeling of several key cash flow items that are components of the Cash Flow from Operations (CFO). Specifically, it covers cash flows related to salaries, employee benefits, advance payments from customers (contract liabilities), payments to suppliers (trade payables), and prepayments for future expenses.

Cash Paid for Salaries For simplicity, we assume the cash paid for salaries in a period is exactly equal to the 'Salary Expense' recognised in the P&L. This cash outflow reduces the 'Accrued Salaries' liability on the balance sheet.

$$\text{Cash Paid for Salaries}_t = \text{Salary Expense}_t \quad (104)$$

Cash Paid for DC Plan Similarly, the cash paid for defined contribution plans is assumed to be equal to the 'DC Plan Expense' from the P&L. This cash outflow affects the 'Employee Benefits Payable' liability on the balance sheet.

$$\text{Cash Paid for DC Plan}_t = \text{DC Plan Expense}_t \quad (105)$$

Cash from Advance Payments (Contract Liability) This represents cash received from customers before the related revenue is recognised, which increases the ‘Contract Liability’ balance. We assume the amount of cash received from such advance payments follows a Log-Normal process. This is modeled by assuming its logarithm follows a linear AR(1) process:

$$\log(\text{Cash from Advance Payments}_t) = c_{adv} + \phi_{adv} \log(\text{Cash from Advance Payments}_{t-1}) + w_{adv,t} \quad (106)$$

Cash Paid to Suppliers (Trade Payables) This cash outflow reduces the ‘Trade Payables’ balance. We model it as a ratio of the opening balance of trade payables for the period. The settlement ratio, $r_{settle,t}$, is defined as:

$$r_{settle,t} = \frac{\text{Cash Paid to Suppliers}_t}{\text{Trade Payables}_{t-1}} \quad (107)$$

Since this ratio represents the fraction of outstanding payables that are settled, it is bounded in $(0, 1)$. Its modeling follows the methodology in Appendix C, using a logit transformation and a linear Gaussian AR(1) process.

Cash Paid for Prepayments This cash outflow represents payments made for goods or services that will be consumed in the future, increasing the ‘Prepayments’ asset on the balance sheet. We assume the amount of ‘Cash Paid for Prepayments’ follows a Log-Normal process, modeled by specifying a linear AR(1) process for its logarithm:

$$\log(\text{Cash Paid for Prepayments}_t) = c_{pre} + \phi_{pre} \log(\text{Cash Paid for Prepayments}_{t-1}) + w_{pre,t} \quad (108)$$

Cash Spent on Services For simplicity, we assume that services are paid for in the same period the expense is recognised. Therefore, the cash outflow is equal to the ‘Cost on Services’ from the P&L.

$$\text{Cash Spent on Services}_t = \text{Cost on Services}_t \quad (109)$$

Tax Paid Similarly, we assume taxes are paid in the period they are incurred. The cash outflow for taxes is equal to the ‘Tax Expense’ from the P&L.

$$\text{Tax Paid}_t = \text{Tax Expense}_t \quad (110)$$

Cash Collection Note that cash collections from customers, which reduce trade receivables, are discussed in the relevant section for that asset.

16.10 Cash Management, Debt Issuance, and Dividends

This section discusses the decision-making process for new debt issuance and dividend payments. It presents a simplified model to determine the company’s financing and payout actions at time t , based on its cash position. A key assumption is that all external financing is raised through debt, not by issuing new shares.

Step 1: Intermediate Cash Calculation The first step is to compute an intermediate cash variable, **cash remained**, which represents the cash available before any new financing or dividend payout decisions.

$$\begin{aligned} \text{cash remained}_t = & \text{cash}_{t-1} + \text{CFO}_t + \text{CFI}_t \\ & - \text{total coupon payment}_t - \text{principal paid}_t \end{aligned} \quad (111)$$

Step 2: Decision Logic The company's actions are governed by its cash adequacy, which is determined by comparing the **cash remained** to a pre-defined minimum cash requirement, denoted cash (where cash > 0). The logic is as follows:

- If **cash remained** is sufficient (i.e., above the minimum requirement), the company may pay dividends.
- If **cash remained** is insufficient (i.e., below the minimum requirement), the company will issue new debt to cover the shortfall.
- The model assumes that the company will not simultaneously pay dividends and issue new debt in the same period.

Step 3: Formalizing the Logic The decision rule is implemented through the following conditional formulas:

- **Dividend Payments:** When $\text{cash remained}_t > \text{cash}$, the company pays a dividend.

$$\text{Dividend}_t = \text{dividend ratio}_t \times (\text{cash remained}_t - \text{cash}) \quad (112)$$

Here, the **dividend ratio**_t is a stochastic variable bounded in (0, 1). Its dynamic behavior is modeled following the methodology in Appendix C (i.e., using a logit transformation and an AR(1) process). This ensures that the company retains a portion of its surplus cash as a buffer, rather than paying it all out.

- **New Debt Issuance:** When $\text{cash remained}_t < \text{cash}$, the company issues new debt.

$$\text{New Issuance}_t = \text{new issuance ratio}_t \times (\text{cash} - \text{cash remained}_t) \quad (113)$$

The **new issuance ratio**_t is defined as $1 + r_t$, where r_t is another stochastic ratio bounded in (0, 1) and modeled as per Appendix C. This construction ensures that the issuance ratio is always greater than 1, so the company raises enough cash not only to cover the deficit but also to establish an additional cash buffer.

A key simplification arises from our earlier assumption that the coupon rate for new debt equals the EIR; under this condition, the cash proceeds from the new issuance are exactly equal to the new par value issued, requiring no present value calculation.

$$\text{New Issuance Cash Received}_t = \text{New Par Value}_t$$

17 Classification of Drivers

In the model specification, we distinguish between three types of variables. Their definitions and roles are summarized as follows:

- **Flow Variables** ($F_{i,t}$): These correspond to items in the cash flow statement and income statement. They satisfy accounting identities and are calculated deterministically.
- **Stock Variables** ($S_{i,t}$): These correspond to items on the balance sheet. Like flow variables, they are deterministic consequences of accounting rules.
- **Drivers** ($D_{i,t}$): These are the model primitives that are not typically found on the financial statements. They are the only variables subject to stochastic processes.

To formalize the simulation logic, we further classify the drivers ($D_{i,t}$) into two distinct categories based on their temporal properties:

- **State Variables** ($x_{i,t}$): These are **backward-looking** variables. Their values are predetermined at time t .
- **Control Variables** ($c_{i,t}$): These are **forward-looking** variables. They represent the active decisions made by the firm at time t .

17.1 Backward-Looking Variables (State Variables)

Backward-looking variables, interchangeably referred to as predetermined variables or **state variables**, represent quantities whose values are already fixed at the beginning of period t . The firm cannot modify these values instantaneously at time t ; they are inherited from history. We denote the vector of state variables as $x_{i,t}$. Within this category, we distinguish between **uncontrolled** and **controlled** state variables.

17.1.1 Uncontrolled State Variables

Uncontrolled state variables follow exogenous stochastic processes. The firm has no influence on their current values nor their future evolution. These variables capture external shocks and market conditions.

A primary example is the vector of macro variables M_t . It evolves according to an independent stochastic process:

$$M_t = \Phi_M M_{t-1} + \eta_t \quad (114)$$

Similarly, certain firm-specific drivers evolve stochastically but are exogenous to the firm's immediate decision-making logic (though they may be sensitive to macro conditions). For instance, consider the product demand (quantity sold) $Q_{i,t}$. Its evolution might be modeled as:

$$\log Q_{i,t} = \rho_Q \log Q_{i,t-1} + \tau \log P_{i,t-1} + \beta_Q M_{t-1} + \epsilon_{i,t}^Q \quad (115)$$

where $P_{i,t}$ is the price, M_t represents the macro influence, and $\epsilon_{i,t}^Q$ is an idiosyncratic shock.

17.1.2 Controlled State Variables

Controlled state variables are also fixed at time t (predetermined), but their future values ($t+1$) are determined by the firm's actions at time t . Thus, the evolution of a controlled state variable is deterministic conditional on the firm's decisions.

A canonical example from DSGE modeling is physical capital $K_{i,t}$. The capital stock at t is fixed, but the capital at $t+1$ is determined by the investment decision $I_{i,t}$ made at t :

$$K_{i,t+1} = (1 - \delta)K_{i,t} + I_{i,t} \quad (116)$$

where δ is the depreciation rate.

In the context of our accounting simulator, a simplified example involving debt structure illustrates this logic. Assuming a debt maturity of one period, the outstanding debt to be repaid at $t+1$ is entirely determined by the new debt issuance at t :

$$\text{Debt}_{i,t+1} = \text{New Debt Issuance}_{i,t} \quad (117)$$

17.2 Forward-Looking Variables (Control Variables)

The second major class of drivers consists of **forward-looking variables**, specifically **control variables**. These represent the active decisions made by the firm's management at time t . It is crucial to distinguish these from controlled state variables; while the state variable accumulates the history, the control variable represents the flow of new decisions.

Examples of control variables in our framework include Capital Expenditure (CAPEX), dividend payments, and new debt issuance.

A key characteristic of control variables in this framework is that they can be expressed as a deterministic function (a policy function) of the current state variables. Let $c_{i,t}$ denote the vector of control variables at time t . We posit that the firm's decision rule takes the form:

$$c_{i,t} = \Psi(x_{i,t}, M_t) \quad (118)$$

where $x_{i,t}$ represents the firm-specific states and M_t represents the macro states. Since $x_{i,t}$ and M_t are predetermined at time t , the decision $c_{i,t}$ is automatically determined once the state is known. This implies that we do not need to solve for future expectations endogenously within the simulation step; the optimal policy is encoded in the function $\Psi(\cdot)$.

Consequently, the firm's decision-making process is fundamentally deterministic conditional on the state. However, to capture heterogeneity in implementation or unmodeled managerial idiosyncrasies, we can introduce stochasticity into the realized control variable $\tilde{c}_{i,t}$:

$$\tilde{c}_{i,t} = c_{i,t} + \epsilon_{i,t}^c \quad (119)$$

where $\epsilon_{i,t}^c$ represents the implementation error or discretionary deviation from the standard policy rule.

17.3 Notation Corresponding to Markov Decision Processes (MDP)

The framework and notations introduced above is borrowed from DSGE models in economics, which shares the same spirit with the Markov Decision Process (MDP) framework used in Reinforcement Learning. Both paradigms describe an **Agent** (the firm) operating within a specific **Environment** (macroeconomic and microeconomic conditions). The agent observes a **State**, executes an **Action** (Control), and thereby influences the future evolution of the system. This section reformulates our model using standard MDP notation.

The foundational assumption of our modeling approach is the **Markov Property**.

17.3.1 State, Action, and Policy

We define the core components of the process as follows. In our context, both States and Actions correspond to specific types of **drivers**:

Notation	Definition	Examples in Simulator
$S_t \in \mathcal{S}$	State (Backward-looking)	- Macroeconomic variables (M_t) - $\log(\text{Price})$, $\log(\text{Quantity})$
$A_t \in \mathcal{A}(S_t)$	Action (Forward-looking)	- Debt Issuance - Capital Expenditure (CAPEX)

Table 66: Mapping of MDP Components to Model Drivers

The behavior of the firm is defined by a policy $\pi_t(a|s)$, which represents the probability that $A_t = a$ given $S_t = s$. In our baseline simulation, we assume the firm follows a deterministic decision rule. Consequently, the policy is degenerate:

$$\pi_t(a|s) = \begin{cases} 1 & \text{if } a = \Psi(s) \\ 0 & \text{otherwise} \end{cases} \quad (120)$$

where $\Psi(\cdot)$ is the deterministic function mapping states to decisions.

17.3.2 State Transitions

One time step later, partly as a consequence of its action $A_t = a$, the agent transitions to a new state S_{t+1} .

The dynamics are described by the transition probability $p(s'|s, a)$. We consider two cases:

Case 1: Exogenous Transitions (Uncontrolled State). The state evolution is independent of the firm's actions. Mathematically:

$$p(s'|s, a) = p(s'|s) \quad (121)$$

An autoregressive process satisfies this condition. For example, a macro variable M_t follows:

$$M_{t+1} = \Phi M_t + \epsilon_{t+1} \quad (122)$$

Case 2: Deterministic Transitions (Controlled State). The next state is a deterministic consequence of the current state and action. The probability mass is concentrated on a single outcome:

$$p(s'|s, a) = \begin{cases} 1 & \text{if } s' = f(s, a) \\ 0 & \text{otherwise} \end{cases} \quad (123)$$

For example, the dynamics of the capital stock is deterministic:

$$K_{t+1} = (1 - \delta)K_t + I_t \quad (124)$$

17.3.3 Classification of Drivers within the MDP Framework

Given the specifications above, we categorize every driver in our model by evaluating the following three dimensions:

1. **Stochastic or Deterministic?** Does the variable's evolution contain a random shock?
2. **Determined by Current or Past Information?** Is the value fixed by information at $t - 1$ (predetermined) or t ?
3. **Controlled by Firm?**

Based on these criteria, there are 8 categories in total but we only utilize three distinct categories of drivers in the simulator:

1. Uncontrolled State Variables (e.g., M_t , $\log P_t$)

- *Stochastic* (Subject to shocks ϵ).
- Determined by *Past* ($t - 1$) information.
- *Not Controlled* by the firm.

2. Controlled State Variables (e.g., Debt Stock)

- *Deterministic* (Given history and past actions).
- Determined by *Past* ($t - 1$) information.
- *Controlled* (Indirectly via past actions).

3. Action (e.g., New Debt, CAPEX)

- *Stochastic* (Result of $\pi_t(a|s)$).
- Determined by *Current* (t) information.
- *Controlled* (Directly by the firm).

18 Simulator Test Plan

To ensure the integrity and validity of the simulation model built upon the described framework, a comprehensive test plan must be implemented. The following key areas must be verified:

1. Reconciliation of Cash Flow from Operations (CFO) The simulator must compute CFO using both the direct method (cash receipts from customers minus cash payments to suppliers, employees, etc.) and the indirect method (starting with Net Income and adjusting for non-cash items and changes in working capital). A core test is to ensure that both methods yield identical results for every simulation period.

2. Double-Entry Principle for Flow Variables Every flow variable generated by the model (e.g., Interest Income, ECL Expense, Salary Expense) must be shown to correctly link the income statement or cash flow statement to the corresponding change in a balance sheet account. For example, Interest Income must be reflected in both the P&L and the change in the Carrying Amount of the respective financial asset.

3. Validation of Debt at Par Assumption The amortised cost of each financial liability cohort must be tested to ensure its carrying amount converges to its face value at maturity, consistent with the debt at par assumption. This involves verifying that the EIR method is correctly implemented and that the carrying amount evolves as expected over time, particularly for cohorts with varying coupon rates and issuance dates.

4. Stationarity of Stochastic Processes All latent stochastic processes, particularly the AR(1) processes for log-ratios, log-values, and interest rates, should be tested for stationarity (e.g., by ensuring $|\phi| < 1$). This is crucial to prevent the model from producing explosive or non-sensical long-term forecasts.

5. Balance Sheet Equation Integrity The most fundamental accounting check must be performed for every period in every simulation run: the balance sheet must balance.

$$\text{Total Assets}_t = \text{Total Liabilities}_t + \text{Total Equity}_t \quad (125)$$

6. Adherence to Variable Constraints and Reasonableness The simulator's output must be monitored to ensure all variables remain within their logical and defined ranges. Key constraints to verify include:

- **Non-negativity:** Key variables such as expenses, assets, liabilities, and equity components must remain greater than or equal to zero.
- **Ratio Constraints:** Ratios defined to be within $(0, 1)$, such as the 'dividend ratio' or ' r_{ECL} ', must not breach these bounds.
- **Depreciation and Amortisation:** The total accumulated depreciation or amortisation for an asset must not exceed its original carrying amount.
- **Debt Repayment:** The principal paid on a maturing debt cohort must not exceed the carrying amount of that debt.
- **Dividend Payout Constraints:** Dividends paid must satisfy two conditions: they cannot exceed the available surplus cash ($\text{cash remained}_t - \text{cash}$), and they cannot exceed the sum of the previous period's retained earnings plus the current period's net income.

- **Sensible Parameters and results:** Define a set of sensible values for each parameters for testing. We need to test two things: what values of parameters can be regarded as 'sensible' and what values of results are 'sensible'.

7. Variable-Equations consistency We need to make sure the number of variables is consistent with the number of equations. For example, if we have 10 flow variables, we should have 10 corresponding equations that define them. This ensures that the system is neither underdetermined nor overdetermined.

8. No circularity The model must be tested to ensure that there are no circular dependencies among variables. For instance, if variable A depends on variable B, and variable B depends on variable A, this would create a circular reference that could lead to computational issues. The test should verify that the dependency graph of variables is acyclic. In application, we can use topological sorting to check for circularity. If the topological sort fails, it indicates the presence of a cycle in the dependency graph. If not, then we have a Direct Acyclic Graph (DAG) which is suitable for our model.

9. Single Graph We need to ensure that all variables are connected in a single graph. This means that there should be a path between any two variables in the model. If there are disconnected subgraphs, it may indicate that some variables are not properly integrated into the model, which could lead to incomplete or inaccurate simulations.

19 Clarifications on relationship between stock, flow and drivers

This section addresses specific modeling nuances regarding variable classification and calculation logic.

19.1 Flow Calculations

The following equations define how flow variables are derived from historical state variables.

- **Left-Hand Side:** Flow variables recorded on financial statements.
- **Right-Hand Side:** Deterministic drivers derived from subsidiary ledgers (memory).

$$\text{Depreciation}_t = \sum_i \delta_{t-i} \cdot \text{Gross PPE}_{t-i} \quad (126)$$

$$\text{Principal Payment}_t = \sum_{i \in \Omega_t} \text{Face Value of Debt Issued}_{t-i} \quad (127)$$

$$\text{Coupon Paid}_t = \sum_i \text{Coupon Rate}_{t-i} \cdot \text{Face Value of Debt Issued}_{t-i} \quad (128)$$

$$\text{Interest Expense}_t = \sum_i \text{EIR}_{t-i} \cdot \text{Carrying Amount of Debt Issued}_{t-i} \quad (129)$$

$$\text{COGS}_t = \sum_j \text{Unit Cost}_j \cdot \text{Quantity Sold}_j \quad (130)$$

where Ω_t represents the set of debt vintages that reach maturity at time t , and j represents specific inventory batches based on FIFO.

19.2 Variable Classification Nuances

Let me clarify two trivial but formally distinct classification issues:

- **Stocks that influence Drivers:** One questions we discussed today is stock variable on balance sheet can affect driver. My answer is yes if the driver does not enter VAR(1) or SSM. For example, certain balance sheet items act as deterministic inputs (drivers) for calculating current flows.

$$\text{Driver for Interest Expense}_t = \text{Carrying Amount of Debt Issued}_{t-i}$$

In fact, in this case, Carrying Amount of Debt Issued_{t-i} simultaneously belongs to both stock and driver. If written in code:

$$\text{driver.carrying_amount_debt}_{t-i} = \text{stock.carrying_amount_debt}_{t-i} \quad (131)$$

- **Decisions to be Flows simultaneously:** Variables like *New Debt Issuance* and *CAPEX* function simultaneously as control drivers (decisions) and financial statement flows.

The algorithmic relationship is expressed as:

$$\begin{aligned} \text{driver.new_debt_issuance}_t &= f(\text{state}_t, \text{macro}_t) \\ \text{flow.new_debt_issuance}_t &= \text{driver.new_debt_issuance}_t \end{aligned} \quad (132)$$

For higher granularity (aggregating by maturity):

$$\text{flow.new_debt}_t = \sum_m \text{driver.new_debt_maturity}_m \quad (133)$$

20 Specification for Reduced-Form Macro Variable Generator

20.1 Overview

This document outlines the mathematical specification for a reduced-form macro variable generator. The objective of this model is to simulate the dynamics of three core macroeconomic variables:

1. **GDP Growth** (ΔGDP_t)
2. **Inflation** (π_t)
3. **Nominal Risk-Free Rate** (i_t^f)

The model departs from complex DSGE structures by adopting a reduced-form state space approach. The core dynamics are driven by two distinct structural shocks—Supply Shocks and Demand Shocks—which propagate through stochastic cycles to generate the observed macroeconomic path.

20.2 Variable Definitions

We define the vector of observable variables y_t at time t as:

$$y_t = \begin{pmatrix} \Delta GDP_t \\ \pi_t \end{pmatrix}$$

Note that the nominal interest rate i_t^f is derived as a policy reaction function to these variables and is calculated recursively.

20.3 Model Construction

20.3.1 Structural Shocks

The economic fluctuations are driven by two orthogonal structural shocks:

- s_t : **Supply Shock**
- d_t : **Demand Shock**

These shocks are modelled as first-order autoregressive (AR(1)) processes to capture persistence:

$$\begin{aligned} s_{t+1} &= \phi_s s_t + \eta_t^{(s)}, & \eta_t^{(s)} &\sim \mathcal{N}(0, \sigma_s^2) \\ d_{t+1} &= \phi_d d_t + \eta_t^{(d)}, & \eta_t^{(d)} &\sim \mathcal{N}(0, \sigma_d^2) \end{aligned}$$

where $0 < \phi_s, \phi_d < 1$ are persistence parameters.

20.3.2 Cycle Dynamics

The cyclical components of GDP and Inflation, denoted as $c_t^{(g)}$ and $c_t^{(\pi)}$, follow a stochastic harmonic process similar to the formulation in *Linear State Space Models* (Section 3.2.4).

To simplify the framework and isolate the impact of structural shocks, we set the auxiliary noise variance to zero ($\omega^* = 0$). The cycles are driven by a linear combination of the structural shocks ω_t .

The general form for the cycle transition is:

$$\begin{pmatrix} c_{t+1} \\ c_{t+1}^* \end{pmatrix} = \rho \begin{pmatrix} \cos \lambda & \sin \lambda \\ -\sin \lambda & \cos \lambda \end{pmatrix} \begin{pmatrix} c_t \\ c_t^* \end{pmatrix} + \begin{pmatrix} \omega_t \\ 0 \end{pmatrix}$$

Impact Logic We define the driving force ω_t based on economic theory:

- **Supply Shock (s_t)**: Increases GDP, Decreases Inflation.
- **Demand Shock (d_t)**: Increases GDP, Increases Inflation.

Thus, the specific drivers for each variable are:

$$\begin{aligned} \omega_t^{(g)} &= s_t + d_t \\ \omega_t^{(\pi)} &= d_t - s_t \end{aligned}$$

20.3.3 Observation Equation

We assume the observations are composed of a time-invariant trend (mean) and the cyclical component. The measurement error is assumed to be negligible ($\sigma_\varepsilon \rightarrow 0$).

$$\begin{pmatrix} \Delta GDP_t \\ \pi_t \end{pmatrix} = \begin{pmatrix} \bar{g} \\ \bar{\pi} \end{pmatrix} + \begin{pmatrix} c_t^{(g)} \\ c_t^{(\pi)} \end{pmatrix}$$

where:

- $\bar{g}$: Long-term potential GDP growth rate.
- $\bar{\pi}$: Long-term inflation target.

20.3.4 Nominal Interest Rate Rule

The nominal risk-free rate i_t^f follows a modified Taylor Rule. Instead of a fixed equilibrium real rate intercept, we employ an interest rate smoothing term $\rho_r i_{t-1}^f$.

$$i_t^f = \rho_r i_{t-1}^f + \beta_\pi(\pi_t - \bar{\pi}) + \beta_g(\Delta GDP_t - \bar{g})$$

Substituting the definitions from the observation equation (where $\pi_t - \bar{\pi} = c_t^{(\pi)}$ and $\Delta GDP_t - \bar{g} = c_t^{(g)}$), the rule simplifies to a function of the previous rate and current states:

$$i_t^f = \rho_r i_{t-1}^f + \beta_\pi c_t^{(\pi)} + \beta_g c_t^{(g)}$$

20.4 State Space Representation

The system can be written in the standard state space form:

$$\begin{aligned} y_t &= \mu + Z\alpha_t \\ \alpha_{t+1} &= T\alpha_t + R\eta_t \end{aligned}$$

20.4.1 State Vector

The state vector α_t contains the cycle components, their auxiliary variables, and the structural shocks:

$$\alpha_t = \left(c_t^{(g)} \quad c_t^{*(g)} \quad c_t^{(\pi)} \quad c_t^{*(\pi)} \quad s_t \quad d_t \right)^\top$$

20.4.2 System Matrices

1. Mean Vector (μ) and Observation Matrix (Z):

$$\mu = \begin{pmatrix} \bar{g} \\ \bar{\pi} \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

2. Transition Matrix (T): The matrix T captures the rotation of the cycles and the injection of shocks s_t and d_t into the next period's cycle state.

Let $C_i = \cos(\lambda_i)$ and $S_i = \sin(\lambda_i)$.

$$T = \begin{pmatrix} \rho_g C_g & \rho_g S_g & 0 & 0 & 1 & 1 \\ -\rho_g S_g & \rho_g C_g & 0 & 0 & 0 & 0 \\ 0 & 0 & \rho_\pi C_\pi & \rho_\pi S_\pi & -1 & 1 \\ 0 & 0 & -\rho_\pi S_\pi & \rho_\pi C_\pi & 0 & 0 \\ 0 & 0 & 0 & 0 & \phi_s & 0 \\ 0 & 0 & 0 & 0 & 0 & \phi_d \end{pmatrix}$$

Explanation of Rows 1 and 3:

- Row 1 corresponds to $c_{t+1}^{(g)}$. The coefficients 1 and 1 in the last two columns represent $\omega_t^{(g)} = s_t + d_t$.
- Row 3 corresponds to $c_{t+1}^{(\pi)}$. The coefficients -1 and 1 in the last two columns represent $\omega_t^{(\pi)} = -s_t + d_t$.

3. Disturbance Matrix (R): Only the structural shock processes receive new random innovations.

$$R = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \eta_t = \begin{pmatrix} \eta_t^{(s)} \\ \eta_t^{(d)} \end{pmatrix}$$

20.5 Direct Implementation Algorithm

For practical coding purposes, it is often more intuitive to implement the generator iteratively without explicitly constructing the sparse state space matrices. The following steps outline the sequential logic for generating time series data from $t = 0$ to $t = T$.

20.5.1 Step 0: Initialization ($t = 0$)

Initialize the structural shocks s_0 and d_0 by sampling from their stationary distributions. For an AR(1) process $x_t = \phi x_{t-1} + \eta_t$ where $\eta_t \sim \mathcal{N}(0, \sigma^2)$, the stationary variance is $\frac{\sigma^2}{1-\phi^2}$.

$$s_0 \sim \mathcal{N}\left(0, \frac{\sigma_s^2}{1-\phi_s^2}\right)$$

$$d_0 \sim \mathcal{N}\left(0, \frac{\sigma_d^2}{1-\phi_d^2}\right)$$

Initialize the cycle components and auxiliary variables (often set to 0 or sampled similarly if steady-state distributions are derived):

$$c_0^{(g)} = 0, \quad c_0^{*(g)} = 0, \quad c_0^{(\pi)} = 0, \quad c_0^{*(\pi)} = 0$$

Initialize the interest rate (assuming equilibrium):

$$i_0^f = \bar{r} \quad (\text{or a specified initial value})$$

20.5.2 Step 1: Generate Structural Shocks

For each time step $t \geq 0$, generate the structural shocks for the *next* period ($t + 1$) using the AR(1) process. Note that these values represent the state of supply and demand drivers that will impact the cycle transition.

$$\eta_{t+1}^{(s)} \sim \mathcal{N}(0, \sigma_s^2)$$

$$\eta_{t+1}^{(d)} \sim \mathcal{N}(0, \sigma_d^2)$$

$$s_{t+1} = \phi_s s_t + \eta_{t+1}^{(s)}$$

$$d_{t+1} = \phi_d d_t + \eta_{t+1}^{(d)}$$

20.5.3 Step 2: Calculate Cycle Impulses

Compute the combined impulses (ω_t) that will drive the GDP and Inflation cycles. These impulses are derived from the current state of structural shocks.

$$\omega_t^{(g)} = s_t + d_t$$

$$\omega_t^{(\pi)} = d_t - s_t$$

20.5.4 Step 3: Update Cycle States

Update the cycle components for $t + 1$ using the rotation dynamics and the impulses calculated in Step 2.

For GDP Growth:

$$\begin{aligned} c_{t+1}^{(g)} &= \rho_g \left(c_t^{(g)} \cos \lambda_g + c_t^{*(g)} \sin \lambda_g \right) + \omega_t^{(g)} \\ c_{t+1}^{*(g)} &= \rho_g \left(-c_t^{(g)} \sin \lambda_g + c_t^{*(g)} \cos \lambda_g \right) \end{aligned}$$

For Inflation:

$$\begin{aligned} c_{t+1}^{(\pi)} &= \rho_\pi \left(c_t^{(\pi)} \cos \lambda_\pi + c_t^{*(\pi)} \sin \lambda_\pi \right) + \omega_t^{(\pi)} \\ c_{t+1}^{*(\pi)} &= \rho_\pi \left(-c_t^{(\pi)} \sin \lambda_\pi + c_t^{*(\pi)} \cos \lambda_\pi \right) \end{aligned}$$

20.5.5 Step 4: Generate Observable Variables

Calculate the observed macro variables at $t + 1$ by adding the long-term means to the cycle components.

$$\begin{aligned} \Delta GDP_{t+1} &= \bar{g} + c_{t+1}^{(g)} \\ \pi_{t+1} &= \bar{\pi} + c_{t+1}^{(\pi)} \end{aligned}$$

20.5.6 Step 5: Determine Nominal Interest Rate

Calculate the nominal risk-free rate for $t + 1$ using the smoothed Taylor Rule.

$$i_{t+1}^f = \rho_r i_t^f + (1 - \rho_r) [\bar{r} + \beta_\pi (\pi_{t+1} - \bar{\pi}) + \beta_g (\Delta GDP_{t+1} - \bar{g})]$$

Note: The process repeats from Step 1 for the next time increment.

20.6 Parameter Summary

To fully instantiate the model, the following parameters must be defined prior to simulation:

20.6.1 Cycle Parameters

- ρ_g, ρ_π : Damping factors ($0 < \rho < 1$, typically near 1).
- λ_g, λ_π : Cycle frequencies (in radians, determined by $2\pi/\text{Period}$).

20.6.2 Shock Process Parameters

- ϕ_s, ϕ_d : Autoregressive persistence coefficients for supply and demand shocks.
- σ_s, σ_d : Standard deviations of the shock innovations ($\eta^{(s)}, \eta^{(d)}$).

20.6.3 Structural and Policy Parameters

- $\bar{g}$: Long-term potential GDP growth rate (mean of ΔGDP_t).
- $\bar{\pi}$: Long-term inflation target (mean of π_t).
- $\bar{r}$: Initial equilibrium nominal interest rate (used for i_0^f).
- ρ_r : Interest rate smoothing parameter (policy inertia).
- β_π, β_g : Central bank policy sensitivities to inflation and growth gaps.

20.7 Deterministic Cycles: Setting Initial Conditions When Innovations Are Zero

When the user sets $\sigma_s = \sigma_d = 0$, all structural shock innovations disappear. In this case, the shocks follow deterministic decay paths:

$$\begin{aligned} s_{t+1} &= \phi_s s_t \\ d_{t+1} &= \phi_d d_t \end{aligned}$$

Without innovations, if the initial conditions $s_0 = d_0 = 0$ and all cycle components are initialized at zero, the system will remain flat at the long-term means $(\bar{g}, \bar{\pi}, \bar{r})$ for all time periods. To generate smooth cyclical fluctuations instead of a flat line, one must manually set non-zero initial conditions for the cycle components.

20.7.1 Initial Condition for Cycle Level (c_0)

The initial cycle level $c_0^{(g)}$ or $c_0^{(\pi)}$ determines the starting position within the cycle:

- **For GDP Growth ($c_0^{(g)}$):**
 - $c_0^{(g)} > 0$: The economy starts in an expansion phase (above potential growth).
 - $c_0^{(g)} < 0$: The economy starts in a recession phase (below potential growth).
 - $c_0^{(g)} = 0$: The economy starts at the trend growth rate.
- **For Inflation ($c_0^{(\pi)}$):**
 - $c_0^{(\pi)} > 0$: Inflation starts above the long-term target.
 - $c_0^{(\pi)} < 0$: Inflation starts below the long-term target.
 - $c_0^{(\pi)} = 0$: Inflation starts at the target level.

20.7.2 Initial Condition for Cycle Momentum (c_0^*)

The auxiliary variable $c_0^{*(g)}$ or $c_0^{*(\pi)}$ represents the momentum or velocity of the cycle:

- $c_0^* > 0$: The cycle is improving (moving upward in the next period). The economy is accelerating toward a peak.
- $c_0^* < 0$: The cycle is deteriorating (moving downward in the next period). The economy is decelerating toward a trough.
- $c_0^* = 0$: The cycle is at a turning point—either a peak (if $c_0 > 0$) or a trough (if $c_0 < 0$).

20.7.3 Example: Setting a Smooth Deterministic Cycle

To generate a smooth sinusoidal cycle for GDP growth starting at an expansion phase with upward momentum:

$$\begin{aligned} c_0^{(g)} &= A \quad (\text{where } A > 0) \\ c_0^{*(g)} &= 0 \quad (\text{starting at a peak}) \end{aligned}$$

Alternatively, to start at the zero-crossing with maximum upward momentum:

$$\begin{aligned} c_0^{(g)} &= 0 \\ c_0^{*(g)} &= A \quad (\text{where } A > 0) \end{aligned}$$

These initial conditions will produce a deterministic, smooth oscillation driven solely by the rotation dynamics of the stochastic cycle model, even in the absence of random shocks.

21 Dynamics of Debt Term Structure

21.1 Continuous-Time Framework

Let $f(t, \tau)$ denote the debt density at calendar time t with remaining maturity τ . The evolution of the debt distribution is governed by the following transport equation with a source term (expressed in differential form):

$$df(t, \tau) = \left(\iota(t, \tau) + \frac{\partial f(t, \tau)}{\partial \tau} \right) dt$$

where:

- $\iota(t, \tau)$ represents the density of new debt issuance at time t with maturity τ .
- The term $\frac{\partial f}{\partial \tau}$ captures the aging effect, representing the flux of debt from longer maturities to shorter maturities.

21.2 Discretization of Maturity Intervals

We discretize the continuous maturity dimension into intervals $[x_{k-1}, x_k]$. Let f_t^k be the average debt density (or normalized total mass) within the k -th interval:

$$f_t^k = \frac{1}{x_k - x_{k-1}} \int_{x_{k-1}}^{x_k} f(t, \tau) d\tau$$

By applying a finite volume method with a first-order upwind scheme, the dynamics of f_t^k are given by:

$$df_t^k = \left(\iota_t^k + \frac{f_t^{k+1} - f_t^k}{x_k - x_{k-1}} \right) dt$$

Here, the term $(f_t^{k+1} - f_t^k)$ represents the net flow of debt mass: inflow from the longer maturity bucket ($k+1$) minus outflow to the shorter maturity bucket (k).

21.3 Unit Intervals and Exact Shift Dynamics

We now assume uniform maturity intervals of unit length ($\Delta x = x_k - x_{k-1} = 1$) and a discrete time step of unit length ($\Delta t = 1$). Substituting these values into the discretized equation yields:

$$f_{t+1}^k - f_t^k = \left(\iota_t^k + \frac{f_t^{k+1} - f_t^k}{1} \right) \cdot 1$$

Simplifying the expression by canceling f_t^k on both sides leads to the following recursive relation:

$$f_{t+1}^k = f_t^{k+1} + \iota_t^k$$

Interpretation and Exactness

This equation describes a **perfect shift** mechanism:

- The outstanding debt with maturity k at time $t+1$ is exactly the debt that had maturity $k+1$ at time t , plus any new issuance.
- **Exactness:** When the time step equals the maturity interval ($\Delta t = \Delta \tau$), the discretization satisfies the Courant-Friedrichs-Lewy (CFL) condition with a Courant number of 1. Consequently, the numerical solution is **exact** and free from numerical diffusion or approximation errors.

22 Discrete Debt Term Structural in Accounting

22.1 Framework and Notation

We define a discrete term structure with maturity intervals ($k \in \{0, 1, 2, 3, 4, 5\}$). The 0Y bucket represents matured debt pending repayment. For each maturity interval k at time t , we identify 6 distinct state variables, resulting in a total of 30 variables to fully characterize the debt profile:

- Carrying Amount $_t^k$
- Face Value $_t^k$
- Coupon Rate $_t^k$
- Effective Interest Rate $_t^k$
- New Issuance Face Value $_t^k$
- New Issuance Carrying Amount $_t^k$

22.2 Visualizing the Shift Mechanism

The transition from time t to $t + 1$ involves a deterministic shift of existing debt to the next shorter maturity bucket, combined with new issuances.

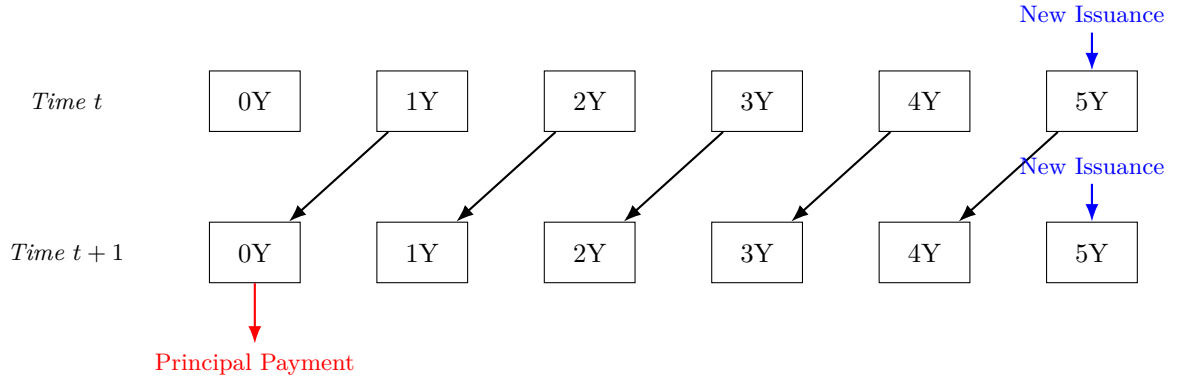


Figure Description: At time $t + 1$, the debt that was in the k -year bucket at time t shifts to the $(k - 1)$ -year bucket. Specifically, the debt in the 1Y bucket at time t shifts to the 0Y bucket at time $t + 1$, where it is subsequently repaid (Principal Payment). The 5Y bucket at time $t + 1$ is populated entirely by new issuance.

22.3 Dynamics of Face Value

Face Value follows the exact shift dynamics, as the principal amount remains constant over time. For a 2-year maturity at $t + 1$:

$$\text{Face Value}_{t+1}^{2Y} = \text{Face Value}_t^{3Y} + \text{New Issuance Face Value}_{t+1}^{2Y}$$

The Face Value at $t + 1$ consists of the Face Value carried over from the 3Y bucket at time t , plus any new issuance of 2Y debt at $t + 1$.

22.4 Dynamics of Carrying Amount

The Carrying Amount evolves according to the amortized cost principle. The balance increases by the effective interest expense and decreases by the coupon payments.

$$\begin{aligned} \text{Carrying Amount}_{t+1}^{2Y} = & \underbrace{\text{Carrying Amount}_t^{3Y}}_{\text{Opening Balance}} \\ & + \underbrace{\text{Carrying Amount}_t^{3Y} \cdot \text{Effective Interest Rate}_t^{3Y}}_{\text{Interest Expense}} \\ & - \underbrace{\text{Face Value}_t^{3Y} \cdot \text{Coupon Rate}_t^{3Y}}_{\text{Coupon Payment}} \\ & + \underbrace{\text{New Issuance Carrying Amount}_{t+1}^{2Y}}_{\text{New Issuance}} \end{aligned}$$

Note that the interest expense and coupon payments are calculated based on the variables from the previous period (Time t , 3Y bucket) before the debt shifts to the 2Y bucket.

22.5 Dynamics of Coupon Rate

The Coupon Rate is an intensive variable and must be updated using a weighted average based on the Face Values of the mixing debt components.

$$\begin{aligned} & \text{Face Value Weighted Coupon Rate}_{t+1}^{2Y} \\ &= \text{Face Value Weighted Coupon Rate}_t^{3Y} \cdot \left(\frac{\text{Face Value}_t^{3Y}}{\text{Face Value}_t^{3Y} + \text{New Issuance Face Value}_{t+1}^{2Y}} \right) \\ &+ \text{New Issuance Coupon Rate}_{t+1}^{2Y} \cdot \left(\frac{\text{New Issuance Face Value}_{t+1}^{2Y}}{\text{Face Value}_t^{3Y} + \text{New Issuance Face Value}_{t+1}^{2Y}} \right) \end{aligned}$$

This formula ensures that the new aggregate coupon rate reflects the proportional contribution of the legacy debt (shifted from 3Y) and the newly issued 2Y debt.

22.6 Dynamics of Effective Interest Rate (EIR)

Determining the new aggregate Effective Interest Rate (EIR) is more complex than other variables. There are two primary approaches: an approximation using weighted averages and an exact method based on the Internal Rate of Return (IRR).

Method 1: Weighted Average Approximation

A common practical approach is to calculate the weighted average of the existing EIR and the new issuance EIR. Unlike the Coupon Rate, the appropriate weight here is the **Carrying Amount**, as interest expense is driven by the carrying value.

$$\begin{aligned} & \text{Carrying Amount Weighted EIR}_{t+1}^{2Y} \\ &\approx \text{Carrying Amount Weighted EIR}_t^{3Y} \cdot \left(\frac{\text{Carrying Amount}_t^{3Y}}{\text{Carrying Amount}_t^{3Y} + \text{New Issuance Carrying Amount}_{t+1}^{2Y}} \right) \\ &+ \text{New Issuance Effective Interest Rate}_{t+1}^{2Y} \cdot \left(\frac{\text{New Issuance Carrying Amount}_{t+1}^{2Y}}{\text{Carrying Amount}_t^{3Y} + \text{New Issuance Carrying Amount}_{t+1}^{2Y}} \right) \end{aligned}$$

Here, New Issuance Effective Interest Rate $_{t+1}^{2Y}$ and New Issuance Coupon Rate $_{t+1}^{2Y}$ are treated as exogenously given market variables. While this method is computationally efficient, it introduces a second-order approximation error because the compounding dynamics of the two debt components differ.

Method 2: Exact Solution via NPV (IRR)

The most accurate method is to solve for the IRR that equates the total current carrying amount with the present value of future cash flows.

First, we determine the total initial Carrying Amount and Face Value at $t + 1$:

$$\text{Total CA}_{t+1}^{2Y} = \text{Carrying Amount}_t^{3Y} + \text{New Issuance Carrying Amount}_{t+1}^{2Y}$$

$$\text{Total FV}_{t+1}^{2Y} = \text{Face Value}_t^{3Y} + \text{New Issuance Face Value}_{t+1}^{2Y}$$

Using the updated aggregate Coupon Rate $_{t+1}^{2Y}$ derived in the previous section, we project the future cash flows for this 2-year debt bucket:

- Year 1 Cash Flow (CF_1): $\text{Total FV}_{t+1}^{2Y} \cdot \text{Coupon Rate}_{t+1}^{2Y}$
- Year 2 Cash Flow (CF_2): $\text{Total FV}_{t+1}^{2Y} \cdot (1 + \text{Coupon Rate}_{t+1}^{2Y})$

The exact Effective Interest Rate $_{t+1}^{2Y}$ is the solution r to the following polynomial equation (setting NPV to zero):

$$\text{Total CA}_{t+1}^{2Y} = \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2}$$

For longer maturities, this requires finding the root of a high-degree polynomial.

22.6.1 Proof of Approximation Error (Jensen's Inequality)

To demonstrate why the weighted average EIR is an approximation, consider a simplified scenario with zero-coupon debt, no new issuance, and no repayment over two periods ($t = 0$ to $t = 2$).

Assume two distinct debt tranches, A and B, with equal initial carrying amounts C but different effective interest rates $r_A \neq r_B$.

Step 1: Initial State ($t = 0$)

$$\text{Total CA}_0 = 2C$$

$$\text{Weighted Average EIR } (\bar{r}) = \frac{C \cdot r_A + C \cdot r_B}{2C} = \frac{r_A + r_B}{2}$$

Step 2: Update to $t = 1$ Under the zero-coupon assumption, the carrying amount grows by $(1 + r)$.

- **Exact Calculation:** $CA_1^{\text{Exact}} = C(1 + r_A) + C(1 + r_B) = 2C + C(r_A + r_B)$
- **Approximation (using $\bar{r}$):** $CA_1^{\text{Approx}} = 2C(1 + \bar{r}) = 2C + C(r_A + r_B)$

At $t = 1$, the results are identical because the growth function is linear in r .

Step 3: Update to $t = 2$ (Compounding Effect)

- **Exact Calculation:**

$$\begin{aligned} CA_2^{\text{Exact}} &= C(1 + r_A)^2 + C(1 + r_B)^2 \\ &= 2C + 2C(r_A + r_B) + C(r_A^2 + r_B^2) \end{aligned}$$

- **Approximation:**

$$\begin{aligned} CA_2^{\text{Approx}} &= 2C(1 + \bar{r})^2 = 2C \left(1 + \frac{r_A + r_B}{2} \right)^2 \\ &= 2C + 2C(r_A + r_B) + \frac{C}{2}(r_A^2 + 2r_A r_B + r_B^2) \end{aligned}$$

Step 4: The Difference Subtracting the approximation from the exact value:

$$\begin{aligned}\text{Diff} &= CA_2^{\text{Exact}} - CA_2^{\text{Approx}} \\ &= C(r_A^2 + r_B^2) - \frac{C}{2}(r_A^2 + 2r_A r_B + r_B^2) \\ &= \frac{C}{2}(r_A^2 - 2r_A r_B + r_B^2) = \frac{C}{2}(r_A - r_B)^2\end{aligned}$$

Since $(r_A - r_B)^2 > 0$, the exact carrying amount is strictly greater than the approximation. This confirms that using a weighted average EIR **underestimates** the final liability due to the convexity of compound interest.

22.7 More Accurate Approximation of Exact EIR

In large-scale debt simulations, solving the NPV polynomial via iterative methods for every maturity bucket can be computationally expensive. We propose more efficient strategies to achieve precision.

Method 3: Trained Functional Mapping

This approach leverages the scale invariance of the bond pricing formula. The fundamental pricing equation for a debt with maturity T is:

$$CA = \sum_{t=1}^T \frac{FV \cdot c}{(1+r)^t} + \frac{FV}{(1+r)^T}$$

Normalizing by the Face Value (FV), we obtain a dimensionless pricing ratio P :

$$P = \frac{CA}{FV} = \sum_{t=1}^T \frac{c}{(1+r)^t} + \frac{1}{(1+r)^T}$$

This normalization demonstrates that for a fixed maturity T , the EIR (r) is solely a function of the price ratio P and the coupon rate c . We can therefore pre-train a high-precision mapping function f_T :

$$r = f_T(P, c)$$

To maintain maximum efficiency and accuracy, we implement the following:

- **Separate Models per Maturity:** Instead of including T as a feature, a dedicated model (e.g., a shallow Neural Network or a multivariate polynomial regression) is trained for each discrete maturity interval $T \in \{1, 2, \dots, 5\}$.
- **Input/Output Definition:**
 - **Input Vector (X):** $\left[\frac{\text{Total Carrying Amount}}{\text{Total Face Value}}, \text{Weighted Coupon Rate}\right]$
 - **Output Scalar (Y):** $[\text{Effective Interest Rate}]$

This mapping converts an implicit iterative problem into an explicit evaluation, significantly reducing the computational burden while preserving high precision.

Method 4: Newton-Raphson Iteration (Duration Adjustment)

The Newton-Raphson method is an efficient alternative for refinement when a good initial guess is available. Given the target aggregate Carrying Amount (Total CA_{target}), we define the objective function $f(r)$ as:

$$f(r) = V(r) - \text{Total } CA_{target} = 0$$

The valuation function $V(r)$ represents the present value of future cash flows, consisting of recurring coupon payments and the final face value repayment:

$$V(r) = \underbrace{\sum_{t=1}^T \frac{\text{Total FV} \cdot \text{Coupon Rate}}{(1+r)^t}}_{\text{Coupon Cash Flows}} + \underbrace{\frac{\text{Total FV}}{(1+r)^T}}_{\text{Face Value Repayment}}$$

The iterative update rule is defined as:

$$r_{n+1} = r_n - \frac{f(r_n)}{f'(r_n)}$$

where the derivative $f'(r)$ is related to the **Dollar Duration** of the debt tranche:

$$f'(r) = V'(r) = - \sum_{t=1}^T \frac{t \cdot (\text{Total FV} \cdot \text{Coupon Rate})}{(1+r)^{t+1}} - \frac{T \cdot \text{Total FV}}{(1+r)^{T+1}}$$

The robustness of this method is guaranteed by the properties of $V(r)$, which is strictly monotonic ($V'(r) < 0$) and strictly convex ($V''(r) > 0$). In practice, using the weighted average EIR as the starting point (r_0), a simple iterative loop in Python (e.g., 1 to 2 iterations) is sufficient to converge to an exact solution for each maturity bucket.

22.8 Deriving New Issuance Face Value from Decisions

If the firm treats the **New Issuance Carrying Amount** (i.e., net proceeds raised) as the primary decision variable, the corresponding **New Issuance Face Value** must be derived endogenously. Given the exogenous market rates (New EIR and New Coupon Rate) and the debt maturity M , the relationship is governed by the standard bond pricing formula.

The **New Issuance Face Value** is calculated as:

$$\text{New Issuance Face Value}_{t+1}^M = f \left(\begin{array}{l} \text{New Issuance Carrying Amount}_{t+1}^M, \\ \text{New Issuance Effective Interest Rate}_{t+1}^M, \\ \text{New Issuance Coupon Rate}_{t+1}^M, \\ \text{Maturity } M \end{array} \right)$$

This equation ensures that the face value recorded on the balance sheet is mathematically consistent with the firm's financing decision and prevailing market conditions.

22.9 Dynamics of the 0Y Bucket (Repayment & Derecognition)

The 0Y bucket represents the state of debt immediately prior to and after repayment at time $t + 1$. This bucket captures the final amortization, repayment, and any gain or loss from derecognition.

State Before Repayment

The variables shift from the 1Y bucket at time t to the 0Y bucket at time $t + 1$. We do not track the Coupon Rate or EIR for this expired debt, nor is there any new issuance.

$$\text{Face Value Before Repayment}_{t+1}^{0Y} = \text{Face Value}_t^{1Y}$$

$$\begin{aligned} \text{Carrying Amount Before Repayment}_{t+1}^{0Y} &= \text{Carrying Amount}_t^{1Y} \\ &\quad + \text{Carrying Amount}_t^{1Y} \cdot \text{Effective Interest Rate}_t^{1Y} \\ &\quad - \text{Face Value}_t^{1Y} \cdot \text{Coupon Rate}_t^{1Y} \end{aligned}$$

Income Statement Impact (Systematic Loss)

The difference between the book value of the debt and the actual repayment amount is recognized as income or expense in the Income Statement.

$$\text{Income} = \text{Carrying Amount Before Repayment}_{t+1}^{0Y} - \text{Face Value Before Repayment}_{t+1}^{0Y}$$

The weighted average approximation systematically underestimates the growth of the Carrying Amount. Consequently, we expect:

$$\text{Carrying Amount Before Repayment}_{t+1}^{0Y} \leq \text{Face Value Before Repayment}_{t+1}^{0Y}$$

This implies that $\text{Income} \leq 0$. Therefore, the adjustment upon derecognition typically manifests as a systematic **Loss** (Expense), reducing Net Income and consequently Retained Earnings.

Final State and Cash Flow

After the repayment event, the debt is fully extinguished.

$$\text{Final Carrying Amount}_{t+1}^{0Y} = 0, \quad \text{Final Face Value}_{t+1}^{0Y} = 0$$

The actual cash outflow is determined strictly by the contractual face value:

$$\text{Cash Flow (Repayment)}_{t+1} = \text{Face Value Before Repayment}_{t+1}^{0Y}$$

Balance Sheet Equilibrium

The balance sheet remains balanced after repayment because the reduction in Retained Earnings (driven by the systematic Loss) perfectly offsets the discrepancy between the reduction in Assets and Liabilities.

- **Assets:** Decrease by the full repayment amount.

$$\Delta \text{Assets} = -\text{Face Value Before Repayment}_{t+1}^{0Y}$$

- **Liabilities:** Decrease by the book value of the debt.

$$\Delta \text{Liabilities} = -\text{Carrying Amount Before Repayment}_{t+1}^{0Y}$$

- **Equity:** Decreases due to the recognized Loss.

$$\Delta \text{Equity} = \text{Carrying Amount Before Repayment}_{t+1}^{0Y} - \text{Face Value Before Repayment}_{t+1}^{0Y}$$

Verification of the accounting identity ($\Delta \text{Assets} = \Delta \text{Liabilities} + \Delta \text{Equity}$):

$$-\text{FV} = -\text{CA} + (\text{CA} - \text{FV}) \implies -\text{FV} = -\text{FV}$$

23 Modeling Balance Sheet Dynamics

In the context of financial statement simulation, balance sheet line items are updated using two primary mechanisms.

23.1 Update Mechanisms

The two methods are:

Method 1: Direct Flow Update The variable is updated recursively via flow variables. A typical example is the Cash account:

$$\text{Cash}_t = \text{Cash}_{t-1} + \text{CFO}_t + \text{CFI}_t + \text{CFF}_t$$

Method 2: Component Calculation The variable is derived from other concurrent state variables rather than being updated directly. Key examples include:

$$\text{Net PPE}_t = \text{Gross PPE}_t - \text{Accumulated Depreciation}_t$$

$$\text{Net Trade Receivables}_t = \text{Gross Trade Receivables}_t - \text{ECL Allowance}_t$$

The underlying components in Method 2 (Gross values and Contra accounts) follow their own stock-flow evolution rules consistent with double-entry principles. Taking Trade Receivables (TR) as the primary example, the internal updates are modeled as:

$$\begin{aligned} \text{Gross TR}_t &= \text{Gross TR}_{t-1} + \underbrace{\text{Credit Sales}_t}_{\text{P\&L}} - \underbrace{\text{Cash Collection}_t}_{\text{SCF}} \\ \text{ECL Allowance}_t &= \text{ECL Allowance}_{t-1} + \underbrace{\text{ECL Expense}_t}_{\text{P\&L}} \end{aligned}$$

23.2 Modeling Objectives

The simulator must satisfy two requirements:

1. **Accounting Identity:** The ability to verify **Assets = Liabilities + Equity**
2. **Initialization:** The ability to start with initial values for lag terms (such as Gross TR_{t-1} and ECL_{t-1}) to support proper recursive updates, either from synthetic data or real life data

23.3 Proposed Architectures

Three approaches are considered to handle component-based variables:

Option 1: Net Asset in Balance Sheet In this approach, **Net Trade Receivables** is the primary state variable recorded on the balance sheet. The **ECL Allowance** is maintained as a contra account, while **Gross Trade Receivables** is treated as an auxiliary category to support flow calculations.

- **Verification:** The standard identity $A = L + E$ is directly verifiable as the net value is stored in the asset vector.
- **Implication:** Requires an account for auxiliary variables like **Gross Trade Receivables** to calculate underlying flows during each iteration.

Option 2: Gross Asset in Balance Sheet In this approach, **Gross Trade Receivables** is classified as the Asset, and **ECL Allowance** is classified as a Contra-Asset. The **Net Trade Receivables** is treated as an **auxiliary variable** generated only for financial reporting views.

- **Benefit:** Ensures robust stock-flow consistency since the state vector tracks variables directly influenced by operational drivers.
- **Verification:** The identity check is adjusted to account for contra-assets:

$$(\text{Assets} - \text{Contra Assets}) = (\text{Liabilities} - \text{Contra Liabs}) + (\text{Equity} - \text{Contra Eq})$$

Option 3: Direct Net Asset Update One can derive a direct update formula for the net asset by combining the underlying stock-flow rules of its components. For instance:

$$\text{Net TR}_t = \text{Net TR}_{t-1} + \text{Credit Sales}_t - \text{Cash Collection}_t - \text{ECL Expense}_t$$

- **Risk:** While mathematically sound, this creates a disconnect between the gross asset modeling logic and the net asset state update.
- **Necessity:** We still need to keep track of gross variables. For example, we need Gross PPE to calculate Depreciation.
- **Graph Separation:** In a computational model, this results in two disjoint graphs—one for gross component logic and another for net balance updates—rather than a single, fully integrated and interconnected graph.

24 Literature Review on Managerial Decision-Making

Although the work of [Fields et al. \(2001\)](#) suggests that empirical research in the 1990s suffered from methodological limitations and yielded somewhat immature results, their synthesis remains a foundational framework for understanding firm behavior.

24.1 Literature Review

24.1.1 The Informativeness and Impact of Accounting Choice

The fundamental question addressed in accounting research is whether accounting choices actually “matter” in an economic sense. In a world of complete and perfect markets, financial disclosures would be redundant, as all information would already be impounded in prices. However, in the presence of market imperfections, such as agency costs, information asymmetries, and externalities, accounting becomes a vital mechanism for contractual efficiency and information dissemination.

Existing research suggests that accounting choices are far from cosmetic. They serve as a primary vehicle through which management intent is reflected. When managers exercise discretion, they are often conveying private information to the market about the firm’s future cash flows, risk profiles, and strategic direction. Thus, the informativeness of accounting output is deeply tied to how these choices are utilized to mitigate or exploit market frictions.

24.1.2 Defining Accounting Choice

[Fields et al. \(2001\)](#) provide a broad definition of accounting choice: it encompasses any decision whose primary purpose is to influence the output of the accounting system, whether in form or substance. This definition extends beyond the simple selection of GAAP-compliant methods

(such as LIFO vs. FIFO) to include the deliberate structuring of transactions (e.g., lease design) and “real” business decisions made to manipulate accounting figures.

The critical criterion for identifying an accounting choice is *managerial intent*. The distinction lies in the impetus behind a decision. For instance, if a firm reduces its Research and Development (R&D) expenditures because it anticipates lower future returns on those investments, the act is a pure “real” business decision. Conversely, if the reduction is primarily driven by a desire to decrease current period expenses and “beautify” the earnings report, it constitutes an accounting choice. This distinction highlights that firm decisions are often a blend of operational necessity and reporting strategy.

24.1.3 The Rationale for Accounting Discretion

A central theme in the literature is the necessity of providing management with discretion, despite the inherent risk of opportunistic behavior. There are several reasons why a rigid, rule-based system without choices is infeasible:

- **Infeasibility of Universal Rules:** It is impossible to devise specific rules for all facts and circumstances. As new economic instruments and complex business models emerge, accounting flexibility allows the system to adapt more quickly than regulators can issue new standards.
- **Information Signaling:** Accounting choices are informative precisely because they require judgment. If managers were forced into a single, rigid method, the private information they could have signaled through their choice would be lost to the market.
- **Mitigation of Costly Real Decisions:** If accounting rules were entirely inflexible, managers might be incentivized to take even more costly “real” actions (such as postponing vital maintenance or capital expenditures) to achieve desired accounting outcomes.

24.1.4 Agency Costs and Executive Compensation

When managerial compensation is tied to current earnings, agency costs often manifest through opportunistic accounting decisions. If bonus plans depend on reported profits, managers are incentivized to adopt income-increasing methods, such as First-In, First-Out (FIFO) inventory valuation or specific depreciation schedules that defer expenses. Conversely, they may choose methods that reduce accounting profit to minimize tax liabilities when compensation ties are weak.

Furthermore, the structure of bonus caps significantly influences behavior; if earnings exceed the maximum threshold for a bonus, managers may utilize accruals to defer profits to future periods. Beyond pure accounting methods, managers may also engage in “real” earnings management by cutting R&D expenditures to boost short-term profits, suggesting that we must account for the systematic under-investment in innovation when incentives are tied to bottom-line performance.

24.1.5 Debt Covenants and External Agency Conflicts

Firms employ accounting choices as a defensive mechanism when dealing with banks and creditors. To protect their interests and prevent financial distress, creditors often embed debt covenants in loan agreements, such as maintaining a debt-to-equity ratio below 60% or an interest coverage ratio above a specific threshold. Violating these covenants grants creditors the right to demand immediate repayment or impose higher interest rates, both of which are extremely costly for the firm.

Consequently, managers are motivated to select accounting methods that inflate reported income or increase net assets to ensure they do not cross these “red lines.”

24.1.6 Asset Pricing and Market Valuation

Since capital markets are not perfectly efficient, accounting choices can have a direct impact on asset prices. Managers often act as if they can influence investor perceptions by manipulating the timing and magnitude of reported earnings. To impact share prices favorably, managers may adopt several strategies: maximizing profits in specific periods (e.g., prior to equity offerings), smoothing earnings over time to reduce perceived volatility, or managing accruals specifically to avoid reporting losses or year-over-year earnings declines.

24.1.7 Taxation and Value Maximization

As value maximizers, firm managers have a strong incentive to select accounting methods that minimize the present value of tax outflows, thereby increasing the firm's net cash flows. Taxation is generally considered a "first-order effect" in the selection of accounting methods because tax savings represent tangible economic benefits. Unlike cosmetic accounting changes, these decisions directly impact cash inflows and outflows, and managers rarely forgo the substantive benefits associated with aggressive tax avoidance.

24.2 Firm Decision-Making and Accounting Choices across Scenarios

Based on the theoretical motivations discussed in the literature, firms typically adopt the following decision-making patterns under specific economic pressures:

- **Scenario 1: Compensation Performance Evaluation**
 - **Business Decisions:** Reduction of discretionary expenses, including R&D expenditures, advertising, and maintenance costs.
 - **Accounting Decisions:** Capitalization of expenses into intangible assets to defer cost recognition and boost current profits.
- **Scenario 2: Debt Covenant Pressures ("Red Lines")**
 - **Business Decisions:** Strategic sale or divestiture of assets to reduce leverage or increase liquidity.
 - **Accounting Decisions:** Aggressive capitalization of expenditures to reduce current-period expenses and premature recognition of revenue to improve solvency ratios.
- **Scenario 3: Taxation and Tax Pressure**
 - **Accounting Decisions:** Recognition of asset impairment or write-downs; deliberate deferral of revenue recognition; and the cessation of expense capitalization to minimize taxable income.
- **Scenario 4: Stock Price Management (e.g., Seasonal Equity Offerings or M&A)**
 - **Strategic Decisions:** Both business and accounting decisions follow the logic of Scenario 1 (reducing expenses and maximizing capitalization) to present a more robust earnings profile and attract higher valuations.

24.3 Accounting Choices and Business Models: Evidence from European Listed Companies

This subsection examines the relationship between firm-specific business models and accounting policy selections, drawing on findings from [Lassini et al. \(2016\)](#)

24.3.1 Uniformity of Accounting Choice across Diverse Business Models

Empirical analysis of European listed companies suggests that business models do not significantly dictate the nature of accounting choices. Despite the inherent differences in how firms create and capture value, research indicates a high degree of uniformity in accounting selections. These choices appear to remain largely unaffected by various firm-specific factors, including:

- ownership structure
- company size
- supply chain relationships
- levels of internationalization
- R&D
- economic performance

24.3.2 Persistence and Common Practices under the IFRS/IAS Framework

Under the regulatory framework of International Financial Reporting Standards (IFRS) and International Accounting Standards (IAS), firm behavior regarding accounting choice exhibits distinct persistence. Generally, companies do not easily or frequently change their established accounting choices, reflecting a preference for consistency in financial reporting. According to the analysis of European firms, a overwhelming majority of companies converge on specific methodologies:

1. **Cash Flow Statement Preparation:** Most firms utilize the indirect method for calculating Cash Flow from Operations (CFO), rather than the direct method.
2. **Measurement of Long-term Assets:** In the treatment of both investment property and Property, Plant, and Equipment (PPE), companies consistently prefer the cost model over the fair value model for subsequent measurement.

25 Financial Frictions and Corporate Policies

In the presence of financial constraints, firms cannot evaluate their investment, cash management, and financing decisions in isolation. This study focuses on the framework developed by [Almeida et al. \(2011\)](#), which emphasizes that corporate policies are endogenously determined to manage intertemporal liquidity frictions.

25.1 Literature Review

25.1.1 Theoretical Framework: Forward-Looking Conservatism

The authors construct a stylized three-period model (*Date* 0, 1, 2) to illustrate that corporate decision-making is fundamentally forward-looking. A firm's optimal policy today is not merely a reaction to current cash flow, but an anticipation of its ability to borrow in the future. To mitigate the "risk of being cash-starved"—the probability that the firm will lack funds to finance high-NPV projects at *Date* 1—constrained firms adopt a conservative stance at *Date* 0. This involves hoarding more cash, avoiding high-risk investments, and prioritizing assets that are liquid.

25.1.2 Core Contribution: Integrated Decisions and Asset Complementarity

The primary contribution of this research is the integration of diverse corporate policies into a unified intertemporal optimization problem. While previous literature often treated investment, cash, and debt as separate topics, this model allows firms to select from an "investment menu" defined by multiple dimensions: risk, payback timing, and liquidity.

A prediction of this model is the **complementarity between asset safety and liquidity**. The model demonstrates that constrained firms are most likely to distort the risk profile of their most liquid assets. Since illiquid assets have little impact on future financing capacity, their riskiness is less of a concern to the firm than the riskiness of assets intended to provide future liquidity.

25.1.3 Investment and Financial Distortions under Constraints

When financing frictions are binding or expected to bind, firms systematically deviate from first-best investment rules:

- **Preference for Stability and Short Payback:** Firms favor projects with stable cash flows and shorter payback periods. This ensures that internal funds are replenished quickly to fund future opportunities.
- **Liquidity Hoarding:** Firms shift their asset composition toward high-liquidity assets. When cash is treated as an investment asset, constrained firms maintain higher cash reserves to ensure future financial flexibility.
- **Precautionary Motive:** These distortions are driven by a precautionary desire to hedge against the risk of underinvestment. By sacrificing current NPV for liquidity, the firm protects its future growth potential.
- **Deviation from NPV Rule:** Consequently, firms do not follow the simple *NPV* maximization principle. Instead, the timing, safety, and pledgeability of cash flows become critical criteria for project selection.

25.1.4 Addressing Long-Standing Corporate Finance Puzzles

The model provides theoretical resolutions to several empirical anomalies and puzzles:

1. **The Payback Method:** It explains why managers rely on the "payback method" rather than *NPV*.
2. **The Anti-Risk-Shifting Result:** Contrary to the classic "gamble for resurrection" theory (Jensen and Meckling, 1976), this model shows that distressed firms often reduce risk to increase their survival probability and secure future funding.
3. **Debt Conservatism (Underleverage):** Firms may maintain lower leverage than predicted by tradeoff models to avoid the indirect distress costs associated with investment distortions.
4. **Cash Management Policy:** It characterizes how firms decide on portfolio liquidity and provides a theoretical basis for the "cash flow sensitivity of cash" as a proxy for constraints.
5. **Operational Hedging:** When financial derivatives are unavailable, managers engage in "operational hedging" by making real investments safer, even if such actions are production-costly.
6. **Cross-Country Differences:** It accounts for why firms in weak investor-protection environments (higher k) prefer tangible, short-term, and low-risk assets.

7. **Financial Development and Productivity:** It suggests that financial development promotes growth not just by increasing the quantity of capital, but by improving the quality of investment—allowing firms to shift from "safe defensive" assets to productive innovation.

25.1.5 The Nature and Origins of Financial Constraints

The authors highlight that financial constraints stem from both institutional and firm-level factors:

- **Institutional Roots:** Legal and institutional frameworks are fundamental sources of constraints. When analyzing cross-country corporate behavior, different levels of constraints must be considered based on the quality of investor protection.
- **Firm-Level Amplifiers:** Financial leverage and the risk of financial distress significantly amplify these constraints. Distress is measured by the gap between available cash flows and required debt payments.
- **The Role of Anticipation:** Financial distress occurs only when cash flows are insufficient to meet debt obligations. However, even if a firm is currently solvent, the anticipation of potential future distress influences its present investment behavior. Forward-looking firms adjust their policies today to mitigate the risk of future insolvency.

25.1.6 Investment Decisions

Under financing constraints, the model yields specific predictions for investment allocation:

1. **Liquidity Bias:** Higher constraints lead to a higher proportion of investment in liquid assets.
2. **Safety Bias within Liquidity:** The more liquid an asset is, the more likely the firm is to choose a safer project within that category.
3. **Internal Cash Flow Effect:** A higher level of internal cash flow from operations reduces the firm's tendency to distort investment towards liquidity and safety.
4. **Distress and Tangibility:** In states of financial distress, firms favor safer investments. In real investment terms, this manifests as a reduction in R&D and an increase in Property, Plant, and Equipment (PPE), which possesses higher collateral value. In financial terms, firms shift toward safer financial instruments.

25.1.7 Strategic Cash Management and Liquidity Trade-offs

Corporate cash policy is defined by a strategic balancing act:

- **Allocation Choices:** Firms manage liquidity by deciding how much cash flow to retain as cash versus how much to invest in other liquid assets.
- **Liquidity vs. Returns:** While other assets may offer higher production returns, cash provides absolute liquidity and immediate funding capacity.
- **External Funding Pressure:** When the cost of obtaining external financing increases, firms prioritize hoarding cash as a precautionary buffer to ensure future project funding.

25.1.8 Capital Structure and the Pursuit of Financial Flexibility

The model explains the gap between empirical observations and traditional financial theories through the need for flexibility:

- **Debt Conservatism:** In reality, many profitable and stable firms maintain debt levels significantly lower than those suggested by standard capital structure models.
- **Financial Flexibility:** Flexibility is defined as the firm's ability to obtain capital at a fair price in the future. By restricting current debt, firms preserve debt capacity to support future investment opportunities.

25.1.9 Operational Hedging as a Strategic Alternative

When firms lack access to specialized financial derivatives or hedging instruments, they resort to **operational hedging**. This is achieved by intentionally biasing the real investment portfolio toward safe assets. Although this reduces pure production efficiency, it serves as a vital risk-mitigation tool when financial markets are incomplete.

25.1.10 Institutional Determinants and Cross-Country Evidence

In countries where external financing is costly due to weak investor protection, firms exhibit a strong preference for short-term, safe, and tangible assets. Empirical evidence supports this:

- Firms in high-cost countries maintain **higher cash reserves**.
- In countries with lower financial development, cash holdings are **more sensitive to internal cash flow**.
- Firms in developing nations possess **more tangible assets (PPE)** but fewer intangible assets (unpatented knowledge, brands) and lower R&D expenditure compared to those in developed nations.

25.2 Corporate Strategic Decisions under Different Scenarios

The table below summarizes the strategic policy adjustments made by constrained firms across various institutional and financial scenarios.

Scenario	Investment Decision	Cash Management	Financing Decision
Poor Legal Protection	• Prefer PPE • Reduce R&D and intangible assets	• Maintain high cash holdings	• Rely on collateralized debt • Restricted access to unsecured credit
High Leverage or Near Debt Limit	• Invest in safe projects	• Maximize internal fund retention	• Reduce leverage
High Market Interest Rates	• Reduce CapEx • Prefer liquid assets	• Increase cash holdings • Increase cash buffer	• Minimize new debt issuance
Lack of Financial Hedging Tools	• Invest in safe projects	• Increase cash holdings • Increase cash buffer	• No significant change

Table 67: Corporate Strategic Policy Matrix under Constraints

26 Theoretical Framework and Corporate Strategy in Dynamic Finance

26.1 Literature Review

26.1.1 Contingent Claims Models and Capital Structure Dynamics

This section reviews the continuous-time contingent claims models as discussed in [Strebulaev et al. \(2012\)](#). While traditional static models often provide conclusions that are considered unreliable due to their neglect of intertemporal decision-making and their reliance on market values—as opposed to the book values used in accounting reports—they remain a foundational reference for understanding financial frictions.

Optimal Investment and Static Foundations According to the real options approach (Section 2.2.2), firms typically choose to wait longer before committing to projects than suggested by simple NPV rules. In the realm of static capital structure (Section 2.3), default is treated not as a sudden catastrophe but as an endogenous real option exercised by equityholders (Section 2.3.2). Notably, the absolute size of a firm does not dictate its capital structure choice due to model homogeneity, while the optimal dividend rate is found to be directly proportional to cash flow levels (Section 2.3.4). Furthermore, Section 2.3.5 identifies several determinants of leverage: higher growth rates lead to higher debt levels to capture tax shields (though with lower coupon rates); higher interest rates lead to a passive increase in leverage as equity values decline; and increased volatility significantly shifts the optimal balance.

Dynamic Capital Structure and the Inaction Region From Section 2.4 onwards, the framework incorporates forward-looking decision-making. The introduction of adjustment costs leads to several key behaviors:

- **Inaction Region:** Leverage is not continuously adjusted. Within a certain range, cash flow fluctuations do not trigger active financial changes.
- **Threshold Effects:** Firms refinance when cash flows reach an upper bound and default when they fall below a lower bound. Higher cash flow growth generally correlates with lower default probability.
- **Conservative Leverage:** Dynamic models predict more conservative leverage than static ones. This stems from the interaction between refinancing and default options; firms preserve debt capacity today to maintain flexibility in future "good states" and avoid premature default or high adjustment costs.
- **Lumpiness:** Active refinancing typically involves issuing a large, discrete amount of debt rather than incremental changes.

Empirical Conflicts and Reconciliations Section 2.5 highlights the tension between Pecking Order Theory (predicting a negative profitability-leverage correlation due to internal funding) and Trade-off Theory (predicting a positive correlation due to tax benefits). While empirical data robustly shows a negative correlation—often seen as a failure of the Trade-off Theory—dynamic models reconcile this by showing that the correlation is indeed positive if one observes firms only at their active refinancing points.

Macroeconomic Impacts and Strategic Timing Macroeconomic conditions (Section 2.6) dictate that during booms, growth is higher and volatility lower, while the risk premium rises in recessions due to higher correlation with consumption.

- **Timing:** Firms refinancing in recessions choose lower leverage due to poor growth prospects. Those refinancing in booms take on more debt, leading to "default clustering" in subsequent downturns.
- **Cyclical:** While optimal leverage at the point of refinancing is pro-cyclical, the actual cross-sectional dynamics appear counter-cyclical because equity values drop faster than debt values during recessions, causing a passive rise in leverage.

Finally, the impact of temporary shocks leads to lower leverage compared to permanent shocks (Section 2.8). Strategically, firms may accelerate investment if they anticipate high future funding costs, or use "over-investment" and high debt as signaling mechanisms to demonstrate quality to the market (Section 2.9).

26.1.2 Discrete-Time Investment Models with Financial Frictions

Moving to the discrete-time investment models (Section 3), firms internalize investment decisions alongside financial constraints. Even in the basic dynamic model (Section 3.1), the presence of fixed adjustment costs results in an inaction region where no financing or investment rebalancing occurs.

Costly External Finance and Investment Volatility When external equity financing is costly (Section 3.2), firms exhibit a strong preference for internal funds. A key result is that high external financing costs actually reduce investment volatility; because firms cannot easily raise capital to expand, they become less responsive to positive productivity shocks.

Precautionary Savings and Cash Holdings The introduction of cash as a state variable (Section 3.3) reveals a substitution effect between financial and physical assets. Firms increase investment and decrease cash holdings when productivity improves. Firms engage in precautionary saving behavior, where any factor increasing the probability of needing future external finance raises the shadow value of cash. Specifically:

- **Depreciation:** Faster depreciation necessitates more frequent equipment replacement, leading to higher cash holdings.
- **Volatility:** Higher or more persistent profit volatility increases the risk of funding shortfalls, driving firms to hold more cash on average.
- **Adjustment Costs:** Firms with high fixed investment costs exhibit "lumpy" investment patterns and accumulate large cash reserves to fund discrete investment bursts.

Debt Capacity and Financial Flexibility With risk-free debt (Section 3.4), firms facing high external financing costs choose lower leverage to maintain "debt capacity" for future opportunities. Highly profitable firms, possessing abundant internal cash flows, face lower marginal benefits from debt and thus maintain lower leverage ratios.

The simultaneous holding of cash and debt (Section 3.5) is explained by fixed costs associated with rebalancing debt. Firms use cash as a buffer to avoid these costs. When in financial distress (low profitability), firms reduce leverage by accumulating cash reserves rather than paying down debt directly. Similarly, when increasing leverage, firms typically deplete their cash levels first.

Risky Debt, Endogenous Default, and Payout Policy Allowing for risky debt (Section 3.6) endogenizes the default decision. Firms exercise the default option when equity value falls to zero. As default costs increase, firms actively reduce leverage to avoid these deadweight

losses. Furthermore, higher capital depreciation reduces the firm's debt capacity because it diminishes the unit capital's ability to generate value, leading to lower average leverage.

Finally, Section 3.7 notes that firms strategically smooth dividends to manage agency conflicts or market expectations. Across the macroeconomy, leverage exhibits cyclicity that is deeply intertwined with broader economic cycles and shocks.

26.2 Empirical Facts

This section outlines several significant empirical observations that highlight the limitations of current dynamic theoretical frameworks. These facts represent "puzzles" or data patterns that the models discussed in the previous sections either fail to replicate or explicitly contradict.

- **The Low-Leverage Puzzle:** A major discrepancy exists between theoretical predictions and observed capital structures. While standard trade-off models and dynamic refinancing frameworks often predict optimal leverage ratios exceeding 50% or even 70%, empirical data from the COMPUSTAT universe shows that the average quasi-market leverage for firms consistently remains in the 20% to 25% range.
- **The Existence of Negative Cash Flows:** Most continuous-time models rely on geometric Brownian motion (GBM) to describe cash flow dynamics, which by definition cannot take negative values. However, empirical evidence indicates that approximately 17% to 25% of quarterly firm observations involve negative cash flows, suggesting that the non-negativity assumption in basic models is a significant abstraction from reality.
- **Magnitude of External Financing:** Simulated data from discrete-time investment models often suggests that firms raise external capital equivalent to roughly 20% of their assets. In contrast, actual corporate data shows that annual external financing is much more modest, typically staying below 10% of total assets, indicating that models may underestimate the frictions or costs associated with accessing capital markets.
- **Strategic and Operational Cash Demand:** While models often treat cash as a precautionary buffer against financing shocks, they frequently omit other primary empirical drivers of liquidity. For instance, firms hold significant cash reserves for working capital management (such as paying suppliers) or for strategic purposes like the acquisition of patented technologies, which are not captured in the financial friction-centric frameworks.
- **Dividend Smoothing (The Lintner Fact):** A robust empirical finding in corporate finance is that firms go to great lengths to smooth dividend payments over time. However, this behavior is largely absent from the core dynamic capital structure and investment models, which tend to treat payouts as a residual rather than a strategic smoothing target.
- **Legal and Institutional Barriers to Renegotiation:** Theoretical models of endogenous default often assume that debt can be strategically restructured. In reality, legal constraints such as the Trust Indenture Act of 1939 in the U.S. make the modification of core public bond terms practically impossible without unanimous consent, a rigid institutional fact that most general models do not incorporate.

26.3 Corporate Strategy under Different Conditions

26.3.1 Structural Thresholds and Timing Logic

Based on the contingent claims and dynamic investment frameworks, firm behavior is characterized by periods of inaction followed by discrete adjustments.

- **Timing of Financing:** In the absence of immediate needs or when shocks are small, firms do not engage in active financing. Only when cash flows hit critical upper thresholds do they initiate discrete, large-scale refinancing. This "lumpy" adjustment avoids the frequent incurrence of fixed transaction costs.
- **Timing of Investment:** Similar to financing, investment is not continuous. Due to fixed adjustment costs and the real option to wait, firms may remain idle for extended periods. They suddenly undertake significant investment only when productivity shocks are sufficiently positive to offset the cost of exercising the investment option and the associated adjustment expenses.
- **Pre-emptive Strategies:** Firms may "over-invest" or accelerate the timing of their investment and debt issuance if they anticipate a significant increase in future capital costs or wish to signal their superior quality to the market.

26.3.2 Strategic Responses to Financial and Operational Frictions

The following table summarizes how firms adjust their physical investment, leverage, and cash holdings in response to various financial frictions and operational environments.

Table 68: Strategic Corporate Responses under Different Conditions

Condition	Strategic Response
High external financing costs	Decrease physical investment and reduce leverage levels.
Positive productivity shock	Increase physical investment while depleting cash reserves.
Rapid asset depreciation	Increase cash holdings and maintain lower average leverage.
High profit volatility	Accumulate precautionary cash and lower leverage ratios.
High fixed cost	Exist investment inaction interval.
Financial distress / Low profit	Reduce leverage by increasing cash reserves (rather than direct debt repayment).
High default cost	Proactively maintain significantly lower leverage levels.

27 Modeling Corporate Operational Decisions

This section outlines the framework for corporate decision-making, focusing on operational decisions such as investment and financing. We define the decision process as a conditional policy mapping $\pi(a_t|s_t)$, where s_t represents the state variables and a_t represents the actions.

27.1 State and Action Variables

The decision environment is characterized by a vector of state variables $s_t \in \mathbb{R}^N$ and a set of actions a_t .

27.1.1 State Variables (s_t)

The state space is partitioned into macro-economic conditions and firm-specific characteristics, as summarized in Table 1.

Table 69: Classification of State Variables

Category	Variables	Notation
Macro States	Risk-free rate, Inflation, GDP growth	$i_t^f, \pi_t, \Delta GDP_t$
	Investment return baseline	$R_{base,t}$
Firm-Specific States	Profit level, Profit volatility, Productivity	Π_t, σ_Π, z_t
	Current leverage (Debt-to-Equity / $\log D - \log E$)	L_t
	Investment return, Financing interest rate	i_t^I, i_t^d
	Production fixed cost, Financing fixed cost	C_f, C_{fin}
	Depreciation rate, Default cost, Savings rate	δ, Ψ, i_t^s

27.1.2 Action Space (a_t)

The firm selects an action vector a_t spanning real and financial decisions. The actions and their corresponding notations are detailed in Table 2.

Table 70: Action Space and Notations

Category	Action	Notation
Investment	Capital Expenditure (CAPEX)	$a_{inv,t}$
	Sale of Property, Plant, and Equipment (PPE)	$a_{sale,t}$
	Research & Development (R&D)	$a_{rd,t}$
Financing	Debt issuance (by maturity m)	$a_{debt,m,t}$
	Dividend payouts	$a_{div,t}$
Liquidity	Cash buffer adjustments	$a_{cash,t}$
	Savings from excessive cash flow	$a_{sav,t}$

27.2 State Interdependencies and Interest Rates

The model assumes that market rates and returns are driven by the risk-free rate adjusted by state-dependent components:

- **Financing Interest Rate:**

$$i_t^d = i_t^f + RP(s_t)$$

- **Saving Interest Rate:**

$$i_t^s = i_t^f$$

- **Investment Return:**

$$i_t^I = i_t^f + ER(s_t)$$

27.3 Sequential Decision-Making and Decomposition

Decisions follow a sequential hierarchy where investment choices impact the financing requirements. We decompose each action $a_{i,t}$ into a binary participation indicator $D_{i,t} \in \{0, 1\}$ and a continuous quantity $Q_{i,t} \in \mathbb{R}^+$.

27.3.1 Participation Choice (Binary)

The probability $p_{i,t} = P(D_{i,t} = 1 | \mathcal{F}_{t-1})$ is modeled using a Logit link:

$$\ln \left(\frac{p_{i,t}}{1 - p_{i,t}} \right) = x'_t \beta_p \implies p_{i,t} = \frac{\exp(x'_t \beta_p)}{1 + \exp(x'_t \beta_p)}$$

27.3.2 Quantity Choice (Continuous)

Conditional on $D_{i,t} = 1$, the quantity $Q_{i,t}$ follows a conditional distribution $g(\lambda_{i,t})$. The following distributional assumptions are illustrative examples:

- **Positive Scaling (e.g., Investment):** For actions where values must be strictly positive, such as CAPEX, a Log-Normal distribution can be employed:

$$Q_{i,t} | (D_{i,t} = 1) \sim \text{Log-Normal}(\mu(x'_t \beta_\lambda), \sigma^2)$$

- **Debt Issuance Quantity:** The quantity of debt issued is modeled using a Lower-Truncated Normal distribution:

$$Q_{debt,t} | (D_{debt,t} = 1) \sim \mathcal{TN}(\mu_t, \sigma^2, L_t, \infty)$$

The lower truncation point L_t captures the firm's mandatory financing needs. Specifically, if the net cash flow before financing is insufficient to cover the required cash level (minimum cash plus buffer), the firm must issue at least enough debt to bridge this gap. This is defined as:

$$L_t = \max(0, -(CF_{pre,t} - (C_{min} + a_{cash,t})))$$

27.4 Discrete vs. Continuous Specifications for Actions

In this framework, actions are distinguished by whether they require a two-stage modeling process involving an explicit participation decision $D_{i,t}$.

27.4.1 Actions with Discrete Participation Decisions

The following actions are modeled using the decomposition $a_{i,t} = D_{i,t} \cdot Q_{i,t}$:

- **Capital Expenditures:** Investment in CAPEX ($a_{inv,t}$) and the sale of PPE ($a_{sale,t}$).
- **Financing Actions:** Debt issuance across various maturities ($a_{debt,m,t}$) and dividend payouts ($a_{div,t}$).
- **Research & Development:** Strategic R&D expenditures ($a_{rd,t}$).

27.4.2 Actions with Direct Quantity Modeling

Certain actions are modeled directly by their continuous quantity Q_t without a preceding discrete participation stage:

- **Cash Buffer:** The level of the cash buffer ($a_{cash,t}$) is determined directly as a continuous quantity based on liquidity requirements and pre-financing cash flow states.

27.5 Interdependencies and Dynamics of State Variables

The state vector s_t is defined as the union of macro-economic and firm-specific variables. Although s_t encompasses all potential factors, the following discussion focuses on the subset of states most critical to the current policy functions.

27.5.1 Exogenous Dynamics

All macro-economic states (risk-free rate i_t^f , inflation π_t , and GDP growth ΔGDP_t) are treated as purely exogenous. Similarly, several firm-specific states are assumed to be exogenous for this modeling step: productivity (z_t), production fixed cost (C_f), financing fixed cost (C_{fin}), depreciation rate (δ), and default cost (Ψ).

27.5.2 Derived and Estimated States

Given the exogenous states, other key variables are derived or estimated based on realized data and theoretical relationships:

1. **Profit Level:** The realized profit Π_t is determined through the firm's income statement once outcomes for period t are finalized.
2. **Profit Volatility:** The volatility $\sigma_{\Pi,t}$ is a function of current macro-economic conditions:

$$\sigma_{\Pi,t} = f_{\sigma}(\text{Macro}_t, \text{Historical realizations})$$

3. **Interest Rates and Returns:**

- **Investment Return:** $i_t^I = i_t^f + f^I(s_t)$
- **Financing Interest Rate:** $i_t^d = i_t^f + f^d(s_t)$
- **Saving Interest Rate:** $i_t^s = i_t^f$

4. **Leverage:** The current leverage L_t is defined using the log-difference between liabilities and equity:

$$L_t = \log(\text{Liability}_t) - \log(\text{Equity}_t)$$

27.6 Operational Action Policies via Conditional Decomposition

The firm's operational decisions involve the simultaneous determination of four primary actions: the cash buffer ($a_{cash,t}$), investment ($a_{inv,t}$), dividends ($a_{div,t}$), and financing ($a_{fin,t}$). While these decisions are determined jointly, the joint policy can be decomposed into a sequence of conditional decisions using the probability chain rule.

Let $\mathcal{F}_t$ denote the information set at time t , defined as:

$$\mathcal{F}_t = \{s_{t-1}, CFO_t, \text{repayment}_t\}$$

where s_{t-1} represents the state variables from the previous period, CFO_t is the current cash flow from operations, and repayment_t denotes mandatory debt repayments.

27.6.1 Sequential Decomposition of the Joint Policy

The joint conditional distribution of the four actions can be decomposed as follows:

$$\begin{aligned} P(a_{cash,t}, a_{inv,t}, a_{div,t}, a_{fin,t} | \mathcal{F}_t) &= P(a_{cash,t} | \mathcal{F}_t) \\ &\quad \times P(a_{inv,t} | \mathcal{F}_t, a_{cash,t}) \\ &\quad \times P(a_{div,t} | \mathcal{F}_t, a_{cash,t}, a_{inv,t}) \\ &\quad \times P(a_{fin,t} | \mathcal{F}_t, a_{cash,t}, a_{inv,t}, a_{div,t}) \end{aligned}$$

This sequential structure implies that the firm first determines its liquidity target ($a_{cash,t}$), then decides on its investment scale, followed by the distribution of profits to shareholders, and finally determines the required external financing to close any remaining funding gap.

27.6.2 Expanded Hurdle Decomposition

By incorporating the discrete-continuous decomposition ($a = D \cdot Q$) into the sequence, the probability density is further expanded:

$$\begin{aligned} P(a_{cash,t}, \dots, a_{fin,t} | \mathcal{F}_t) &= P(a_{cash,t} | \mathcal{F}_t) \\ &\times P(D_{inv,t} = 1 | \mathcal{F}_t, a_{cash,t}) \cdot P(Q_{inv,t} | D_{inv,t} = 1, \mathcal{F}_t, a_{cash,t}) \\ &\times P(D_{div,t} = 1 | \mathcal{F}_t, a_{cash,t}, a_{inv,t}) \cdot P(Q_{div,t} | D_{div,t} = 1, \mathcal{F}_t, a_{cash,t}, a_{inv,t}) \\ &\times P(D_{fin,t} = 1 | \mathcal{F}_t, a_{cash,t}, a_{inv,t}, a_{div,t}) \cdot P(Q_{fin,t} | D_{fin,t} = 1, \mathcal{F}_t, a_{cash,t}, a_{inv,t}, a_{div,t}) \end{aligned}$$

27.6.3 Modeling the Cash Buffer ($a_{cash,t}$)

The first component of the decision sequence is the determination of the cash buffer, defined as the target ratio of the cash buffer to total assets. Unlike other operational actions, $a_{cash,t}$ is modeled as a purely continuous variable, as firms continuously recalibrate their liquidity levels without a discrete activation hurdle.

The policy for $a_{cash,t}$ is determined by the information set $\mathcal{F}_t$:

$$a_{cash,t} = f(i_{t-1}^s, L_{t-1}, \delta, \sigma_{\Pi,t}, \Pi_t, z_t)$$

The relationship between $a_{cash,t}$ and its driving state variables is characterized by the following directions of influence:

- **Positive Relationships (+):**

- *Saving interest rate* (i_{t-1}^s): Higher returns on savings increase the incentive to hold liquidity.
- *Leverage* (L_{t-1}): Higher debt burdens necessitate larger precautionary buffers to mitigate default risk.
- *Depreciation rate* (δ): Increased asset depreciation implies greater future replacement needs, requiring more liquidity.
- *Profit volatility* ($\sigma_{\Pi,t}$): Higher earnings uncertainty promotes precautionary cash holdings.

- **Negative Relationships (-):**

- *Profit level* (Π_t): Robust current profitability reduces the immediate pressure to maintain external liquidity reserves.
- *Productivity* (z_t): Greater efficiency allows the firm to manage resources with lower idle cash requirements.

27.6.4 Modeling Investment ($a_{inv,t}$)

The second stage in the decision sequence is the investment policy (CAPEX), conditional on the information set $\mathcal{F}_t$ and the previously determined cash buffer $a_{cash,t}$. The investment action is decomposed into a binary participation choice $D_{inv,t}$ and a continuous quantity $Q_{inv,t}$, both driven by a common set of states:

$$\begin{aligned} D_{inv,t} &= f_D(i_t^I, C_f, z_t, a_{cash,t}, \Delta GDP_t, \pi_t, CFO_t - \text{repayment}_t, \sigma_{\Pi,t}, L_{t-1}) \\ Q_{inv,t} &= f_Q(i_t^I, C_f, z_t, a_{cash,t}, \Delta GDP_t, \pi_t, CFO_t - \text{repayment}_t, \sigma_{\Pi,t}, L_{t-1}) \end{aligned}$$

The directions of influence for the investment policy are as follows:

- **Positive Relationships (+):**

- *Investment return* (i_t^I): +
- *Productivity* (z_t): +
- *GDP growth* (ΔGDP_t): +
- *Net operating cash flow* ($CFO_t - repayment_t$): +

- **Negative Relationships (-):**

- *Production fixed cost* (C_f): -
- *Cash buffer* ($a_{cash,t}$): -
- *Inflation* (π_t): -
- *Profit volatility* ($\sigma_{\Pi,t}$): -
- *Lagged leverage* (L_{t-1}): -

27.6.5 Modeling Dividends ($a_{div,t}$)

The third stage in the decision sequence is the dividend policy, which is determined conditional on the information set $\mathcal{F}_t$, the target cash buffer $a_{cash,t}$, and the investment decision $a_{inv,t}$. This structure accounts for the hierarchical nature of profit allocation. The dividend action is decomposed into a binary payout choice $D_{div,t}$ and a payout quantity $Q_{div,t}$:

$$D_{div,t} = f_D(a_{cash,t}, D_{inv,t}, Q_{inv,t}, \pi_t, L_t, a_{div,t-1}, CFO_t - repayment_t)$$

$$Q_{div,t} = f_Q(a_{cash,t}, D_{inv,t}, Q_{inv,t}, \pi_t, L_t, a_{div,t-1}, CFO_t - repayment_t)$$

The relationship between the dividend policy and its driving variables is characterized by:

- **Positive Relationships (+):**

- *Inflation* (π_t): + (May reflect nominal payout adjustments or a preference for current returns during inflationary periods).
- *Lagged dividends* ($a_{div,t-1}$): + (Reflects dividend smoothing or the "stickiness" of payout policies).
- *Net operating cash flow* ($CFO_t - repayment_t$): + (Higher available internal funds increase payout capacity).

- **Negative Relationships (-):**

- *Cash buffer* ($a_{cash,t}$): - (Increased liquidity retention competes with dividend distribution).
- *Investment decision* ($D_{inv,t}, Q_{inv,t}$): - (Higher real investment expenditures crowd out profit distribution to shareholders).
- *Current leverage* (L_t): - (Higher debt levels increase interest burdens and may trigger restrictive covenants, limiting dividends).

27.6.6 Modeling Dividends ($a_{div,t}$)

The third stage in the decision sequence is the dividend policy, which is determined conditional on the information set $\mathcal{F}_t$, the target cash buffer $a_{cash,t}$, and the investment decision $a_{inv,t}$. The dividend action is decomposed into a binary payout choice $D_{div,t}$ and a payout quantity $Q_{div,t}$:

$$P(D_{div,t} = 1) = f_P(a_{cash,t}, a_{inv,t}, \pi_t, L_t, a_{div,t-1}, CFO_t - repayment_t)$$

$$Q_{div,t}|(D_{div,t} = 1) = f_Q(a_{cash,t}, a_{inv,t}, \pi_t, L_t, a_{div,t-1}, CFO_t - repayment_t)$$

The relationship between the dividend policy and its driving variables is characterized by:

- **Positive Relationships (+):**
 - *Inflation* (π_t): +
 - *Lagged dividends* ($a_{div,t-1}$): +
 - *Net operating cash flow* ($CFO_t - repayment_t$): +
- **Negative Relationships (-):**
 - *Cash buffer* ($a_{cash,t}$): -
 - *Investment decision* ($a_{inv,t}$): -
 - *Current leverage* (L_t): -

27.6.7 Modeling Financing ($a_{fin,t}$)

The final stage of the sequential decision process is the financing policy, which acts as the residual closure for the firm's budget. We first define the pre-financing excess cash (EC_t), which accounts for the previous cash balance, current operations, and all preceding decisions:

$$EC_t = Cash_{t-1} + CFO_t - repayment_t - Q_{inv,t} - Q_{div,t} - \text{TargetAmount}(a_{cash,t})$$

where $\text{TargetAmount}(a_{cash,t})$ represents the cash required to achieve the target buffer.

The conditional probability of debt issuance, $P(D_{fin,t} = 1 | \mathcal{F}_t, a_{cash,t}, a_{inv,t}, a_{div,t})$, is modeled as a piecewise function of the funding gap:

$$P(D_{fin,t} = 1 | \dots) = \begin{cases} 1 & \text{if } EC_t < 0 \\ f_P(i_t^d, C_{fin}, \delta, \sigma_{\Pi,t}, L_{t-1}, EC_t) & \text{if } EC_t > 0 \end{cases}$$

In the case of discretionary issuance ($EC_t > 0$), the firm weighs the benefits of additional liquidity against the costs and risks. The conditional quantity of debt issued follows a similar mapping:

$$Q_{fin,t} | (D_{fin,t} = 1) = f_Q(i_t^d, C_{fin}, \delta, \sigma_{\Pi,t}, L_{t-1}, EC_t)$$

The relationship between both the participation and quantity decisions and their driving state variables is summarized as follows:

- **Negative Relationships (-):**
 - *Financing interest rate* (i_t^d): - (Higher marginal borrowing costs).
 - *Financing fixed cost* (C_{fin}): - (Increases the friction of debt issuance).
 - *Depreciation rate* (δ): - (Potentially lower collateral value or shrinking capacity).
 - *Profit volatility* ($\sigma_{\Pi,t}$): - (Increases financial distress risk and lending spreads).
 - *Lagged leverage* (L_{t-1}): - (Constraints from existing debt overhang).
 - *Pre-financing excess cash* (EC_t): - (Lower internal need for external funds).

Part III

Work Summary

28 Current Work

28.1 Big Picture: The Financial Statement Simulator

The core of this framework is a high-fidelity simulator designed to generate projected financial statements, including the Balance Sheet, Income Statement, and Statement of Cash Flows. Unlike traditional static models, this simulator is built upon two fundamental pillars:

- **Accounting Integrity:** The simulated statements strictly adhere to all fundamental accounting identities (e.g., $\text{Assets} = \text{Liabilities} + \text{Equity}$). Every transaction and policy decision flows through the three statements consistently, ensuring structural coherence.
- **Stochastic Distributions:** Rather than providing a single-point estimate or a "best-case" forecast, the simulator produces a full distribution of potential future outcomes. This probabilistic approach allows users to quantify risk and understand the likelihood of various financial states.

28.1.1 Target User Groups

The simulator serves a diverse set of stakeholders in the financial ecosystem:

- **Firm Managers:** For strategic planning, capital budgeting, and assessing the impact of internal policy shifts on long-term firm value and liquidity.
- **Commercial Banks:** To evaluate credit risk, perform stress tests on loan portfolios, and monitor the probability of covenant breaches.
- **Investment Banks:** For M&A advisory, capital structure optimization, and valuation under complex corporate restructuring scenarios.
- **Secondary Market Investors:** To perform robust valuation, risk assessment, and style-based investment analysis by looking beyond historical data into potential future distributions.

28.1.2 Key Applications: Scenario Analysis

The primary application of the framework is advanced scenario analysis, which allows users to disentangle the drivers of corporate performance across two dimensions:

Environmental Scenarios (Exogenous) This involves testing how the firm's financial health evolves when external conditions change, such as shifts in macroeconomic indicators (e.g., GDP growth, inflation, interest rates) or market-specific shocks. Users can observe how the entire distribution of financial outcomes shifts in response to a changing environment.

Policy Scenarios (Managerial Decisions) This focuses on the impact of active managerial choices. The framework decomposes corporate strategy into four critical decision pillars, each involving fundamental economic trade-offs:

1. **Cash Management Decisions:** In each period, managers decide how much excess cash to retain versus deploy. *Trade-off:* Retaining cash serves as a precautionary saving to hedge against future idiosyncratic shocks and preserves "dry powder" for future investment opportunities, but it comes at the cost of forgoing immediate investment returns.
2. **Financing Decisions:** Managers determine whether to issue debt, the total amount of issuance, and the optimal maturity structure. *Trade-off:* While debt can fund high-return investment opportunities and provide tax shields, it introduces mandatory interest and principal repayments, creating liquidity pressure and increasing default risk.
3. **Investment Decisions:** Firms must allocate capital between safe assets and risky assets. *Trade-off:* Safe assets provide high liquidity and stable cash flows but offer lower returns. Risky assets drive growth and higher long-term value but introduce significant volatility and the potential for large losses.

4. **Dividend Decisions:** Managers choose whether to distribute profits to shareholders. *Trade-off:* Payouts can improve the manager's utility or signal financial stability to the market. However, dividends reduce internal funds, potentially forcing the firm to forgo future growth-enhancing investments or rely more heavily on costly external financing.

28.1.3 Predictive Applications

Beyond scenario analysis, the framework integrates with predictive modeling (as detailed in subsequent sections) to provide forward-looking insights into firm performance, credit migration, and market valuation based on current state variables and historical policy trends:

- **Equity Valuation:** Predicting the distribution of future performance → Calculating Free Cash Flow (FCF) → Deriving the distribution of future stock prices.
- **Risk Management:** Assessing future performance distributions → Identifying tail risks → Calculating Value at Risk (VaR) for commercial bank credit risk assessment.
- **Investment Banking:** Identifying firms likely to require future financing to assist in deal sourcing and client acquisition.
- **Fraud Detection:** Identifying discrepancies between actual reported statements and predicted distributions → Flagging potential fraud or high default risk.

28.2 The Simplified Three-Statement Model

28.2.1 Balance Sheet (BS)

Represents the state of the firm at a point in time. Assets must always equal Liabilities plus Equity ($A = L + E$).

Step-by-Step Incorporation Example Step 1: Equity Injection (Initial Funding)

Assets	Value	Liabilities & Equity	Value
Cash	100	Equity	100
Total	100	Total	100

Step 2: Inventory Purchase (Asset Exchange)

Assets	Value	Liabilities & Equity	Value
Cash	70	Equity	100
Inventory	30		
Total	100	Total	100

Step 3: Debt Issuance to Purchase PPE

Assets	Value	Liabilities & Equity	Value
Cash	70	Equity	100
Inventory	30	Debt	50
PPE	50		
Total	150	Total	150

28.2.2 Profit and Loss (P&L) and Statement of Cash Flows (SCF)

- **P&L (Income Statement):** Based on the **accrual basis**. Revenue and costs are recognized when a transaction occurs, regardless of cash timing.
- **SCF (Statement of Cash Flows):** Reflects actual **cash-based** transactions.

28.3 Interconnectivity and Double-Entry Principle

The three statements are linked via the double-entry principle, ensuring the Balance Sheet remains balanced. Below are some examples:

1. CAPEX (Capital Expenditure)

SCF: Cash Outflow (Investing)

BS: ↓ Cash, ↑ PPE

2. Sale of PPE

SCF: Cash Inflow (Investing)

BS: ↑ Cash, ↓ PPE

3. Revenue Recognition

P&L: ↑ Revenue, ↑ Net Income

BS: ↑ Accounts Receivable, ↑ Equity (Retained Earnings)

4. Debt Issuance

SCF: Cash Inflow (Financing)

BS: ↑ Cash, ↑ Liabilities (Debt)

28.4 Dynamics and Variable Types

Given the items on the P&L and SCF, the dynamics of Balance Sheet items are determined. For example, the evolution of Property, Plant, and Equipment (PPE) is:

$$PPE_{t+1} = PPE_t + \underbrace{CAPEX_{t+1}}_{(Cash_{t+1}\downarrow)} - \underbrace{Sales_{t+1}}_{(Cash_{t+1}\uparrow)} - \underbrace{Depreciation_{t+1}}_{(Equity_{t+1}\downarrow)}$$

28.4.1 Stock vs. Flow Variables

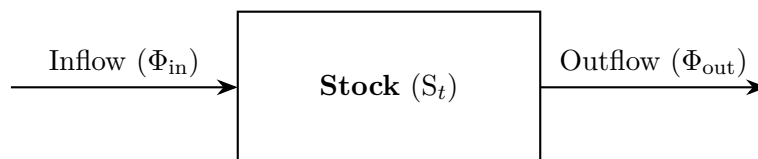
- **Stock Variables:** Balance Sheet items (representing a state at time t).
- **Flow Variables:** P&L and SCF items (representing activities over the period $[t, t + 1]$).

28.5 Modeling Flow Variables and Structural Dynamics

In financial simulation, the core challenge is ensuring that stock variables evolve in a way that respects accounting identities. This requires a shift from modeling states directly to modeling the flows that connect them.

28.5.1 Conceptual Framework: Stocks and Flows

Stock variables represent the state of a firm at a point in time, while flow variables represent the changes over a period. This relationship is visualized below:



The fundamental dynamic is given by:

$$\text{Stock}_{t+1} = \text{Stock}_t + \Phi_{\text{in},t+1} - \Phi_{\text{out},t+1}$$

28.5.2 The Problem with Direct Stock Modeling

Consider a simplified initial Balance Sheet at time t :

Assets (t)	Value	Liabilities & Equity (t)	Value
Cash	70	Equity	100
Inventory	30		
Total Assets	100	Total L&E	100

If we apply a direct predictive model to each stock item independently for time $t + 1$, we might get the following "unbalanced" result where the structural identity is violated:

Assets ($t + 1$)	Value	Liabilities & Equity ($t + 1$)	Value
Cash _{Predicted}	75	Equity _{Predicted}	105
Inventory _{Predicted}	28		
Total Assets	103	Total L&E	105
Error: Assets $\neq$ Liabilities + Equity (Gap = 2)			

28.5.3 The Correct Approach: Flow-Based Evolution

To maintain structural integrity, the simulator models the individual flow components. Assume the following relationships:

$$\text{Net Income}_{t+1} = \text{Revenue}_{t+1} - \text{COGS}_{t+1}$$

$$\text{Equity}_{t+1} = \text{Equity}_t + \text{Net Income}_{t+1}$$

$$\text{Cash}_{t+1} = \text{Cash}_t + \text{Revenue}_{t+1}$$

$$\text{Inventory}_{t+1} = \text{Inventory}_t - \text{COGS}_{t+1}$$

If we simulate flows where $\text{Revenue}_{t+1} = 10$ and $\text{COGS}_{t+1} = 5$:

- $\text{Net Income}_{t+1} = 10 - 5 = 5$
- $\text{Equity}_{t+1} = 100 + 5 = 105$
- $\text{Cash}_{t+1} = 70 + 10 = 80$
- $\text{Inventory}_{t+1} = 30 - 5 = 25$

The resulting Balance Sheet is perfectly balanced:

Assets ($t + 1$)	Value	Liabilities & Equity ($t + 1$)	Value
Cash	80	Equity	105
Inventory	25		
Total Assets	105	Total L&E	105

By modeling the flows that drive these identities, the simulator ensures that all accounting constraints are met by construction across any simulated horizon.

28.6 Driver-Based Modeling and State-Space Framework

Flow variables are not independent; they are determined by underlying **drivers** external to the financial statements. While a State Space Model (SSM) provides a general framework, we focus on specific mathematical determinants.

28.6.1 Deterministic Drivers

Flow variables are deterministically calculated once drivers are modeled:

- **Sales Revenue:** $Revenue = \sum(Price \times Quantity)$.
- **Financing:** Modeling the firm's financing *decision* rather than the debt issuance item itself.

28.6.2 Reinforcement Learning Scope

A financing decision can be viewed as an **action** (a) taken within a specific **state** (s):

$$A^{fin} \sim \pi(a^{fin}|s)$$

Where π is the policy distribution and s represents underlying state variables (firm-level or macro-level).

28.6.3 General Stochastic Framework

The overall evolution follows a joint distribution of actions and states:

$$(A_{t+1}, S_{t+1}) \sim \Pr(a_{t+1}, s_{t+1}|a_t, s_t)$$

Actions (policy functions) can be determined simultaneously or sequentially.

28.7 Methodologies for Driver Prediction

- **Traditional Statistical Methods:** Leveraging existing academic and financial literature.
- **Large Language Models (LLMs):** Using LLMs to process annual reports and qualitative data to generate quantitative driver forecasts.

28.8 Code Design Objectives

The technical implementation of the Financial Statement Simulator is guided by specific objectives that ensure the model is not only commercially viable for users but also robust and maintainable for developers.

28.8.1 Core Functional Requirements

To provide high-utility financial insights, the simulator must meet the following functional criteria:

- **Stochastic Multi-Firm Simulation:** Given the current state of a set of firms, the model must be capable of simulating the full future probability distributions of their financial statements over multiple horizons.
- **Inference Speed:** The simulation engine must achieve high-speed inference to facilitate real-time decision-making and large-scale Monte Carlo analysis.
- **Parallel Processing:** The architecture should support massive parallelization, allowing simulations for thousands of distinct firms to be performed simultaneously.
- **Trainability:** The model should be differentiable or structured in a way that allows it to be trained and optimized using historical financial data to improve its predictive accuracy.

28.8.2 User-Centric Features

From the perspective of the end-user (managers, banks, or investors), the system is designed for flexibility and ease of use:

- **Autonomous Default Simulation:** In the absence of specific user inputs, the simulator should employ a default generative process to project financial outcomes based on historical firm behavior and learned priors.
- **Exogenous Data Integration:** Users should be able to input exogenous time series data (e.g., macroeconomic forecasts, industry trends). These inputs serve as dynamic state variables that influence the managerial decision-making logic within the model.
- **Policy Intervention Scenarios:** The interface must allow users to manually select or override specific corporate decisions—such as choosing a more aggressive debt issuance policy—to observe the resulting shifts in the firm’s financial distribution.

28.8.3 Developer and System Robustness

A critical design pillar for developers is ensuring the structural integrity of the generated data to facilitate debugging and scaling:

- **Automated Structural Integrity Checks:** The system must automatically verify that the accounting identity ($\text{Assets} = \text{Liabilities} + \text{Equity}$) holds true in every simulated step.
- **Logical vs. Numerical Verification:** Verification must go beyond simple numerical checks (e.g., $\text{Assets} - \text{Liabilities} - \text{Equity} < \epsilon$). A small numerical error (e.g., 0.5) is insufficient for debugging because it does not identify the source of the logical failure.
- **Error Localization:** The simulator is designed to perform localized tracking of every flow. If an imbalance occurs, the system should automatically identify exactly which component or equation (e.g., a missing depreciation flow or a miscalculated debt repayment) caused the error.
- **Scalability and Debugging:** By ensuring that every accounting identity is respected through explicit logical structures, the model becomes significantly easier to debug, maintain, and scale to incorporate more complex financial instruments.

28.9 Graph-Based Solution and Symbolic Checking

The pipeline utilizes a symbolic-first approach to manage complexity:

SymPy Equations $\rightarrow$ Symbolic Checking $\rightarrow$ TensorFlow Operations $\rightarrow$ Numerical Checking

28.9.1 Structural Verification via Graph Theory

By representing equations as a graph, we can perform the following:

- **DAG Validation:** Confirming the system is a Directed Acyclic Graph.
- **Topological Sorting:** Automatically determining the required execution order of equations.
- **Leaf Node Dynamics:** Defining Balance Sheet items at $t+1$ as leaf nodes of the system.
- **Symbolic Balance Check:** Verifying that if $A_t = L_t + E_t$ and all flow variables cancel out symbolically according to double-entry rules, then $A_{t+1} = L_{t+1} + E_{t+1}$ is guaranteed before any numerical computation.

28.10 TensorFlow Implementation Pipeline

The symbolic framework maps directly to a deep learning architecture for high-performance simulation:

- **Variables:** SymPy symbols are mapped to TensorFlow Tensors.
- **Equations:** Each SymPy equation is converted into a custom Keras Layer. A system of 100 equations results in 100 layers executed sequentially according to the topological sort.
- **Parameters:** Constants or coefficients in equations are treated as trainable weights within the Keras layers.

28.11 Performance Optimization

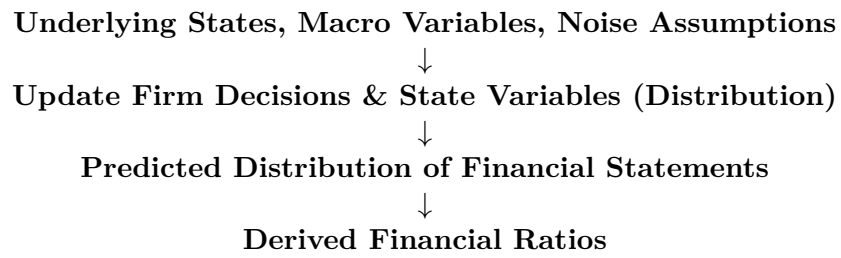
By converting the sequential Keras execution into a **Static Graph** (e.g., via `@tf.function`), the simulator achieves extremely high throughput, allowing for large-scale Monte Carlo simulations of firm distributions across time.

28.12 Limitations and Potential Solutions

1. **Balance Sheet Granularity:** Real-world balance sheets are significantly more complex. For example, PPE is an aggregate of numerous physical assets.
 - *Solution:* While defining more variables/equations is straightforward, obtaining granular asset data is difficult. LLMs can be utilized to extract detailed asset information from annual reports.
2. **Parameter Uncertainty:** Current Monte Carlo simulations incorporate stochastic noise but do not yet account for uncertainty in the model parameters themselves.
3. **Standardization Across Firms:** Different companies report different items using varied nomenclature.
4. *Solution:* Develop an LLM-based dictionary to map heterogeneous reporting items to standardized names.

28.13 Current Capabilities

The current framework enables the following end-to-end simulation flow:



Part IV

Future Work

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Part V

Appendix

A State Space Model and Likelihood Function

We can generalize the system dynamics models used in accounting simulation into a state-space model (SSM) framework. By introducing specific assumptions, the SSM can be simplified for more tractable computation and solution. This structured approach enables us to employ Bayesian estimation techniques to obtain the posterior distributions of the model's parameters, thereby formally capturing uncertainty. A critical component of Bayesian estimation is the calculation of the likelihood function. Therefore, this section provides a detailed discussion of the state-space model architecture and the various methods for computing its likelihood.

A.1 General Form

A.1.1 State Space model

A general framework. The law of motion of the latent states and observations follows:

$$\begin{aligned} X_t &\sim p(x_t|x_{1:t-1}, \theta) \\ Y_t &\sim p(y_t|x_{1:t}, y_{1:t-1}, \theta) \end{aligned}$$

with initial condition

$$X_1 \sim p(x_1|\theta)$$

A.1.2 Joint distribution and likelihood function

To obtain likelihood, we start with joint distribution. Time series $(X_1, X_2, \dots, X_T)$ and $(Y_1, Y_2, \dots, Y_T)$ has joint distribution $p(y_{1:T}, x_{1:T}|\theta)$, which can be decomposed

$$p(y_{1:T}, x_{1:T}|\theta) = p(x_{1:T}|\theta) p(y_{1:T}|x_{1:T}, \theta)$$

The first element is

$$\begin{aligned} p(x_{1:T}|\theta) &= p(x_1|\theta) p(x_2|x_1, \theta) p(x_3|x_{1:2}, \theta) \cdots p(x_T|x_{1:T-1}, \theta) \\ &= p(x_1|\theta) \prod_{t=2}^T p(x_t|x_{1:t-1}, \theta) \end{aligned}$$

The second element is

$$\begin{aligned} p(y_{1:T}|x_{1:T}, \theta) &= p(y_1|x_1, \theta) p(y_2|x_{1:2}, y_1, \theta) p(y_3|x_{1:3}, y_{1:2}, \theta) \cdots p(y_T|x_{1:T}, y_{1:T-1}, \theta) \\ &= \prod_{t=1}^T p(y_t|x_{1:t}, y_{1:t-1}, \theta) \end{aligned}$$

So the joint distribution can be expressed as:

$$p(y_{1:T}, x_{1:T}|\theta) = p(x_1|\theta) \prod_{t=2}^T p(x_t|x_{1:t-1}, \theta) \prod_{t=1}^T p(y_t|x_{1:t}, y_{1:t-1}, \theta)$$

The likelihood is obtained through integrating over $x_{1:T}$:

$$\begin{aligned} p(y_{1:T}|\theta) &= \int p(y_{1:T}, x_{1:T}|\theta) dx_{1:T} \\ &= \int \left[p(x_1|\theta) \prod_{t=2}^T p(x_t|x_{1:t-1}, \theta) \prod_{t=1}^T p(y_t|x_{1:t}, y_{1:t-1}, \theta) \right] dx_{1:T} \end{aligned} \quad (134)$$

A.1.3 Chain rule and likelihood function

The previous likelihood generated from joint distribution is difficult to compute, so we start with another method, the chain rule. The likelihood is

$$p(y_{1:T}|\theta) = \prod_{t=1}^T p(y_t|y_{1:t-1}, \theta)$$

Further decomposing each element leads to:

$$\begin{aligned} p(y_t|y_{1:t-1}, \theta) &= \int p(y_t|x_{1:t}, y_{1:t-1}, \theta) p(x_{1:t}|y_{1:t-1}, \theta) dx_{1:t} \\ p(y_t|y_{1:t-1}, \theta) &= \int p(y_t|x_{1:t}, y_{1:t-1}, \theta) p(x_t|x_{1:t-1}, y_{1:t-1}, \theta) p(x_{1:t-1}|y_{1:t-1}, \theta) dx_{1:t} \end{aligned} \quad (135)$$

with initial condition

$$p(y_1|\theta) = \int p(y_1|x_1, \theta) p(x_1|\theta) dx_1$$

A.2 Markov Property and Conditional Independence

A.2.1 A simplification

Let's introduce two assumptions in the DGP:

Markov Property The latent state X_t only depends on its previous state X_{t-1} , which is

$$p(x_t|x_{1:t-1}, y_{1:t-1}, \theta) = p(x_t|x_{t-1}, \theta)$$

Conditional Independence Observation Y_t only depends on current state X_t , which is

$$p(y_t|x_{1:t}, y_{1:t-1}, \theta) = p(y_t|x_t, \theta)$$

With these two conditions, equation (134) will be simplified to

$$p(y_{1:T}|\theta) = \int \left[p(x_1|\theta) \prod_{t=2}^T p(x_t|x_{t-1}, \theta) \prod_{t=1}^T p(y_t|x_t, \theta) \right] dx_{1:T}$$

and equation (135) is rewritten as:

$$p(y_t|y_{1:t-1}, \theta) = \int p(y_t|x_t, \theta) p(x_t|x_{t-1}, \theta) p(x_{1:t-1}|y_{1:t-1}, \theta) dx_{1:t}$$

135 will now be the focused case. Continue,

$$\begin{aligned} p(y_t|y_{1:t-1}, \theta) &= \int \left[\int p(y_t|x_t, \theta) p(x_t|x_{t-1}, \theta) dx_t \right] p(x_{1:t-1}|y_{1:t-1}, \theta) dx_{1:t-1} \\ &= \int \left[\int p(y_t|x_t, \theta) p(x_t|x_{t-1}, \theta) dx_t \right] p(x_{t-1}|y_{1:t-1}, \theta) p(x_{1:t-2}|x_{t-1}, y_{1:t-1}, \theta) dx_{1:t-1} \end{aligned}$$

Because $p(x_{1:t-2}|x_{t-1}, y_{1:t-1}, \theta)$ is a conditional pdf, we can Integrating over $x_{1:t-2}$ first, and

$$\int p(x_{1:t-2}|x_{t-1}, y_{1:t-1}, \theta) dx_{1:t-2} = 1$$

Why? The definition of conditional probability is

$$p(x_{1:t-2}|x_{t-1}, y_{1:t-1}, \theta) = \frac{p(x_{1:t-2}, x_{t-1}, y_{1:t-1}|\theta)}{p(x_{t-1}, y_{1:t-1}|\theta)}$$

Taking integral,

$$\begin{aligned}\int p(x_{1:t-2}|x_{t-1}, y_{1:t-1}, \theta) dx_{1:t-2} &= \int \frac{p(x_{1:t-2}, x_{t-1}, y_{1:t-1}|\theta)}{p(x_{t-1}, y_{1:t-1}|\theta)} dx_{1:t-2} \\ &= \frac{p(x_{t-1}, y_{1:t-1}|\theta)}{p(x_{t-1}, y_{1:t-1}|\theta)} = 1\end{aligned}$$

So that

$$\begin{aligned}p(y_t|y_{1:t-1}, \theta) &= \int \left[\int p(y_t|x_t, \theta) p(x_t|x_{t-1}, \theta) dx_t \right] p(x_{t-1}|y_{1:t-1}, \theta) dx_{t-1} \\ &= \int p(y_t|x_t, \theta) \left[\int p(x_t|x_{t-1}, \theta) p(x_{t-1}|y_{1:t-1}, \theta) dx_{t-1} \right] dx_t \\ &= \int p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta) dx_t\end{aligned}$$

A.2.2 Application to likelihood

How do we make use of $p(y_t|y_{1:t-1}, \theta) = \int p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta) dx_t$? We already have $p(y_t|x_t, \theta)$ from conditional independence. So we need to deal with $p(x_t|y_{1:t-1}, \theta)$. Let's start with the equation below:

$$p(y_t|y_{1:t-1}, \theta) = \int p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta) dx_t$$

and a forward algorithm is applied.

The initial condition is (at $t = 1$)

$$p(x_1|y_{1:0}, \theta) = p(x_1|\theta)$$

$$p(y_1|\theta) = \int p(y_1|x_1, \theta) p(x_1|\theta) dx_1$$

Then, according to Bayesian theorem,

$$p(x_1|y_1, \theta) = \frac{p(y_1|x_1, \theta) p(x_1|\theta)}{p(y_1|\theta)}$$

Starting with $t = 2$, we will have the following three equations

$$\begin{aligned}p(x_2|y_1, \theta) &= \int p(x_2|x_1, \theta) p(x_1|y_1, \theta) dx_1 \\ p(y_2|y_1, \theta) &= \int p(y_2|x_2, \theta) p(x_2|y_1, \theta) dx_2 \\ p(x_2|y_{1:2}, \theta) &= \frac{p(y_2|x_2, \theta) p(x_2|y_1, \theta)}{p(y_2|y_1, \theta)}\end{aligned}$$

With $t \geq 2$,

$$p(x_t|y_{1:t-1}, \theta) = \int p(x_t|x_{t-1}, \theta) p(x_{t-1}|y_{1:t-1}, \theta) dx_{t-1} \quad (136)$$

$$p(y_t|y_{1:t-1}, \theta) = \int p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta) dx_t \quad (137)$$

$$p(x_t|y_{1:t}, \theta) = \frac{p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta)}{p(y_t|y_{1:t-1}, \theta)} \quad (138)$$

A summary is that, during the derivation, we have two products:

1. The likelihood $p(y_t|y_{1:t-1}, \theta)$
2. Estimation of hidden state X_t given observations $Y_{1:t}$

A.3 Analytically Estimate Likelihood

Loglikelihood generated from either joint distribution or chain rule requires computing integral. “Analytically” means computing this with a closed-form expression—no numerical approximations or sampling. First, let’s keep **Markov Property** and **Conditional Independence**. The difficulty of analytically estimating the log likelihood lies in (136) and (137), which requires computing the infinite integral.

A.3.1 Linear Gaussian Model

To make these two infinite integral has closed form solution, three new assumptions are introduced:

1. initial state X_1 is gaussian.
2. The state space model is linear form
3. Noise is gaussian.

Formally,

$$\begin{aligned} X_t &= AX_{t-1} + W_t, & W_t &\sim N(0, Q) \\ Y_t &= CX_t + V_t, & V_t &\sim N(0, R) \end{aligned}$$

with initial state

$$X_1 \sim N(\mu_0, \Sigma_0)$$

A.3.2 Derivation

At $t = 1$ (136) is

$$p(x_1|y_{1:0}, \theta) = p(x_1|\theta) = N(x_1|\mu_0, \Sigma_0)$$

(137) is

$$p(y_1|y_{1:0}, \theta) = p(y_1|\theta) = \int p(y_1|x_1, \theta) p(x_1|\theta) dx_1$$

Since $Y_1 = CX_1 + V_1$, with X_1 and V_1 being Gaussian, Y_1 is Gaussian as well. Specifically,

$$\begin{aligned} E[Y_1] &= E[CX_1 + V_1] = C\mu_0 \\ Cov[Y_1] &= Cov[CX_1 + V_1] = C\Sigma_0C^T + R \end{aligned}$$

so that

$$p(y_1|\theta) = N(y_1|C\mu_0, C\Sigma_0C^T + R)$$

and

$$p(y_1|x_1, \theta) = N(y_1|Cx_1, R)$$

Lastly, (138) becomes

$$\begin{aligned} p(x_1|y_1, \theta) &= \frac{p(y_1|x_1, \theta) p(x_1, \theta)}{p(y_1|\theta)} \\ &= \frac{N(y_1|Cx_1, R)N(x_1|\mu_0, \Sigma_0)}{N(y_1|C\mu_0, C\Sigma_0C^T + R)} \\ &= N(x_1|\mu_1, \Sigma_1) \end{aligned}$$

because of the property of Gaussian distribution, and μ_1 and Σ_1 are obtained by the following:

$$\begin{aligned} N(y_1|Cx_1, R) &= \frac{1}{(2\pi)^{d_y/2} |R|^{1/2}} \exp\left(-\frac{1}{2} (y_1 - Cx_1)^T R^{-1} (y_1 - Cx_1)\right) \\ N(x_1|\mu_0, \Sigma_0) &= \frac{1}{(2\pi)^{d_x/2} |\Sigma_0|^{1/2}} \exp\left(-\frac{1}{2} (x_1 - \mu_0)^T \Sigma_0^{-1} (x_1 - \mu_0)\right) \\ N(y_1|C\mu_0, C\Sigma_0C^T + R) &= \frac{1}{(2\pi)^{d_y/2} |C\Sigma_0C^T + R|^{1/2}} \exp\left(-\frac{1}{2} (y_1 - C\mu_0)^T (C\Sigma_0C^T + R)^{-1} (y_1 - C\mu_0)\right) \end{aligned}$$

$N(y_1|C\mu_0, C\Sigma_0C^T + R)$ is constant relative to x_t . Neglecting other constant terms,

$$\begin{aligned} p(x_1|y_1, \theta) &\propto \exp\left(-\frac{1}{2}(y_1 - Cx_1)^T R^{-1}(y_1 - Cx_1) - \frac{1}{2}(x_1 - \mu_0)^T \Sigma_0^{-1}(x_1 - \mu_0)\right) \\ &= \exp\left(-\frac{1}{2}(x_1 - \mu_1)^T \Sigma_1^{-1}(x_1 - \mu_1)\right) \end{aligned}$$

which gives rise to

$$\begin{aligned} \Sigma_1 &= (\Sigma_0^{-1} + C^T R^{-1} C)^{-1} \\ \mu_1 &= \Sigma_1 C^T R^{-1} y_1 + \Sigma_1 \Sigma_0^{-1} \mu_0 \end{aligned}$$

It can be proved that **Kalman Filter** will yield identical result:

1. $e_1 = y_1 - C\mu_0$
2. $S_1 = C\Sigma_0C^T + R$
3. $K_1 = \Sigma_0C^TS_1^{-1}$
4. $\mu_1 = \mu_0 + K_1e_1$
5. $\Sigma_1 = \Sigma_0 - K_1C\Sigma_0$

Continue with $t \geq 2$ We already know, at $t - 1$, we have

$$p(x_{t-1}|y_{1:t-1}, \theta) = N(x_{t-1}|\mu_{t-1}, \Sigma_{t-1})$$

(136) now gives

$$p(x_t|y_{1:t-1}, \theta) = \int p(x_t|x_{t-1}, \theta) p(x_{t-1}|y_{1:t-1}, \theta) dx_{t-1}$$

with

$$\begin{aligned} p(x_t|x_{t-1}, \theta) &= N(x_t|Ax_{t-1}, Q) \\ p(x_{t-1}|y_{1:t-1}, \theta) &= N(x_{t-1}|\mu_{t-1}, \Sigma_{t-1}) \end{aligned}$$

Since $X_t = AX_{t-1} + W_t$, with X_{t-1} and W_t being Gaussian, X_t is Gaussian as well. Specifically,

$$p(x_t|y_{1:t-1}, \theta) = N(x_t|\hat{\mu}_t, \hat{\Sigma}_t)$$

with

$$\begin{aligned} \hat{\mu}_t &= E[X_t] = E[AX_{t-1} + W_t] = A\mu_{t-1} \\ \hat{\Sigma}_t &= Cov[X_t] = Cov[AX_{t-1} + W_t] = A\Sigma_{t-1}A^T + Q \end{aligned}$$

Next, (137) gives

$$p(y_t|y_{1:t-1}, \theta) = \int p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta) dx_t$$

with

$$\begin{aligned} p(y_t|x_t, \theta) &= N(y_t|Cx_t, R) \\ p(x_t|y_{1:t-1}, \theta) &= N(x_t|\hat{\mu}_t, \hat{\Sigma}_t) \end{aligned}$$

The first Gaussian term comes from the fact that $Y_t = CX_t + V_t$, and the second Gaussian comes from $X_t = AX_{t-1} + W_t$. Therefore, the likelihood is also Gaussian and specifically,

$$p(y_t|y_{1:t-1}, \theta) = N(y_t|C\hat{\mu}_t, C\hat{\Sigma}_tC^T + R)$$

Finally, to calculate (138), which is

$$p(x_t|y_{1:t}, \theta) = \frac{p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta)}{p(y_t|y_{1:t-1}, \theta)}$$

Kalman Filter can be applied as in the case of $t = 1$:

1. $e_t = y_t - C\hat{\mu}_t$
2. $S_t = C\hat{\Sigma}_t C^T + R$
3. $K_t = \hat{\Sigma}_t C^T S_t^{-1}$
4. $\mu_t = \hat{\mu}_t + K_t e_t$
5. $\Sigma_t = \hat{\Sigma}_t - K_t C \hat{\Sigma}_t$

Iteration During each iteration, we will have two products:

1. The likelihood $p(y_t|y_{1:t-1}, \theta)$
2. Estimation of hidden state X_t given observations $Y_{1:t}$, $p(x_t|y_{1:t}, \theta)$.

A.4 Numerically Estimate Likelihood

A.3.1 introduces three assumptions that make likelihood able to be estimated analytically. Without any of which the likelihood needs to be numerically calculated.

Now let's keep Markov Property and conditional independence, we have the following:

The initial condition at $t = 1$,

$$p(x_1|y_{1:0}, \theta) = p(x_1|\theta) \quad (139)$$

$$p(y_1|\theta) = \int p(y_1|x_1, \theta) p(x_1|\theta) dx_1 \quad (140)$$

$$p(x_1|y_1, \theta) = \frac{p(y_1|x_1, \theta) p(x_1|\theta)}{p(y_1|\theta)} \quad (141)$$

With $t \geq 2$,

$$p(x_t|y_{1:t-1}, \theta) = \int p(x_t|x_{t-1}, \theta) p(x_{t-1}|y_{1:t-1}, \theta) dx_{t-1} \quad (142)$$

$$p(y_t|y_{1:t-1}, \theta) = \int p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta) dx_t \quad (143)$$

$$p(x_t|y_{1:t}, \theta) = \frac{p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta)}{p(y_t|y_{1:t-1}, \theta)} \quad (144)$$

I will first discuss some properties of Dirac delta function, which will be used in later derivation.

Then, I will discuss the sampling process and the computation of loglikelihood $\log p(y_{1:T}|\theta)$ in SSM by using particle filtering without resampling.

Lastly, I will discuss the above under the case of using particle filtering with resampling.

A.4.1 Dirac function

This section discusses some preliminary on Dirac function. Dirac delta function is a generalized function defined as

$$\delta(x - x_0) = \begin{cases} +\infty & x = x_0 \\ 0 & x \neq x_0 \end{cases}$$

with property

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(x - x_0) dx &= 1 \\ \int_{-\infty}^{\infty} f(x) \delta(x - x_0) dx &= f(x_0) \end{aligned} \quad (145)$$

A.4.2 Particle Filter without resampling

Particle filter can be applied to above equations and numerically estimate likelihood and hidden states.

A derivation for PF without resampling

t=1: At $t = 1$, we can sample N particles from (139) to approximate it:

$$x_1^{(i)} \sim p(x_1|\theta), \quad i = 1, 2, \dots, N$$

so that

$$p(x_1|\theta) \approx \frac{1}{N} \sum_{i=1}^N \delta(x_1 - x_1^{(i)})$$

and (141) is approximated as

$$p(x_1|y_1, \theta) \approx \sum_{i=1}^N w_1^{(i)} \delta(x_1 - x_1^{(i)})$$

which means we are using particles with weights $(x_1^{(i)}, w_1^{(i)})$ to approximate the distribution of x_1 after observing y_1 . Where does $w_1^{(i)}$ come from? According to (141),

$$\begin{aligned} p(x_1|y_1, \theta) &= \frac{p(y_1|x_1, \theta) p(x_1|\theta)}{p(y_1|\theta)} = \frac{p(y_1|x_1, \theta) p(x_1|\theta)}{\int p(y_1|x_1, \theta) p(x_1|\theta) dx_1} \\ &\approx \frac{\frac{1}{N} \sum_{i=1}^N \delta(x_1 - x_1^{(i)}) p(y_1|x_1^{(i)}, \theta)}{\frac{1}{N} \sum_{j=1}^N p(y_1|x_1^{(j)}, \theta)} \\ &= \frac{\sum_{i=1}^N \delta(x_1 - x_1^{(i)}) p(y_1|x_1^{(i)}, \theta)}{\sum_{j=1}^N p(y_1|x_1^{(j)}, \theta)} \\ &= \sum_{i=1}^N \frac{p(y_1|x_1^{(i)}, \theta)}{\sum_{j=1}^N p(y_1|x_1^{(j)}, \theta)} \delta(x_1 - x_1^{(i)}) \end{aligned}$$

where

$$w_1^{(i)} = \frac{p(y_1|x_1^{(i)}, \theta)}{\sum_{j=1}^N p(y_1|x_1^{(j)}, \theta)}$$

and loglikelihood is approximated as

$$p(y_1|\theta) = \int p(y_1|x_1, \theta) p(x_1|\theta) dx_1 \approx \frac{1}{N} \sum_{j=1}^N p(y_1|x_1^j, \theta)$$

t>1: When $t \geq 2$, we make use of (142), (143), (144), and the property of Dirac function. First, we have obtained

$$p(x_{t-1}|y_{1:t-1}, \theta) \approx \sum_{i=1}^N w_{t-1}^{(i)} \delta(x_{t-1} - x_{t-1}^{(i)})$$

(142) can be approximated using particles,

$$\begin{aligned}
p(x_t|y_{1:t-1}, \theta) &= \int p(x_t|x_{t-1}, \theta) p(x_{t-1}|y_{1:t-1}, \theta) dx_{t-1} \\
&\approx \int p(x_t|x_{t-1}, \theta) \sum_{i=1}^N w_{t-1}^{(i)} \delta(x_{t-1} - x_{t-1}^{(i)}) dx_{t-1} \\
&= \sum_{i=1}^N w_{t-1}^{(i)} \int p(x_t|x_{t-1}, \theta) \delta(x_{t-1} - x_{t-1}^{(i)}) dx_{t-1}
\end{aligned}$$

Applying (145),

$$p(x_t|y_{1:t-1}, \theta) \approx \sum_{i=1}^N w_{t-1}^{(i)} p(x_t|x_{t-1}^{(i)}, \theta)$$

For each particle, we can sample it from $p(x_t|x_{t-1}^{(i)}, \theta)$:

$$x_t^{(i)} \sim p(x_t|x_{t-1}^{(i)}, \theta)$$

Therefore, (142) is finally approximated as

$$p(x_t|y_{1:t-1}, \theta) \approx \sum_{i=1}^N w_{t-1}^{(i)} \delta(x_t - x_t^{(i)})$$

Next, we apply (143)

$$\begin{aligned}
p(y_t|y_{1:t-1}, \theta) &= \int p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta) dx_t \\
&\approx \int \sum_{i=1}^N w_{t-1}^{(i)} p(y_t|x_t^{(i)}, \theta) \delta(x_t - x_t^{(i)}) dx_t \\
&= \sum_{i=1}^N w_{t-1}^{(i)} p(y_t|x_t^{(i)}, \theta)
\end{aligned}$$

where the last equation again uses the property of Dirac delta function in (145). Lastly, we apply (144),

$$\begin{aligned}
p(x_t|y_{1:t}, \theta) &= \frac{p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta)}{p(y_t|y_{1:t-1}, \theta)} \\
&\approx \frac{\sum_{i=1}^N w_{t-1}^{(i)} \delta(x_t - x_t^{(i)}) p(y_t|x_t^{(i)}, \theta)}{\sum_{j=1}^N w_{t-1}^{(j)} p(y_t|x_t^{(j)}, \theta)} \\
&= \sum_{i=1}^N \frac{w_{t-1}^{(i)} p(y_t|x_t^{(i)}, \theta)}{\sum_{j=1}^N w_{t-1}^{(j)} p(y_t|x_t^{(j)}, \theta)} \delta(x_t - x_t^{(i)})
\end{aligned}$$

so that we have a new weight

$$w_t^{(i)} = \frac{w_{t-1}^{(i)} p(y_t|x_t^{(i)}, \theta)}{\sum_{j=1}^N w_{t-1}^{(j)} p(y_t|x_t^{(j)}, \theta)}$$

Compute log likelihood through particle filter without resampling The log likelihood is

$$\log(y_{1:T}|\theta) = \sum_{t=1}^T \log p(y_t|y_{1:t-1}, \theta)$$

which can be iteratively calculated.

The workflow goes as below:

0. Initialize $l_0 = 0$
1. Initialize weight $w_0^{(i)} = 1/N$, sample $x_1^{(i)} \sim p(x_1|\theta)$
2. Obtain the first loglikelihood

$$\log p(y_1|\theta) = \log \left[\frac{1}{N} \sum_{j=1}^N p(y_1|x_1^{(j)}, \theta) \right]$$

$$l_1 \leftarrow l_0 + \log p(y_1|\theta)$$

3. Update w_0 to w_1 and

$$p(x_1|y_1, \theta) \approx \sum_{i=1}^N w_1^{(i)} \delta(x_1 - x_1^{(i)})$$

with

$$w_1^{(i)} = \frac{p(y_1|x_1^{(i)}, \theta)}{\sum_{j=1}^N p(y_1|x_1^{(j)}, \theta)}$$

4. For $t \geq 2$, sample $x_t^{(i)} \sim p(x_t|x_{t-1}^{(i)}, \theta)$
5. Obtain the likelihood

$$\log p(y_t|y_{1:t-1}, \theta) = \log \left(\sum_{i=1}^N w_{t-1}^{(i)} p(y_t|x_t^{(i)}, \theta) \right)$$

$$l_t \leftarrow l_{t-1} + \log p(y_t|y_{1:t-1}, \theta)$$

6. Update the weight $w_t^{(i)}$ and the distribution of x_t conditional on observation $y_{1:t}$.

$$p(x_t|y_{1:t}, \theta) \approx \sum_{i=1}^N w_t^{(i)} \delta(x_t - x_t^{(i)})$$

$$w_t^{(i)} = \frac{w_{t-1}^{(i)} p(y_t|x_t^{(i)}, \theta)}{\sum_{j=1}^N w_{t-1}^{(j)} p(y_t|x_t^{(j)}, \theta)}$$

7. Iterate from step 4 to 6 until periods T is reached, with l_T as the final log likelihood.

A.4.3 Particle filters with resampling

Full step of particle filter includes an extra step, resampling. The reason is particle degeneracy. Specifically, during each iteration,

$$w_t^{(i)} = \frac{w_{t-1}^{(i)} p(y_t|x_t^{(i)}, \theta)}{\sum_{j=1}^N w_{t-1}^{(j)} p(y_t|x_t^{(j)}, \theta)}$$

will lead to that weight of one particle j become close to 1, $w_t^j \approx 1$, and weights of all other particles become close to zero: $w_t^{i \neq j} \approx 0$.

Resampling Resampling particles in each iteration helps mitigate this problem, but the computation of likelihood function $p(y_t|y_{1:t-1}, \theta)$ will be different. Specifically, resampling takes a set of particles $\{x_t^{(i)}\}_{i=1}^N$ with weights $\{w_t^{(i)}\}_{i=1}^N$ (where $\sum_{i=1}^N w_t^{(i)} = 1$), and produces a new set of particles $\{x_t^{*(i)}\}_{i=1}^N$ with equal weights $\{w_t^{*(i)} = \frac{1}{N}\}_{i=1}^N$. The workflow of traditional resampling is as below:

1. Input $\{x_t^{(i)}\}_{i=1}^N$ with weights $\{w_t^{(i)}\}_{i=1}^N$
2. Compute cumulative sum of weights $s_j = \sum_{i=1}^j w_t^{(i)}$ for $j = 1, 2, \dots, N$ with $s_0 = 0$ and $s_1 = 1$.
3. Sampling indices: first, draw a single uniform random variable $u \sim U[0, 1/N]$. Then, compute $u_k = u + \frac{k-1}{N}$ for k from 1 to N . Then, find the smallest s_j such that $s_j \geq u_k$. Finally, select $x_t^{*(k)} = x_t^{(j)}$.
4. Output $\{x_t^{*(i)}\}_{i=1}^N$ with equal weights $\{w_t^{*(i)} = \frac{1}{N}\}_{i=1}^N$

Note that this is only one method of resampling and there are other techniques for resampling.

Iteration steps and likelihood function Let's now consider the case in which particles will be resampled after we have obtained the normalized weight, $w_t^{(i)}$, every iteration.

At $t = 1$,

0. Initialize $l_0 = 0$
1. Sample $x_1^{(i)} \sim p(x_1|\theta)$
2. Calculate

$$p(y_1|\theta) = \frac{1}{N} \sum_{j=1}^N p(y_1|x_1^{(j)}, \theta)$$

$$l_1 \leftarrow l_0 + \log p(y_1|\theta)$$

3. Update w_0 to w_1 and

$$p(x_1|y_1, \theta) \approx \sum_{i=1}^N w_1^{(i)} \delta(x_1 - x_1^{(i)})$$

with

$$w_1^{(i)} = \frac{p(y_1|x_1^{(i)}, \theta)}{\sum_{j=1}^N p(y_1|x_1^{(j)}, \theta)}$$

4. Resampling: select particles $x_1^{*(i)}$ according to $w_1^{(i)}$, and generate same numbers of particles. Reset all particles weight to $w_1^{*(i)} = 1/N$.

5. After resampling,

$$p(x_1^*|y_1, \theta) \approx \frac{1}{N} \sum_{i=1}^N \delta(x_1 - x_1^{*(i)})$$

At $t \geq 2$,

1. Sample $x_t^{(i)} \sim p(x_t|x_{t-1}^{*(i)}, \theta)$, which means we are using the resampled particles from last period to generate current particles that is yet to be resampled. The weight for each $x_t^{(i)}$ is the same as $x_{t-1}^{*(i)}$ and is $1/N$. This weight will be used in step 3 below, where we apply equation (144).

2. Since $w_{t-1}^{*(i)} = 1/N$, the likelihood is calculated as

$$\begin{aligned}
p(y_t|y_{1:t-1}, \theta) &= \int p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta) dx_t \\
&\approx \int \sum_{i=1}^N w_{t-1}^{*(i)} p(y_t|x_t^{(i)}, \theta) \delta(x_t - x_t^{(i)}) dx_t \\
&= \frac{1}{N} \sum_{i=1}^N \int p(y_t|x_t^{(i)}, \theta) \delta(x_t - x_t^{(i)}) dx_t \\
&= \frac{1}{N} \sum_{i=1}^N p(y_t|x_t^{(i)}, \theta) \\
l_t &\leftarrow l_{t-1} + \log p(y_t|y_{1:t-1}, \theta)
\end{aligned}$$

3. Update the weight $w_t^{(i)}$ and the distribution of x_t conditional on observation $y_{1:t}$ according to equation (144).

$$\begin{aligned}
p(x_t|y_{1:t}, \theta) &= \frac{p(y_t|x_t, \theta) p(x_t|y_{1:t-1}, \theta)}{p(y_t|y_{1:t-1}, \theta)} \\
&\approx \frac{\frac{1}{N} \sum_{i=1}^N \delta(x_t - x_t^{(i)}) p(y_t|x_t^{(i)}, \theta)}{\frac{1}{N} \sum_{j=1}^N p(y_t|x_t^{(j)}, \theta)} \\
&= \sum_{i=1}^N \frac{p(y_t|x_t^{(i)}, \theta)}{\sum_{j=1}^N p(y_t|x_t^{(j)}, \theta)} \delta(x_t - x_t^{(i)}) \\
&= \sum_{i=1}^N w_t^{(i)} \delta(x_t - x_t^{(i)}) \\
w_t^{(i)} &= \frac{p(y_t|x_t^{(i)}, \theta)}{\sum_{j=1}^N p(y_t|x_t^{(j)}, \theta)}
\end{aligned}$$

4. Resampling: select particles $x_t^{*(i)}$ according to $w_t^{(i)}$, and generate same numbers of particles. Reset all particles weight to $w_t^{*(i)} = 1/N$.

5. After resampling

$$p(x_t^*|y_{1:t}, \theta) \approx \frac{1}{N} \sum_{i=1}^N \delta(x_t - x_t^{*(i)})$$

6. Iterate from step 1 to 5 until periods T is reached, with l_T as the final log likelihood.

A brief summary In the case with resampling, the spirit of calculating the likelihood is similar to the case without resampling. The subtle difference to be noticed is that at time t , the weight from last iteration becomes $w_{t-1}^{*(i)} = 1/N$, while we still need to use particles before resampling at this period $x_t^{(i)}$ to calculate the likelihood, rather than $x_t^{*(i)}$.

Also, the resampling process breaks the continuity of log likelihood in θ , because a minor change in θ will abruptly change the resampled particles from the last period, leading to a different likelihood:

$$\begin{aligned}
p(y_t|y_{1:t-1}, \theta) &= \frac{1}{N} \sum_{i=1}^N p(y_t|x_t^{(i)}, \theta) \\
x_t^{(i)} &\sim p(x_t|x_{t-1}^{*(i)}, \theta)
\end{aligned}$$

As a result, $\log p(y_{1:T}|\theta)$ becomes not differentiable with respect to θ . Therefore, we can use MCMC to estimate the posterior, or further extend the estimation using HMC with differentiable particle filters.

B Equations for Experimental Simulator

B.1 The Dynamics of Variables Following Accounting Identity

Variables on the left-hand side of the equations follow accounting identities. Variables on the right-hand side are controlled by the firm or affected by the macroeconomy. Specifically, variables in red are considered as behaviors of a firm, and needs to be estimated. The dynamics are:

$$\begin{aligned}
MA_t &= MA_{t-1} + I_t^{MA} - S_t^{MA} - EDEP_t^{MA} \\
BU_t &= BU_{t-1} + I_t^{BU} - EDEP_t^{BU} \\
OFA_t &= OFA_{t-1} + dofa_t \\
CA_t &= CA_{t-1} + dca_t \\
SC_t &= SC_{t-1} + dsc_t \\
RR_t &= RR_{t-1} + drr_t \\
OUR_t &= OUR_{t-1} + dour_t \\
ASD_t &= ASD_{t-1} + (TDEP_t^{MA} - EDEP_t^{MA}) \\
CMA_t &= MA_t - ASD_t \\
p_t^{allo} &= \max[0, \min(p_t^{allo}, \eta \times pbase_t)] \\
MPA_t &= \max[0, \eta \times pbase_t] \\
dmpa_t &= MPA_t - p_t^{allo} \\
ddmpa_t &= dmpa_t - dmpa_{t-1} \\
PF_t &= PF_{t-1} + p_t^{allo} - zpf_t \\
LL_t &= LL_{t-1} + dll_t \\
CL_t &= CL_{t-1} + dcl_t \\
EBT_t &= OIBD_t - EDEP_t^{BU} + FI_t - FE_t - TDEP_t^{MA} - p_t^{allo} + zpf_t + OA_t \\
TA_t &= OTA_t - TDEP_t^{BU} - OL_{t-1} \\
TAX_t &= \tau \max[0, (EBT_t - TL_t + TA_t)] \\
FTAX_t &= TAX_t - ROT_t \\
NBI_t &= EBT_t - FTAX_t \\
OL_t &= \text{abs}(\min[0, (EBT_t - TL_t + TA_t)]) + OL_{t-1} \\
cashfl_t &= OIBD_t + FI_t - FE_t + OA_t - FTAX_t - DIV_{t-1} + dsc_t + dcl_t \\
&\quad + dll_t + dour_t - I_t^{MA} + S_t^{MA} - I_t^{BU} - dofa_t - dca_t \\
URE_t &= URE_{t-1} + NBI_t - DIV_{t-1} - drr_t - cashfl_t \\
mcash_t &= URE_{t-1} + NBI_t - drr_t \\
dmcash_t &= mcash_t - cashfl_t \\
ddmcash_t &= dmcash_t - dmcash_{t-1} \\
dmtdm_t &= MTDM_t - TDEP_t^{MA} \\
ddmtdm_t &= dmtdm_t - dmtdm_{t-1} \\
DIV_t &= \max[0, \min(cashfl_t, mcash_t)]
\end{aligned}$$

B.2 Variable Estimation

For the variables that are estimated, we use the following methods respectively

Huber-Schweppes Robust Estimation This is a one-stage estimation method. The idea is to estimate the parameters using a robust loss function.

Tobit This is a one stage estimation method. The idea is to estimate the parameters using a Tobit model.

Two stage Logistic with complementary log-log link This is a two stage estimation method. The first stage is to estimate the probability of behavior. In some cases, the behavior can be positive or negative, while in other cases, it can only be positive. The second stage is to estimate the parameters using a logistic model.

Two stage Logistic with cunmulative logit link This is a two stage estimation method. The first stage is to estimate the probability of behavior. The behavior can be positive or negative. In some cases, the behavior can be positive or negative, while in other cases, it can only be positive. The second stage is to estimate the parameters using a logistic model.

Multinomial Logit with complementary log-log link This is a two stage estimation method. The first stage is to estimate the probability of behavior. The behavior can be positive or negative. The third stage is to estimate the parameters using a multinomial logistic model.

The table below summarizes the methods used for each variable:

Dependent	Method	States	First state	Noise of first state	Second state	Noise of second state
EDEPMA (T1)	LLG	Two	Logistic	clog-log	Huber-Schweppes	Heavy tail
SMA (T2)	MUNO	Three	Multinomial	clog-log	Huber-Schweppes	Heavy tail
IMA (T3)	Tobit	One	Tobit 1	Logistic	N/A	N/A
EDEPBU (T4)	LLG	Two	Logistic	clog-log	Huber-Schweppes	Heavy tail
IBU (T5)	LLG	Two	Logistic	clog-log	Huber-Schweppes	Heavy tail
dofa (T6)	LSG	Three	Logistic	clog-log	Huber-Schweppes	Heavy tail
dca (T7)	HS	One	Huber-Schweppes	clog-log	N/A	N/A
dll (T8)	LLN	Two	Logistic	Cumulative Logistic	Huber-Schweppes	Heavy tail
dcl (T9)	HS	One	Huber-Schweppes	Heavy tail	N/A	N/A
dsc (T10)	LSG	Three	Logistic	clog-log	Huber-Schweppes	Heavy tail
drr (T11)	LLG	Two	Logistic	clog-log	Huber-Schweppes	Heavy tail
OIBD (T12)	HS	One	Huber-Schweppes	Heavy tail	N/A	N/A
FI (T13)	LLG	Two	Logistic	clog-log	Huber-Schweppes	Heavy tail
FE (T14)	LLG	Two	Logistic	clog-log	Huber-Schweppes	Heavy tail
TDEPMA (T15)	Tobit	One	Tobit 1	Logistic	N/A	N/A
ZPF (T16)	LLG	Two	Logistic	clog-log	Huber-Schweppes	Heavy tail
dour (T17)	MUNO	Three	Multinomial	clog-log	Huber-Schweppes	Heavy tail
GC (T18)	LSG	Three	Logistic	clog-log	Huber-Schweppes	Heavy tail
OA (T19)	MUNO	Three	Multinomial	clog-log	Huber-Schweppes	Heavy tail
TL (T20)	LLN	Two	Logistic	Cumulative Logistic	Huber-Schweppes	Heavy tail
OTA (T21)	MUNO	Three	Multinomial	clog-log	Huber-Schweppes	Heavy tail
TDEPBU (T22)	Tobit	One	Tobit 1	Logistic	N/A	N/A
Pallo (T23)	HS	One	Huber-Schweppes	Heavy tail	N/A	N/A
ROT (T24)	LLG	Two	Logistic	clog-log	Huber-Schweppes	Heavy tail

B.3 Dynamic of Behaviors

B.3.1 EDEPMA

First stage:

$$\begin{aligned}\eta_{EDEPMA} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 TDEP_{t-1}^{MA} + \beta_4 MA_{t-1} + \beta_5 I_{t-1}^{MA} + \beta_6 (I_{t-1}^{MA})^2 \\ & + \beta_7 EDEP_{t-1}^{BU} + \beta_8 (EDEP_{t-1}^{BU})^2 + \beta_9 ddmtdm_{t-1} + \beta_{10} (ddmtdm_{t-1})^2 + \beta_{11} dca_{t-1} \\ & + \beta_{12} ddmpa_{t-1} + \beta_{13} (ddmpa_{t-1})^2 + \beta_{14} dcl_{t-1} + \beta_{15} dgnpt + \beta_{16} FAAB_t \\ & + \beta_{17} Public_t + \beta_{18} ruralare_t + \beta_{19} largcity_t + \beta_{20} market_t + \beta_{21} marketw_t\end{aligned}$$

Second stage:

$$\begin{aligned}EDEP_t^{MA} = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 TDEP_{t-1}^{MA} + \gamma_4 MA_{t-1} + \gamma_5 I_{t-1}^{MA} + \gamma_6 (I_{t-1}^{MA})^2 \\ & + \gamma_7 EDEP_{t-1}^{BU} + \gamma_8 (EDEP_{t-1}^{BU})^2 + \gamma_9 ddmtdm_{t-1} + \gamma_{10} (ddmtdm_{t-1})^2 + \gamma_{11} dca_{t-1} \\ & + \gamma_{12} ddmpa_{t-1} + \gamma_{13} (ddmpa_{t-1})^2 + \gamma_{14} dcl_{t-1} + \gamma_{15} dgnpt + \gamma_{16} FAAB_t \\ & + \gamma_{17} Public_t + \gamma_{18} ruralare_t + \gamma_{19} largcity_t + \gamma_{20} market_t + \gamma_{21} marketw_t\end{aligned}$$

B.3.2 SMA

First stage:

$$\begin{aligned}\eta = & Intercept_1 + Intercept_2 + \beta_{1,1} ddmcash_{t-1} + \beta_{2,1} dcash_{t-1} + \beta_{3,1} TDEP_{t-1}^{MA} + \beta_{4,1} EDEP_t^{MA} \\ & + \beta_{5,1} (EDEP_t^{MA})^2 + \beta_{6,1} MA_{t-1} + \beta_{7,1} I_{t-1}^{BU} + \beta_{8,1} (I_{t-1}^{BU})^2 \\ & + \beta_{9,1} EDEP_{t-1}^{BU} + \beta_{10,1} (EDEP_{t-1}^{BU})^2 + \beta_{11,1} ddmtdm_{t-1} + \beta_{12,1} (ddmtdm_{t-1})^2 \\ & + \beta_{13,1} dca_{t-1} + \beta_{14,1} ddmpa_{t-1} + \beta_{15,1} (ddmpa_{t-1})^2 + \beta_{16,1} dcl_{t-1} \\ & + \beta_{17,1} dgnpt + \beta_{18,1} FAAB_t + \beta_{19,1} Public_t + \beta_{20,1} ruralare_t \\ & + \beta_{21,1} largcity_t + \beta_{22,1} market_t + \beta_{23,1} marketw_t\end{aligned}$$

Second stage for positive sales:

$$\begin{aligned}S_t^{MA} = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 EDEP_t^{MA} + \gamma_4 MA_{t-1} + \gamma_5 I_{t-1}^{BU} + \gamma_6 (I_{t-1}^{BU})^2 \\ & + \gamma_7 EDEP_{t-1}^{BU} + \gamma_8 (EDEP_{t-1}^{BU})^2 + \gamma_9 ddmtdm_{t-1} + \gamma_{10} dca_{t-1} + \gamma_{11} ddmpa_{t-1} \\ & + \gamma_{12} (ddmpa_{t-1})^2 + \gamma_{13} dcl_{t-1} + \gamma_{14} (dcl_{t-1})^2 + \gamma_{15} dgnpt + \gamma_{16} FAAB_t \\ & + \gamma_{17} Public_t + \gamma_{18} ruralare_t + \gamma_{19} largcity_t + \gamma_{20} market_t + \gamma_{21} marketw_t\end{aligned}$$

Second stage for negative sales:

$$\begin{aligned}S_t^{MA} = & \delta_0 + \delta_1 ddmcash_{t-1} + \delta_2 dcash_{t-1} + \delta_3 dcat_{t-1} + \delta_4 dcl_{t-1} + \delta_5 TDEP_{t-1}^{MA} + \delta_6 EDEP_t^{MA} \\ & + \delta_7 (EDEP_t^{MA})^2 + \delta_8 MA_{t-1} + \delta_9 I_{t-1}^{BU} + \delta_{10} EDEP_{t-1}^{BU} + \delta_{11} (EDEP_{t-1}^{BU})^2 \\ & + \delta_{12} ddmtdm_{t-1} + \delta_{13} ddmpa_{t-1} + \delta_{14} dgnpt + \delta_{15} FAAB_t + \delta_{16} Public_t \\ & + \delta_{17} ruralare_t + \delta_{18} largcity_t + \delta_{19} market_t + \delta_{20} marketw_t\end{aligned}$$

B.3.3 IMA

$$\begin{aligned}I_t^{MA*} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 S_t^{MA} + \beta_4 I_{t-1}^{BU} + \beta_5 EDEP_{t-1}^{BU} + \beta_6 (EDEP_{t-1}^{BU})^2 \\ & + \beta_7 EDEP_t^{MA} + \beta_8 TDEP_{t-1}^{MA} + \beta_9 (TDEP_{t-1}^{MA})^2 + \beta_{10} ddmtdm_{t-1} + \beta_{11} dca_{t-1} \\ & + \beta_{12} ddmpa_{t-1} + \beta_{13} (ddmpa_{t-1})^2 + \beta_{14} dcl_{t-1} + \beta_{15} dgnpt + \beta_{16} FAAB_t \\ & + \beta_{17} Public_t + \beta_{18} ruralare_t + \beta_{19} largcity_t + \beta_{20} market_t + \beta_{21} marketw_t \\ I_t^{MA} = & \max(0, I_t^{MA*})\end{aligned}$$

B.3.4 EDEPB

First stage:

$$\begin{aligned}\eta_{EDEPB} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 EDEP_t^{MA} + \beta_4 (EDEP_t^{MA})^2 + \beta_5 S_t^{MA} + \beta_6 I_t^{MA} \\ & + \beta_7 BU_{t-1} + \beta_8 (BU_{t-1})^2 + \beta_9 dca_{t-1} + \beta_{10} ddmpa_{t-1} + \beta_{11} dcl_{t-1} \\ & + \beta_{12} dgnp_t + \beta_{13} FAAB_t + \beta_{14} Public_t + \beta_{15} ruralare_t + \beta_{16} largcity_t \\ & + \beta_{17} market_t + \beta_{18} marketw_t\end{aligned}$$

Second stage

$$\begin{aligned}EDEP_t^{BU} = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 dcat_{t-1} + \gamma_4 dcl_{t-1} + \gamma_5 EDEP_t^{MA} + \gamma_6 (EDEP_t^{MA})^2 \\ & + \gamma_7 S_t^{MA} + \gamma_8 (S_t^{MA})^2 + \gamma_9 I_t^{MA} + \gamma_{10} BU_{t-1} + \gamma_{11} ddmpa_{t-1} \\ & + \gamma_{12} dgnp_t + \gamma_{13} FAAB_t + \gamma_{14} Public_t + \gamma_{15} ruralare_t + \gamma_{16} largcity_t \\ & + \gamma_{17} market_t + \gamma_{18} marketw_t\end{aligned}$$

B.3.5 IBU

First stage

$$\begin{aligned}\eta_{IBU} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 EDEP_t^{MA} + \beta_4 (EDEP_t^{MA})^2 + \beta_5 S_t^{MA} + \beta_6 I_t^{MA} \\ & + \beta_7 EDEP_t^{BU} + \beta_8 (EDEP_t^{BU})^2 + \beta_9 dca_{t-1} + \beta_{10} ddmpa_{t-1} + \beta_{11} (ddmpa_{t-1})^2 \\ & + \beta_{12} dcl_{t-1} + \beta_{13} dgnp_t + \beta_{14} FAAB_t + \beta_{15} Public_t + \beta_{16} ruralare_t \\ & + \beta_{17} largcity_t + \beta_{18} market_t + \beta_{19} marketw_t\end{aligned}$$

Second stage

$$\begin{aligned}I_t^{BU} = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 dcat_{t-1} + \gamma_4 dcl_{t-1} + \gamma_5 EDEP_t^{MA} + \gamma_6 (EDEP_t^{MA})^2 \\ & + \gamma_7 S_t^{MA} + \gamma_8 I_t^{MA} + \gamma_9 EDEP_t^{BU} + \gamma_{10} (EDEP_t^{BU})^2 + \gamma_{11} ddmpa_{t-1} \\ & + \gamma_{12} dgnp_t + \gamma_{13} FAAB_t + \gamma_{14} Public_t + \gamma_{15} ruralare_t + \gamma_{16} largcity_t \\ & + \gamma_{17} market_t + \gamma_{18} marketw_t\end{aligned}$$

B.3.6 dofa

First stage

$$\begin{aligned}\eta_{dofa,pos} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 ddmpa_{t-1} + \beta_4 (ddmpa_{t-1})^2 + \beta_5 (ddmpa_{t-1})^3 \\ & + \beta_6 DI_t^{MA} + \beta_7 DI_t^{BU} + \beta_8 realr_t + \beta_9 FAAB_t + \beta_{10} Public_t \\ & + \beta_{11} ruralare_t + \beta_{12} largcity_t + \beta_{13} market_t + \beta_{14} marketw_t\end{aligned}$$

$$\begin{aligned}\eta_{dofa,neg} = & \delta_0 + \delta_1 ddmcash_{t-1} + \delta_2 dcash_{t-1} + \delta_3 ddmpa_{t-1} + \delta_4 (ddmpa_{t-1})^2 + \delta_5 DI_t^{MA} \\ & + \delta_6 DI_t^{BU} + \delta_7 realr_t + \delta_8 FAAB_t + \delta_9 Public_t + \delta_{10} ruralare_t \\ & + \delta_{11} largcity_t + \delta_{12} market_t + \delta_{13} marketw_t\end{aligned}$$

Second stage for positive level

$$\begin{aligned}dofa_t = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 ddmpa_{t-1} + \gamma_4 (ddmpa_{t-1})^2 + \gamma_5 (ddmpa_{t-1})^3 \\ & + \gamma_6 DI_t^{MA} + \gamma_7 DI_t^{BU} + \gamma_8 realr_t + \gamma_9 FAAB_t + \gamma_{10} Public_t \\ & + \gamma_{11} ruralare_t + \gamma_{12} largcity_t + \gamma_{13} market_t + \gamma_{14} marketw_t\end{aligned}$$

Second stage for negative level

$$\begin{aligned}
dofa_t = & \lambda_0 + \lambda_1 ddmcash_{t-1} + \lambda_2 dcash_{t-1} + \lambda_3 ddmpa_{t-1} + \lambda_4 (ddmpa_{t-1})^2 + \lambda_5 DI_t^{MA} \\
& + \lambda_6 DI_t^{BU} + \lambda_7 realr_t + \lambda_8 FAAB_t + \lambda_9 Public_t + \lambda_{10} ruralare_t \\
& + \lambda_{11} largcity_t + \lambda_{12} market_t + \lambda_{13} marketw_t
\end{aligned}$$

B.3.7 dca

$$\begin{aligned}
dca_t = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 EDEP_t^{MA} + \gamma_4 (EDEP_t^{MA})^2 + \gamma_5 S_t^{MA} + \gamma_6 I_t^{MA} \\
& + \gamma_7 EDEP_t^{BU} + \gamma_8 (EDEP_t^{BU})^2 + \gamma_9 I_t^{BU} + \gamma_{10} (I_t^{BU})^2 + \gamma_{11} (I_t^{BU})^3 \\
& + \gamma_{12} dcl_t + \gamma_{13} ddmpa_{t-1} + \gamma_{14} (ddmpa_{t-1})^2 + \gamma_{15} dgnp_t + \gamma_{16} FAAB_t \\
& + \gamma_{17} Public_t + \gamma_{18} ruralare_t + \gamma_{19} largcity_t + \gamma_{20} market_t + \gamma_{21} marketw_t
\end{aligned}$$

B.3.8 dll

First stage

$$\begin{aligned}
\eta_{dll} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 ddmpa_{t-1} + \beta_4 (ddmpa_{t-1})^2 + \beta_5 (ddmpa_{t-1})^3 \\
& + \beta_6 DI_t^{MA} + \beta_7 DI_t^{BU} + \beta_8 Ddofa_t + \beta_9 realr_t + \beta_{10} FAAB_t \\
& + \beta_{11} Public_t + \beta_{12} ruralare_t + \beta_{13} largcity_t + \beta_{14} market_t + \beta_{15} marketw_t
\end{aligned}$$

Second stage

$$\begin{aligned}
dll_t = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 ddmpa_{t-1} + \gamma_4 (ddmpa_{t-1})^2 + \gamma_5 (ddmpa_{t-1})^3 \\
& + \gamma_6 DI_t^{MA} + \gamma_7 DI_t^{BU} + \gamma_8 Ddofa_t + \gamma_9 realr_t + \gamma_{10} FAAB_t \\
& + \gamma_{11} Public_t + \gamma_{12} ruralare_t + \gamma_{13} largcity_t + \gamma_{14} market_t + \gamma_{15} marketw_t
\end{aligned}$$

B.3.9 dcl

$$\begin{aligned}
dcl_t = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 EDEP_t^{MA} + \gamma_4 (EDEP_t^{MA})^2 + \gamma_5 S_t^{MA} + \gamma_6 I_t^{MA} \\
& + \gamma_7 EDEP_t^{BU} + \gamma_8 (EDEP_t^{BU})^2 + \gamma_9 I_t^{BU} + \gamma_{10} (I_t^{BU})^2 + \gamma_{11} ddmpa_{t-1} \\
& + \gamma_{12} (ddmpa_{t-1})^2 + \gamma_{13} (ddmpa_{t-1})^3 + \gamma_{14} dca_t + \gamma_{15} dgnp_t + \gamma_{16} FAAB_t \\
& + \gamma_{17} Public_t + \gamma_{18} ruralare_t + \gamma_{19} largcity_t + \gamma_{20} market_t + \gamma_{21} marketw_t
\end{aligned}$$

B.3.10 dsc

First stage

$$\begin{aligned}
\eta_{dsc,pos} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 ddmpa_{t-1} + \beta_4 (ddmpa_{t-1})^2 + \beta_5 (ddmpa_{t-1})^3 \\
& + \beta_6 DI_t^{MA} + \beta_7 DI_t^{BU} + \beta_8 Ddofa_t + \beta_9 Ddll_t + \beta_{10} realr_t \\
& + \beta_{11} FAAB_t + \beta_{12} Public_t + \beta_{13} ruralare_t + \beta_{14} largcity_t + \beta_{15} market_t + \beta_{16} marketw_t
\end{aligned}$$

$$\begin{aligned}
\eta_{dsc,neg} = & \delta_0 + \delta_1 ddmcash_{t-1} + \delta_2 dcash_{t-1} + \delta_3 ddmpa_{t-1} + \delta_4 (ddmpa_{t-1})^2 + \delta_5 (ddmpa_{t-1})^3 \\
& + \delta_6 DI_t^{MA} + \delta_7 DI_t^{BU} + \delta_8 Ddofa_t + \delta_9 Ddll_t + \delta_{10} realr_t \\
& + \delta_{11} FAAB_t + \delta_{12} Public_t + \delta_{13} ruralare_t + \delta_{14} largcity_t + \delta_{15} market_t + \delta_{16} marketw_t
\end{aligned}$$

Second stage for positive level

$$\begin{aligned}
dsc_t = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 ddmpa_{t-1} + \gamma_4 (ddmpa_{t-1})^2 + \gamma_5 DI_t^{MA} \\
& + \gamma_6 DI_t^{BU} + \gamma_7 Ddofa_t + \gamma_8 Ddll_t + \gamma_9 realr_t + \gamma_{10} FAAB_t \\
& + \gamma_{11} Public_t + \gamma_{12} ruralare_t + \gamma_{13} largcity_t + \gamma_{14} market_t + \gamma_{15} marketw_t
\end{aligned}$$

Second stage for negative level

$$\begin{aligned} dsc_t = & \lambda_0 + \lambda_1 ddmcash_{t-1} + \lambda_2 dcash_{t-1} + \lambda_3 ddmpa_{t-1} + \lambda_4 DI_t^{MA} + \lambda_5 DI_t^{BU} \\ & + \lambda_6 Ddofa_t + \lambda_7 Ddll_t + \lambda_8 realr_t + \lambda_9 FAAB_t + \lambda_{10} Public_t \\ & + \lambda_{11} ruralare_t + \lambda_{12} largcity_t + \lambda_{13} market_t + \lambda_{14} marketw_t \end{aligned}$$

B.3.11 drr

First stage

$$\begin{aligned} \eta_{drr} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 (ddmcash_{t-1})^2 + \beta_3 DI_t^{MA} + \beta_4 DI_t^{BU} + \beta_5 Ddofa_t \\ & + \beta_6 Ddll_t + \beta_7 Ddsc_t + \beta_8 realr_t + \beta_9 FAAB_t + \beta_{10} Public_t \\ & + \beta_{11} ruralare_t + \beta_{12} largcity_t + \beta_{13} market_t + \beta_{14} marketw_t \end{aligned}$$

Second stage

$$\begin{aligned} drr_t = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 (ddmcash_{t-1})^2 + \gamma_3 DI_t^{MA} + \gamma_4 DI_t^{BU} + \gamma_5 Ddofa_t \\ & + \gamma_6 Ddll_t + \gamma_7 Ddsc_t + \gamma_8 realr_t + \gamma_9 FAAB_t + \gamma_{10} Public_t \\ & + \gamma_{11} ruralare_t + \gamma_{12} largcity_t + \gamma_{13} market_t + \gamma_{14} marketw_t \end{aligned}$$

B.3.12 OIBD

$$\begin{aligned} OIBD_t = & \gamma_0 + \gamma_1 CA_{t-1} + \gamma_2 CL_{t-1} + \gamma_3 MA_{t-1} + \gamma_4 I_t^{MA} + \gamma_5 S_t^{MA} \\ & + \gamma_6 EDEP_t^{MA} + \gamma_7 (EDEP_t^{MA})^2 + \gamma_8 BU_{t-1} + \gamma_9 I_t^{BU} + \gamma_{10} EDEP_t^{BU} \\ & + \gamma_{11} (EDEP_t^{BU})^2 + \gamma_{12} dca_t + \gamma_{13} (dca_t)^2 + \gamma_{14} ddmpa_{t-1} + \gamma_{15} (ddmpa_{t-1})^2 \\ & + \gamma_{16} (ddmpa_{t-1})^3 + \gamma_{17} dcash_{t-1} + \gamma_{18} (dcash_{t-1})^2 + \gamma_{19} dcl_t + \gamma_{20} dgnp_t \\ & + \gamma_{21} FAAB_t + \gamma_{22} Public_t + \gamma_{23} ruralare_t + \gamma_{24} largcity_t + \gamma_{25} market_t + \gamma_{26} marketw_t \end{aligned}$$

B.3.13 FI

First stage

$$\begin{aligned} \eta_{FI} = & \beta_0 + \beta_1 I_t^{BU} + \beta_2 EDEP_t^{MA} + \beta_3 (EDEP_t^{MA})^2 + \beta_4 S_t^{MA} + \beta_5 I_t^{MA} \\ & + \beta_6 (I_t^{MA})^2 + \beta_7 EDEP_t^{BU} + \beta_8 (EDEP_t^{BU})^2 + \beta_9 dca_t + \beta_{10} (dca_t)^2 \\ & + \beta_{11} dofa_t + \beta_{12} OFA_{t-1} + \beta_{13} CA_{t-1} + \beta_{14} MA_{t-1} + \beta_{15} BU_{t-1} \\ & + \beta_{16} realr_t + \beta_{17} FAAB_t + \beta_{18} Public_t + \beta_{19} ruralare_t + \beta_{20} largcity_t \\ & + \beta_{21} market_t + \beta_{22} marketw_t \end{aligned}$$

Second stage

$$\begin{aligned} FI_t = & \gamma_0 + \gamma_1 I_t^{BU} + \gamma_2 EDEP_t^{MA} + \gamma_3 (EDEP_t^{MA})^2 + \gamma_4 S_t^{MA} + \gamma_5 I_t^{MA} \\ & + \gamma_6 (I_t^{MA})^2 + \gamma_7 EDEP_t^{BU} + \gamma_8 (EDEP_t^{BU})^2 + \gamma_9 dca_t + \gamma_{10} (dca_t)^2 \\ & + \gamma_{11} dofa_t + \gamma_{12} OFA_{t-1} + \gamma_{13} CA_{t-1} + \gamma_{14} MA_{t-1} + \gamma_{15} BU_{t-1} \\ & + \gamma_{16} realr_t + \gamma_{17} FAAB_t + \gamma_{18} Public_t + \gamma_{19} ruralare_t + \gamma_{20} largcity_t \\ & + \gamma_{21} market_t + \gamma_{22} marketw_t \end{aligned}$$

B.3.14 FE

First stage

$$\begin{aligned}\eta_{FE} = & \beta_0 + \beta_1 I_t^{BU} + \beta_2 EDEP_t^{MA} + \beta_3 S_t^{MA} + \beta_4 I_t^{MA} + \beta_5 EDEP_t^{BU} \\ & + \beta_6 OFA_{t-1} + \beta_7 MA_{t-1} + \beta_8 BU_{t-1} + \beta_9 LL_{t-1} + \beta_{10} CA_{t-1} \\ & + \beta_{11} CL_{t-1} + \beta_{12} dca_t + \beta_{13} dcl_t + \beta_{14} dofa_t + \beta_{15} dll_t \\ & + \beta_{16} realr_t + \beta_{17} FAAB_t + \beta_{18} Public_t + \beta_{19} ruralare_t + \beta_{20} largcity_t \\ & + \beta_{21} market_t + \beta_{22} marketw_t\end{aligned}$$

Second stage

$$\begin{aligned}FE_t = & \gamma_0 + \gamma_1 I_t^{BU} + \gamma_2 EDEP_t^{MA} + \gamma_3 S_t^{MA} + \gamma_4 I_t^{MA} + \gamma_5 EDEP_t^{BU} \\ & + \gamma_6 OFA_{t-1} + \gamma_7 MA_{t-1} + \gamma_8 BU_{t-1} + \gamma_9 LL_{t-1} + \gamma_{10} CA_{t-1} \\ & + \gamma_{11} CL_{t-1} + \gamma_{12} dca_t + \gamma_{13} dcl_t + \gamma_{14} dofa_t + \gamma_{15} dll_t \\ & + \gamma_{16} realr_t + \gamma_{17} FAAB_t + \gamma_{18} Public_t + \gamma_{19} ruralare_t + \gamma_{20} largcity_t \\ & + \gamma_{21} market_t + \gamma_{22} marketw_t\end{aligned}$$

B.3.15 TDEPMA

$$\begin{aligned}TDEP_t^{MA*} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 S_t^{MA} + \beta_4 EDEP_t^{MA} + \beta_5 (EDEP_t^{MA})^2 \\ & + \beta_6 I_t^{MA} + \beta_7 (I_t^{MA})^2 + \beta_8 ddmpa_{t-1} + \beta_9 (ddmpa_{t-1})^2 + \beta_{10} (ddmpa_{t-1})^3 \\ & + \beta_{11} realr_t + \beta_{12} FAAB_t + \beta_{13} Public_t + \beta_{14} ruralare_t + \beta_{15} largcity_t \\ & + \beta_{16} market_t + \beta_{17} marketw_t \\ TDEP_t^{MA} = & \max(0, TDEP_t^{MA*})\end{aligned}$$

B.3.16 zpf

First stage

$$\begin{aligned}\eta_{ZPF} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 p_{t-1}^{allo} + \beta_4 ddmpa_{t-1} + \beta_5 (ddmpa_{t-1})^2 \\ & + \beta_6 (ddmpa_{t-1})^3 + \beta_7 DTDEP_t^{MA} + \beta_8 realr_t + \beta_9 FAAB_t + \beta_{10} Public_t \\ & + \beta_{11} ruralare_t + \beta_{12} largcity_t + \beta_{13} market_t + \beta_{14} marketw_t\end{aligned}$$

Second stage

$$\begin{aligned}zpf_t = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 p_{t-1}^{allo} + \gamma_4 ddmpa_{t-1} + \gamma_5 DTDEP_t^{MA} \\ & + \gamma_6 realr_t + \gamma_7 FAAB_t + \gamma_8 Public_t + \gamma_9 ruralare_t + \gamma_{10} largcity_t \\ & + \gamma_{11} market_t + \gamma_{12} marketw_t\end{aligned}$$

B.3.17 dour

First stage

$$\begin{aligned}\eta_1 = & Intercept_1 + Intercept_2 + \beta_{1,1} ddmcash_{t-1} + \beta_{2,1} dcash_{t-1} + \beta_{3,1} ddmpa_{t-1} + \beta_{4,1} (ddmpa_{t-1})^2 \\ & + \beta_{5,1} (ddmpa_{t-1})^3 + \beta_{6,1} DTDEP_t^{MA} + \beta_{7,1} Dzpf_t + \beta_{8,1} realr_t \\ & + \beta_{9,1} FAAB_t + \beta_{10,1} Public_t + \beta_{11,1} ruralare_t + \beta_{12,1} largcity_t \\ & + \beta_{13,1} market_t + \beta_{14,1} marketw_t\end{aligned}$$

Second stage for positive

$$\begin{aligned}dour_t = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 ddmpa_{t-1} + \gamma_4 (ddmpa_{t-1})^2 + \gamma_5 DTDEP_t^{MA} \\ & + \gamma_6 Dzpf_t + \gamma_7 realr_t + \gamma_8 FAAB_t + \gamma_9 Public_t + \gamma_{10} ruralare_t \\ & + \gamma_{11} largcity_t + \gamma_{12} market_t + \gamma_{13} marketw_t\end{aligned}$$

Second stage for negative

$$\begin{aligned}dour_t = & \delta_0 + \delta_1 ddmcash_{t-1} + \delta_2 dcash_{t-1} + \delta_3 ddmpa_{t-1} + \delta_4 DTDEP_t^{MA} + \delta_5 Dzpf_t \\ & + \delta_6 realr_t + \delta_7 FAAB_t + \delta_8 Public_t + \delta_9 ruralare_t + \delta_{10} largcity_t \\ & + \delta_{11} market_t + \delta_{12} marketw_t\end{aligned}$$

B.3.18 GC

First stage

$$\begin{aligned}\eta_{GC,pos} = & \beta_0 + \beta_1 OIBD_t + \beta_2 (OIBD_t)^2 + \beta_3 (OIBD_t)^3 + \beta_4 FI_t + \beta_5 FE_t \\ & + \beta_6 TDEP_t^{MA} + \beta_7 (TDEP_t^{MA})^2 + \beta_8 EDEP_t^{BU} + \beta_9 (EDEP_t^{BU})^2 + \beta_{10} zpf_t \\ & + \beta_{11} dour_t + \beta_{12} dgnp_t + \beta_{13} FAAB_t + \beta_{14} Public_t + \beta_{15} ruralare_t \\ & + \beta_{16} largcity_t + \beta_{17} market_t + \beta_{18} marketw_t\end{aligned}$$

$$\begin{aligned}\eta_{GC,neg} = & \delta_0 + \delta_1 OIBD_t + \delta_2 (OIBD_t)^2 + \delta_3 FI_t + \delta_4 FE_t + \delta_5 TDEP_t^{MA} \\ & + \delta_6 (TDEP_t^{MA})^2 + \delta_7 EDEP_t^{BU} + \delta_8 (EDEP_t^{BU})^2 + \delta_9 zpf_t + \delta_{10} dour_t \\ & + \delta_{11} dgnp_t + \delta_{12} FAAB_t + \delta_{13} Public_t + \delta_{14} ruralare_t + \delta_{15} largcity_t \\ & + \delta_{16} market_t + \delta_{17} marketw_t\end{aligned}$$

Second stage for positive

$$\begin{aligned}GC_t = & \gamma_0 + \gamma_1 OIBD_t + \gamma_2 (OIBD_t)^2 + \gamma_3 (OIBD_t)^3 + \gamma_4 FI_t + \gamma_5 FE_t \\ & + \gamma_6 TDEP_t^{MA} + \gamma_7 (TDEP_t^{MA})^2 + \gamma_8 EDEP_t^{BU} + \gamma_9 (EDEP_t^{BU})^2 + \gamma_{10} zpf_t \\ & + \gamma_{11} dour_t + \gamma_{12} dgnp_t + \gamma_{13} FAAB_t + \gamma_{14} Public_t + \gamma_{15} ruralare_t \\ & + \gamma_{16} largcity_t + \gamma_{17} market_t + \gamma_{18} marketw_t\end{aligned}$$

Second stage for negative

$$\begin{aligned}GC_t = & \lambda_0 + \lambda_1 OIBD_t + \lambda_2 (OIBD_t)^2 + \lambda_3 FI_t + \lambda_4 FE_t + \lambda_5 TDEP_t^{MA} \\ & + \lambda_6 (TDEP_t^{MA})^2 + \lambda_7 EDEP_t^{BU} + \lambda_8 (EDEP_t^{BU})^2 + \lambda_9 zpf_t + \lambda_{10} dour_t \\ & + \lambda_{11} dgnp_t + \lambda_{12} FAAB_t + \lambda_{13} Public_t + \lambda_{14} ruralare_t + \lambda_{15} largcity_t \\ & + \lambda_{16} market_t + \lambda_{17} marketw_t\end{aligned}$$

B.3.19 OA

First stage

$$\begin{aligned}\eta_1 = & Intercept_1 + Intercept_2 + \beta_{0,1} + \beta_{1,1} dour_t + \beta_{2,1} GC_t + \beta_{3,1} DTDEP_t^{MA} + \beta_{4,1} Dzpf_t \\ & + \beta_{5,1} realr_t + \beta_{6,1} FAAB_t + \beta_{7,1} Public_t + \beta_{8,1} ruralare_t \\ & + \beta_{9,1} largcity_t + \beta_{10,1} market_t + \beta_{11,1} marketw_t\end{aligned}$$

Second stage for positive

$$\begin{aligned}OA_t = & \gamma_0 + \gamma_1 dour_t + \gamma_2 GC_t + \gamma_3 DTDEP_t^{MA} + \gamma_4 Dzpf_t + \gamma_5 realr_t \\ & + \gamma_6 FAAB_t + \gamma_7 Public_t + \gamma_8 ruralare_t + \gamma_9 largcity_t + \gamma_{10} market_t + \gamma_{11} marketw_t\end{aligned}$$

Second stage for negative

$$\begin{aligned}OA_t = & \delta_0 + \delta_1 dour_t + \delta_2 GC_t + \delta_3 DTDEP_t^{MA} + \delta_4 Dzpf_t + \delta_5 realr_t \\ & + \delta_6 FAAB_t + \delta_7 Public_t + \delta_8 ruralare_t + \delta_9 largcity_t + \delta_{10} market_t + \delta_{11} marketw_t\end{aligned}$$

B.3.20 TL

First stage

$$\begin{aligned}\eta_{TL} = & \beta_0 + \beta_1 OIBD_t + \beta_2 (OIBD_t)^2 + \beta_3 FI_t + \beta_4 (FI_t)^2 + \beta_5 FE_t \\ & + \beta_6 (FE_t)^2 + \beta_7 TDEP_t^{MA} + \beta_8 (TDEP_t^{MA})^2 + \beta_9 EDEP_t^{BU} + \beta_{10} (EDEP_t^{BU})^2 \\ & + \beta_{11} dour_t + \beta_{12} (dour_t)^2 + \beta_{13} zpf_t + \beta_{14} p_{t-1}^{allo} + \beta_{15} dgnp_t \\ & + \beta_{16} FAAB_t + \beta_{17} Public_t + \beta_{18} ruralare_t + \beta_{19} largcity_t + \beta_{20} market_t + \beta_{21} marketw_t\end{aligned}$$

Second stage

$$\begin{aligned}TL_t = & \gamma_0 + \gamma_1 OIBD_t + \gamma_2 (OIBD_t)^2 + \gamma_3 FI_t + \gamma_4 (FI_t)^2 + \gamma_5 FE_t \\ & + \gamma_6 (FE_t)^2 + \gamma_7 TDEP_t^{MA} + \gamma_8 (TDEP_t^{MA})^2 + \gamma_9 EDEP_t^{BU} + \gamma_{10} (EDEP_t^{BU})^2 \\ & + \gamma_{11} dour_t + \gamma_{12} (dour_t)^2 + \gamma_{13} zpf_t + \gamma_{14} p_{t-1}^{allo} + \gamma_{15} dgnp_t \\ & + \gamma_{16} FAAB_t + \gamma_{17} Public_t + \gamma_{18} ruralare_t + \gamma_{19} largcity_t + \gamma_{20} market_t + \gamma_{21} marketw_t\end{aligned}$$

B.3.21 OTA

First stage

$$\begin{aligned}\eta_1 = & Intercept_1 + Intercept_2 + \beta_{1,1} p_{t-1}^{allo} + \beta_{2,1} zpf_t + \beta_{3,1} TDEP_t^{MA} + \beta_{4,1} (TDEP_t^{MA})^2 \\ & + \beta_{5,1} OIBD_t + \beta_{6,1} (OIBD_t)^2 + \beta_{7,1} EDEP_t^{BU} + \beta_{8,1} (EDEP_t^{BU})^2 \\ & + \beta_{9,1} dour_t + \beta_{10,1} TL_t + \beta_{11,1} FI_t + \beta_{12,1} FE_t \\ & + \beta_{13,1} dgnp_t + \beta_{14,1} FAAB_t + \beta_{15,1} Public_t + \beta_{16,1} ruralare_t \\ & + \beta_{17,1} largcity_t + \beta_{18,1} market_t + \beta_{19,1} marketw_t\end{aligned}$$

Second stage for positive

$$\begin{aligned}OTA_t = & \gamma_0 + \gamma_1 p_{t-1}^{allo} + \gamma_2 zpf_t + \gamma_3 TDEP_t^{MA} + \gamma_4 (TDEP_t^{MA})^2 + \gamma_5 OIBD_t \\ & + \gamma_6 (OIBD_t)^2 + \gamma_7 EDEP_t^{BU} + \gamma_8 (EDEP_t^{BU})^2 + \gamma_9 dour_t + \gamma_{10} TL_t \\ & + \gamma_{11} FI_t + \gamma_{12} FE_t + \gamma_{13} dgnp_t + \gamma_{14} FAAB_t + \gamma_{15} Public_t \\ & + \gamma_{16} ruralare_t + \gamma_{17} largcity_t + \gamma_{18} market_t + \gamma_{19} marketw_t\end{aligned}$$

Second stage for negative

$$\begin{aligned}OTA_t = & \delta_0 + \delta_1 p_{t-1}^{allo} + \delta_2 zpf_t + \delta_3 TDEP_t^{MA} + \delta_4 OIBD_t + \delta_5 (OIBD_t)^2 \\ & + \delta_6 EDEP_t^{BU} + \delta_7 (EDEP_t^{BU})^2 + \delta_8 dour_t + \delta_9 TL_t + \delta_{10} FI_t \\ & + \delta_{11} FE_t + \delta_{12} dgnp_t + \delta_{13} FAAB_t + \delta_{14} Public_t + \delta_{15} ruralare_t \\ & + \delta_{16} largcity_t + \delta_{17} market_t + \delta_{18} marketw_t\end{aligned}$$

B.3.22 TDEPBU

$$\begin{aligned}TDEP_t^{BU*} = & \beta_0 + \beta_1 ddmcash_{t-1} + \beta_2 dcash_{t-1} + \beta_3 EDEP_t^{MA} + \beta_4 (EDEP_t^{MA})^2 + \beta_5 S_t^{MA} \\ & + \beta_6 I_t^{MA} + \beta_7 BU_{t-1} + \beta_8 (BU_{t-1})^2 + \beta_9 dca_t + \beta_{10} (dca_t)^2 \\ & + \beta_{11} dcl_t + \beta_{12} ddmpa_{t-1} + \beta_{13} (ddmpa_{t-1})^2 + \beta_{14} (ddmpa_{t-1})^3 + \beta_{15} dgnp_t \\ & + \beta_{16} FAAB_t + \beta_{17} Public_t + \beta_{18} ruralare_t + \beta_{19} largcity_t + \beta_{20} market_t + \beta_{21} marketw_t \\ TDEP_t^{BU} = & \max(0, TDEP_t^{BU*})\end{aligned}$$

B.3.23 pallo

$$\begin{aligned} p_t^{allo} = & \gamma_0 + \gamma_1 ddmcash_{t-1} + \gamma_2 dcash_{t-1} + \gamma_3 zpf_t + \gamma_4 dmpa_{t-1} + \gamma_5 MPA_t \\ & + \gamma_6 realr_t + \gamma_7 FAAB_t + \gamma_8 Public_t + \gamma_9 ruralare_t + \gamma_{10} largcity_t \\ & + \gamma_{11} market_{t-1} + \gamma_{12} marketw_{t-1} \end{aligned}$$

B.3.24 ROT

First stage

$$\begin{aligned} \eta_{ROT} = & \beta_0 + \beta_1 p_t^{allo} + \beta_2 zpf_t + \beta_3 TDEP_t^{MA} + \beta_4 (TDEP_t^{MA})^2 + \beta_5 OIBD_t \\ & + \beta_6 (OIBD_t)^2 + \beta_7 EDEP_t^{BU} + \beta_8 (EDEP_t^{BU})^2 + \beta_9 OTA_t + \beta_{10} (OTA_t)^2 \\ & + \beta_{11} TDEP_t^{BU} + \beta_{12} (TDEP_t^{BU})^2 + \beta_{13} dour_t + \beta_{14} TL_t + \beta_{15} FI_t \\ & + \beta_{16} FE_t + \beta_{17} dgnp_t + \beta_{18} FAAB_t + \beta_{19} Public_t + \beta_{20} ruralare_t \\ & + \beta_{21} largcity_t + \beta_{22} market_t + \beta_{23} marketw_t \end{aligned}$$

Second stage

$$\begin{aligned} ROT_t = & \gamma_0 + \gamma_1 p_t^{allo} + \gamma_2 zpf_t + \gamma_3 OIBD_t + \gamma_4 EDEP_t^{BU} + \gamma_5 TDEP_t^{BU} \\ & + \gamma_6 dour_t + \gamma_7 TL_t + \gamma_8 FI_t + \gamma_9 FE_t + \gamma_{10} dgnp_t \\ & + \gamma_{11} FAAB_t + \gamma_{12} Public_t + \gamma_{13} ruralare_t + \gamma_{14} largcity_t + \gamma_{15} market_t + \gamma_{16} marketw_t \end{aligned}$$

B.4 Noise simulation

Suppose we have four moments: Mean, Variance, Skewness and Kurtosis. We want to find a distribution to fit these four moments. Two distribution are suggested: Gaussian mixture and JohnsonSU.

B.4.1 Definition of Moments

$$\begin{aligned} mean &= E(X) = \mu \\ Var &= E(X - \mu)^2 = \sigma^2 \\ Skew &= \frac{E(X - \mu)^3}{\sigma^3} \\ Kurt &= \frac{E(X - \mu)^4}{\sigma^4} \end{aligned}$$

B.4.2 JohnsonSU

Random variable X follows JohnsonSU distribution if it follows

$$Z = \gamma + \delta \sinh^{-1} \left(\frac{x - \xi}{\lambda} \right)$$

where $Z \sim N(0, 1)$. It has 4 parameters: ξ for location, λ for scale, γ and δ for shape,

Parameter	
ξ	location
λ	scale
γ	shape
δ	shape

We have four inputs, $\mu(x)$, $\sigma(x)$, $Skew$, $Kurt$. The algorithm is as follows

1. Define a variable called ω , whose value can be searched from an given interval. Define $\beta_2 = Kurt, \beta_1 = Skew^2$.

2. Find a lower boundary ω_1 , which is a positive root of $\omega^4 + 2\omega^3 + 3\omega^2 - 3 = \beta_2$. The solution is

$$\begin{aligned} D &= (3 + \beta_2) (16\beta_2^2 + 87\beta_2 + 171) / 27 \\ d &= -1 + \sqrt[3]{7 + 2\beta_2 + 2\sqrt{D}} - \sqrt[3]{2\sqrt{D} - 7 - 2\beta_2} \\ \omega_1 &= \frac{1}{2} \times \left(-1 + \sqrt{d} + \sqrt{\frac{4}{\sqrt{d}} - d - 3} \right) \end{aligned}$$

3. Define $f(\omega)$ as

$$\begin{aligned} m(\omega) &= -2 + \sqrt{4 + 2 \left[\omega^2 - \frac{\beta_2 + 3}{\omega^2 + 2\omega + 3} \right]} \\ f(\omega) &= (\omega - 1 - m(\omega)) \left(\omega + 2 + \frac{1}{2}m(\omega) \right)^2 \end{aligned}$$

4. If $(\omega_1 - 1)(\omega_1 + 2)^2 \leq \beta_1$, there is no solution, otherwise go to next step.

5. $\omega_2 = \sqrt{-1 + \sqrt{2(\beta_2 - 1)}}$.

6. Use a root finding procedure on the interval $(\omega_1, \omega_2]$ to find the desired ω^* that satisfies:

$$f(\omega^*) = \beta_1$$

7. With ω^* , we can recover the parameters according to the following:

$$\begin{aligned} m &= m(\omega^*) \\ \Omega &= -\text{sgn}(Skew) \sinh^{-1} \sqrt{\frac{\omega^* - 1}{2\omega^*} \left(\frac{\omega^* - 1}{m} - 1 \right)} \\ \delta &= 1/\sqrt{\ln \omega^*} \\ \gamma &= \Omega/\sqrt{\ln \omega^*} \\ \lambda &= \frac{\sigma(x)}{\omega^* - 1} \sqrt{\frac{2m}{\omega^* + 1}} \\ \xi &= \mu(x) - \text{sgn}(Skew) \frac{\sigma(x)}{\omega^* - 1} \sqrt{\omega^* - 1 - m} \end{aligned}$$

where

$$\text{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

B.4.3 Gaussian mixture

The JohnsonSU is the quickest way, but it only applies to cases with $(\omega_1 - 1)(\omega_1 + 2)^2 \leq \beta_1$. For the rest of the cases, we fit the following distribution

$$p(x|\theta) = \sum_{i=1}^K w_i N(x|\mu_i, \sigma_i^2)$$

where $\theta = (w_1, \dots, w_k, \mu_1, \dots, \mu_k, \sigma_1, \dots, \sigma_k)$. We minimize the loss:

$$\left\{ \begin{array}{c} \text{logit}(w_1), \dots, \text{logit}(w_{K-1}) \\ \mu_1, \dots, \mu_L \\ \sigma_1^2, \dots, \sigma_K^2 \end{array} \right\} = \arg \min \left[\left(\frac{\text{FittedSkew} - \text{TargetSkew}}{\text{FittedSkew}} \right)^2 + \left(\frac{\text{FittedKurt} - \text{TargetKurt}}{\text{FittedKurt}} \right)^2 \right]$$

and obtain $g(x)$.

Then, we do a linear transformation such that

$$y = ax + b$$

where

$$\begin{aligned} a &= \frac{\sigma_{\text{target}}}{\sigma_x} \\ b &= \mu_{\text{target}} - a\mu_x \end{aligned}$$

to make sure we reach our desired mean and variance. The final distribution to be used is

$$p_{\text{final}}(x|\theta) = a \times \sum_{i=1}^K w_i N(x|\mu_i, \sigma_i^2) + b$$

B.4.4 Gaussian mixture linear transformation proof

Note that we only need to fit Skewness and Kurtosis, and do a linear transform of the generated desired results without changing the Skewness and Kurtosis. Suppose Mean and Std of X is μ_x, σ_x , $y = ax + b$, $a > 0$

Proof:

Step 1: Obviously, with $\mu_y = a\mu_x + b$, $\sigma_y^2 = a^2\sigma_x^2$, $\sigma_y = |a|\sigma_x$, we know the target mean and variance are matched.

$$E(y) = E[ax + b] = a\mu_x + (\mu_{\text{target}} - a\mu_x) = \mu_{\text{target}}$$

$$\text{Var}(y) = \text{Var}(ax) = a^2 \text{Var}(x) = \frac{\sigma_{\text{target}}^2}{\sigma_x^2} \sigma_x^2 = \sigma_{\text{target}}^2$$

Step 2: For $n \geq 2$, higher moments are

$$\mu_{n,y} = E[(y - \mu_y)^n] = E[(ax + b - (a\mu_x + b))^n] = E[(a(x - \mu_x))^n] = a^n E[(x - \mu_x)^n] = a^n \mu_{n,x}$$

Step 3: Skew and Kurt

$$\text{skew}(y) = \frac{\mu_{3,y}}{\sigma_y^3} = \frac{a^3 \mu_{3,x}}{|a|^3 \sigma_x^3} = \text{sgn}(a) \text{skew}(x)$$

$$\text{kurt}(y) = \frac{\mu_{4,y}}{\sigma_y^4} = \frac{a^4 \mu_{4,x}}{|a|^4 \sigma_x^4} = \text{kurt}(x)$$

Step 4: Since $a = \sigma_{\text{target}}/\sigma_x > 0$, the results hold.

Step 5: y is still a random variable and follows a valid distribution. The CDF of y is

$$F_y(y) = P(y \leq y_0) = P(ax + b \leq y_0) = P\left(x \leq \frac{y_0 - b}{a}\right) = F_x\left(\frac{y_0 - b}{a}\right)$$

with pdf

$$f_y(y) = \frac{1}{a} f_x\left(\frac{y - b}{a}\right)$$

Let $z = \frac{y-b}{a}$, then $y = az + b$, $dy = adz$

$$\int_{-\infty}^{\infty} \frac{1}{a} f_x\left(\frac{y - b}{a}\right) dy = \int_{-\infty}^{\infty} \frac{1}{a} f_x\left(\frac{y - b}{a}\right) adz = \int_{-\infty}^{\infty} f_x(z) dz = 1$$

B.5 Simulation Methods

The simulation methods are used to either generate the synthetic data, or run the simulation model given the parameters. Suppose we have initial data at $t = 0$, we can recursively update the variables. When perform updates, we use different methods for each variable.

For Huber-Schweppes robust estimation model:

1. Generate error $\tilde{\epsilon}_t$ using Gaussian mixture or JohnsonSU
2. Compute

$$\tilde{y}_i = X_i' \hat{\beta} + \tilde{\epsilon}_t$$

For Tobit model:

1. Generate error $\tilde{\epsilon}_t = scale \times \ln\left(\frac{U}{1-U}\right)$ where $U \sim Uniform(0, 1)$
2. Compute latent variable

$$\tilde{y}_i^* = X_i' \hat{\beta} + \tilde{\epsilon}_t$$

3. Censor variable to obtain the final prediction

$$\tilde{y}_i = \max(0, \tilde{y}_i^*)$$

For logistic with cloglog model (LLG):

1. Generate error $\tilde{\epsilon}_t$ using Gaussian mixture or JohnsonSU
2. Check if a firm will make decision:

$$\hat{\eta} = X_i' \hat{\beta}_{1st}$$

$$\hat{P} = 1 - e^{-e^{\hat{\eta}}}$$

$$U \sim Uniform(0, 1)$$

- If $\hat{P} > U$, then make decision
 - If $\hat{P} < U$, then not make decision, and set = 0
3. Generate data for firms that make positive decision

$$\tilde{y}^* = X' \hat{\beta}_{2nd} + \epsilon_t$$

4. Censor variable to obtain the final prediction

$$\tilde{y} = \max(0, \tilde{y}^*)$$

For logistic with cumulative logistic model (LLN):

1. Generate error $\tilde{\epsilon}_t$ using Gaussian mixture or JohnsonSU
2. Check if a firm will make decision:

$$\hat{\eta} = X_i' \hat{\beta}_{1st}$$

$$\hat{P} = \frac{\exp(\hat{\eta})}{1 + \exp(\hat{\eta})}$$

$$U \sim Uniform(0, 1)$$

- If $\hat{P} > U$, then make decision
 - If $\hat{P} < U$, then not make decision, and set = 0
3. Generate data for firms that make positive decision

$$\tilde{y}^* = X' \hat{\beta}_{2nd} + \epsilon_t$$

4. Censor variable to obtain the final prediction

$$\tilde{y} = \max(0, \tilde{y}^*)$$

For multinomial logit with cloglog model (MUNO):

1. Generate error $\tilde{\epsilon}_t$ using Gaussian mixture or JohnsonSU
2. Create the thresholds for firms and decide firms' action

$$\hat{\eta}_1 = Intercept_1 + X' \hat{\beta}_{1st}$$

$$\hat{\eta}_2 = Intercept_2 + X' \hat{\beta}_{1st}$$

$$\hat{P}_1 = 1 - e^{-e^{\hat{\eta}_1}}$$

$$\hat{P}_2 = 1 - e^{-e^{\hat{\eta}_2}}$$

$$U \sim Uniform(0, 1)$$

where $Intercept_1$ and $Intercept_2$ come from the same estimation result table.

- If $U < \hat{P}_1$, then take negative action
 - If $\hat{P}_1 < U < \hat{P}_2$, then does not take any action, and set = 0
 - If $\hat{P}_2 < U$, then take positive action
3. Generate data for firms that make either positive or negative decision

$$\tilde{y}_i^* = X_i' \hat{\beta}_{2nd} + \epsilon_{it}$$

where $i = positive, negative$

4. Censor variable to obtain the final prediction

$$\tilde{y}_{pos} = \max(0, \tilde{y}_{pos}^*)$$

$$\tilde{y}_{neg} = \min(0, \tilde{y}_{neg}^*)$$

For logistic with cloglog model (LSG):

1. Generate error $\tilde{\epsilon}_t$ using Gaussian mixture or JohnsonSU
2. Create the thresholds for firms and decide firms' action

$$\hat{\eta}_1 = Intercept_{pos} + X'_{pos} \hat{\beta}_{1st,pos}$$

$$\hat{\eta}_2 = Intercept_{neg} + X'_{neg} \hat{\beta}_{1st,neg}$$

$$\hat{P}_1 = 1 - e^{-e^{\hat{\eta}_1}}$$

$$\hat{P}_2 = 1 - e^{-e^{\hat{\eta}_2}}$$

$$U \sim Uniform(0, 1)$$

where $Intercept_{pos}$ and X_{pos} come from one estimation result table and $Intercept_{neg}$ and X_{neg} come from the other estimation result table.

- If $U < \hat{P}_1$, then take positive action
 - If $\hat{P}_1 < U < \hat{P}_2$, then does not take any action, and set = 0
 - If $\hat{P}_2 < U$, then take negative action
3. Generate data for firms that make either positive or negative decision

$$\tilde{y}_i^* = X_i' \hat{\beta}_{2nd} + \epsilon_{it}$$

where $i = positive, negative$

4. Censor variable to obtain the final prediction

$$\tilde{y}_{pos} = \max(0, \tilde{y}_{pos}^*)$$

$$\tilde{y}_{neg} = \min(0, \tilde{y}_{neg}^*)$$

C Modelling ratio variable

C.1 Problem Statement

We aim to model a time-series variable, denoted as r_t , which represents a ratio or proportion. This variable is strictly bounded between 0 and 1, i.e., $r_t \in (0, 1)$. Our modeling objective is to satisfy the following two critical requirements:

1. The underlying dynamic process of the variable should be representable as a first-order autoregressive (AR(1)) process that is both **linear** and driven by **Gaussian noise**. This structure is essential for the direct application of standard, efficient algorithms like the Kalman filter.
2. The model's predictions and filtered estimates for the ratio, after any necessary transformations, must strictly adhere to the $(0, 1)$ bounds.

Directly modeling r_t as a standard AR(1) process, i.e., $r_t = c + \phi r_{t-1} + w_t$, is inappropriate because the Gaussian noise term w_t can push the value of r_t outside the $(0, 1)$ interval. Therefore, a transformation is required.

C.2 The Logit Transformation Method

The standard and most effective method to address this challenge is to perform the modeling in a transformed space. We use the **Logit function** to map the bounded interval $(0, 1)$ to the entire real line $(-\infty, +\infty)$. After establishing a linear Gaussian model in this unbounded space, we use the inverse transformation, the **Logistic (or Sigmoid) function**, to map the results back to the original $(0, 1)$ interval.

C.2.1 Step 1: The Logit Transformation

First, we define a new variable, x_t , which is the logit transformation of our original ratio r_t . This new variable x_t is often referred to as the *log-odds*.

$$x_t = \text{logit}(r_t) = \log\left(\frac{r_t}{1 - r_t}\right) \quad (146)$$

This function maps any value in $(0, 1)$ to a unique value in $(-\infty, +\infty)$. This transformed variable x_t can now be appropriately modeled by a process with Gaussian noise without violating any bounds.

C.2.2 Step 2: Modeling in the Transformed Space (The AR(1) Process)

Now that we have an unbounded variable x_t , we can posit that its dynamics follow a standard linear Gaussian AR(1) process. This x_t can be interpreted as the latent (unobserved) state in a State-Space Model.

The state transition equation is given by:

$$x_t = c + \phi x_{t-1} + w_t, \quad \text{where } w_t \sim \mathcal{N}(0, Q) \quad (147)$$

Here, c is a constant, ϕ is the autoregressive coefficient, and w_t is the process noise, which is assumed to be Gaussian with variance Q . This equation precisely satisfies our first requirement: it is linear, and the noise is normally distributed. This is the core equation that would be used in the prediction step of a Kalman filter.

If we assume we can observe the transformed ratio with some measurement error, the observation equation would be:

$$y_t^{\text{transformed}} = x_t + v_t, \quad \text{where } v_t \sim \mathcal{N}(0, R) \quad (148)$$

where $y_t^{\text{transformed}} = \text{logit}(r_t^{\text{observed}})$. The combination of these two equations forms a standard Linear Gaussian State-Space Model (LGSSM), which can be optimally solved by the Kalman filter.

C.2.3 Step 3: The Inverse Transformation (Logistic Function)

After running the Kalman filter, we obtain filtered estimates or forecasts for the latent state x_t (e.g., the posterior mean $\mathbb{E}[x_t|\text{data}]$). To get our final result for the ratio r_t , we must apply the inverse of the logit function, which is the logistic (sigmoid) function.

$$r_t = \text{logit}^{-1}(x_t) = \frac{1}{1 + e^{-x_t}} \quad (149)$$

This function takes any real number x_t and maps it back to a value strictly between 0 and 1, thus satisfying our second requirement.

C.3 Summary of the Workflow

The complete workflow for modeling the ratio r_t is as follows:

1. **Data Pre-processing:** Transform the entire time series of observed ratios $\{r_t^{\text{obs}}\}$ into the log-odds space using the logit function (Equation 146) to get $\{y_t^{\text{transformed}}\}$.
2. **State-Space Modeling:** Define a linear Gaussian state-space model where the state equation describes the AR(1) dynamics of the latent log-odds x_t (Equation 147), and the observation equation links x_t to the transformed observations $y_t^{\text{transformed}}$.
3. **Filtering and Forecasting:** Apply the Kalman filter and related algorithms to the transformed data to obtain filtered estimates, smoothed estimates, and forecasts for the latent state x_t . These results (e.g., $\hat{x}_{t|t}, \hat{x}_{t+k|t}$) will be in the form of means and variances of Gaussian distributions.
4. **Back-transformation:** Take the estimated or forecasted mean of the state, $\hat{x}_t$, and apply the logistic function (Equation 149) to transform it back into the original $(0, 1)$ space to get the final estimate for the ratio, $\hat{r}_t$.

This procedure ensures that all calculations are performed within a valid linear Gaussian framework, while guaranteeing that the final results for the ratio are always constrained within their meaningful $(0, 1)$ bounds.

D Stationary Distributions for AR(1) Processes

D.1 AR(1) Process with Constant Term

Consider the AR(1) process

$$x_t = c + \phi x_{t-1} + \epsilon_t,$$

where $|\phi| < 1$ for stationarity and $\epsilon_t \sim N(0, \sigma_\epsilon^2)$ are i.i.d. normal random variables.

D.1.1 Wold Representation

The stationary solution for x_t is

$$x_t = \frac{c}{1 - \phi} + \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}.$$

The infinite sum converges because $|\phi| < 1$, so the weights ϕ^j decay geometrically.

D.1.2 Normality of x_t

Each ϵ_{t-j} is normal and independent. A linear combination of independent normal random variables is also normal; therefore x_t is normally distributed.

D.1.3 Mean via Wold Representation

Taking expectations,

$$\mathbb{E}[x_t] = \mathbb{E} \left[\frac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j} \right] = \frac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j \mathbb{E}[\epsilon_{t-j}].$$

Since $\mathbb{E}[\epsilon_{t-j}] = 0$ for all j , we obtain

$$\mathbb{E}[x_t] = \frac{c}{1-\phi}.$$

D.1.4 Variance via Wold Representation

For the variance,

$$\text{Var}(x_t) = \text{Var} \left(\sum_{j=0}^{\infty} \phi^j \epsilon_{t-j} \right).$$

Because the ϵ_{t-j} are independent,

$$\text{Var}(x_t) = \sum_{j=0}^{\infty} (\phi^j)^2 \text{Var}(\epsilon_{t-j}) = \sum_{j=0}^{\infty} \phi^{2j} \sigma_{\epsilon}^2 = \sigma_{\epsilon}^2 \sum_{j=0}^{\infty} \phi^{2j}.$$

The sum is geometric,

$$\sum_{j=0}^{\infty} \phi^{2j} = \frac{1}{1-\phi^2},$$

so

$$\text{Var}(x_t) = \frac{\sigma_{\epsilon}^2}{1-\phi^2}.$$

D.2 AR(1) with Exogenous AR(1) Input

Now consider the bivariate system

$$\begin{aligned} x_t &= c + \phi x_{t-1} + \epsilon_t, \\ y_t &= a + \rho_y y_{t-1} + \rho_x x_{t-1} + v_t, \end{aligned}$$

where $|\phi| < 1$, $|\rho_y| < 1$ for stationarity, $\epsilon_t \sim N(0, \sigma_{\epsilon}^2)$ and $v_t \sim N(0, \sigma_v^2)$ are i.i.d. and mutually independent.

D.2.1 Normality of y_t

Both x_t and y_t are linear functions of normal random variables, hence y_t is normally distributed.

D.2.2 Mean of y_t

Taking expectations in the y_t equation,

$$\mathbb{E}[y_t] = a + \rho_y \mathbb{E}[y_{t-1}] + \rho_x \mathbb{E}[x_{t-1}] + \mathbb{E}[v_t].$$

In stationarity $\mathbb{E}[y_t] = \mathbb{E}[y_{t-1}] = \mu_y$ and $\mathbb{E}[v_t] = 0$, so

$$\mu_y = a + \rho_y \mu_y + \rho_x \mu_x.$$

From the AR(1) for x_t we have

$$\mu_x = \frac{c}{1 - \phi}.$$

Thus

$$\mu_y(1 - \rho_y) = a + \rho_x \frac{c}{1 - \phi},$$

and therefore

$$\mu_y = \frac{a(1 - \phi) + \rho_x c}{(1 - \rho_y)(1 - \phi)}.$$

D.2.3 Variance and Covariance

Denote $\text{Var}(x_t) = \text{Var}(x)$ and $\text{Var}(y_t) = \text{Var}(y)$, $\text{Cov}(y_t, x_t) = \text{Cov}(y, x)$ under stationarity. The variance of y_t satisfies

$$\text{Var}(y_t) = \rho_y^2 \text{Var}(y_{t-1}) + \rho_x^2 \text{Var}(x_{t-1}) + 2\rho_x \rho_y \text{Cov}(y_{t-1}, x_{t-1}) + \sigma_v^2.$$

The covariance between y_t and x_t is

$$\text{Cov}(y_t, x_t) = \rho_y \phi \text{Cov}(y_{t-1}, x_{t-1}) + \rho_x \phi \text{Var}(x_{t-1}).$$

Under stationarity this becomes

$$\begin{aligned} \text{Var}(y) &= \rho_y^2 \text{Var}(y) + \rho_x^2 \text{Var}(x) + 2\rho_x \rho_y \text{Cov}(y, x) + \sigma_v^2, \\ \text{Cov}(y, x) &= \rho_y \phi \text{Cov}(y, x) + \rho_x \phi \text{Var}(x). \end{aligned}$$

Solving these equations yields the closed forms

$$\text{Cov}(y, x) = \frac{\rho_x \phi \text{Var}(x)}{1 - \rho_y \phi},$$

and

$$\text{Var}(y) = \frac{\rho_x^2 \text{Var}(x) + 2\rho_x \rho_y \text{Cov}(y, x) + \sigma_v^2}{1 - \rho_y^2}.$$

These formulas describe the stationary joint normal distribution of (x_t, y_t) , summarizing the long-run effects of shocks to both processes without explicitly writing the corresponding infinite-order moving-average representation.